

Question ID 259ab62a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 259ab62a

If $6n = 12$, what is the value of $n + 4$?

ID: 259ab62a Answer

Correct Answer: 6

Rationale

The correct answer is **6**. Dividing both sides of the equation $6n = 12$ by **6** yields $n = 2$. Substituting **2** for n in the expression $n + 4$ yields $2 + 4$, or **6**.

Question Difficulty: Medium

Question ID ec9fddca

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: ec9fddca

Line t in the xy -plane has a slope of $-\frac{1}{3}$ and passes through the point $(9, 10)$. Which equation defines line t ?

- A. $y = 13x - \frac{1}{3}$
- B. $y = 9x + 10$
- C. $y = -\frac{x}{3} + 10$
- D. $y = -\frac{x}{3} + 13$

ID: ec9fddca Answer

Correct Answer: D

Rationale

Choice D is correct. The equation that defines line t in the xy -plane can be written in slope-intercept form $y = mx + b$, where m is the slope of line t and $(0, b)$ is its y -intercept. It's given that line t has a slope of $-\frac{1}{3}$. Therefore, $m = -\frac{1}{3}$. Substituting $-\frac{1}{3}$ for m in the equation $y = mx + b$ yields $y = -\frac{1}{3}x + b$, or $y = -\frac{x}{3} + b$. It's also given that line t passes through the point $(9, 10)$. Substituting 9 for x and 10 for y in the equation $y = -\frac{x}{3} + b$ yields $10 = -\frac{9}{3} + b$, or $10 = -3 + b$. Adding 3 to both sides of this equation yields $13 = b$. Substituting 13 for b in the equation $y = -\frac{x}{3} + b$ yields $y = -\frac{x}{3} + 13$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This equation defines a line that has a slope of 9 , not $-\frac{1}{3}$, and passes through the point $(0, 10)$, not $(9, 10)$.

Choice C is incorrect. This equation defines a line that passes through the point $(0, 10)$, not $(9, 10)$.

Question Difficulty: Medium

Question ID ec8fc70e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: ec8fc70e

$y < -4x + 4$

Which point (x, y) is a solution to the given inequality in the xy -plane?

- A. $(-4, 0)$
- B. $(0, 5)$
- C. $(2, 1)$
- D. $(2, -1)$

ID: ec8fc70e Answer

Correct Answer: A

Rationale

Choice A is correct. For a point (x, y) to be a solution to the given inequality in the xy -plane, the value of the point's y -coordinate must be less than the value of $-4x + 4$, where x is the value of the x -coordinate of the point. This is true of the point $(-4, 0)$ because $0 < -4(-4) + 4$, or $0 < 20$. Therefore, the point $(-4, 0)$ is a solution to the given inequality.

Choice B is incorrect. This point isn't a solution to the given inequality because the point's y -coordinate is greater than the value of $-4x + 4$ for the point's x -coordinate.

Choice C is incorrect. This point isn't a solution to the given inequality because the point's y -coordinate is greater than the value of $-4x + 4$ for the point's x -coordinate.

Choice D is incorrect. This point isn't a solution to the given inequality because the point's y -coordinate is greater than the value of $-4x + 4$ for the point's x -coordinate.

Question Difficulty: Medium

Question ID 337be703

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 337be703

A geologist needs to collect at least **67** samples of lava from a volcano. If the geologist has already collected **63** samples from the volcano, what is the minimum number of additional samples the geologist needs to collect?

- A. **130**
- B. **63**
- C. **4**
- D. **0**

ID: 337be703 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the geologist has already collected **63** samples from the volcano. Let x represent the number of additional samples the geologist needs to collect. After collecting x additional samples, the geologist will have collected a total of $63 + x$ samples. It's given that the geologist needs to collect at least **67** samples. Therefore, $63 + x \geq 67$. Subtracting **63** from each side of this inequality yields the inequality $x \geq 4$. Thus, the geologist needs to collect a minimum of **4** additional samples.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of samples the geologist has already collected, rather than the minimum number of additional samples the geologist needs to collect.

Choice D is incorrect. If the geologist collects **0** additional samples, the geologist will have collected a total of **63** samples, which is less than **67** samples.

Question Difficulty: Easy

Question ID afcbd26d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: afcbd26d

The length, y , of a white whale was **162 centimeters (cm)** when it was born and increased an average of **4.8 cm** per month for the first **12** months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in **cm**, of the whale?

- A. $y = 162x$
- B. $y = 162x + 162$
- C. $y = 4.8x + 4.8$
- D. $y = 4.8x + 162$

ID: afcbd26d Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the length of the whale was **162 cm** when it was born and that its length increased an average of **4.8 cm** per month for the first **12** months after it was born. Since x represents the number of months after the whale was born, the total increase in the whale's length, in **cm**, is **4.8** times x , or **4.8x**. The length of the whale y , in **cm**, can be found by adding the whale's length at birth, **162 cm**, to the total increase in length, **4.8x cm**. Therefore, the equation that best represents this situation is $y = 4.8x + 162$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 09dd5dc0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 09dd5dc0

The triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. If a triangle has side lengths of **6** and **12**, which inequality represents the possible lengths, x , of the third side of the triangle?

- A. $x < 18$
- B. $x > 18$
- C. $6 < x < 18$
- D. $x < 6$ or $x > 18$

ID: 09dd5dc0 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that a triangle has side lengths of **6** and **12**, and x represents the length of the third side of the triangle. It's also given that the triangle inequality theorem states that the sum of any two sides of a triangle must be greater than the length of the third side. Therefore, the inequalities $6 + x > 12$, $6 + 12 > x$, and $12 + x > 6$ represent all possible values of x . Subtracting **6** from both sides of the inequality $6 + x > 12$ yields $x > 12 - 6$, or $x > 6$. Adding **6** and **12** in the inequality $6 + 12 > x$ yields $18 > x$, or $x < 18$. Subtracting **12** from both sides of the inequality $12 + x > 6$ yields $x > 6 - 12$, or $x > -6$. Since all x -values that satisfy the inequality $x > 6$ also satisfy the inequality $x > -6$, it follows that the inequalities $x > 6$ and $x < 18$ represent the possible values of x . Therefore, the inequality $6 < x < 18$ represents the possible lengths, x , of the third side of the triangle.

Choice A is incorrect. This inequality gives the upper bound for x but does not include its lower bound.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f3e46b78

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: f3e46b78

$y = 70x + 8$

Which table gives three values of x and their corresponding values of y for the given equation?

- A.

x	y
0	8
2	148
4	288
- B.

x	y
0	70
2	78
4	86
- C.

x	y
0	70
2	140
4	280
- D.

x	y
0	8
2	132
4	272

ID: f3e46b78 Answer

Correct Answer: A

Rationale

Choice A is correct. Each of the given choices gives three values of x : **0**, **2**, and **4**. Substituting **0** for x in the given equation yields $y = 70(0) + 8$, or $y = 8$. Therefore, when $x = 0$, the corresponding value of y for the given equation is **8**. Substituting **2** for x in the given equation yields $y = 70(2) + 8$, or $y = 148$. Therefore, when $x = 2$, the corresponding value of y for the given equation is **148**. Substituting **4** for x in the given equation yields $y = 70(4) + 8$, or $y = 288$. Therefore, when $x = 4$, the corresponding value of y for the given equation is **288**. Thus, if the three values of x are **0**, **2**, and **4**, then their corresponding values of y are **8**, **148**, and **288**, respectively, for the given equation.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x + 70$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID fe28a56a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: fe28a56a

$$f(x) = 4x + b$$

For the linear function f , b is a constant and $f(7) = 28$. What is the value of b ?

- A. 0
- B. 1
- C. 4
- D. 7

ID: fe28a56a Answer

Correct Answer: A

Rationale

Choice A is correct. For the linear function f , it's given that $f(7) = 28$. Substituting 7 for x and 28 for $f(x)$ in the given function yields $28 = 4(7) + b$, or $28 = 28 + b$. Subtracting 28 from each side of this equation yields $0 = b$. Therefore, the value of b is 0.

Choice B is incorrect. Substituting 1 for b in the given function yields $f(x) = 4x + 1$. For this function, when the value of x is 7, the value of $f(x)$ is 29, not 28.

Choice C is incorrect. Substituting 4 for b in the given function yields $f(x) = 4x + 4$. For this function, when the value of x is 7, the value of $f(x)$ is 32, not 28.

Choice D is incorrect. Substituting 7 for b in the given function yields $f(x) = 4x + 7$. For this function, when the value of x is 7, the value of $f(x)$ is 35, not 28.

Question Difficulty: Medium

Question ID e3672617

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: e3672617

The equation $46 = 2a + 2b$ gives the relationship between the side lengths a and b of a certain parallelogram. If $a = 9$, what is the value of b ?

ID: e3672617 Answer

Correct Answer: 14

Rationale

The correct answer is **14**. It's given that the equation $46 = 2a + 2b$ gives the relationship between the side lengths a and b of a certain parallelogram. Substituting **9** for a in the given equation yields $46 = 2(9) + 2b$, or $46 = 18 + 2b$. Subtracting **18** from both sides of this equation yields $28 = 2b$. Dividing both sides of this equation by **2** yields $14 = b$. Therefore, if $a = 9$, the value of b is **14**.

Question Difficulty: Easy

Question ID 13b10306

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 13b10306

$$d = 16 - \frac{x}{30}$$

The equation shown gives the estimated amount of diesel d , in gallons, that remains in the gas tank of a truck after being driven x miles, where $0 \leq x \leq 480$. What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when $x = 300$?

- A. 0
- B. 6
- C. 14
- D. 16

ID: 13b10306 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $d = 16 - \frac{x}{30}$ gives the estimated amount of diesel d , in gallons, that remains in the gas tank of the truck after being driven x miles. Substituting 300 for x in the given equation yields $d = 16 - \frac{300}{30}$, which is equivalent to $d = 16 - 10$, or $d = 6$. Therefore, the estimated amount of diesel that remains in the gas tank of the truck when $x = 300$ is 6 gallons.

Choice A is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 480$, not when $x = 300$.

Choice C is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 60$, not when $x = 300$.

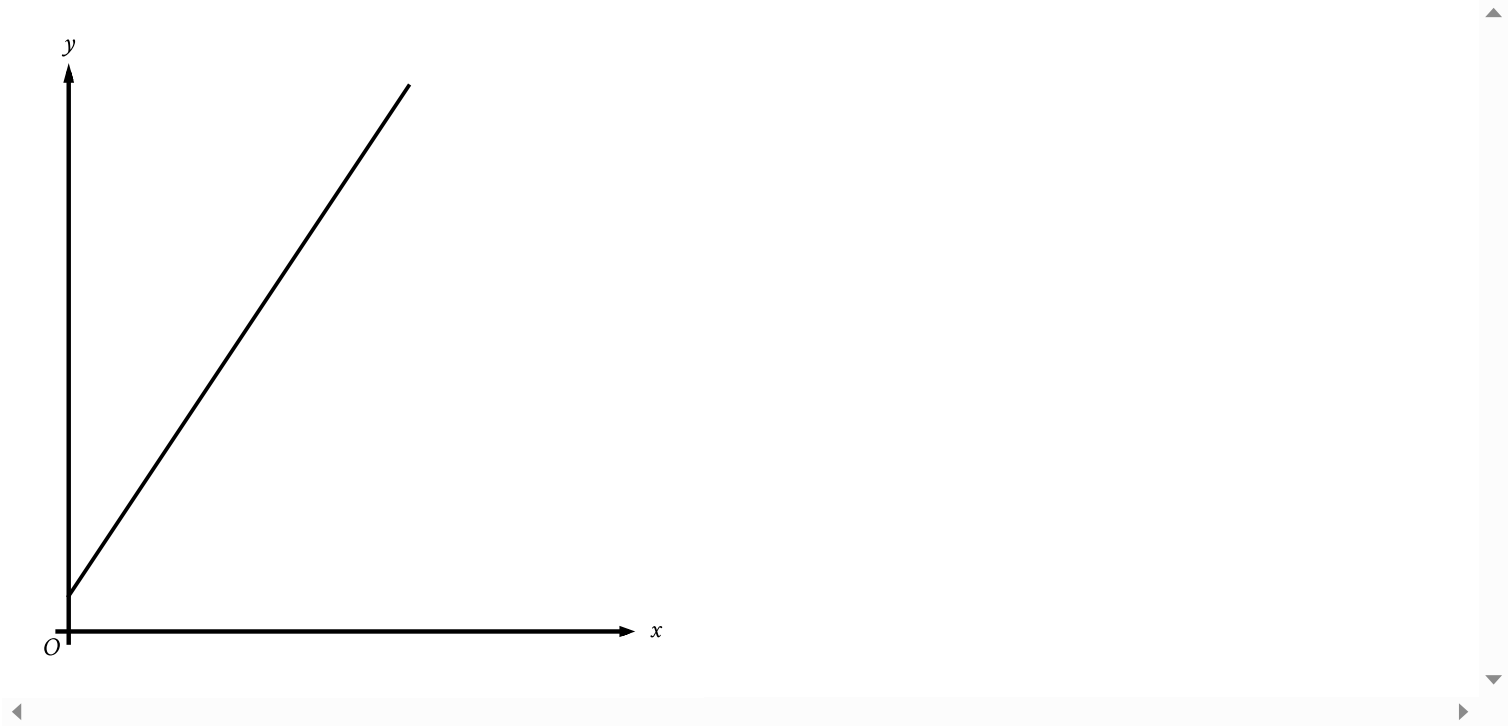
Choice D is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 0$, not when $x = 300$.

Question Difficulty: Easy

Question ID 85c1496b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 85c1496b



The graph represents the total charge, in dollars, by an electrician for x hours of work. The electrician charges a onetime fee plus an hourly rate. What is the best interpretation of the slope of the graph?

- A. The electrician’s hourly rate
- B. The electrician’s onetime fee
- C. The maximum amount that the electrician charges
- D. The total amount that the electrician charges

ID: 85c1496b Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the electrician charges a onetime fee plus an hourly rate. It’s also given that the graph represents the total charge, in dollars, for x hours of work. This graph shows a linear relationship in the xy -plane. Thus, the total charge y , in dollars, for x hours of work can be represented as $y = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of the equation in the xy -plane. Since the given graph represents the total charge, in dollars, by an electrician for x hours of work, it follows that its slope is m , or the electrician’s hourly rate.

Choice B is incorrect. The electrician’s onetime fee is represented by the y -coordinate of the y -intercept, not the slope, of the graph.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID 8f4965b5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 8f4965b5

A certain township consists of a **5**-hectare industrial park and a **24**-hectare neighborhood. The total number of trees in the township is **4,529**. The equation $5x + 24y = 4,529$ represents this situation. Which of the following is the best interpretation of x in this context?

- A. The average number of trees per hectare in the industrial park
- B. The average number of trees per hectare in the neighborhood
- C. The total number of trees in the industrial park
- D. The total number of trees in the neighborhood

ID: 8f4965b5 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a certain township consists of a **5**-hectare industrial park and a **24**-hectare neighborhood and that the total number of trees in the township is **4,529**. It's also given that the equation $5x + 24y = 4,529$ represents this situation. Since the total number of trees for a given area can be determined by taking the size of the area, in hectares, times the average number of trees per hectare, the best interpretation of $5x$ is the number of trees in the industrial park and the best interpretation of $24y$ is the number of trees in the neighborhood. Since **5** is the size of the industrial park, in hectares, the best interpretation of x is the average number of trees per hectare in the industrial park.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID d9069ff3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: d9069ff3

$$f(x) = 14 + 4x$$

The function f represents the total cost, in dollars, of attending an arcade when x games are played. How many games can be played for a total cost of \$58?

ID: d9069ff3 Answer

Correct Answer: 11

Rationale

The correct answer is **11**. It's given that the function $f(x) = 14 + 4x$ represents the total cost, in dollars, of attending an arcade when x games are played. Substituting **58** for $f(x)$ in the given equation yields $58 = 14 + 4x$. Subtracting **14** from each side of this equation yields $44 = 4x$. Dividing each side of this equation by **4** yields $11 = x$. Therefore, **11** games can be played for a total cost of \$58.

Question Difficulty: Medium

Question ID 9eaba458

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 9eaba458

If $2x + 3 = 9$, what is the value of $6x - 1$?

ID: 9eaba458 Answer

Correct Answer: 17

Rationale

The correct answer is **17**. It's given that $2x + 3 = 9$. Multiplying each side of this equation by **3** yields $3(2x + 3) = 3(9)$, or $6x + 9 = 27$. Subtracting **10** from each side of this equation yields $6x + 9 - 10 = 27 - 10$, or $6x - 1 = 17$. Therefore, the value of $6x - 1$ is **17**.

Question Difficulty: Easy

Question ID 37bbfa3d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 37bbfa3d

One gallon of stain will cover **170** square feet of a surface. A yard has a total fence area of w square feet. Which equation represents the total amount of stain S , in gallons, needed to stain the fence in this yard twice?

- A. $S = \frac{w}{170}$
- B. $S = 170w$
- C. $S = 340w$
- D. $S = \frac{w}{85}$

ID: 37bbfa3d Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that w represents the total fence area, in square feet. Since the fence will be stained twice, the amount of stain, in gallons, will need to cover $2w$ square feet. It's also given that one gallon of stain will cover **170** square feet. Dividing the total area, in square feet, of the surface to be stained by the number of square feet covered by one gallon of stain gives the number of gallons of stain that will be needed. Dividing $2w$ by **170** yields $\frac{2w}{170}$, or $\frac{w}{85}$. Therefore, the equation that represents the total amount of stain S , in gallons, needed to stain the fence of the yard twice is $S = \frac{w}{85}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 61bbd673

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 61bbd673

A model predicts that a certain animal weighed **241** pounds when it was born and that the animal gained **3** pounds per day in its first year of life. This model is defined by an equation in the form $f(x) = a + bx$, where $f(x)$ is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. What is the value of a ?

ID: 61bbd673 Answer

Correct Answer: 241

Rationale

The correct answer is **241**. For a certain animal, it's given that a model predicts the animal weighed **241** pounds when it was born and gained **3** pounds per day in its first year of life. It's also given that this model is defined by an equation in the form $f(x) = a + bx$, where $f(x)$ is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. It follows that a represents the predicted weight, in pounds, of the animal when it was born and b represents the predicted rate of weight gain, in pounds per day, in its first year of life. Thus, the value of a is **241**.

Question Difficulty: Medium

Question ID 365b9585

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 365b9585

The minimum value of x is **12** less than **6** times another number n . Which inequality shows the possible values of x ?

- A. $x \leq 6n - 12$
- B. $x \geq 6n - 12$
- C. $x \leq 12 - 6n$
- D. $x \geq 12 - 6n$

ID: 365b9585 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the minimum value of x is **12** less than **6** times another number n . Therefore, the possible values of x are all greater than or equal to the value of **12** less than **6** times n . The value of **6** times n is given by the expression **$6n$** . The value of **12** less than **$6n$** is given by the expression **$6n - 12$** . Therefore, the possible values of x are all greater than or equal to **$6n - 12$** . This can be shown by the inequality **$x \geq 6n - 12$** .

Choice A is incorrect. This inequality shows the possible values of x if the maximum, not the minimum, value of x is **12** less than **6** times n .

Choice C is incorrect. This inequality shows the possible values of x if the maximum, not the minimum, value of x is **6** times n less than **12**, not **12** less than **6** times n .

Choice D is incorrect. This inequality shows the possible values of x if the minimum value of x is **6** times n less than **12**, not **12** less than **6** times n .

Question Difficulty: Hard

Question ID 3617a434

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 3617a434

$5x + 7y = 1$

$ax + by = 1$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

- A. $10x + 7y = 1$
 $ax - 2by = 1$
- B. $10x + 7y = 1$
 $ax + 2by = 1$
- C. $10x + 7y = 1$
 $2ax + by = 1$
- D. $5x - 7y = 1$
 $ax + by = 1$

ID: 3617a434 Answer

Correct Answer: B

Rationale

Choice B is correct. Two lines are perpendicular if their slopes are negative reciprocals, meaning that the slope of the first line is equal to -1 divided by the slope of the second line. Each equation in the given pair of equations can be written in slope-intercept form, $y = mx + b$, where m is the slope of the graph of the equation in the xy -plane and $(0, b)$ is the y -intercept. For the first equation, $5x + 7y = 1$, subtracting $5x$ from both sides gives $7y = -5x + 1$, and dividing both sides of this equation by 7 gives $y = -\frac{5}{7}x + \frac{1}{7}$. Therefore, the slope of the graph of this equation is $-\frac{5}{7}$. For the second equation, $ax + by = 1$, subtracting ax from both sides gives $by = -ax + 1$, and dividing both sides of this equation by b gives $y = -\frac{a}{b}x + \frac{1}{b}$. Therefore, the slope of the graph of this equation is $-\frac{a}{b}$. Since the graph of the given pair of equations is a pair of perpendicular lines, the slope of the graph of the second equation, $-\frac{a}{b}$, must be the negative reciprocal of the slope of the graph of the first equation, $-\frac{5}{7}$. The negative reciprocal of $-\frac{5}{7}$ is $\frac{-1}{(-\frac{5}{7})}$, or $\frac{7}{5}$. Therefore, $-\frac{a}{b} = \frac{7}{5}$, or $\frac{a}{b} = -\frac{7}{5}$. Similarly, rewriting the equations in choice B in slope-intercept form yields $y = -\frac{10}{7}x + \frac{1}{7}$ and $y = -\frac{a}{2b}x + \frac{1}{2b}$. It follows that the slope of the graph of the first equation in choice B is $-\frac{10}{7}$ and the slope of the graph of the second equation in choice B is $-\frac{a}{2b}$. Since $\frac{a}{b} = -\frac{7}{5}$, $-\frac{a}{2b}$ is equal to $-\left(\frac{1}{2}\right)\left(-\frac{7}{5}\right)$, or $\frac{7}{10}$. Since $\frac{7}{10}$ is the negative reciprocal of $-\frac{10}{7}$, the pair of equations in choice B represents a pair of perpendicular lines.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID c55794d8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: c55794d8

A sample of a certain alloy has a total mass of **50.0** grams and is **50.0%** silicon by mass. The sample was created by combining two pieces of different alloys. The first piece was **30.0%** silicon by mass and the second piece was **80.0%** silicon by mass. What was the mass, in grams, of the silicon in the second piece?

- A. **9.0**
- B. **16.0**
- C. **20.0**
- D. **30.0**

ID: c55794d8 Answer

Correct Answer: B

Rationale

Choice B is correct. Let x represent the total mass, in grams, of the first piece, and let y represent the total mass, in grams, of the second piece. It's given that the sample has a total mass of **50.0** grams. Therefore, the equation $x + y = 50.0$ represents this situation. It's also given that the sample is **50.0%** silicon by mass. Therefore, the total mass of the silicon in the sample is **0.500(50.0)**, or **25.0**, grams. It's also given that the first piece was **30.0%** silicon by mass and the second piece was **80.0%** silicon by mass. Therefore, the masses, in grams, of the silicon in the first and second pieces can be represented by the expressions **0.300x** and **0.800y**, respectively. Since the sample was created by combining the first and second pieces, and the total mass of the silicon in the sample is **25.0** grams, the equation **0.300x + 0.800y = 25.0** represents this situation. Subtracting y from both sides of the equation $x + y = 50.0$ yields $x = 50.0 - y$. Substituting $50.0 - y$ for x in the equation **0.300x + 0.800y = 25.0** yields **0.300(50.0 - y) + 0.800y = 25.0**. Distributing **0.300** on the left-hand side of this equation yields **15.0 - 0.300y + 0.800y = 25.0**. Combining like terms on the left-hand side of this equation yields **15.0 + 0.500y = 25.0**. Subtracting **15.0** from both sides of this equation yields **0.500y = 10.0**. Dividing both sides of this equation by **0.500** yields $y = 20.0$. Substituting **20.0** for y in the expression representing the mass, in grams, of the silicon in the second piece, **0.800y**, yields **0.800(20.0)**, or **16.0**. Therefore, the mass, in grams, of the silicon in the second piece is **16.0**.

Choice A is incorrect. This is the mass, in grams, of the silicon in the first piece, not the second piece.

Choice C is incorrect. This is the total mass, in grams, of the second piece, not the mass, in grams, of the silicon in the second piece.

Choice D is incorrect. This is the total mass, in grams, of the first piece, not the mass, in grams, of the silicon in the second piece.

Question Difficulty: Hard

Question ID 26c46450

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 26c46450

A line in the xy -plane has a slope of $\frac{1}{9}$ and passes through the point $(0, 14)$. Which equation represents this line?

- A. $y = -\frac{1}{9}x - 14$
- B. $y = -\frac{1}{9}x + 14$
- C. $y = \frac{1}{9}x - 14$
- D. $y = \frac{1}{9}x + 14$

ID: 26c46450 Answer

Correct Answer: D

Rationale

Choice D is correct. The equation of a line in the xy -plane can be written as $y = mx + b$, where m represents the slope of the line and $(0, b)$ represents the y -intercept of the line. It's given that the slope of the line is $\frac{1}{9}$. It follows that $m = \frac{1}{9}$. It's also given that the line passes through the point $(0, 14)$. It follows that $b = 14$. Substituting $\frac{1}{9}$ for m and 14 for b in $y = mx + b$ yields $y = \frac{1}{9}x + 14$. Thus, the equation $y = \frac{1}{9}x + 14$ represents this line.

Choice A is incorrect. This equation represents a line with a slope of $-\frac{1}{9}$ and a y -intercept of $(0, -14)$.

Choice B is incorrect. This equation represents a line with a slope of $-\frac{1}{9}$ and a y -intercept of $(0, 14)$.

Choice C is incorrect. This equation represents a line with a slope of $\frac{1}{9}$ and a y -intercept of $(0, -14)$.

Question Difficulty: Medium

Question ID 3b0116a2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 3b0116a2

A machine makes large boxes or small boxes, one at a time, for a total of **700** minutes each day. It takes the machine **10** minutes to make a large box or **5** minutes to make a small box. Which equation represents the possible number of large boxes, x , and small boxes, y , the machine can make each day?

- A. $5x + 10y = 700$
- B. $10x + 5y = 700$
- C. $(x + y)(10 + 5) = 700$
- D. $(10 + x)(5 + y) = 700$

ID: 3b0116a2 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that it takes the machine **10** minutes to make a large box. It's also given that x represents the possible number of large boxes the machine can make each day. Multiplying **10** by x gives $10x$, which represents the amount of time spent making large boxes. It's given that it takes the machine **5** minutes to make a small box. It's also given that y represents the possible number of small boxes the machine can make each day. Multiplying **5** by y gives $5y$, which represents the amount of time spent making small boxes. Combining the amount of time spent making x large boxes and y small boxes yields $10x + 5y$. It's given that the machine makes boxes for a total of **700** minutes each day. Therefore $10x + 5y = 700$ represents the possible number of large boxes, x , and small boxes, y , the machine can make each day.

Choice A is incorrect and may result from associating the time of **10** minutes with small, rather than large, boxes and the time of **5** minutes with large, rather than small, boxes.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 69a5295b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 69a5295b

$2x - y > 883$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
440	0
441	-2
442	-4

B.

x	y
440	0
442	-2
441	-4

C.

x	y
442	0
440	-2
441	-4

D.

x	y
442	0
441	-2
440	-4

ID: 69a5295b Answer

Correct Answer: D

Rationale

Choice D is correct. All the tables in the choices have the same three values of x , **440**, **441**, and **442**, so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting **440** for x in the given inequality yields $2(440) - y > 883$, or $880 - y > 883$. Subtracting **880** from both sides of this inequality yields $-y > 3$. Dividing both sides of this inequality by -1 yields $y < -3$. Therefore, when $x = 440$, the corresponding value of y must be less than -3 . Substituting **441** for x in the given inequality yields $2(441) - y > 883$, or $882 - y > 883$. Subtracting

882 from both sides of this inequality yields $-y > 1$. Dividing both sides of this inequality by -1 yields $y < -1$. Therefore, when $x = 441$, the corresponding value of y must be less than -1 . Substituting 442 for x in the given inequality yields $2(442) - y > 883$, or $884 - y > 883$. Subtracting 884 from both sides of this inequality yields $-y > -1$. Dividing both sides of this inequality by -1 yields $y < 1$. Therefore, when $x = 442$, the corresponding value of y must be less than 1. For the table in choice D, when $x = 440$, the corresponding value of y is -4 , which is less than -3 ; when $x = 441$, the corresponding value of y is -2 , which is less than -1 ; when $x = 442$, the corresponding value of y is 0, which is less than 1. Therefore, the table in choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. When $x = 440$, the corresponding value of y in this table is 0, which isn't less than -3 .

Choice B is incorrect. When $x = 440$, the corresponding value of y in this table is 0, which isn't less than -3 .

Choice C is incorrect. When $x = 440$, the corresponding value of y in this table is -2 , which isn't less than -3 .

Question Difficulty: Hard

Question ID 5f1897a1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 5f1897a1

If the graph of $27x + 33y = 297$ is shifted down 5 units in the xy -plane, what is the y -intercept of the resulting graph?

- A. (0, 4)
- B. (0, 6)
- C. (0, 14)
- D. (0, 28)

ID: 5f1897a1 Answer

Correct Answer: A

Rationale

Choice A is correct. When the graph of an equation in the form $Ax + By = C$, where A , B , and C are constants, is shifted down k units in the xy -plane, the resulting graph can be represented by the equation $Ax + B(y + k) = C$. It's given that the graph of $27x + 33y = 297$ is shifted down 5 units in the xy -plane. Therefore, the resulting graph can be represented by the equation $27x + 33(y + 5) = 297$, or $27x + 33y + 165 = 297$. Subtracting 165 from both sides of this equation yields $27x + 33y = 132$. The y -intercept of the graph of an equation in the xy -plane is the point where the line intersects the y -axis, represented by the point $(0, y)$. Substituting 0 for x in the equation $27x + 33y = 132$ yields $27(0) + 33y = 132$, or $33y = 132$. Dividing both sides of this equation by 33 yields $y = 4$. Therefore, if the graph of $27x + 33y = 297$ is shifted down 5 units, the y -intercept of the resulting graph is $(0, 4)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the y -intercept of the graph of $27x + 33y = 297$ shifted up, not down, 5 units.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID da845af8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: da845af8

$7(2x - 3) = 63$

Which equation has the same solution as the given equation?

- A. $2x - 3 = 9$
- B. $2x - 3 = 56$
- C. $2x - 21 = 63$
- D. $2x - 21 = 70$

ID: da845af8 Answer

Correct Answer: A

Rationale

Choice A is correct. Dividing each side of the given equation by 7 yields $\frac{7(2x-3)}{7} = \frac{63}{7}$, or $2x - 3 = 9$. Therefore, the equation $2x - 3 = 9$ is equivalent to the given equation and has the same solution.

Choice B is incorrect. This equation is equivalent to $7(2x - 3) = 392$, not $7(2x - 3) = 63$.

Choice C is incorrect. Distributing 7 on the left-hand side of the given equation yields $14x - 21 = 63$, not $2x - 21 = 63$.

Choice D is incorrect. Distributing 7 on the left-hand side of the given equation yields $14x - 21 = 63$, not $2x - 21 = 70$.

Question Difficulty: Medium

Question ID 19e7dacc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 19e7dacc

A company that provides whale-watching tours takes groups of **21** people at a time. The company’s revenue is **80** dollars per adult and **60** dollars per child. If the company’s revenue for one group consisting of adults and children was **1,440** dollars, how many people in the group were children?

- A. **3**
- B. **9**
- C. **12**
- D. **18**

ID: 19e7dacc Answer

Correct Answer: C

Rationale

Choice C is correct. Let x represent the number of children in a whale-watching tour group. Let y represent the number of adults in this group. Because it's given that **21** people are in a group and the group consists of adults and children, it must be true that $x + y = 21$. Since the company's revenue is **60** dollars per child, the total revenue from x children in this group was $60x$ dollars. Since the company's revenue is **80** dollars per adult, the total revenue from y adults in this group was $80y$ dollars. Because it's given that the total revenue for this group was **1,440** dollars, it must be true that $60x + 80y = 1,440$. The equations $x + y = 21$ and $60x + 80y = 1,440$ form a linear system of equations that can be solved to find the value of x , which represents the number of children in the group, using the elimination method. Multiplying both sides of the equation $x + y = 21$ by **80** yields $80x + 80y = 1,680$. Subtracting $60x + 80y = 1,440$ from $80x + 80y = 1,680$ yields $(80x + 80y) - (60x + 80y) = 1,680 - 1,440$, which is equivalent to $80x - 60x + 80y - 80y = 240$, or $20x = 240$. Dividing both sides of this equation by **20** yields $x = 12$. Therefore, **12** people in the group were children.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of adults in the group, not the number of children in the group.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8b5f8787

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 8b5f8787

x	y
-34	t
-17	$t + 23$
0	$t + 46$

For a linear relationship between x and y , the table gives three values of x and their corresponding values of y , where t is a constant. Which equation represents this relationship?

- A. $y = -2x + t + 23$
- B. $y = 2x + t + 23$
- C. $y = -\frac{23}{17}x + t + 46$
- D. $y = \frac{23}{17}x + t + 46$

ID: 8b5f8787 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that for a linear relationship between x and y , the table gives three values of x and their corresponding values of y . A linear equation can be written in the form $y = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of the equation in the xy -plane. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(-34, t)$ and $(-17, t + 23)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{(t + 23) - t}{(-17) - (-34)}$, which is equivalent to $m = \frac{t + 23 - t}{-17 + 34}$, or $m = \frac{23}{17}$. Based on the table, the graph of this relationship passes through the point $(0, t + 46)$. It follows that $b = t + 46$. Substituting $\frac{23}{17}$ for m and $t + 46$ for b in the equation $y = mx + b$ yields $y = \frac{23}{17}x + t + 46$. Therefore, the equation $y = \frac{23}{17}x + t + 46$ represents this relationship.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID d60b089f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: d60b089f

The function f is defined by $f(x) = 3x - 8$. What is the value of $f(7)$?

- A. 29
- B. 13
- C. -5
- D. -29

ID: d60b089f Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 3x - 8$. The value of $f(7)$ is the value of $f(x)$ when $x = 7$. Substituting 7 for x in the given equation yields $f(7) = 3(7) - 8$, which is equivalent to $f(7) = 21 - 8$, or $f(7) = 13$.

Choice A is incorrect. This is the value of $f(7)$ when $f(x) = 3x + 8$, rather than $f(x) = 3x - 8$.

Choice C is incorrect. This is the value of $f(1)$, rather than $f(7)$.

Choice D is incorrect. This is the value of $f(-7)$, rather than $f(7)$.

Question Difficulty: Easy

Question ID df59edf3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: df59edf3

A bus is traveling at a constant speed along a straight portion of road. The equation $d = 30t$ gives the distance d , in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be **2** seconds after passing the marker?

- A. **30**
- B. **32**
- C. **60**
- D. **90**

ID: df59edf3 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that t represents the number of seconds after the bus passes the marker. Substituting **2** for t in the given equation $d = 30t$ yields $d = 30(2)$, or $d = 60$. Therefore, the bus will be **60** feet from the marker **2** seconds after passing it.

Choice A is incorrect. This is the distance, in feet, the bus will be from the marker **1** second, not **2** seconds, after passing it.

Choice B is incorrect and may result from conceptual or calculation errors.

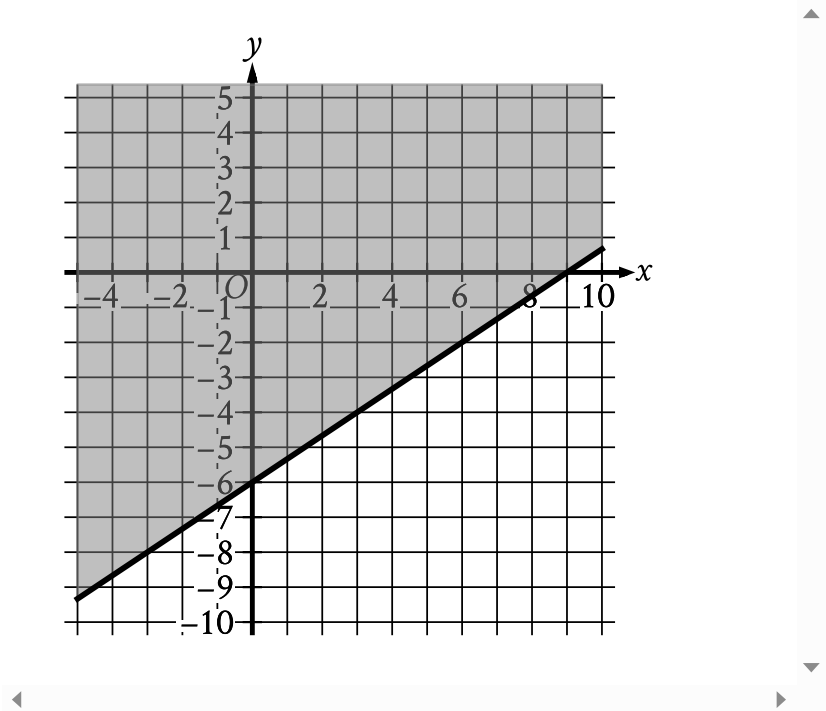
Choice D is incorrect. This is the distance, in feet, the bus will be from the marker **3** seconds, not **2** seconds, after passing it.

Question Difficulty: Easy

Question ID a0016bd6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: a0016bd6



The shaded region shown represents the solutions to which inequality?

- A. $y \geq \frac{2}{3}x - 6$
- B. $y \geq \frac{2}{3}x + 6$
- C. $y \geq \frac{2}{3}x - 9$
- D. $y \geq \frac{2}{3}x + 9$

ID: a0016bd6 Answer

Correct Answer: A

Rationale

Choice A is correct. The equation for the line representing the boundary of the shaded region can be written in slope-intercept form $y = mx + b$, where m is the slope and $(0, b)$ is the y-intercept of the line. For the graph shown, the boundary line passes through the points $(0, -6)$ and $(9, 0)$. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, -6)$ and $(9, 0)$ for (x_1, y_1) and (x_2, y_2) , respectively, in this equation yields $m = \frac{0 - (-6)}{9 - 0}$, which is equivalent to $m = \frac{6}{9}$, or $m = \frac{2}{3}$. Since the point $(0, -6)$ represents the y-intercept, it follows that $b = -6$. Substituting $\frac{2}{3}$ for m and -6 for b in the equation $y = mx + b$ yields $y = \frac{2}{3}x - 6$ as the equation of the boundary line. Since the shaded region represents all the points on and above this boundary line, it follows that the shaded region shown represents the solutions to the inequality $y \geq \frac{2}{3}x - 6$.

Choice B is incorrect. This inequality represents a region whose boundary line has a y-intercept of $(0, 6)$, not $(0, -6)$.

Choice C is incorrect. This inequality represents a region whose boundary line has a y-intercept of $(0, -9)$, not $(0, -6)$.

Choice D is incorrect. This inequality represents a region whose boundary line has a y -intercept of $(0, 9)$, not $(0, -6)$.

Question Difficulty: Easy

Question ID b781b8cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: b781b8cc

$$\begin{aligned} 4x - 9y &= 9y + 5 \\ hy &= 2 + 4x \end{aligned}$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h ?

- A. -9
- B. 0
- C. 9
- D. 18

ID: b781b8cc Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are distinct and parallel. The graphs of two lines in the xy -plane represented by equations in the form $Ax + By = C$, where A , B , and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation. The first equation in the given system can be written in the form $Ax + By = C$ by subtracting $9y$ from both sides of the equation to yield $4x - 18y = 5$. The second equation in the given system can be written in the form $Ax + By = C$ by subtracting $4x$ from both sides of the equation to yield $-4x + hy = 2$. The coefficient of x in this second equation, -4 , is -1 times the coefficient of x in the first equation, 4 . For the lines to be parallel, the coefficient of y in the second equation, h , must also be -1 times the coefficient of y in the first equation, -18 . Thus, $h = -1(-18)$, or $h = 18$. Therefore, if the given system has no solution, the value of h is 18 .

Choice A is incorrect. If the value of h is -9 , then the given system would have one solution, rather than no solution.

Choice B is incorrect. If the value of h is 0 , then the given system would have one solution, rather than no solution.

Choice C is incorrect. If the value of h is 9 , then the given system would have one solution, rather than no solution.

Question Difficulty: Hard

Question ID 9b145590

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 9b145590

An employee at a restaurant prepares sandwiches and salads. It takes the employee **1.5** minutes to prepare a sandwich and **1.9** minutes to prepare a salad. The employee spends a total of **46.1** minutes preparing x sandwiches and y salads. Which equation represents this situation?

- A. $1.9x + 1.5y = 46.1$
- B. $1.5x + 1.9y = 46.1$
- C. $x + y = 46.1$
- D. $30.7x + 24.3y = 46.1$

ID: 9b145590 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the employee takes **1.5** minutes to prepare a sandwich. Multiplying **1.5** by the number of sandwiches, x , yields $1.5x$, the amount of time the employee spends preparing x sandwiches. It's also given that the employee takes **1.9** minutes to prepare a salad. Multiplying **1.9** by the number of salads, y , yields $1.9y$, the amount of time the employee spends preparing y salads. It follows that the total amount of time, in minutes, the employee spends preparing x sandwiches and y salads is $1.5x + 1.9y$. It's given that the employee spends a total of **46.1** minutes preparing x sandwiches and y salads. Thus, the equation $1.5x + 1.9y = 46.1$ represents this situation.

Choice A is incorrect. This equation represents a situation where it takes the employee **1.9** minutes, rather than **1.5** minutes, to prepare a sandwich and **1.5** minutes, rather than **1.9** minutes, to prepare a salad.

Choice C is incorrect. This equation represents a situation where it takes the employee **1** minute, rather than **1.5** minutes, to prepare a sandwich and **1** minute, rather than **1.9** minutes, to prepare a salad.

Choice D is incorrect. This equation represents a situation where it takes the employee **30.7** minutes, rather than **1.5** minutes, to prepare a sandwich and **24.3** minutes, rather than **1.9** minutes, to prepare a salad.

Question Difficulty: Easy

Question ID d76c37db

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: d76c37db

For a training program, Juan rides his bike at an average rate of **5.7** minutes per mile. Which function *m* models the number of minutes it will take Juan to ride *x* miles at this rate?

- A. $m(x) = \frac{x}{5.7}$
- B. $m(x) = x + 5.7$
- C. $m(x) = x - 5.7$
- D. $m(x) = 5.7x$

ID: d76c37db Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that Juan rides his bike at an average rate of **5.7** minutes per mile. The number of minutes it will take Juan to ride *x* miles can be determined by multiplying his average rate by the number of miles, *x*, which yields **5.7x**. Therefore, the function $m(x) = 5.7x$ models the number of minutes it will take Juan to ride *x* miles.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID b1916547

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: b1916547

$y = -4x + 40$

Which table gives three values of x and their corresponding values of y for the given equation?

- A.

x	y
0	0
1	-4
2	-8
- B.

x	y
0	40
1	44
2	48
- C.

x	y
0	40
1	36
2	32
- D.

x	y
0	0
1	4
2	8

ID: b1916547 Answer

Correct Answer: C

Rationale

Choice C is correct. Each of the given choices gives three values of x : **0**, **1**, and **2**. Substituting **0** for x in the given equation yields $y = -4(0) + 40$, or $y = 40$. Therefore, when $x = 0$, the corresponding value of y for the given equation is **40**. Substituting **1** for x in the given equation yields $y = -4(1) + 40$, or $y = 36$. Therefore, when $x = 1$, the corresponding value of y for the given equation is **36**. Substituting **2** for x in the given equation yields $y = -4(2) + 40$, or $y = 32$. Therefore, when $x = 2$, the corresponding value of y for the given equation is **32**. Choice C gives three values of x , **0**, **1**, and **2**, and their corresponding values of y , **40**, **36**, and **32**, respectively, for the given equation.

- Choice A is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -4x$.
- Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x + 40$.
- Choice D is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = 4x$.
- Question Difficulty: Easy

Question ID 468de4be

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 468de4be

If $9x + 4 = 67$, what is the value of $90x + 40$?

- A. 7
- B. 70
- C. 130
- D. 670

ID: 468de4be Answer

Correct Answer: D

Rationale

Choice D is correct. Multiplying both sides of the given equation by 10 yields $10(9x + 4) = 10(67)$, or $90x + 40 = 670$. Therefore, the value of $90x + 40$ is 670.

Choice A is incorrect. This is the value of x , not $90x + 40$.

Choice B is incorrect and may result from conceptual or calculation errors.

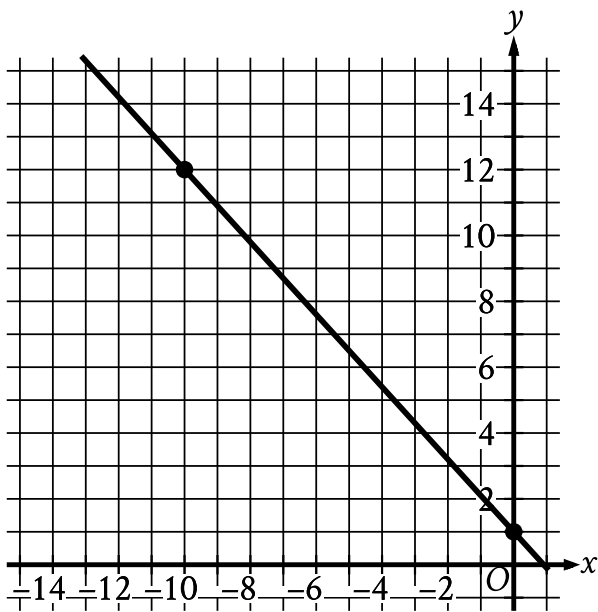
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID ac0b1093

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: ac0b1093



The graph of line g is shown in the xy -plane. Line k is defined by $165x + py = w$, where p and w are constants. If line k is graphed in this xy -plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of $p + w$?

ID: ac0b1093 Answer

Correct Answer: 300

Rationale

The correct answer is **300**. It's given that the graph of line g is shown in the xy -plane and that line k is defined by $165x + py = w$, where p and w are constants. It's also given that if line k is graphed in this xy -plane, the resulting system of two linear equations will have infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Therefore, line g and line k have the same slope and the same y -intercept. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph shows that line g passes through the points $(-10, 12)$ and $(0, 1)$. Substituting $(-10, 12)$ and $(0, 1)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{1 - 12}{0 - (-10)}$, or $m = -\frac{11}{10}$. An equation representing a linear relationship can be written in the form $y = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the line in the xy -plane. Since line g passes through the point $(0, 1)$, it follows that $b = 1$. Thus, the equation of line g is $y = -\frac{11}{10}x + 1$. For line k , the equation $165x + py = w$ can be rewritten in $y = mx + b$ form by isolating y . Subtracting $165x$ from both sides of this equation yields $py = -165x + w$. Dividing both sides of this equation by p yields $y = \frac{-165}{p}x + \frac{w}{p}$. It follows that in the xy -plane, line k has a slope of $\frac{-165}{p}$ and $(0, \frac{w}{p})$ is the y -intercept. Since line g and line k have the same slope and the same y -intercept, it follows that $-\frac{165}{p} = -\frac{11}{10}$ and $\frac{w}{p} = 1$. Multiplying both sides of $-\frac{165}{p} = -\frac{11}{10}$ by $-10p$ yields $1,650 = 11p$. Dividing both sides of this equation by 11 yields $150 = p$. Substituting 150 for p in the equation $\frac{w}{p} = 1$ yields $\frac{w}{150} = 1$. Multiplying both sides of this equation by 150 yields $w = 150$. Substituting 150 for p and 150 for w in the expression $p + w$ yields $150 + 150$, which is equivalent to **300**. Therefore, the value of $p + w$ is **300**.

Question ID 24c175fc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 24c175fc

If $9(4 - 3x) + 2 = 8(4 - 3x) + 18$, what is the value of $4 - 3x$?

- A. -16
- B. -4
- C. 4
- D. 16

ID: 24c175fc Answer

Correct Answer: D

Rationale

Choice D is correct. The value of $4 - 3x$ can be found by isolating this expression in the given equation. Subtracting 2 from both sides of the given equation yields $9(4 - 3x) = 8(4 - 3x) + 16$. Subtracting $8(4 - 3x)$ from both sides of this equation yields $9(4 - 3x) - 8(4 - 3x) = 16$, which gives $1(4 - 3x) = 16$, or $4 - 3x = 16$. Therefore, the value of $4 - 3x$ is 16 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not $4 - 3x$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 159b76a4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 159b76a4

A line segment that has a length of **115 centimeters (cm)** is divided into three parts. One part is **47 cm** long. The other two parts have lengths that are equal to each other. What is the length, in **cm**, of one of the other two parts of equal length?

ID: 159b76a4 Answer

Correct Answer: 34

Rationale

The correct answer is **34**. It's given that a line segment has a length of **115 cm** and is divided into three parts, where one part is **47 cm** long and the other two parts have lengths that are equal. If x represents the length, in cm, of each of the two parts of equal length, then the equation $47 + x + x = 115$, or $47 + 2x = 115$, represents this situation. Subtracting **47** from each side of this equation yields $2x = 68$. Dividing each side of this equation by **2** yields $x = 34$. Therefore, the length, in cm, of one of the two parts of equal length is **34**.

Question Difficulty: Easy

Question ID ae5abcba

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: ae5abcba

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by **2.10** kelvins, by how much did the temperature increase, in degrees Fahrenheit?

- A. **3.78**
- B. **35.78**
- C. **487.89**
- D. **519.89**

ID: ae5abcba Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the function $F(x) = \frac{9}{5}(x - 273.15) + 32$ gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. A temperature that increased by **2.10** kelvins means that the value of x increased by **2.10** kelvins. It follows that an increase in x by **2.10** increases $F(x)$ by $\frac{9}{5}(2.10)$, or **3.78**. Therefore, if a temperature increased by **2.10** kelvins, the temperature increased by **3.78** degrees Fahrenheit.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 8d0a9044

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 8d0a9044

Two customers purchased the same kind of bread and eggs at a store. The first customer paid **12.45** dollars for **1** loaf of bread and **2** dozen eggs. The second customer paid **19.42** dollars for **4** loaves of bread and **1** dozen eggs. What is the cost, in dollars, of **1** dozen eggs?

- A. **3.77**
- B. **3.88**
- C. **4.15**
- D. **4.34**

ID: 8d0a9044 Answer

Correct Answer: D

Rationale

Choice D is correct. Let b represent the cost, in dollars, of **1** loaf of bread, and let d represent the cost, in dollars, of **1** dozen eggs. It's given that the first customer paid **12.45** dollars for **1** loaf of bread and **2** dozen eggs. Therefore, the first customer's purchase can be represented by the equation $b + 2d = 12.45$. It's also given that the second customer paid **19.42** dollars for **4** loaves of bread and **1** dozen eggs. Therefore, the second customer's purchase can be represented by the equation $4b + d = 19.42$. The equations $b + 2d = 12.45$ and $4b + d = 19.42$ form a system of linear equations, which can be solved by elimination to find the value of d . Multiplying the first equation in the system by -4 yields $-4b - 8d = -49.8$. Adding $-4b - 8d = -49.8$ to the second equation, $4b + d = 19.42$, yields $(-4b + 4b) + (-8d + d) = (-49.8 + 19.42)$, which is equivalent to $-7d = -30.38$. Dividing both sides of this equation by -7 yields $d = 4.34$. Therefore, the cost, in dollars, of **1** dozen eggs is **4.34**.

Choice A is incorrect. This is the cost, in dollars, of **1** loaf of bread.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a504b146

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: a504b146

A museum rents tablets to visitors. The museum earns revenue of **\$14** for each tablet rented for the day. On Wednesday, the museum earned **\$406** in profit from renting tablets after paying daily expenses of **\$112**. How many tablets did the museum rent on Wednesday? (**profit = total revenue – total expenses**)

ID: a504b146 Answer

Correct Answer: 37

Rationale

The correct answer is **37**. It's given that the museum earns revenue of **\$14** for each tablet rented for the day. It's also given that on Wednesday, the museum earned **\$406** in profit from renting tablets after paying daily expenses of **\$112**. Let ***x*** represent the number of tablets the museum rented on Wednesday. It follows that the total revenue can be represented by the expression **14*x***. Because **profit = total revenue – total expenses**, the equation **406 = 14*x* – 112** represents this situation. Adding **112** to both sides of this equation yields **14*x* = 518**. Dividing both sides of this equation by **14** yields ***x* = 37**. Therefore, the museum rented **37** tablets on Wednesday.

Question Difficulty: Medium

Question ID 3adf7966

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 3adf7966

$$\begin{aligned}y &= -2x \\ 3x + y &= 40\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

ID: 3adf7966 Answer

Correct Answer: 40

Rationale

The correct answer is **40**. It's given in the first equation of the system that $y = -2x$. Substituting $-2x$ for y in the second equation of the system yields $3x + (-2x) = 40$. Combining like terms on the left-hand side of this equation yields $x = 40$. Therefore, the value of x is **40**.

Question Difficulty: Medium

Question ID 14a65be1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 14a65be1

The function f is defined by $f(x) = 5x + 8$. For what value of x does $f(x) = 58$?

- A. 10
- B. 13
- C. 50
- D. 298

ID: 14a65be1 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function f is defined by $f(x) = 5x + 8$. Substituting 58 for $f(x)$ in this equation yields $58 = 5x + 8$. Subtracting 8 from both sides of this equation yields $50 = 5x$. Dividing both sides of this equation by 5 yields $10 = x$. Therefore, the value of x when $f(x) = 58$ is 10.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $f(58)$, not the value of x when $f(x) = 58$.

Question Difficulty: Medium

Question ID 73807c8b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 73807c8b

Keenan made **32** cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

- A. The number of large jars Keenan filled
- B. The number of small jars Keenan filled
- C. The total number of cups of vegetable broth in the large jars
- D. The total number of cups of vegetable broth in the small jars

ID: 73807c8b Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the equation $3x + 5y = 32$ represents the situation where Keenan filled x small jars and y large jars with all the vegetable broth he made, which was **32** cups. Therefore, $3x$ represents the total number of cups of vegetable broth in the small jars and $5y$ represents the total number of cups of vegetable broth in the large jars.

Choice A is incorrect. The number of large jars Keenan filled is represented by y , not $5y$.

Choice B is incorrect. The number of small jars Keenan filled is represented by x , not $5y$.

Choice D is incorrect. The total number of cups of vegetable broth in the small jars is represented by $3x$, not $5y$.

Question Difficulty: Hard

Question ID d791004d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: d791004d

In the xy -plane, the graph of the linear function f contains the points $(0, 3)$ and $(7, 31)$. Which equation defines f , where $y = f(x)$?

- A. $f(x) = 28x + 34$
- B. $f(x) = 3x + 38$
- C. $f(x) = 4x + 3$
- D. $f(x) = 7x + 3$

ID: d791004d Answer

Correct Answer: C

Rationale

Choice C is correct. In the xy -plane, an equation of the graph of a linear function can be written in the form $f(x) = mx + b$, where m represents the slope and $(0, b)$ represents the y -intercept of the graph of $y = f(x)$. It's given that the graph of the linear function f , where $y = f(x)$, in the xy -plane contains the point $(0, 3)$. Thus, $b = 3$. The slope of the graph of a line containing any two points (x_1, y_1) and (x_2, y_2) can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $(0, 3)$ and $(7, 31)$, it follows that the slope of the graph of the line containing these points is $m = \frac{31 - 3}{7 - 0}$, or $m = 4$. Substituting 4 for m and 3 for b in $f(x) = mx + b$ yields $f(x) = 4x + 3$.

Choice A is incorrect. This function represents a graph with a slope of 28 and a y -intercept of $(0, 34)$.

Choice B is incorrect. This function represents a graph with a slope of 3 and a y -intercept of $(0, 38)$.

Choice D is incorrect. This function represents a graph with a slope of 7 and a y -intercept of $(0, 3)$.

Question Difficulty: Medium

Question ID ff117182

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: ff117182

$7x - 4y = -84$

For the given equation, which table gives three values of x and their corresponding values of y ?

- A.

x	0	4	8
y	21	28	35
- B.

x	0	4	8
y	35	28	21
- C.

x	21	28	35
y	0	4	8
- D.

x	21	28	35
y	8	4	0

ID: ff117182 Answer

Correct Answer: A

Rationale

Choice A is correct. To verify which table represents this linear relationship, the values in each table can be checked by substituting them into the given equation. The table in choice A shows that when $x = 0, y = 21$. Substituting these values into the given equation yields $7(0) - 4(21) = -84$, or $-84 = -84$, which is true. Additionally, the table in choice A shows that when $x = 4, y = 28$. Substituting these values into the given equation yields $7(4) - 4(28) = -84$, or $-84 = -84$, which is true. Finally, the table in choice A shows that when $x = 8, y = 35$. Substituting these values into the given equation yields $7(8) - 4(35) = -84$, or $-84 = -84$, which is true. Therefore, the table in choice A gives three values of x and their corresponding values of y .

Choice B is incorrect. The table in choice B shows that when $x = 0, y = 35$. Substituting these values into the given equation yields $7(0) - 4(35) = -84$, or $-140 = -84$, which is not true.

Choice C is incorrect. The table in choice C shows that when $x = 21, y = 0$. Substituting these values into the given equation yields $7(21) - 4(0) = -84$, or $147 = -84$, which is not true.

Choice D is incorrect. The table in choice D shows that when $x = 21, y = 8$. Substituting these values into the given equation yields $7(21) - 4(8) = -84$, or $115 = -84$, which is not true.

Question Difficulty: Medium

Question ID abb3d9cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: abb3d9cc

When line n is graphed in the xy -plane, it has an x -intercept of $(-4, 0)$ and a y -intercept of $(0, \frac{86}{3})$. What is the slope of line n ?

- A. $\frac{3}{344}$
- B. $\frac{6}{43}$
- C. $\frac{43}{6}$
- D. $\frac{344}{3}$

ID: abb3d9cc Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that when line n is graphed in the xy -plane, it has an x -intercept of $(-4, 0)$ and a y -intercept of $(0, \frac{86}{3})$. The slope, m , of a line can be found using any two points on the line, (x_1, y_1) and (x_2, y_2) , and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(-4, 0)$ and $(0, \frac{86}{3})$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{\frac{86}{3} - 0}{0 - (-4)}$, or $m = \frac{43}{6}$. Therefore, the slope of line n is $\frac{43}{6}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the slope of a line that has an x -intercept of $(\frac{86}{3}, 0)$ and a y -intercept of $(0, -4)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 5ebcd502

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 5ebcd502

A contract for a certain service requires a onetime activation cost of **\$35** and a monthly cost of **\$23**. Which equation represents this situation, where c is the total cost, in dollars, of this service contract for t months?

- A. $c = \frac{t}{23} + 35$
- B. $c = \frac{t}{35} + 23$
- C. $c = 23t + 35$
- D. $c = 35t + 23$

ID: 5ebcd502 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that this service contract requires a monthly cost of **\$23**. A monthly cost of **\$23** for t months results in a cost of **$23t$** . It's also given that this service contract requires a onetime activation cost of **\$35**. Adding the onetime activation cost to the monthly cost of the service contract for t months yields the total cost c , in dollars, of this service contract for t months. Therefore, this situation can be represented by the equation $c = 23t + 35$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c8038d24

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: c8038d24

A producer is creating a video with a length of **70** minutes. The video will consist of segments that are **1** minute long and segments that are **3** minutes long. Which equation represents this situation, where ***x*** represents the number of **1**-minute segments and ***y*** represents the number of **3**-minute segments?

- A. **$4xy = 70$**
- B. **$4(x + y) = 70$**
- C. **$3x + y = 70$**
- D. **$x + 3y = 70$**

ID: c8038d24 Answer

Correct Answer: D

Rationale

Choice D is correct. Since ***x*** represents the number of **1**-minute segments and ***y*** represents the number of **3**-minute segments, the total length of the video is **$1 \cdot x + 3 \cdot y$** , or **$x + 3y$** , minutes. Since the video is **70** minutes long, the equation **$x + 3y = 70$** represents this situation.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 311bf2f1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 311bf2f1

A certain apprentice has enrolled in **85** hours of training courses. The equation $10x + 15y = 85$ represents this situation, where x is the number of on-site training courses and y is the number of online training courses this apprentice has enrolled in. How many more hours does each online training course take than each on-site training course?

ID: 311bf2f1 Answer

Correct Answer: 5

Rationale

The correct answer is **5**. It's given that the equation $10x + 15y = 85$ represents the situation, where x is the number of on-site training courses, y is the number of online training courses, and **85** is the total number of hours of training courses the apprentice has enrolled in. Therefore, $10x$ represents the number of hours the apprentice has enrolled in on-site training courses, and $15y$ represents the number of hours the apprentice has enrolled in online training courses. Since x is the number of on-site training courses and y is the number of online training courses the apprentice has enrolled in, **10** is the number of hours each on-site course takes and **15** is the number of hours each online course takes. Subtracting these numbers gives $15 - 10$, or **5** more hours each online training course takes than each on-site training course.

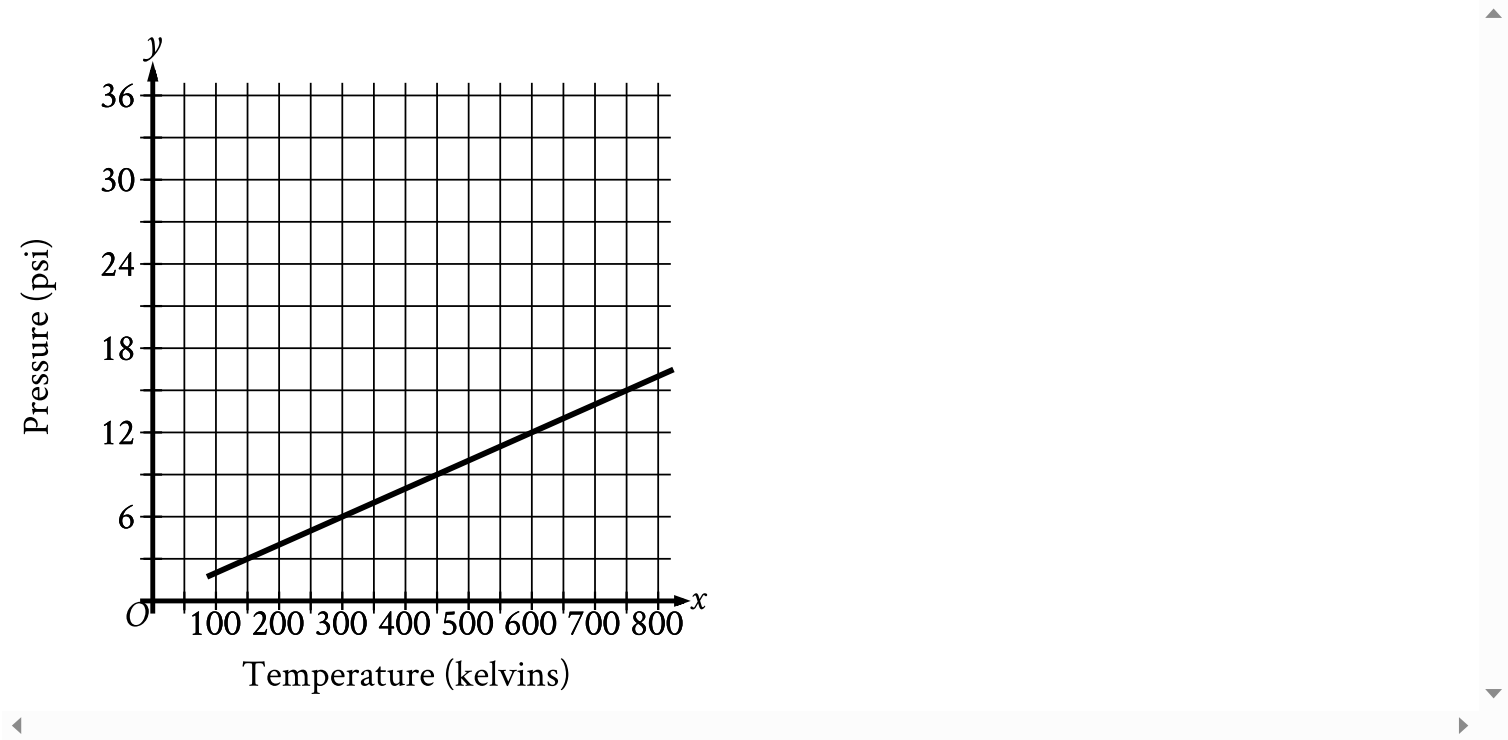
Question Difficulty: Hard

Question ID fa6dad4f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: fa6dad4f

Argon is placed inside a container with a constant volume. The graph shows the estimated pressure y , in **pounds per square inch (psi)**, of the argon when its temperature is x kelvins.



What is the estimated pressure of the argon, in **psi**, when the temperature is **600** kelvins?

- A. 6
- B. 12
- C. 300
- D. 600

ID: fa6dad4f Answer

Correct Answer: B

Rationale

Choice B is correct. For the graph shown, the x -axis represents temperature, in kelvins, and the y -axis represents the estimated pressure, in **pounds per square inch (psi)**. The estimated pressure of the argon when the temperature is **600** kelvins can be found by locating the point on the graph where the value of x is equal to **600**. The graph passes through the point **(600, 12)**. This means that when the temperature is **600** kelvins, the estimated pressure is **12 psi**.

Choice A is incorrect. This is the estimated pressure, in **psi**, of the argon when the temperature is **300** kelvins, not **600** kelvins.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the temperature, in kelvins, of the argon.

Question ID 362c4d27

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 362c4d27

$$f(x) = x + \frac{8}{11}$$

The function f is defined by the given equation. What is the value of $f(x)$ when $x = \frac{3}{11}$?

ID: 362c4d27 Answer

Correct Answer: 1

Rationale

The correct answer is **1**. It's given that the function f is defined by $f(x) = x + \frac{8}{11}$. Substituting $\frac{3}{11}$ for x in the given function yields $f(\frac{3}{11}) = \frac{3}{11} + \frac{8}{11}$, which gives $f(\frac{3}{11}) = \frac{11}{11}$, or $f(\frac{3}{11}) = 1$. Therefore, when $x = \frac{3}{11}$, the value of $f(x)$ is **1**.

Question Difficulty: Easy

Question ID 1d1e6588

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 1d1e6588

The equation $46 = 2x + 2y$ gives the perimeter of a rectangular rug that has length x , in feet, and width y , in feet. The width of the rug is 8 feet. What is the length, in feet, of the rug?

ID: 1d1e6588 Answer

Correct Answer: 15

Rationale

The correct answer is 15. It's given that the equation $46 = 2x + 2y$ gives the perimeter of a rectangular rug that has length x , in feet, and width y , in feet. It's also given that the width of the rug is 8 feet. Substituting 8 for y in the equation $46 = 2x + 2y$ yields $46 = 2x + 2(8)$, or $46 = 2x + 16$. Subtracting 16 from both sides of this equation yields $30 = 2x$. Dividing both sides of this equation by 2 yields $15 = x$. Since x represents the length, in feet, of the rug, it follows that the length of the rug is 15 feet.

Question Difficulty: Medium

Question ID d4b059b1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: d4b059b1

The equation $7g + 7b = 840$ represents the number of blue tiles, b , and the number of green tiles, g , an artist needs for an ~~840~~-square-inch tile project. The artist needs ~~71~~ blue tiles for the project. How many green tiles does he need?

ID: d4b059b1 Answer

Correct Answer: 49

Rationale

The correct answer is **49**. It's given that the equation $7g + 7b = 840$ represents the number of blue tiles, b , and the number of green tiles, g , an artist needs for an ~~840~~-square-inch tile project. It's also given that the artist needs ~~71~~ blue tiles for the project. Substituting **71** for b in the equation $7g + 7b = 840$ yields $7g + 7(71) = 840$, or $7g + 497 = 840$. Subtracting **497** from both sides of this equation yields $7g = 343$. Dividing both sides of this equation by **7** yields $g = 49$. Therefore, the artist needs **49** green tiles for the project.

Question Difficulty: Medium

Question ID 26fdb3e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 26fdb3e

The graph of $7x + 2y = -31$ in the xy -plane has an x -intercept at $(a, 0)$ and a y -intercept at $(0, b)$, where a and b are constants. What is the value of $\frac{b}{a}$?

- A. $-\frac{7}{2}$
- B. $-\frac{2}{7}$
- C. $\frac{2}{7}$
- D. $\frac{7}{2}$

ID: 26fdb3e Answer

Correct Answer: D

Rationale

Choice D is correct. The x -coordinate a of the x -intercept $(a, 0)$ can be found by substituting 0 for y in the given equation, which gives $7x + 2(0) = -31$, or $7x = -31$. Dividing both sides of this equation by 7 yields $x = -\frac{31}{7}$. Therefore, the value of a is $-\frac{31}{7}$. The y -coordinate b of the y -intercept $(0, b)$ can be found by substituting 0 for x in the given equation, which gives $7(0) + 2y = -31$, or $2y = -31$. Dividing both sides of this equation by 2 yields $y = -\frac{31}{2}$. Therefore, the value of b is $-\frac{31}{2}$. It follows that the value of $\frac{b}{a}$ is $\frac{-\frac{31}{2}}{-\frac{31}{7}}$, which is equivalent to $(\frac{31}{2})(\frac{7}{31})$, or $\frac{7}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID be74dbbf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: be74dbbf

$$f(x) = 2x + 3$$

For the given function f , the graph of $y = f(x)$ in the xy -plane is parallel to line j . What is the slope of line j ?

ID: be74dbbf Answer

Correct Answer: 2

Rationale

The correct answer is **2**. It's given that function f is defined by $f(x) = 2x + 3$. Therefore, the equation representing the graph of $y = f(x)$ in the xy -plane is $y = 2x + 3$, and the graph is a line. For a linear equation in the form $y = mx + b$, m represents the slope of the line. Since the value of m in the equation $y = 2x + 3$ is **2**, the slope of the line defined by function f is **2**. It's given that line j is parallel to the line defined by function f . The slopes of parallel lines are equal. Therefore, the slope of line j is also **2**.

Question Difficulty: Medium

Question ID 73596e60

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 73596e60

A linear function f gives a company’s profit, in dollars, for selling x items. The company’s profit is \$220 when it sells 8 items, and its profit is \$320 when it sells 10 items. Which equation defines f ?

- A. $f(x) = 150x - 320$
- B. $f(x) = 32x$
- C. $f(x) = 50x - 10x$
- D. $f(x) = 50x - 180$

ID: 73596e60 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the relationship between x and $f(x)$ is linear. A linear function can be written in the form $f(x) = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane. Given two points on a line, (x_1, y_1) and (x_2, y_2) , the slope of the line can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It’s given that the company’s profit is \$220 when it sells 8 items and the profit is \$320 when it sells 10 items. Since $f(x)$ represents the company's profit, in dollars, for selling x items, the graph of $y = f(x)$ in the xy-plane passes through the points $(8, 220)$ and $(10, 320)$. Substituting $(8, 220)$ and $(10, 320)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{320 - 220}{10 - 8}$, which gives $m = \frac{100}{2}$, or $m = 50$. Substituting 50 for m , 8 for x , and 220 for $f(x)$ in $f(x) = mx + b$ yields $220 = (50)(8) + b$, or $220 = 400 + b$. Subtracting 400 from each side of this equation yields $-180 = b$. Substituting 50 for m and -180 for b in $f(x) = mx + b$ yields $f(x) = 50x - 180$. Therefore, the equation that defines f is $f(x) = 50x - 180$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID fb272309

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: fb272309

$$\begin{aligned}ax + by &= 72 \\ 6x + 2by &= 56\end{aligned}$$

In the given system of equations, a and b are constants. The graphs of these equations in the xy -plane intersect at the point $(4, y)$. What is the value of a ?

- A. 3
- B. 4
- C. 6
- D. 14

ID: fb272309 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the graphs of the given system of equations intersect at the point $(4, y)$. Therefore, $(4, y)$ is the solution to the given system. Multiplying the first equation in the given system by -2 yields $-2ax - 2by = -144$. Adding this equation to the second equation in the system yields $(-2a + 6)x + (-2b + 2b)y = (-144 + 56)$, or $(-2a + 6)x = -88$. Since $(4, y)$ is the solution to the system, the value of a can be found by substituting 4 for x in this equation, which yields $(-2a + 6)(4) = -88$. Dividing both sides of this equation by 4 yields $-2a + 6 = -22$. Subtracting 6 from both sides of this equation yields $-2a = -28$. Dividing both sides of this equation by -2 yields $a = 14$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 5cabd603

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 5cabd603

During a portion of a flight, a small airplane's cruising speed varied between **150** miles per hour and **170** miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

- A. $s \leq 20$
- B. $s \leq 150$
- C. $s \leq 170$
- D. $150 \leq s \leq 170$

ID: 5cabd603 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that during a portion of a flight, a small airplane's cruising speed varied between **150** miles per hour and **170** miles per hour. It's also given that s represents the cruising speed, in miles per hour, during this portion of the flight. It follows that the airplane's cruising speed, in miles per hour, was at least **150**, which means $s \geq 150$, and was at most **170**, which means $s \leq 170$. Therefore, the inequality that best represents this situation is $150 \leq s \leq 170$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 0b52d235

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 0b52d235

During spring migration, a dragonfly traveled a minimum of **1,510** miles and a maximum of **4,130** miles between stopover locations. Which inequality represents this situation, where *d* is a possible distance, in miles, this dragonfly traveled between stopover locations during spring migration?

- A. $d \leq 1,510$
- B. $1,510 \leq d \leq 4,130$
- C. $d \geq 4,130$
- D. $4,130 \leq d \leq 5,640$

ID: 0b52d235 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that during spring migration, a dragonfly traveled a minimum of **1,510** miles and a maximum of **4,130** miles between stopover locations. It's also given that *d* represents a possible distance, in miles, this dragonfly traveled between stopover locations. It follows that the inequality $1,510 \leq d \leq 4,130$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which a dragonfly traveled a maximum of **1,510** miles between stopover locations.

Choice C is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of **4,130** miles between stopover locations.

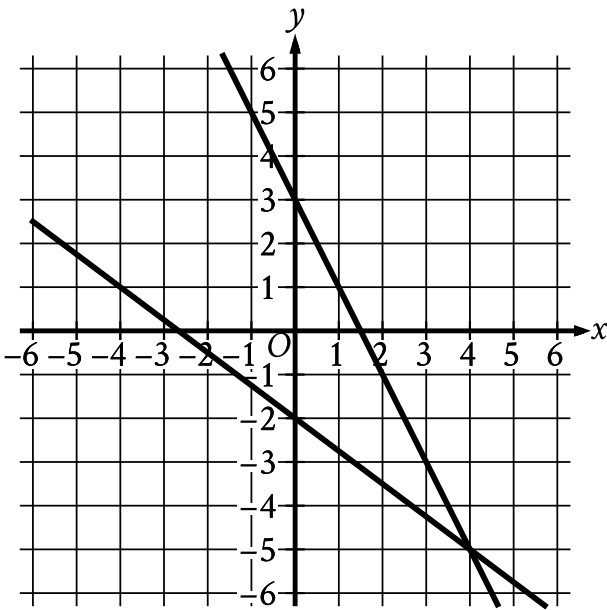
Choice D is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of **4,310** miles and a maximum of **5,640** miles between stopover locations.

Question Difficulty: Easy

Question ID 9dd0e1d5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 9dd0e1d5



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- A. $(4, -5)$
- B. $(0, 3)$
- C. $(0, -2)$
- D. $(-2, 3)$

ID: 9dd0e1d5 Answer

Correct Answer: A

Rationale

Choice A is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 4$ and $y = -5$, or at the point $(4, -5)$. Therefore, the solution (x, y) to the system is $(4, -5)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 4fc5a7d2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 4fc5a7d2

A line in the xy -plane has a slope of $-\frac{1}{2}$ and passes through the point $(0, 3)$. Which equation represents this line?

- A. $y = -\frac{1}{2}x - 3$
- B. $y = -\frac{1}{2}x + 3$
- C. $y = \frac{1}{2}x - 3$
- D. $y = \frac{1}{2}x + 3$

ID: 4fc5a7d2 Answer

Correct Answer: B

Rationale

Choice B is correct. A line in the xy -plane with a slope of m and a y -intercept of $(0, b)$ can be represented by the equation $y = mx + b$. It's given that the line has a slope of $-\frac{1}{2}$. Therefore, $m = -\frac{1}{2}$. It's also given that the line passes through the point $(0, 3)$. Therefore, $b = 3$. Substituting $-\frac{1}{2}$ for m and 3 for b in the equation $y = mx + b$ yields $y = -\frac{1}{2}x + 3$. Therefore, the equation $y = -\frac{1}{2}x + 3$ represents this line.

Choice A is incorrect. This equation represents a line in the xy -plane that passes through the point $(0, -3)$, not $(0, 3)$.

Choice C is incorrect. This equation represents a line in the xy -plane that has a slope of $\frac{1}{2}$, not $-\frac{1}{2}$, and passes through the point $(0, -3)$, not $(0, 3)$.

Choice D is incorrect. This equation represents a line in the xy -plane that has a slope of $\frac{1}{2}$, not $-\frac{1}{2}$.

Question Difficulty: Easy

Question ID 8652a731

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 8652a731

How many solutions does the equation $12(x - 3) = -3(x + 12)$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 8652a731 Answer

Correct Answer: A

Rationale

Choice A is correct. Distributing **12** on the left-hand side and **−3** on the right-hand side of the given equation yields **$12x - 36 = -3x - 36$** . Adding **$3x$** to each side of this equation yields **$15x - 36 = -36$** . Adding **36** to each side of this equation yields **$15x = 0$** . Dividing each side of this equation by **15** yields **$x = 0$** . This means that **0** is the only solution to the given equation. Therefore, the given equation has exactly one solution.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a06c0a64

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: a06c0a64

A number x is at most 17 less than 5 times the value of y . If the value of y is 3, what is the greatest possible value of x ?

ID: a06c0a64 Answer

Correct Answer: -2

Rationale

The correct answer is -2 . It's given that a number x is at most 17 less than 5 times the value of y , or $x \leq 5y - 17$. Substituting 3 for y in this inequality yields $x \leq 5(3) - 17$, or $x \leq -2$. Thus, if the value of y is 3, the greatest possible value of x is -2 .

Question Difficulty: Hard

Question ID 609fc9a7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 609fc9a7

x	y
-18	-48
7	52

The table shows two values of x and their corresponding values of y . In the xy -plane, the graph of the linear equation representing this relationship passes through the point $(\frac{1}{7}, a)$. What is the value of a ?

- A. $-\frac{4}{11}$
- B. $-\frac{4}{77}$
- C. $\frac{4}{7}$
- D. $\frac{172}{7}$

ID: 609fc9a7 Answer

Correct Answer: D

Rationale

Choice D is correct. The linear relationship between x and y can be represented by the equation $y = mx + b$, where m is the slope of the graph of this equation in the xy -plane and b is the y -coordinate of the y -intercept. The slope of a line between any two points (x_1, y_1) and (x_2, y_2) on the line can be calculated using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Based on the table, the graph contains the points $(-18, -48)$ and $(7, 52)$. Substituting $(-18, -48)$ and $(7, 52)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{52 - (-48)}{7 - (-18)}$, which is equivalent to $m = \frac{100}{25}$, or $m = 4$. Substituting 4 for m , -18 for x , and -48 for y in the equation $y = mx + b$ yields $-48 = 4(-18) + b$, or $-48 = -72 + b$. Adding 72 to both sides of this equation yields $24 = b$. Therefore, $m = 4$ and $b = 24$. Substituting 4 for m and 24 for b in the equation $y = mx + b$ yields $y = 4x + 24$. Thus, the equation $y = 4x + 24$ represents the linear relationship between x and y . It's also given that the graph of the linear equation representing this relationship in the xy -plane passes through the point $(\frac{1}{7}, a)$. Substituting $\frac{1}{7}$ for x and a for y in the equation $y = 4x + 24$ yields $a = 4(\frac{1}{7}) + 24$, which is equivalent to $a = \frac{4}{7} + \frac{168}{7}$, or $a = \frac{172}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 552ce561

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 552ce561

$$s = 40 + 3t$$

The equation gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car **5** seconds after it began to accelerate?

- A. **40**
- B. **43**
- C. **45**
- D. **55**

ID: 552ce561 Answer

Correct Answer: D

Rationale

Choice D is correct. In the given equation, s is the speed, in miles per hour, of a certain car t seconds after it began to accelerate. Therefore, the speed of the car, in miles per hour, **5** seconds after it began to accelerate can be found by substituting **5** for t in the given equation, which yields $s = 40 + 3(\mathbf{5})$, or $s = \mathbf{55}$. Thus, the speed of the car **5** seconds after it began to accelerate is **55** miles per hour.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 9767a58f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 9767a58f

A total of **2** squares each have side length **r** . A total of **6** equilateral triangles each have side length **t** . None of these squares and triangles shares a side. The sum of the perimeters of all these squares and triangles is **210**. Which equation represents this situation?

- A. $6r + 24t = 210$
- B. $2r + 6t = 210$
- C. $8r + 18t = 210$
- D. $6r + 2t = 210$

ID: 9767a58f Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that a total of **2** squares each have side length **r** . Therefore, each of the squares has perimeter **$4r$** . Since there are a total of **2** squares, the sum of the perimeters of these squares is **$4r + 4r$** , which is equivalent to **$2(4r)$** , or **$8r$** . It's also given that a total of **6** equilateral triangles each have side length **t** . Therefore, each of the equilateral triangles has perimeter **$3t$** . Since there are a total of **6** equilateral triangles, the sum of the perimeters of these triangles is **$3t + 3t + 3t + 3t + 3t + 3t$** , which is equivalent to **$6(3t)$** , or **$18t$** . Since the sum of the perimeters of the squares is **$8r$** and the sum of the perimeters of the triangles is **$18t$** , the sum of the perimeters of all these squares and triangles is **$8r + 18t$** . It's given that the sum of the perimeters of all these squares and triangles is **210**. Therefore, the equation **$8r + 18t = 210$** represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID cff208ce

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: cff208ce

The function f is defined by $f(x) = 9(2x + 3)$. For what value of x does $f(x) = 63$?

- A. 2
- B. 5
- C. 7
- D. 30

ID: cff208ce Answer

Correct Answer: A

Rationale

Choice A is correct. Substituting **63** for $f(x)$ in the given function yields $63 = 9(2x + 3)$. Dividing both sides of this equation by **9** yields $7 = 2x + 3$. Subtracting **3** from both sides of this equation yields $4 = 2x$. Dividing both sides of this equation by **2** yields $2 = x$. Therefore, $f(x) = 63$ when the value of x is **2**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 60bac251

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 60bac251

$$\begin{aligned}x + 3y &= 29 \\ 3y &= 11\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

ID: 60bac251 Answer

Correct Answer: 18

Rationale

The correct answer is **18**. It's given by the second equation in the system that **$3y = 11$** . Substituting **11** for **$3y$** in the first equation in the system, **$x + 3y = 29$** , yields **$x + 11 = 29$** . Subtracting **11** from both sides of this equation yields **$x = 18$** .

Question Difficulty: Medium

Question ID 2df665b8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 2df665b8

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 2 and has a y -intercept at $(0, -5)$. Which equation defines f ?

- A. $f(x) = \frac{1}{2}x - 5$
- B. $f(x) = -\frac{1}{2}x - 5$
- C. $f(x) = -2x - 5$
- D. $f(x) = 2x - 5$

ID: 2df665b8 Answer

Correct Answer: D

Rationale

Choice D is correct. An equation defining the linear function f can be written in the form $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of $y = f(x)$ in the xy -plane. It's given that the graph of $y = f(x)$ has a slope of 2 . Therefore, $m = 2$. It's also given that the graph of $y = f(x)$ has a y -intercept at $(0, -5)$. Therefore, $b = -5$. Substituting 2 for m and -5 for b in the equation $f(x) = mx + b$ yields $f(x) = 2x - 5$. Thus, the equation that defines f is $f(x) = 2x - 5$.

Choice A is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane has a slope of $\frac{1}{2}$, not 2 .

Choice B is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane has a slope of $-\frac{1}{2}$, not 2 .

Choice C is incorrect. For this function, the graph of $y = f(x)$ in the xy -plane has a slope of -2 , not 2 .

Question Difficulty: Medium

Question ID 862cc556

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 862cc556

<i>x</i>	1	2	3
<i>y</i>	11	16	21

The table shows three values of *x* and their corresponding values of *y*. Which equation represents the linear relationship between *x* and *y*?

- A. $y = 5x + 6$
- B. $y = 5x + 11$
- C. $y = 6x + 5$
- D. $y = 6x + 11$

ID: 862cc556 Answer

Correct Answer: A

Rationale

Choice A is correct. The linear relationship between *x* and *y* can be represented by the equation $y = mx + b$, where *m* is the slope of the line in the *xy*-plane that represents the relationship, and *b* is the *y*-coordinate of the *y*-intercept. The slope can be computed using any two points on the line. The slope of a line between any two points, (*x*₁, *y*₁) and (*x*₂, *y*₂), on the line can be calculated using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. In the given table, each value of *x* and its corresponding value of *y* can be represented by a point (*x*, *y*). In the given table, when the value of *x* is 1, the corresponding value of *y* is 11 and when the value of *x* is 2, the corresponding value of *y* is 16. Therefore, the points (1, 11) and (2, 16) are on the line. Substituting (1, 11) and (2, 16) for (*x*₁, *y*₁) and (*x*₂, *y*₂), respectively, in the slope formula yields $m = \frac{16 - 11}{2 - 1}$, or $m = 5$. Substituting 5 for *m* in the equation $y = mx + b$ yields $y = 5x + b$. Substituting the first value of *x* in the table, 1, and its corresponding value of *y*, 11, for *x* and *y*, respectively, in this equation yields $11 = 5(1) + b$, or $11 = b + 5$. Subtracting 5 from both sides of this equation yields $6 = b$. Substituting 6 for *b* in the equation $y = 5x + b$ yields $y = 5x + 6$. Therefore, the equation $y = 5x + 6$ represents the linear relationship between *x* and *y*.

Choice B is incorrect. For this relationship, when the value of *x* is 1, the corresponding value of *y* is 16, not 11.

Choice C is incorrect. For this relationship, when the value of *x* is 2, the corresponding value of *y* is 17, not 16.

Choice D is incorrect. For this relationship, when the value of *x* is 1, the corresponding value of *y* is 17, not 11.

Question Difficulty: Medium

Question ID b08cf45d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: b08cf45d

$-49x = -98x$

How many solutions does the given equation have?

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

ID: b08cf45d Answer

Correct Answer: B

Rationale

Choice B is correct. Adding $98x$ to each side of the given equation yields $49x = 0$. Dividing each side of this equation by 49 yields $x = 0$. This means that 0 is the only solution to the given equation. Therefore, the given equation has exactly one solution.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 9655c984

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 9655c984

Leo goes to a packing store to buy containers and tape. Leo has **\$15**. Each container costs **\$1.87** and each roll of tape costs **\$2.40**. Which inequality represents the relationship between the number of containers, c , and the number of rolls of tape, t , Leo can buy?

- A. $1.87c + 2.40t \leq 15$
- B. $1.87c + 2.40t \geq 15$
- C. $2.40c + 1.87t \leq 15$
- D. $2.40c + 1.87t \geq 15$

ID: 9655c984 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that each container costs **\$1.87** and c represents the number of containers. It follows that $1.87c$ represents the cost, in dollars, of c containers. It's also given that each roll of tape costs **\$2.40** and t represents the number of rolls of tape. It follows that $2.40t$ represents the cost, in dollars, of t rolls of tape. Therefore, the total cost, in dollars, for c containers and t rolls of tape is $1.87c + 2.40t$. It's given that Leo has **\$15**. It follows that the total cost, in dollars, must be less than or equal to **15**. Therefore, the inequality $1.87c + 2.40t \leq 15$ represents this relationship.

Choice B is incorrect. This inequality represents a relationship where Leo has at least **\$15**, not at most **\$15**.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID dd181f54

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: dd181f54

Naomi bought both rabbit snails and nerite snails for a total of **\$52**. Each rabbit snail costs **\$8** and each nerite snail costs **\$6**. If Naomi bought **2** nerite snails, how many rabbit snails did she buy?

- A. **5**
- B. **12**
- C. **14**
- D. **50**

ID: dd181f54 Answer

Correct Answer: A

Rationale

Choice A is correct. Let x represent the number of rabbit snails that Naomi bought. It's given that each rabbit snail costs **\$8**. Therefore, the total cost, in dollars, of the rabbit snails that Naomi bought can be represented by the expression **$8x$** . It's also given that each nerite snail costs **\$6**, and that Naomi bought **2** nerite snails. Therefore, the total cost, in dollars, of the nerite snails that Naomi bought is **$6(2)$** , or **12**. Since Naomi bought both the rabbit snails and the nerite snails for a total of **\$52**, the equation **$8x + 12 = 52$** can be used to represent the situation. Subtracting **12** from both sides of this equation yields **$8x = 40$** . Dividing both sides of this equation by **8** yields **$x = 5$** . Therefore, Naomi bought **5** rabbit snails.

Choice B is incorrect. This is the total cost, in dollars, of the nerite snails that Naomi bought, not the number of rabbit snails.

Choice C is incorrect. This is the cost, in dollars, of one rabbit snail and one nerite snail, not the number of rabbit snails that Naomi bought.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c4f9f99e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: c4f9f99e

$$\begin{aligned}y &= 4x - 9 \\ y &= 19\end{aligned}$$

What is the solution (x, y) to the given system of equations?

- A. $(4, 19)$
- B. $(7, 19)$
- C. $(19, 4)$
- D. $(19, 7)$

ID: c4f9f99e Answer

Correct Answer: B

Rationale

Choice B is correct. It's given by the second equation in the system that $y = 19$. Substituting 19 for y in the first equation yields $19 = 4x - 9$. Adding 9 to both sides of this equation yields $28 = 4x$. Dividing both sides of this equation by 4 yields $7 = x$. Therefore, since $x = 7$ and $y = 19$, the solution (x, y) to the given system of equations is $(7, 19)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 347a3050

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 347a3050

A line passes through the points $(4, 6)$ and $(15, 24)$ in the xy -plane. What is the slope of the line?

ID: 347a3050 Answer

Correct Answer: 1.636, 18/11

Rationale

The correct answer is $\frac{18}{11}$. For a line that passes through the points (x_1, y_1) and (x_2, y_2) in the xy -plane, the slope of the line can be calculated using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that a line passes through the points $(4, 6)$ and $(15, 24)$ in the xy -plane. Substituting $(4, 6)$ for (x_1, y_1) and $(15, 24)$ for (x_2, y_2) in the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$, yields $m = \frac{24 - 6}{15 - 4}$, or $m = \frac{18}{11}$. Therefore, the slope of the line is $\frac{18}{11}$. Note that 18/11 and 1.636 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID bb0ac882

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: bb0ac882

At how many points do the graphs of the equations $y = x + 20$ and $y = 8x$ intersect in the xy -plane?

- A. 0
- B. 1
- C. 2
- D. 8

ID: bb0ac882 Answer

Correct Answer: B

Rationale

Choice B is correct. Each given equation is written in slope-intercept form, $y = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of the equation in the xy -plane. The graphs of two lines that have different slopes will intersect at exactly one point. The graph of the first equation is a line with slope 1. The graph of the second equation is a line with slope 8. Since the graphs are lines with different slopes, they will intersect at exactly one point.

Choice A is incorrect because two graphs of linear equations have 0 intersection points only if they are parallel and therefore have the same slope.

Choice C is incorrect because two graphs of linear equations in the xy -plane can have only 0, 1, or infinitely many points of intersection.

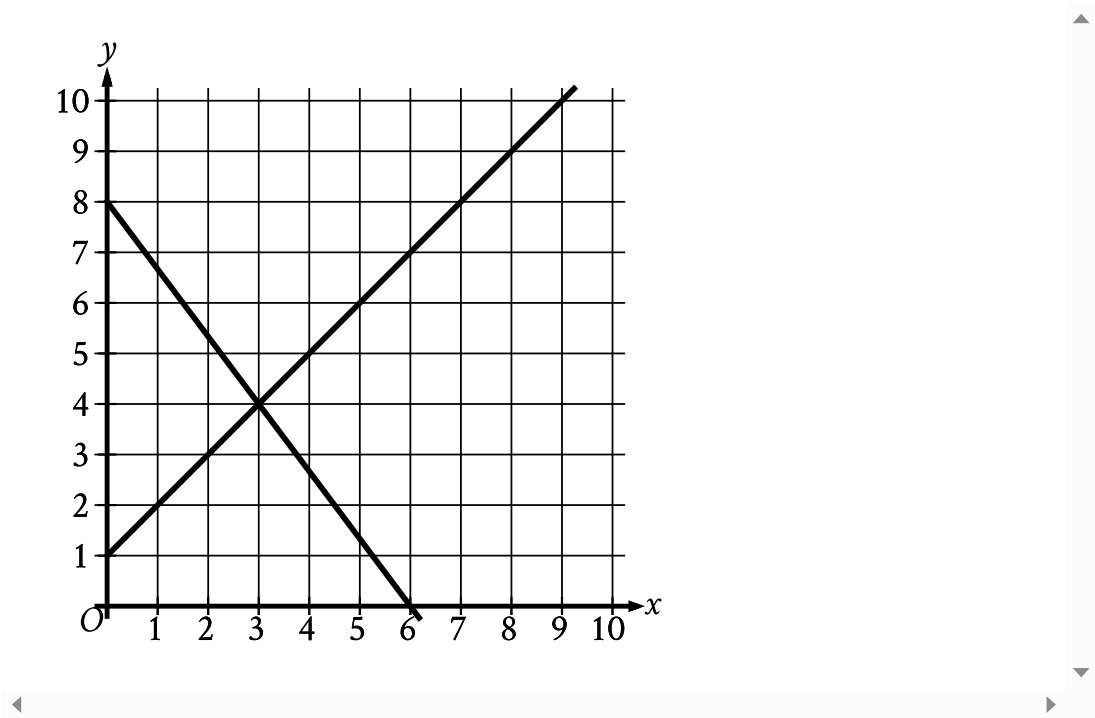
Choice D is incorrect because two graphs of linear equations in the xy -plane can have only 0, 1, or infinitely many points of intersection.

Question Difficulty: Medium

Question ID 4263af26

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 4263af26



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- A. $(2, 3)$
- B. $(3, 4)$
- C. $(4, 5)$
- D. $(5, 6)$

ID: 4263af26 Answer

Correct Answer: B

Rationale

Choice B is correct. If a point (x, y) lies on both lines in the graph of a system of two linear equations, the ordered pair (x, y) is a solution to the system. The graph shown is the graph of a system of two linear equations, where the two lines in the graph intersect at the point $(3, 4)$. Therefore, the point $(3, 4)$ lies on both lines, so the ordered pair $(3, 4)$ is the solution to the system.

Choice A is incorrect. The point $(2, 3)$ lies on one, not both, of the lines in the graph shown.

Choice C is incorrect. The point $(4, 5)$ lies on one, not both, of the lines in the graph shown.

Choice D is incorrect. The point $(5, 6)$ lies on one, not both, of the lines in the graph shown.

Question Difficulty: Easy

Question ID 39dbeef2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 39dbeef2

Line p is defined by $2y + 8x = 11$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

- A. -4
- B. $-\frac{1}{4}$
- C. $\frac{1}{4}$
- D. 4

ID: 39dbeef2 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that line p is defined by the equation $2y + 8x = 11$. The equation can be written in slope-intercept form $y = mx + b$, where m represents the slope of the line. Subtracting $8x$ from both sides of the equation $2y + 8x = 11$ yields $2y = -8x + 11$. Dividing both sides of this equation by 2 yields $y = -4x + \frac{11}{2}$. Therefore, the slope of line p in the xy -plane is -4 . It's also given that line p is perpendicular to line r . Perpendicular lines have slopes that are negative reciprocals of each other, so the slope of line r is $-\frac{1}{(-4)} = \frac{1}{4}$.

Choice A is incorrect. This is the slope of line p , not line r .

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of the slope of line p .

Choice D is incorrect. This is the negative, not the negative reciprocal, of the slope of line p .

Question Difficulty: Hard

Question ID a8c506d4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: a8c506d4

$y = 6x + 18$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

- A. $-6x + y = 18$
- B. $-6x + y = 22$
- C. $-12x + y = 36$
- D. $-12x + y = 18$

ID: a8c506d4 Answer

Correct Answer: B

Rationale

Choice B is correct. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$; and the lines are distinct if the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ or $\frac{E}{B}$. The given equation, $y = 6x + 18$, can be written in standard form by subtracting $6x$ from both sides of the equation to yield $-6x + y = 18$. Therefore, the given equation can be written in the form $Ax + By = C$, where $A = -6$, $B = 1$, and $C = 18$. The equation in choice B, $-6x + y = 22$, is written in the form $Dx + Ey = F$, where $D = -6$, $E = 1$, and $F = 22$. Therefore, $\frac{D}{A} = \frac{-6}{-6}$, which can be rewritten as $\frac{D}{A} = 1$; $\frac{E}{B} = \frac{1}{1}$, which can be rewritten as $\frac{E}{B} = 1$; and $\frac{F}{C} = \frac{22}{18}$, which can be rewritten as $\frac{F}{C} = \frac{11}{9}$. Since $\frac{D}{A} = 1$, $\frac{E}{B} = 1$, and $\frac{F}{C}$ is not equal to 1 , it follows that the given equation and the equation $-6x + y = 22$ are parallel and distinct. Therefore, a system of two linear equations consisting of the given equation and the equation $-6x + y = 22$ has no solution. Thus, the equation in choice B could be the second equation in the system.

Choice A is incorrect. The equation $-6x + y = 18$ and the given equation represent the same line in the xy -plane. Therefore, a system of these linear equations would have infinitely many solutions, rather than no solution.

Choice C is incorrect. The equation $-12x + y = 36$ and the given equation represent lines in the xy -plane that are distinct and not parallel. Therefore, a system of these linear equations would have exactly one solution, rather than no solution.

Choice D is incorrect. The equation $-12x + y = 18$ and the given equation represent lines in the xy -plane that are distinct and not parallel. Therefore, a system of these linear equations would have exactly one solution, rather than no solution.

Question Difficulty: Hard

Question ID 259829c6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 259829c6

$\frac{3}{5}x + \frac{3}{4}y = 7$

Which table gives three values of x and their corresponding values of y for the given equation?

- A.

x	y
1	$\frac{113}{20}$
2	$\frac{101}{20}$
4	$\frac{77}{20}$
- B.

x	y
1	$\frac{47}{5}$
2	$\frac{44}{5}$
4	$\frac{38}{5}$
- C.

x	y
1	$\frac{148}{15}$
2	$\frac{136}{15}$
4	$\frac{112}{15}$
- D.

x	y
1	$\frac{128}{15}$
2	$\frac{116}{15}$
4	$\frac{92}{15}$

ID: 259829c6 Answer

Correct Answer: D

Rationale

Choice D is correct. Each of the tables gives the same three values of x : **1**, **2**, and **4**. Substituting **1** for x in the given equation yields $\left(\frac{3}{5}\right)(1) + \frac{3}{4}y = 7$, or $\frac{3}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{3}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{32}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{128}{15}$. Therefore, when $x = 1$, the corresponding value of y for the given equation is $\frac{128}{15}$. Substituting **2** for x in the given equation yields $\left(\frac{3}{5}\right)(2) + \frac{3}{4}y = 7$, or $\frac{6}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{6}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{29}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{116}{15}$. Therefore, when $x = 2$, the corresponding value of y for the given equation is $\frac{116}{15}$. Substituting **4** for x in the given equation yields $\left(\frac{3}{5}\right)(4) + \frac{3}{4}y = 7$, or

$\frac{12}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{12}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{23}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{92}{15}$. Therefore, when $x = 4$, the corresponding value of y for the given equation is $\frac{92}{15}$. The table in choice D gives x -values of 1, 2, and 4 and corresponding y -values of $\frac{128}{15}$, $\frac{116}{15}$, and $\frac{92}{15}$, respectively. Therefore, the table in choice D gives three values of x and their corresponding values of y for the given equation.

Choice A is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + \frac{3}{4}y = 7$.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + y = 10$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + \frac{3}{4}y = 8$.

Question Difficulty: Hard

Question ID 9f0eba60

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 9f0eba60

The function f is defined by $f(x) = 4x - 3$. What is the value of $f(10)$?

- A. -30
- B. 37
- C. 40
- D. 43

ID: 9f0eba60 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 4x - 3$. Substituting 10 for x in the given function yields $f(10) = 4(10) - 3$, which is equivalent to $f(10) = 40 - 3$, or $f(10) = 37$. Therefore, the value of $f(10)$ is 37 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f(10)$ for the function $f(x) = 4x$, not $f(x) = 4x - 3$.

Choice D is incorrect. This is the value of $f(10)$ for the function $f(x) = 4x + 3$, not $f(x) = 4x - 3$.

Question Difficulty: Easy

Question ID 5602cce0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 5602cce0

If $4x - 28 = -24$, what is the value of $x - 7$?

- A. -24
- B. -22
- C. -6
- D. -1

ID: 5602cce0 Answer

Correct Answer: C

Rationale

Choice C is correct. Dividing all terms in the given equation by 4 yields $\frac{4x}{4} - \frac{28}{4} = -\frac{24}{4}$, or $x - 7 = -6$. Therefore, the value of $x - 7$ is -6 .

Choice A is incorrect. This is the value of $4x - 28$, not $x - 7$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c335e923

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: c335e923

A student council group is selling school posters for a fundraiser. They use the function $p(x) = 5x - 220$ to determine their profit $p(x)$, in dollars, for selling x school posters. In order to earn a profit of \$900, how many school posters must they sell?

ID: c335e923 Answer

Correct Answer: 224

Rationale

The correct answer is **224**. It's given that a student council group uses the function $p(x) = 5x - 220$ to determine their profit $p(x)$, in dollars, for selling x school posters. Substituting **900** for $p(x)$ in the given function yields $900 = 5x - 220$. Adding **220** to each side of this equation yields $1,120 = 5x$. Dividing each side of this equation by **5** yields $224 = x$. Therefore, in order to earn a profit of \$900, they must sell **224** school posters.

Question Difficulty: Medium

Question ID 088777da

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 088777da

An event planner is planning a party. It costs the event planner a onetime fee of **\$35** to rent the venue and **\$10.25** per attendee. The event planner has a budget of **\$200**. What is the greatest number of attendees possible without exceeding the budget?

ID: 088777da Answer

Correct Answer: 16

Rationale

The correct answer is **16**. The total cost of the party is found by adding the onetime fee of the venue to the cost per attendee times the number of attendees. Let x be the number of attendees. The expression $35 + 10.25x$ thus represents the total cost of the party. It's given that the budget is **\$200**, so this situation can be represented by the inequality $35 + 10.25x \leq 200$. The greatest number of attendees can be found by solving this inequality for x . Subtracting **35** from both sides of this inequality gives $10.25x \leq 165$. Dividing both sides of this inequality by **10.25** results in approximately $x \leq 16.098$. Since the question is stated in terms of attendees, rounding x down to the nearest whole number, **16**, gives the greatest number of attendees possible.

Question Difficulty: Medium

Question ID 6c9eeff9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 6c9eeff9

$$\begin{aligned}y &= 4x \\ 5x - 3y &= -14\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

- A. 4
- B. 6
- C. 8
- D. 10

ID: 6c9eeff9 Answer

Correct Answer: D

Rationale

Choice D is correct. The first equation in the given system is $y = 4x$. Substituting $4x$ for y in the second equation yields $5x - 3(4x) = -14$, which is equivalent to $5x - 12x = -14$, or $-7x = -14$. Dividing both sides of this equation by -7 yields $x = 2$. Substituting 2 for x in the equation $y = 4x$ yields $y = 4(2)$, or $y = 8$. Substituting 2 for x and 8 for y in the expression $x + y$ yields $2 + 8$, or 10 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $y - x$, not the value of $x + y$.

Choice C is incorrect. This is the value of y , not the value of $x + y$.

Question Difficulty: Medium

Question ID 02d3857d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 02d3857d

Lily made **36** cups of jam. Lily then filled **x** small containers and **y** large containers with all the jam she made. The equation **$4x + 6y = 36$** represents this situation. Which is the best interpretation of **$6y$** in this context?

- A. The number of large containers Lily filled
- B. The number of small containers Lily filled
- C. The total number of cups of jam in the large containers
- D. The total number of cups of jam in the small containers

ID: 02d3857d Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the equation **$4x + 6y = 36$** represents the situation where Lily filled **x** small containers and **y** large containers with all the jam she made, which was **36** cups. Therefore, **$6y$** represents the total number of cups of jam in the large containers.

Choice A is incorrect. The number of large containers Lily filled is represented by **y** , not **$6y$** .

Choice B is incorrect. The number of small containers Lily filled is represented by **x** , not **$6y$** .

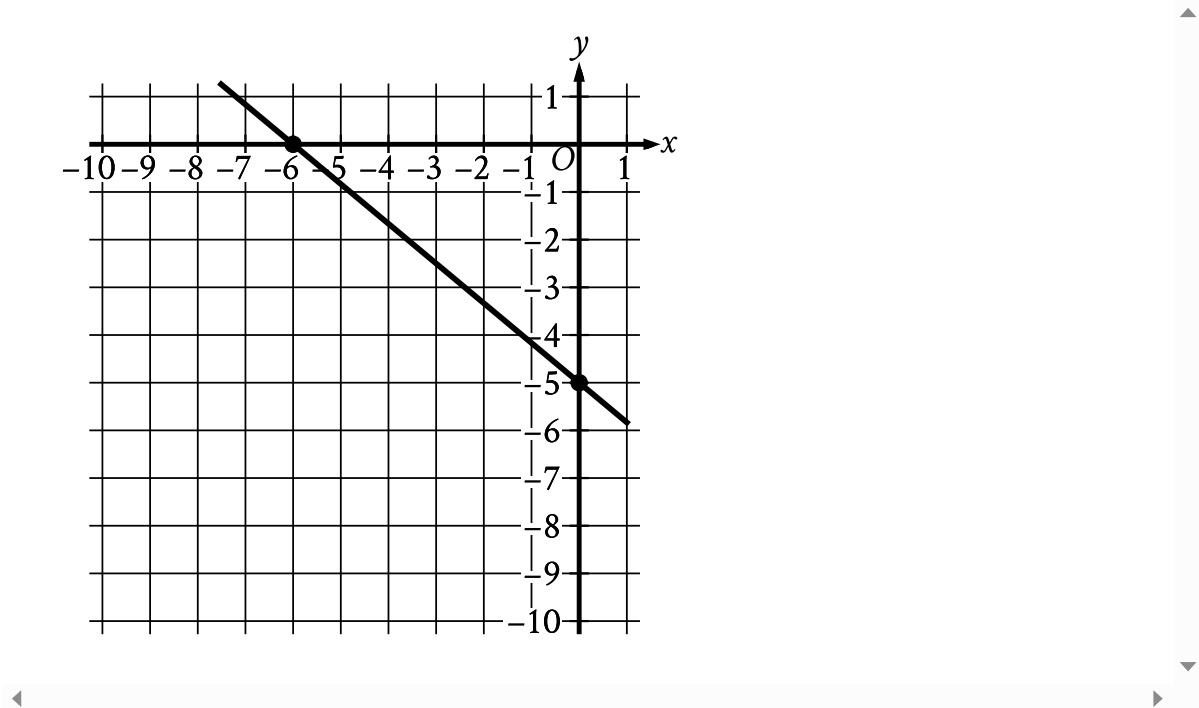
Choice D is incorrect. The total number of cups of jam in the small containers is represented by **$4x$** , not **$6y$** .

Question Difficulty: Medium

Question ID 2f5dd392

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 2f5dd392



Line *k* is shown in the *xy*-plane. Line *j* (not shown) is perpendicular to line *k*. What is the slope of line *j*?

ID: 2f5dd392 Answer

Correct Answer: 1.2, 6/5

Rationale

The correct answer is $\frac{6}{5}$. It's given that line *j* is perpendicular to line *k* in the *xy*-plane. This means that the slope of line *j* is the opposite reciprocal of the slope of line *k*. For a line that passes through the points (x_1, y_1) and (x_2, y_2) in the *xy*-plane, the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. It's shown that line *k* passes through the points $(-6, 0)$ and $(0, -5)$ in the *xy*-plane. Substituting -6 for x_1 , 0 for y_1 , 0 for x_2 , and -5 for y_2 in $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{-5 - 0}{0 - (-6)}$, or $-\frac{5}{6}$. The opposite reciprocal of $-\frac{5}{6}$ is $\frac{6}{5}$. Therefore, the slope of line *j* is $\frac{6}{5}$. Note that 6/5 and 1.2 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID afea4206

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: afea4206

A store sells two different-sized containers of blueberries. The store’s sales of these blueberries totaled **896.86** dollars last month. The equation **$4.51x + 6.07y = 896.86$** represents this situation, where **x** is the number of smaller containers sold and **y** is the number of larger containers sold. According to the equation, what is the price, in dollars, of each smaller container?

ID: afea4206 Answer

Correct Answer: 4.51, 451/100

Rationale

The correct answer is **4.51**. It’s given that the equation **$4.51x + 6.07y = 896.86$** represents this situation, where **x** is the number of smaller containers sold, **y** is the number of larger containers sold, and **896.86** is the store’s total sales, in dollars, of blueberries last month. Therefore, **$4.51x$** represents the store's sales, in dollars, of smaller containers, and **$6.07y$** represents the store's sales, in dollars, of larger containers. Since **x** is the number of smaller containers sold, the price, in dollars, of each smaller container is **4.51**.

Question Difficulty: Hard

Question ID c403eab7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: c403eab7

A computer company pays customers for trading in old computers. The payment p , in dollars, is the original retail price of a computer minus \$150 for each year since the computer was released. Which equation represents this situation for a computer with an original retail price of \$690 when it was released t years ago, where t is a whole number and $1 \leq t \leq 4$?

- A. $p = 150 - 690t$
- B. $p = 690 - 150t$
- C. $t = 150 - 690p$
- D. $t = 690 - 150p$

ID: c403eab7 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the payment p , in dollars, for trading in an old computer is the original retail price of the computer minus \$150 for each year since the computer was released t years ago. It's also given that the original retail price of a computer was \$690. Thus, the equation that represents this situation is $p = 690 - 150t$.

Choice A is incorrect. This equation represents a situation for a computer with an original retail price of \$150 minus \$690 for each year since the computer was released.

Choice C is incorrect and may result from conceptual errors.

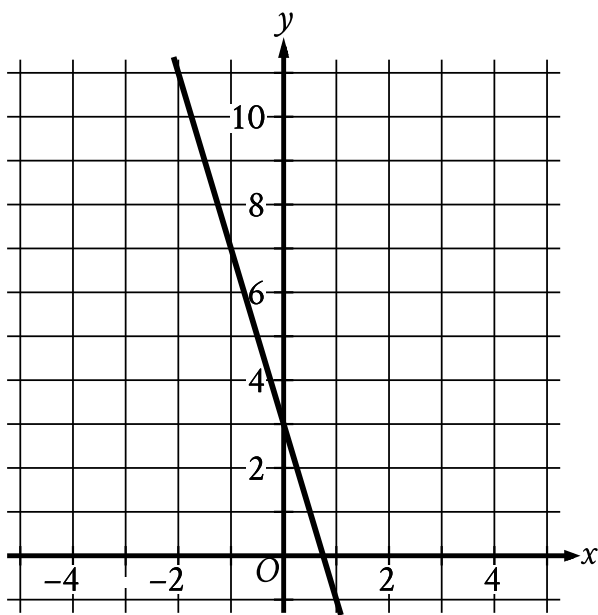
Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 2f1a5f2a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 2f1a5f2a



The graph of the linear function $y = f(x) + 19$ is shown. If c and d are positive constants, which equation could define f ?

- A. $f(x) = -d - cx$
- B. $f(x) = d - cx$
- C. $f(x) = -d + cx$
- D. $f(x) = d + cx$

ID: 2f1a5f2a Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph of the linear function $y = f(x) + 19$ is shown. This means that the graph of $y = f(x) + 19$ can be translated down 19 units to create the graph of $y = f(x)$ and the y-coordinate of every point on the graph of $y = f(x) + 19$ can be decreased by 19 to find the resulting point on the graph of $y = f(x)$. The y-intercept of the graph of $y = f(x) + 19$ is $(0, 3)$. Translating the graph of $y = f(x) + 19$ down 19 units results in a y-intercept of the graph of $y = f(x)$ at the point $(0, 3 - 19)$, or $(0, -16)$. The graph of $y = f(x) + 19$ slants down from left to right, so the slope of the graph is negative. The translation of a linear graph changes its position, but does not change its slope. It follows that the slope of the graph of $y = f(x)$ is also negative. The equation of a linear function f can be written in the form $f(x) = b + mx$, where b is the y-coordinate of the y-intercept and m is the slope of the graph of $y = f(x)$. It's given that c and d are positive constants. Since the y-coordinate of the y-intercept and the slope of the graph of $y = f(x)$ are both negative, it follows that $f(x) = -d - cx$ could define f .

Choice B is incorrect. This could define a linear function where its graph has a positive, not negative, y-intercept.

Choice C is incorrect. This could define a linear function where its graph has a positive, not negative, slope.

Choice D is incorrect. This could define a linear function where its graph has a positive, not negative, y -intercept and a positive, not negative, slope.

Question Difficulty: Hard

Question ID de24df33

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: de24df33

In the linear function h , $h(0) = 41$ and $h(1) = 40$. Which equation defines h ?

- A. $h(x) = -x + 41$
- B. $h(x) = -x$
- C. $h(x) = -41x$
- D. $h(x) = -41$

ID: de24df33 Answer

Correct Answer: A

Rationale

Choice A is correct. An equation defining a linear function can be written in the form $h(x) = ax + b$, where a and b are constants. It's given that $h(0) = 41$. Substituting 0 for x and 41 for $h(x)$ in the equation $h(x) = ax + b$ yields $41 = a(0) + b$, or $b = 41$. Substituting 41 for b in the equation $h(x) = ax + b$ yields $h(x) = ax + 41$. It's also given that $h(1) = 40$. Substituting 1 for x and 40 for $h(x)$ in the equation $h(x) = ax + 41$ yields $40 = a(1) + 41$, or $40 = a + 41$. Subtracting 41 from the left- and right-hand sides of this equation yields $-1 = a$. Substituting -1 for a in the equation $h(x) = ax + 41$ yields $h(x) = -1x + 41$, or $h(x) = -x + 41$.

Choice B is incorrect. Substituting 0 for x and 41 for $h(x)$ in this equation yields $41 = -0$, which isn't a true statement.

Choice C is incorrect. Substituting 0 for x and 41 for $h(x)$ in this equation yields $41 = -41(0)$, or $41 = 0$, which isn't a true statement.

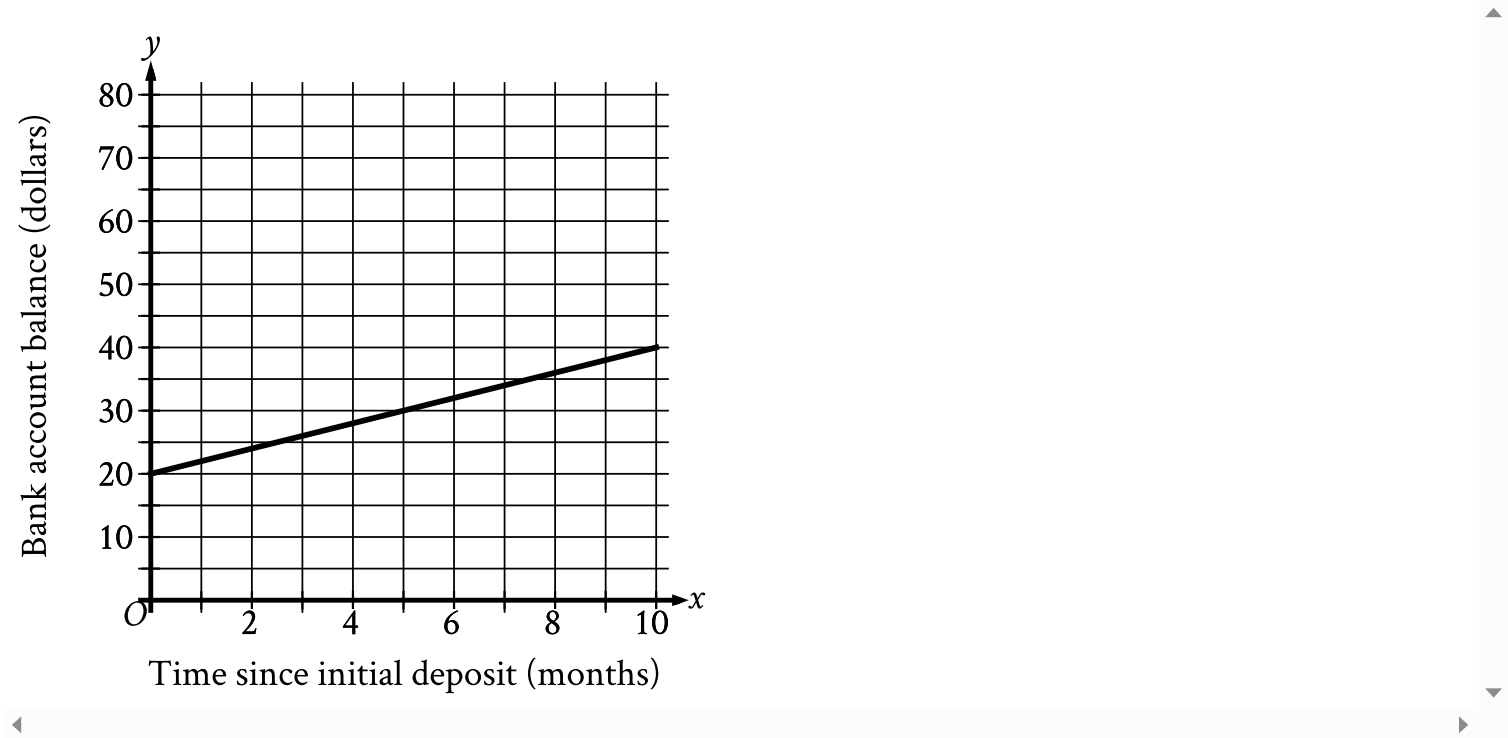
Choice D is incorrect. Substituting 41 for $h(x)$ in this equation yields $41 = -41$, which isn't a true statement.

Question Difficulty: Medium

Question ID 909d7e2b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 909d7e2b



A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

ID: 909d7e2b Answer

Correct Answer: 20

Rationale

The correct answer is **20**. For the graph shown, the x -axis represents the time since the initial deposit, in months, and the y -axis represents the bank account balance, in dollars. The amount of the initial deposit is estimated by the y -coordinate of the point on the graph that represents **0** months since the initial deposit. Therefore, the amount of the initial deposit is estimated by the corresponding y -value for the point when $x = 0$. When $x = 0$, it is estimated that $y = 20$. Thus, the amount of the initial deposit estimated by the graph, to the nearest whole dollar, is **20**.

Question Difficulty: Medium

Question ID a36030f7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: a36030f7

If $\frac{x+6}{3} = \frac{x+6}{13}$, the value of $x + 6$ is between which of the following pairs of values?

- A. -7 and -3
- B. -2 and 2
- C. 2 and 7
- D. 8 and 13

ID: a36030f7 Answer

Correct Answer: B

Rationale

Choice B is correct. Multiplying both sides of the given equation by $(3)(13)$, or 39 , yields $(39)\left(\frac{x+6}{3}\right) = (39)\left(\frac{x+6}{13}\right)$, or $13(x + 6) = 3(x + 6)$. Subtracting $3(x + 6)$ from both sides of this equation yields $10(x + 6) = 0$. Dividing both sides of this equation by 10 yields $x + 6 = 0$. Therefore, if $\frac{x+6}{3} = \frac{x+6}{13}$, then the value of $x + 6$ is 0 . It follows that of the given choices, the value of $x + 6$ is between -2 and 2 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a8c33268

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: a8c33268

A salesperson’s total earnings consist of a base salary of x dollars per year, plus commission earnings of **11%** of the total sales the salesperson makes during the year. This year, the salesperson has a goal for the total earnings to be at least **3** times and at most **4** times the base salary. Which of the following inequalities represents all possible values of total sales s , in dollars, the salesperson can make this year in order to meet that goal?

- A. $2x \leq s \leq 3x$
- B. $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$
- C. $3x \leq s \leq 4x$
- D. $\frac{3}{0.11}x \leq s \leq \frac{4}{0.11}x$

ID: a8c33268 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that a salesperson's total earnings consist of a base salary of x dollars per year plus commission earnings of **11%** of the total sales the salesperson makes during the year. If the salesperson makes s dollars in total sales this year, the salesperson’s total earnings can be represented by the expression $x + 0.11s$. It’s also given that the salesperson has a goal for the total earnings to be at least **3** times and at most **4** times the base salary, which can be represented by the expressions $3x$ and $4x$, respectively. Therefore, this situation can be represented by the inequality $3x \leq x + 0.11s \leq 4x$. Subtracting x from each part of this inequality yields $2x \leq 0.11s \leq 3x$. Dividing each part of this inequality by **0.11** yields $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$. Therefore, the inequality $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$ represents all possible values of total sales s , in dollars, the salesperson can make this year in order to meet their goal.

Choice A is incorrect. This inequality represents a situation in which the total sales, rather than the total earnings, are at least **2** times and at most **3** times, rather than at least **3** times and at most **4** times, the base salary.

Choice C is incorrect. This inequality represents a situation in which the total sales, rather than the total earnings, are at least **3** times and at most **4** times the base salary.

Choice D is incorrect. This inequality represents a situation in which the total earnings are at least **4** times and at most **5** times, rather than at least **3** times and at most **4** times, the base salary.

Question Difficulty: Hard

Question ID bd6a65aa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: bd6a65aa

One gallon of paint will cover **220** square feet of a surface. A room has a total wall area of w square feet. Which equation represents the total amount of paint P , in gallons, needed to paint the walls of the room twice?

- A. $P = \frac{w}{110}$
- B. $P = 440w$
- C. $P = \frac{w}{220}$
- D. $P = 220w$

ID: bd6a65aa Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that w represents the total wall area, in square feet. Since the walls of the room will be painted twice, the amount of paint, in gallons, needs to cover $2w$ square feet. It's also given that one gallon of paint will cover **220** square feet. Dividing the total area, in square feet, of the surface to be painted by the number of square feet covered by one gallon of paint gives the number of gallons of paint that will be needed. Dividing $2w$ by **220** yields $\frac{2w}{220}$, or $\frac{w}{110}$. Therefore, the equation that represents the total amount of paint P , in gallons, needed to paint the walls of the room twice is $P = \frac{w}{110}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from finding the amount of paint needed to paint the walls once rather than twice.

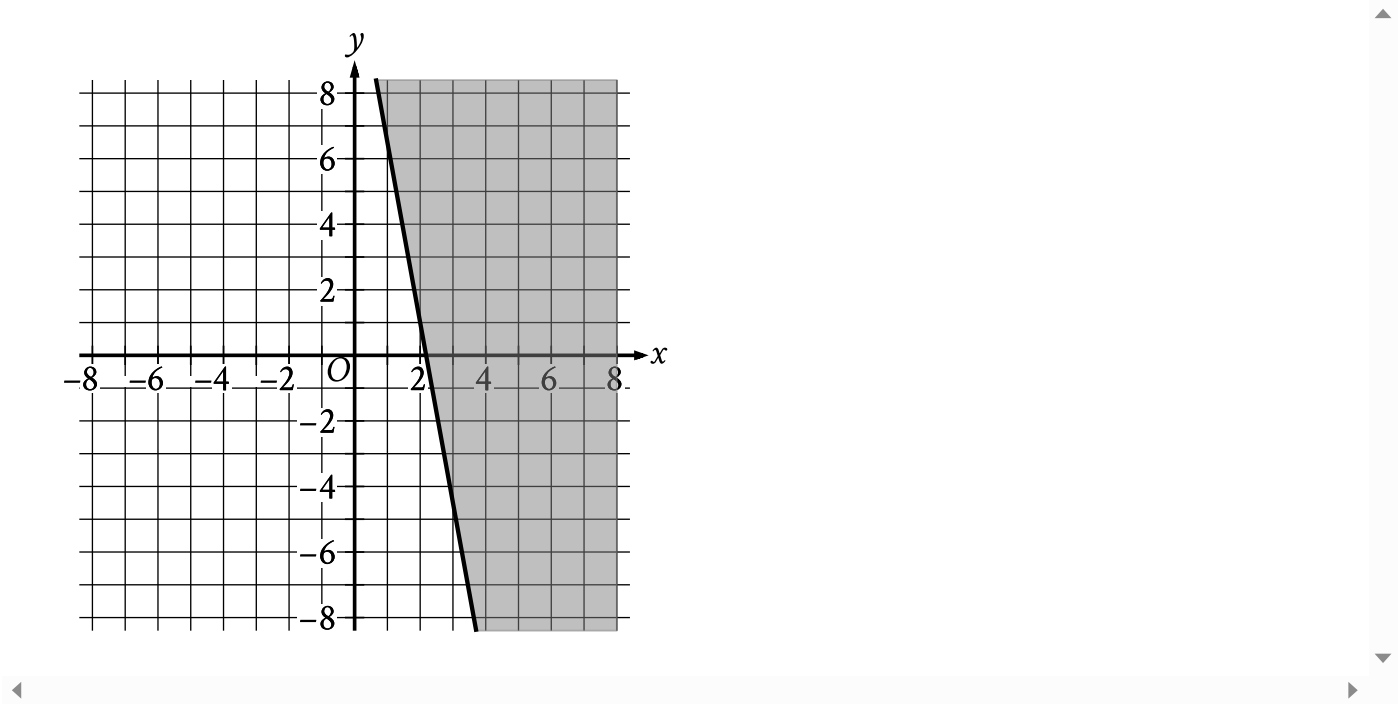
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e7976caa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: e7976caa



The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

- A. $(0, -4)$
- B. $(0, 4)$
- C. $(-4, 0)$
- D. $(4, 0)$

ID: e7976caa Answer

Correct Answer: D

Rationale

Choice D is correct. Since the shaded region shown represents solutions to an inequality, an ordered pair (x, y) is a solution to the inequality if it's represented by a point in the shaded region. Of the given choices, only $(4, 0)$ is represented by a point in the shaded region. Therefore, $(4, 0)$ is a solution to the inequality.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID b3e6012b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: b3e6012b

To repair a refrigerator, a technician charges \$60 per hour for labor plus \$120 for parts. Which function f represents the total amount, in dollars, the technician will charge for this job if it takes x hours?

- A. $f(x) = x + 120$
- B. $f(x) = 60x$
- C. $f(x) = 60x + 120$
- D. $f(x) = 60x - 120$

ID: b3e6012b Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the technician charges \$60 per hour for labor. Therefore, if the job takes x hours, the technician will charge $\left(\frac{\$60}{1 \text{ hour}}\right)(x \text{ hours})$, or $\$60x$, for labor. It's also given that the technician charges \$120 for parts. Therefore, $f(x) = 60x + 120$ represents the total amount, in dollars, the technician will charge for this job if it takes x hours.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This function represents the total amount, in dollars, the technician charges for labor only, not the total amount charged for labor and parts.

Choice D is incorrect. This function represents the total amount, in dollars, the technician would charge if the charge for parts were subtracted from, rather than added to, the charge for labor.

Question Difficulty: Easy

Question ID 2251c98f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 2251c98f

Connor has c dollars and Maria has m dollars. Connor has 4 times as many dollars as Maria, and together they have a total of \$25.00. Which system of equations represents this situation?

- A. $c = 4m$
 $c + m = 25$
- B. $m = 4c$
 $c + m = 25$
- C. $c = 25m$
 $c + m = 4$
- D. $m = 25c$
 $c + m = 4$

ID: 2251c98f Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that Connor has c dollars, Maria has m dollars, and Connor has 4 times as many dollars as Maria. This can be represented by the equation $c = 4m$. It's also given that together, Connor and Maria have a total of \$25.00, which can be represented by the equation $c + m = 25$. Therefore, the system consisting of the equations $c = 4m$ and $c + m = 25$ represents this situation.

Choice B is incorrect. The equation $m = 4c$ represents a situation where Maria has 4 times as many dollars as Connor, rather than the situation where Connor has 4 times as many dollars as Maria.

Choice C is incorrect. The equation $c = 25m$ represents a situation where Connor has 25 times, rather than 4 times, as many dollars as Maria. The equation $c + m = 4$ represents a situation where Connor and Maria together have a total of \$4.00, rather than \$25.00.

Choice D is incorrect. The equation $m = 25c$ represents a situation where Maria has 25 times as many dollars as Connor, rather than the situation where Connor has 4 times as many dollars as Maria. The equation $c + m = 4$ represents a situation where Connor and Maria together have a total of \$4.00, rather than \$25.00.

Question Difficulty: Easy

Question ID e4eb14ec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: e4eb14ec

If $6 + x = 9$, what is the value of $18 + 3x$?

ID: e4eb14ec Answer

Correct Answer: 27

Rationale

The correct answer is **27**. Multiplying both sides of the given equation by **3** yields **$3(6 + x) = 3(9)$** , or **$18 + 3x = 27$** . Therefore, the value of **$18 + 3x$** is **27**.

Question Difficulty: Medium

Question ID b9890078

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: b9890078

A rocket contained **467,000** kilograms (kg) of propellant before launch. Exactly **21** seconds after launch, **362,105** kg of this propellant remained. On average, approximately how much propellant, in kg, did the rocket burn each second after launch?

- A. **4,995**
- B. **17,243**
- C. **39,481**
- D. **104,895**

ID: b9890078 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the rocket contained **467,000 kilograms (kg)** of propellant before launch and had **362,105 kg** remaining exactly **21** seconds after launch. Finding the difference between the amount, in **kg**, of propellant before launch and the remaining amount, in **kg**, of propellant after launch gives the amount, in **kg**, of propellant burned during the **21** seconds: **467,000 — 362,105 = 104,895**. Dividing the amount of propellant burned by the number of seconds yields $\frac{104,895}{21} = 4,995$. Thus, an average of **4,995 kg** of propellant burned each second after launch.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

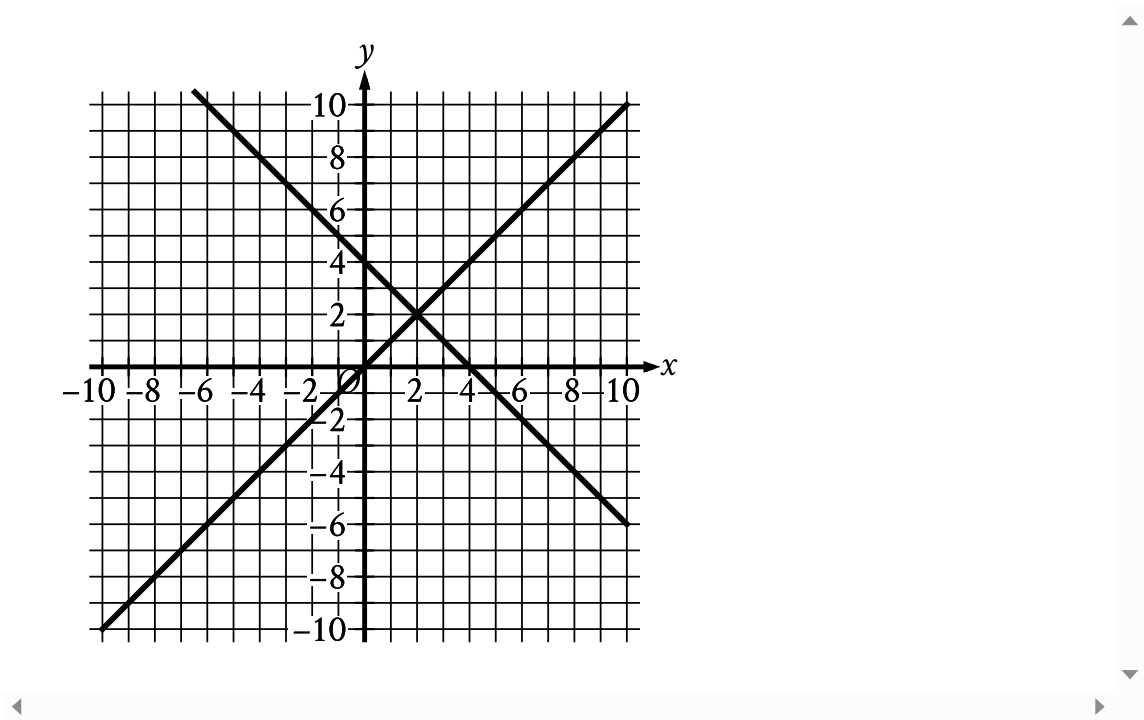
Choice D is incorrect and may result from finding the amount of propellant burned, rather than the amount of propellant burned each second.

Question Difficulty: Medium

Question ID 52a3fa93

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 52a3fa93



The graph of a system of two linear equations is shown. What is the solution (x, y) to the system?

- A. $(0, 4)$
- B. $(2, 2)$
- C. $(4, 0)$
- D. $(4, 4)$

ID: 52a3fa93 Answer

Correct Answer: B

Rationale

Choice B is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 2$ and $y = 2$, or at the point $(2, 2)$. Therefore, the solution (x, y) to the system is $(2, 2)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 336e802a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 336e802a

Scientists collected fallen acorns that each housed a colony of the ant species *P. ohioensis* and analyzed each colony's structure. For any of these colonies, if the colony has x worker ants, the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y , in the colony. If one of these colonies has 58 worker ants, which of the following is closest to the predicted number of larvae in the colony?

- A. 41
- B. 61
- C. 83
- D. 190

ID: 336e802a Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, y , in a colony of ants if the colony has x worker ants. If one of these colonies has 58 worker ants, the predicted number of larvae in that colony can be found by substituting 58 for x in the given equation. Substituting 58 for x in the given equation yields $y = 0.67(58) + 2.6$, or $y = 41.46$. Of the given choices, 41 is closest to the predicted number of larvae in the colony.

Choice B is incorrect. This is closest to the predicted number of larvae in a colony with 87 worker ants.

Choice C is incorrect. This is closest to the number of worker ants for which the predicted number of larvae in a colony is 58.

Choice D is incorrect. This is closest to the predicted number of larvae in a colony with 280 worker ants.

Question Difficulty: Medium

Question ID be8e1f90

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: be8e1f90

Vivian bought party hats and cupcakes for **\$71**. Each package of party hats cost **\$3**, and each cupcake cost **\$1**. If Vivian bought **10** packages of party hats, how many cupcakes did she buy?

ID: be8e1f90 Answer

Correct Answer: 41

Rationale

The correct answer is **41**. The number of cupcakes Vivian bought can be found by first finding the amount Vivian spent on cupcakes. The amount Vivian spent on cupcakes can be found by subtracting the amount Vivian spent on party hats from the total amount Vivian spent. The amount Vivian spent on party hats can be found by multiplying the cost per package of party hats by the number of packages of party hats, which yields **$\$3 \cdot 10$** , or **\$30**. Subtracting the amount Vivian spent on party hats, **\$30**, from the total amount Vivian spent, **\$71**, yields **$\$71 - \$30$** , or **\$41**. Since the amount Vivian spent on cupcakes was **\$41** and each cupcake cost **\$1**, it follows that Vivian bought **41** cupcakes.

Question Difficulty: Medium

Question ID f707b663

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: f707b663

x	y
0	18
1	13
2	8

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A. $y = 18x + 13$
- B. $y = 18x + 18$
- C. $y = -5x + 13$
- D. $y = -5x + 18$

ID: f707b663 Answer

Correct Answer: D

Rationale

Choice D is correct. A linear relationship can be represented by an equation of the form $y = mx + b$, where m and b are constants. It's given in the table that when $x = 0$, $y = 18$. Substituting 0 for x and 18 for y in $y = mx + b$ yields $18 = m(0) + b$, or $18 = b$. Substituting 18 for b in the equation $y = mx + b$ yields $y = mx + 18$. It's also given in the table that when $x = 1$, $y = 13$. Substituting 1 for x and 13 for y in the equation $y = mx + 18$ yields $13 = m(1) + 18$, or $13 = m + 18$. Subtracting 18 from both sides of this equation yields $-5 = m$. Therefore, the equation $y = -5x + 18$ represents the relationship between x and y .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID e81b7277

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: e81b7277

$y < 5x + 6$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	17
5	27
7	37

B.

x	y
3	17
5	35
7	37

C.

x	y
3	25
5	35
7	45

D.

x	y
3	21
5	31
7	41

ID: e81b7277 Answer

Correct Answer: A

Rationale

Choice A is correct. Substituting **3** for x in the given inequality yields $y < 5(3) + 6$, or $y < 21$. Therefore, when $x = 3$, the corresponding value of y is less than **21**. Substituting **5** for x in the given inequality yields $y < 5(5) + 6$, or $y < 31$. Therefore, when $x = 5$, the corresponding value of y is less than **31**. Substituting **7** for x in the given inequality yields $y < 5(7) + 6$, or $y < 41$. Therefore, when $x = 7$, the corresponding value of y is less than **41**. For the table in choice A, when $x = 3$, the corresponding value of y is **17**, which is less than **21**; when $x = 5$, the corresponding value of y is **27**, which is less than **31**; and

when $x = 7$, the corresponding value of y is **37**, which is less than **41**. Therefore, the table in choice A gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 3f6c5272

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: 3f6c5272

A truck can haul a maximum weight of **5,630** pounds. During one trip, the truck will be used to haul a **190**-pound piece of equipment as well as several crates. Some of these crates weigh **25** pounds each and the others weigh **62** pounds each. Which inequality represents the possible combinations of the number of **25**-pound crates, ***x***, and the number of **62**-pound crates, ***y***, the truck can haul during one trip if only the piece of equipment and the crates are being hauled?

- A. $25x + 62y \leq 5,440$
- B. $25x + 62y \geq 5,440$
- C. $62x + 25y \leq 5,630$
- D. $62x + 25y \geq 5,630$

ID: 3f6c5272 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that a truck can haul a maximum of **5,630** pounds. It's also given that during one trip, the truck will be used to haul a **190**-pound piece of equipment as well as several crates. It follows that the truck can haul at most **5,630 — 190**, or **5,440**, pounds of crates. Since ***x*** represents the number of **25**-pound crates, the expression **25*x*** represents the weight of the **25**-pound crates. Since ***y*** represents the number of **62**-pound crates, **62*y*** represents the weight of the **62**-pound crates. Therefore, **25*x* + 62*y*** represents the total weight of the crates the truck can haul. Since the truck can haul at most **5,440** pounds of crates, the total weight of the crates must be less than or equal to **5,440** pounds, or **25*x* + 62*y* ≤ 5,440**.

Choice B is incorrect. This represents the possible combinations of the number of **25**-pound crates, ***x***, and the number of **62**-pound crates, ***y***, the truck can haul during one trip if it can haul a minimum, not a maximum, of **5,630** pounds.

Choice C is incorrect. This represents the possible combinations of the number of **62**-pound crates, ***x***, and the number of **25**-pound crates, ***y***, the truck can haul during one trip if only crates are being hauled.

Choice D is incorrect. This represents the possible combinations of the number of **62**-pound crates, ***x***, and the number of **25**-pound crates, ***y***, the truck can haul during one trip if it can haul a minimum, not a maximum, weight of **5,630** pounds and only crates are being hauled.

Question Difficulty: Hard

Question ID 7b1c35e3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 7b1c35e3

Figure A and figure B are both regular polygons. The sum of the perimeter of figure A and the perimeter of figure B is **63** inches. The equation $3x + 6y = 63$ represents this situation, where x is the number of sides of figure A and y is the number of sides of figure B. Which statement is the best interpretation of **6** in this context?

- A. Each side of figure B has a length of **6** inches.
- B. The number of sides of figure B is **6**.
- C. Each side of figure A has a length of **6** inches.
- D. The number of sides of figure A is **6**.

ID: 7b1c35e3 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that figure A and figure B (not shown) are both regular polygons and the sum of the perimeters of the two figures is **63 inches**. It's also given that x is the number of sides of figure A and y is the number of sides of figure B, and that the equation $3x + 6y = 63$ represents this situation. Thus, $3x$ and $6y$ represent the perimeters, in inches, of figure A and figure B, respectively. Since $6y$ represents the perimeter, in inches, of figure B and y is the number of sides of figure B, it follows that each side of figure B has a length of **6 inches**.

Choice B is incorrect. The number of sides of figure B is y , not **6**.

Choice C is incorrect. Since the perimeter, in inches, of figure A is represented by $3x$, each side of figure A has a length of **3 inches**, not **6 inches**.

Choice D is incorrect. The number of sides of figure A is x , not **6**.

Question Difficulty: Hard

Question ID a75d2a8c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: a75d2a8c

What is the equation of the line that passes through the point $(0, 5)$ and is parallel to the graph of $y = 7x + 4$ in the xy -plane?

- A. $y = 5x$
- B. $y = 7x + 5$
- C. $y = 7x$
- D. $y = 5x + 7$

ID: a75d2a8c Answer

Correct Answer: B

Rationale

Choice B is correct. The equation of a line in the xy -plane can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and $(0, b)$ is its y -intercept. It's given that the line passes through the point $(0, 5)$. Therefore, $b = 5$. It's also given that the line is parallel to the graph of $y = 7x + 4$, which means the line has the same slope as the graph of $y = 7x + 4$. The slope of the graph of $y = 7x + 4$ is 7 . Therefore, $m = 7$. Substituting 7 for m and 5 for b in the equation $y = mx + b$ yields $y = 7x + 5$.

Choice A is incorrect. The graph of this equation passes through the point $(0, 0)$, not $(0, 5)$, and has a slope of 5 , not 7 .

Choice C is incorrect. The graph of this equation passes through the point $(0, 0)$, not $(0, 5)$.

Choice D is incorrect. The graph of this equation passes through the point $(0, 7)$, not $(0, 5)$, and has a slope of 5 , not 7 .

Question Difficulty: Medium

Question ID c717a869

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: c717a869

x	y
18	130
23	160
26	178

For line h , the table shows three values of x and their corresponding values of y . Line k is the result of translating line h down 5 units in the xy -plane. What is the x -intercept of line k ?

- A. $(-\frac{26}{3}, 0)$
- B. $(-\frac{9}{2}, 0)$
- C. $(-\frac{11}{3}, 0)$
- D. $(-\frac{17}{6}, 0)$

ID: c717a869 Answer

Correct Answer: D

Rationale

Choice D is correct. The equation of line h can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the y -intercept of the line. It's given that line h contains the points $(18, 130)$, $(23, 160)$, and $(26, 178)$. Therefore, its slope m can be found as $\frac{160-130}{23-18}$, or 6. Substituting 6 for m in the equation $y = mx + b$ yields $y = 6x + b$. Substituting 130 for y and 18 for x in this equation yields $130 = 6(18) + b$, or $130 = 108 + b$. Subtracting 108 from both sides of this equation yields $22 = b$. Substituting 22 for b in $y = 6x + b$ yields $y = 6x + 22$. Since line k is the result of translating line h down 5 units, an equation of line k is $y = 6x + 22 - 5$, or $y = 6x + 17$. Substituting 0 for y in this equation yields $0 = 6x + 17$. Solving this equation for x yields $x = -\frac{17}{6}$. Therefore, the x -intercept of line k is $(-\frac{17}{6}, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a0755945

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: a0755945

The function $f(x) = 55.20 - 0.16x$ gives the estimated surface water temperature $f(x)$, in degrees Celsius, of a body of water on the x th day of the year, where $220 \leq x \leq 360$. Based on the model, what is the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year?

- A. 55.20
- B. 3.04
- C. -0.16
- D. -52.16

ID: a0755945 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the function $f(x) = 55.20 - 0.16x$ gives the estimated surface water temperature, in degrees Celsius, of a body of water on the x th day of the year. Substituting 326 for x in the given function yields $f(326) = 55.20 - 0.16(326)$, which is equivalent to $f(326) = 55.20 - 52.16$, or $f(326) = 3.04$. Therefore, the estimated surface water temperature, in degrees Celsius, of this body of water on the 326th day of the year is 3.04.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the rate of change, in degrees Celsius per day, of the estimated surface water temperature.

Choice D is incorrect. This is the change, in degrees Celsius, in the estimated surface water temperature over 326 days.

Question Difficulty: Medium

Question ID 481a2c00

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 481a2c00

$s(x) = 133 - 4x$

The function s shown gives the number of spoons, $s(x)$, remaining in a dispenser x minutes after the dispenser had been loaded with spoons. Which statement is the best interpretation of $s(3) = 121$?

- A. The number of spoons remaining in the dispenser decreased by a total of 3 spoons after 121 minutes.
- B. There were 3 spoons remaining in the dispenser 121 minutes after the dispenser had been loaded with spoons.
- C. There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons.
- D. The number of spoons remaining in the dispenser decreased by a total of 121 spoons after 3 minutes.

ID: 481a2c00 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $s(x)$ is the number of spoons remaining in a dispenser x minutes after the dispenser had been loaded with spoons. The expression $s(3)$ represents the value of $s(x)$ when $x = 3$. It follows that $s(3)$ represents the number of spoons remaining in the dispenser 3 minutes after it had been loaded with spoons. Therefore, the best interpretation of the equation $s(3) = 121$ is that there were 121 spoons remaining in the dispenser 3 minutes after it had been loaded with spoons.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This would be the best interpretation of $s(121) = 3$, not $s(3) = 121$.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID aab16dcc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: aab16dcc

To complete a landscaping project, Adam charges a fee of **\$20.00** for his equipment and **\$9.50** per hour spent working on the project. To complete this same landscaping project, Caroline charges a fee of **\$17.00** for her equipment and **\$10.00** per hour spent working on the project. If x represents the number of hours spent working on the landscaping project, what are all the values of x for which Caroline's total charge is greater than Adam's total charge?

- A. $5 \leq x \leq 6$
- B. $6 \leq x \leq 7$
- C. $x < 5$
- D. $x > 6$

ID: aab16dcc Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that Adam charges a fee of **\$20.00** for his equipment and **\$9.50** per hour spent working on the project. For x hours spent working on the project, it follows that Adam's total charge, in dollars, is $20.00 + 9.50x$. It's also given that Caroline charges a fee of **\$17.00** for her equipment and **\$10.00** per hour spent working on the project. For x hours spent working on the project, it follows that Caroline's total charge, in dollars, is $17.00 + 10.00x$. Therefore, if x represents the number of hours spent working on the landscaping project, then Caroline's total charge is greater than Adam's total charge if $17.00 + 10.00x > 20.00 + 9.50x$, or $17 + 10x > 20 + 9.5x$. Subtracting $9.5x$ from both sides of this inequality yields $17 + 0.5x > 20$. Subtracting 17 from both sides of this inequality yields $0.5x > 3$. Dividing both sides of this inequality by 0.5 yields $x > 6$.

Choice A is incorrect. For example, if $x = 5$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(5)$, or **67**, and Adam's total charge, in dollars, would be $20.00 + 9.50(5)$, or **67.50**. Since Caroline's total charge would be less than Adam's total charge, **5** can't be a possible value of x .

Choice B is incorrect. For example, if $x = 6$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(6)$, or **77**, and Adam's total charge, in dollars, would be $20.00 + 9.50(6)$, or **77**. Since Caroline's total charge would be equal to Adam's total charge, **6** can't be a possible value of x .

Choice C is incorrect. For example, if $x = 4$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(4)$, or **57**, and Adam's total charge, in dollars, would be $20.00 + 9.50(4)$, or **58**. Since Caroline's total charge would be less than Adam's total charge, **4** can't be a possible value of x .

Question Difficulty: Hard

Question ID fd5ffee5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: fd5ffee5

$g(x) = 11x + 4$

For the given linear function g , which table shows three values of x and their corresponding values of $g(x)$?

A.

x	$g(x)$
-1	7
0	11
1	15

B.

x	$g(x)$
-1	-4
0	0
1	4

C.

x	$g(x)$
-1	-7
0	4
1	15

D.

x	$g(x)$
-1	-11
0	0
1	11

ID: fd5ffee5 Answer

Correct Answer: C

Rationale

Choice C is correct. Each of the tables shows the same three values of x : -1 , 0 , and 1 . Substituting -1 for x in the given function yields $g(-1) = 11(-1) + 4$, or $g(-1) = -7$. Therefore, when $x = -1$, the corresponding value of $g(x)$ is -7 . Substituting 0 for x in the given function yields $g(0) = 11(0) + 4$, or $g(0) = 4$. Therefore, when $x = 0$, the corresponding value of $g(x)$ is 4 . Substituting 1 for x in the given function yields $g(1) = 11(1) + 4$, or $g(1) = 15$. Therefore, when $x = 1$, the corresponding value of $g(x)$ is 15 . The table in choice C shows -7 , 4 , and 15 as the corresponding value of $g(x)$ for x -values of -1 , 0 , and 1 , respectively. Therefore, the table in choice C shows three values of x and their corresponding values of $g(x)$.

Choice A is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 4x + 11$.

Choice B is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 4x$.

Choice D is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 11x$.

Question Difficulty: Medium

Question ID 85f211b6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 85f211b6

The function f is defined by $f(x) = -3x + 60$. What is the value of $f(x)$ when $x = -8$?

- A. 49
- B. 52
- C. 57
- D. 84

ID: 85f211b6 Answer

Correct Answer: D

Rationale

Choice D is correct. The value of $f(x)$ when $x = -8$ can be found by substituting -8 for x in the given function. This yields $f(-8) = -3(-8) + 60$, or $f(-8) = 84$. Therefore, when $x = -8$, the value of $f(x)$ is 84.

Choice A is incorrect. This is the value of $(-3 + (-8)) + 60$, not $-3(-8) + 60$.

Choice B is incorrect. This is the value of $-8 + 60$, not $-3(-8) + 60$.

Choice C is incorrect. This is the value of $-3 + 60$, not $-3(-8) + 60$.

Question Difficulty: Easy

Question ID 36f22674

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 36f22674

Kaylani used fabric measuring 5 yards in length to make each suit for a men's choir. The relationship between the number of suits that Kaylani made, x , and the total length of fabric that she purchased y , in yards, is represented by the equation $y - 5x = 6$. What is the best interpretation of 6 in this context?

- A. Kaylani made 6 suits.
- B. Kaylani purchased a total of 6 yards of fabric.
- C. Kaylani used a total of 6 yards of fabric to make the suits.
- D. Kaylani purchased 6 yards more fabric than she used to make the suits.

ID: 36f22674 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the equation $y - 5x = 6$ represents the relationship between the number of suits that Kaylani made, x , and the total length of fabric she purchased, y , in yards. Adding $5x$ to both sides of the given equation yields $y = 5x + 6$. Since Kaylani made x suits and used 5 yards of fabric to make each suit, the expression $5x$ represents the total amount of fabric she used to make the suits. Since y represents the total length of fabric Kaylani purchased, in yards, it follows from the equation $y = 5x + 6$ that Kaylani purchased $5x$ yards of fabric to make the suits, plus an additional 6 yards of fabric. Therefore, the best interpretation of 6 in this context is that Kaylani purchased 6 yards more fabric than she used to make the suits.

Choice A is incorrect. Kaylani made a total of x suits, not 6 suits.

Choice B is incorrect. Kaylani purchased a total of y yards of fabric, not a total of 6 yards of fabric.

Choice C is incorrect. Kaylani used a total of $5x$ yards of fabric to make the suits, not a total of 6 yards of fabric.

Question Difficulty: Hard

Question ID 5ed72ba4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 5ed72ba4

If $3x = 30$, what is the value of $3x - 12$?

- A. -2
- B. 18
- C. 22
- D. 42

ID: 5ed72ba4 Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting 12 from each side of the given equation yields $3x - 12 = 30 - 12$, or $3x - 12 = 18$. Therefore, the value of $3x - 12$ is 18 .

Choice A is incorrect. This is the value of $x - 12$, not $3x - 12$.

Choice C is incorrect. This is the value of $x + 12$, not $3x - 12$.

Choice D is incorrect. This is the value of $3x + 12$, not $3x - 12$.

Question Difficulty: Easy

Question ID 6b5c6ffa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 6b5c6ffa

x	y
k	13
$k + 7$	-15

The table gives the coordinates of two points on a line in the xy -plane. The y -intercept of the line is $(k - 5, b)$, where k and b are constants. What is the value of b ?

ID: 6b5c6ffa Answer

Correct Answer: 33

Rationale

The correct answer is **33**. It's given in the table that the coordinates of two points on a line in the xy -plane are $(k, 13)$ and $(k + 7, -15)$. The y -intercept is another point on the line. The slope computed using any pair of points from the line will be the same. The slope of a line, m , between any two points, (x_1, y_1) and (x_2, y_2) , on the line can be calculated using the slope formula, $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$. It follows that the slope of the line with the given points from the table, $(k, 13)$ and $(k + 7, -15)$, is $m = \frac{-15 - 13}{k + 7 - k}$, which is equivalent to $m = \frac{-28}{7}$, or $m = -4$. It's given that the y -intercept of the line is $(k - 5, b)$. Substituting -4 for m and the coordinates of the points $(k - 5, b)$ and $(k, 13)$ into the slope formula yields $-4 = \frac{13 - b}{k - (k - 5)}$, which is equivalent to $-4 = \frac{13 - b}{k - k + 5}$, or $-4 = \frac{13 - b}{5}$. Multiplying both sides of this equation by **5** yields $-20 = 13 - b$. Subtracting **13** from both sides of this equation yields $-33 = -b$. Dividing both sides of this equation by -1 yields $b = 33$. Therefore, the value of b is **33**.

Question Difficulty: Hard

Question ID 5d030e69

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 5d030e69

$$f(x) = 45x + 600$$

The function f gives the monthly fee $f(x)$, in dollars, a facility charges to keep x crates in storage. What is the monthly fee, in dollars, the facility charges to keep 50 crates in storage?

ID: 5d030e69 Answer

Correct Answer: 2850

Rationale

The correct answer is **2,850**. It's given that the function $f(x) = 45x + 600$ gives the monthly fee, in dollars, a facility charges to keep x crates in storage. Substituting 50 for x in this function yields $f(50) = 45(50) + 600$, or $f(50) = 2,850$. Therefore, the monthly fee, in dollars, the facility charges to keep 50 crates in storage is **2,850**.

Question Difficulty: Medium

Question ID a443befa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: a443befa

x	$g(x)$
1	54
2	51
3	48
4	45

For the linear function g , the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b ?

- A. 3
- B. 27
- C. 54
- D. 57

ID: a443befa Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that for the linear function g , the table shows four values of x and their corresponding values of $g(x)$. It's also given that the function can be written as $g(x) = mx + b$, where m and b are constants. The table shows that when the value of x is 1, the corresponding value of $g(x)$ is 54. Substituting 1 for x and 54 for $g(x)$ in $g(x) = mx + b$ yields $54 = m(1) + b$ or, $54 = m + b$. Subtracting b from both sides of this equation yields $54 - b = m$. The table also shows that when the value of x is 2, the corresponding value of $g(x)$ is 51. Substituting 2 for x and 51 for $g(x)$ in $g(x) = mx + b$ yields $51 = m(2) + b$, or $51 = 2m + b$. Substituting $54 - b$ for m in this equation yields $51 = 2(54 - b) + b$. Applying the distributive property to the right-hand side of this equation yields $51 = 108 - 2b + b$, or $51 = 108 - b$. Subtracting 108 from both sides of this equation yields $-57 = -b$. Dividing both sides of this equation by -1 yields $57 = b$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 9940011a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 9940011a

$w + 7 = 357$

What value of w is the solution to the given equation?

- A. 51
- B. 350
- C. 364
- D. 3,577

ID: 9940011a Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting 7 from each side of the given equation yields $w = 350$. Therefore, the value of w that is the solution to the given equation is 350.

Choice A is incorrect. This is the value of w that is the solution to the equation $7w = 357$, not $w + 7 = 357$.

Choice C is incorrect. This is the value of w that is the solution to the equation $w - 7 = 357$, not $w + 7 = 357$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID d8150e94

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: d8150e94

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

- A. -4
- B. 8
- C. 42
- D. 44

ID: d8150e94 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(x) = x + 6$. Substituting 2 for x in the function f yields $f(2) = 2 + 6$, or $f(2) = 8$. It's also given that $g(x) = 6x$. Substituting 2 for x in the function g yields $g(2) = (6)(2)$, or $g(2) = 12$. Substituting 8 for $f(2)$ and 12 for $g(2)$ in the expression $7f(2) - g(2)$ yields $7(8) - 12$, or 44 . Therefore, the value of $7f(2) - g(2)$ is 44 .

Choice A is incorrect. This is the value of $f(2) - g(2)$, not $7f(2) - g(2)$.

Choice B is incorrect. This is the value of $f(2)$, not $7f(2) - g(2)$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8e92a0bc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 8e92a0bc

$$\frac{1}{4}(x + 5) - \frac{1}{3}(x + 5) = -7$$

What value of x is the solution to the given equation?

- A. -12
- B. -5
- C. **79**
- D. 204

ID: 8e92a0bc Answer

Correct Answer: C

Rationale

Choice C is correct. For the given equation, $(x + 5)$ is a factor of both terms on the left-hand side. Therefore, the given equation can be rewritten as $(\frac{1}{4} - \frac{1}{3})(x + 5) = -7$, or $(\frac{3}{12} - \frac{4}{12})(x + 5) = -7$, which is equivalent to $-\frac{1}{12}(x + 5) = -7$. Multiplying both sides of this equation by -12 yields $x + 5 = 84$. Subtracting 5 from both sides of this equation yields $x = 79$.

Choice A is incorrect. This is the value of x for which the left-hand side of the given equation equals $\frac{7}{12}$, not -7 .

Choice B is incorrect. This is the value of x for which the left-hand side of the given equation equals 0 , not -7 .

Choice D is incorrect. This is the value of x for which the left-hand side of the given equation equals $-\frac{209}{12}$, not -7 .

Question Difficulty: Medium

Question ID d4907440

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear inequalities in one or two variables	

ID: d4907440

$$11x + 14y \leq 115$$

Anthony will spend at most \$115 to purchase x small cheese pizzas and y large cheese pizzas for a team dinner. The given inequality represents this situation. Which of the following is the best interpretation of $14y$ in this context?

- A. The amount, in dollars, Anthony will spend on each large cheese pizza
- B. The amount, in dollars, Anthony will spend on each small cheese pizza
- C. The total amount, in dollars, Anthony will spend on large cheese pizzas
- D. The total amount, in dollars, Anthony will spend on small cheese pizzas

ID: d4907440 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that Anthony will spend at most \$115 to purchase x small cheese pizzas and y large cheese pizzas. In the inequality $11x + 14y \leq 115$, y represents the number of large cheese pizzas that Anthony will purchase. This means the coefficient 14 represents the amount, in dollars, Anthony will spend on each large cheese pizza. Therefore, the best interpretation of $14y$ in this context is the total amount, in dollars, Anthony will spend on large cheese pizzas.

Choice A is incorrect. This is the best interpretation of 14 , not $14y$.

Choice B is incorrect. This is the best interpretation of 11 , not $14y$.

Choice D is incorrect. This is the best interpretation of $11x$, not $14y$.

Question Difficulty: Hard

Question ID 29f70e3b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 29f70e3b

$$\begin{aligned}x &= 8 \\ x + 3y &= 26\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: 29f70e3b Answer

Correct Answer: 6

Rationale

The correct answer is **6**. The first equation in the given system is $x = 8$. Substituting **8** for x in the second equation in the given system yields $8 + 3y = 26$. Subtracting **8** from both sides of this equation yields $3y = 18$. Dividing both sides of this equation by **3** yields $y = 6$. Therefore, the value of y is **6**.

Question Difficulty: Easy

Question ID 244273dd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 244273dd

In the xy -plane, line k passes through the points $(0, -5)$ and $(1, -1)$. Which equation defines line k ?

- A. $y = -x + \frac{1}{4}$
- B. $y = \frac{1}{4}x - 5$
- C. $y = -x + 4$
- D. $y = 4x - 5$

ID: 244273dd Answer

Correct Answer: D

Rationale

Choice D is correct. An equation defining a line in the xy -plane can be written in the form $y = mx + b$, where m represents the slope and $(0, b)$ represents the y -intercept of the line. It's given that line k passes through the point $(0, -5)$; therefore, $b = -5$. The slope, m , of a line can be found using any two points on the line, (x_1, y_1) and (x_2, y_2) , and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, -5)$ and $(1, -1)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{(-1 - (-5))}{(1 - 0)}$, or $m = 4$. Substituting 4 for m and -5 for b in the equation $y = mx + b$ yields $y = 4x - 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 28857637

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in one variable	

ID: 28857637

$4x + 6 = 18$

Which equation has the same solution as the given equation?

- A. $4x = 108$
- B. $4x = 24$
- C. $4x = 12$
- D. $4x = 3$

ID: 28857637 Answer

Correct Answer: C

Rationale

Choice C is correct. Subtracting 6 from both sides of the given equation yields $4x = 12$, which is the equation given in choice C. Since this equation is equivalent to the given equation, it has the same solution as the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID b1182278

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: b1182278

The graph of $9x - 10y = 19$ is translated down 4 units in the xy -plane. What is the x -coordinate of the x -intercept of the resulting graph?

ID: b1182278 Answer

Correct Answer: 59/9, 6.555, 6.556

Rationale

The correct answer is $\frac{59}{9}$. When the graph of an equation in the form $Ax + By = C$, where A , B , and C are constants, is translated down k units in the xy -plane, the resulting graph can be represented by the equation $Ax + B(y + k) = C$. It's given that the graph of $9x - 10y = 19$ is translated down 4 units in the xy -plane. Therefore, the resulting graph can be represented by the equation $9x - 10(y + 4) = 19$, or $9x - 10y - 40 = 19$. Adding 40 to both sides of this equation yields $9x - 10y = 59$. The x -coordinate of the x -intercept of the graph of an equation in the xy -plane is the value of x in the equation when $y = 0$. Substituting 0 for y in the equation $9x - 10y = 59$ yields $9x - 10(0) = 59$, or $9x = 59$. Dividing both sides of this equation by 9 yields $x = \frac{59}{9}$. Therefore, the x -coordinate of the x -intercept of the resulting graph is $\frac{59}{9}$. Note that 59/9, 6.555, and 6.556 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 73d572f1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 73d572f1

$$\begin{aligned}(x - 2) - 4(y + 7) &= 117 \\ (x - 2) + 4(y + 7) &= 442\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $6(x - 2)$?

ID: 73d572f1 Answer

Correct Answer: 1677

Rationale

The correct answer is **1,677**. Adding the first equation to the second equation in the given system yields $(x - 2) + (x - 2) + (-4)(y + 7) + 4(y + 7) = 117 + 442$, or $2(x - 2) = 559$. Multiplying both sides of this equation by **3** yields $6(x - 2) = 1,677$. Therefore, the value of $6(x - 2)$ is **1,677**.

Question Difficulty: Hard

Question ID b32ff819

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: b32ff819

A total of **364** paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. What is the best interpretation of $(x, y) = (24, 73)$ in this context?

- A. If **24** triangles were constructed, then **73** rectangles were constructed.
- B. If **24** triangles were constructed, then **73** paper straws were used.
- C. If **73** triangles were constructed, then **24** rectangles were constructed.
- D. If **73** triangles were constructed, then **24** paper straws were used.

ID: b32ff819 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that **364** paper straws of equal length were used to construct triangles and rectangles, where no two polygons had a common side. It's also given that the equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. The equation $(x, y) = (24, 73)$ means that if $x = 24$, then $y = 73$. Substituting **24** for x and **73** for y in $3x + 4y = 364$ yields $3(24) + 4(73) = 364$, or $364 = 364$, which is true. Therefore, in this context, the equation $(x, y) = (24, 73)$ means that if **24** triangles were constructed, then **73** rectangles were constructed.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 55ce77fa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 55ce77fa

The relationship between two variables, x and y , is linear. For every increase in the value of x by 1 , the value of y increases by 8 . When the value of x is 2 , the value of y is 18 . Which equation represents this relationship?

- A. $y = 2x + 18$
- B. $y = 2x + 8$
- C. $y = 8x + 2$
- D. $y = 3x + 26$

ID: 55ce77fa Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the relationship between x and y is linear. An equation representing a linear relationship can be written in the form $y = mx + b$, where m is the slope and b is the y -coordinate of the y -intercept of the graph of the relationship in the xy -plane. It's given that for every increase in the value of x by 1 , the value of y increases by 8 . The slope of a line can be expressed as the change in y over the change in x . Thus, the slope, m , of the line representing this relationship can be expressed as $\frac{8}{1}$, or 8 . Substituting 8 for m in the equation $y = mx + b$ yields $y = 8x + b$. It's also given that when the value of x is 2 , the value of y is 18 . Substituting 2 for x and 18 for y in the equation $y = 8x + b$ yields $18 = 8(2) + b$, or $18 = 16 + b$. Subtracting 16 from each side of this equation yields $2 = b$. Substituting 2 for b in the equation $y = 8x + b$ yields $y = 8x + 2$. Therefore, the equation $y = 8x + 2$ represents this relationship.

Choice A is incorrect. This equation represents a relationship where for every increase in the value of x by 1 , the value of y increases by 2 , not 8 , and when the value of x is 2 , the value of y is 22 , not 18 .

Choice B is incorrect. This equation represents a relationship where for every increase in the value of x by 1 , the value of y increases by 2 , not 8 , and when the value of x is 2 , the value of y is 12 , not 18 .

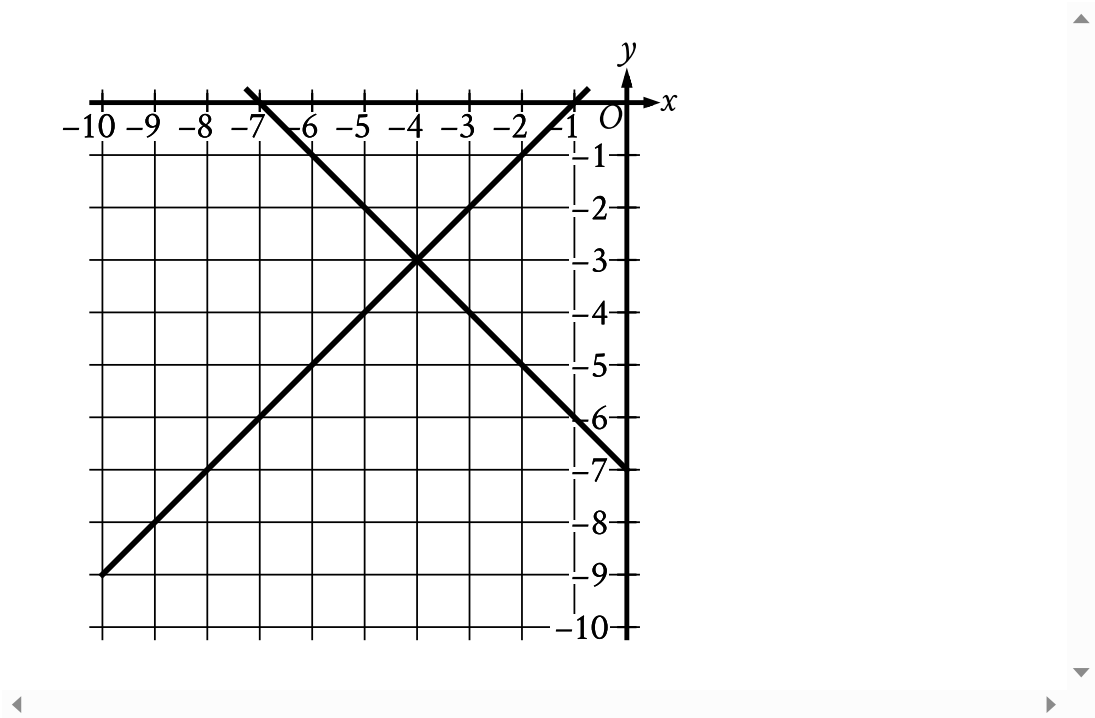
Choice D is incorrect. This equation represents a relationship where for every increase in the value of x by 1 , the value of y increases by 3 , not 8 , and when the value of x is 2 , the value of y is 32 , not 18 .

Question Difficulty: Medium

Question ID 3fb0deb1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: 3fb0deb1



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

- A. $(0, -7)$
- B. $(0, -3)$
- C. $(-4, -3)$
- D. $(-4, 0)$

ID: 3fb0deb1 Answer

Correct Answer: C

Rationale

Choice C is correct. The solution to a system of linear equations is represented by the point that lies on the graph of each equation in the system, or the point where the lines intersect on a graph. On the graph shown, the two lines intersect at the point $(-4, -3)$. Therefore, the solution to the system is $(-4, -3)$.

Choice A is incorrect. This is the y-intercept of the graph of one of the lines shown, not the intersection point of the two lines.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 72e16f81

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear functions	

ID: 72e16f81

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7 and passes through the point $(0, 5)$. Which equation defines f ?

- A. $f(x) = 5x$
- B. $f(x) = 35x$
- C. $f(x) = 7x + 5$
- D. $f(x) = 12x + 5$

ID: 72e16f81 Answer

Correct Answer: C

Rationale

Choice C is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m is the slope of the graph of $y = f(x)$ in the xy -plane and $(0, b)$ is the y -intercept of the graph. It's given that for the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7 . Therefore, $m = 7$. It's also given that the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 5)$. Therefore, the y -intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting 7 for m and 5 for b in the equation $f(x) = mx + b$ yields $f(x) = 7x + 5$.

Choice A is incorrect. This equation defines a function whose graph has a slope of 5 , not 7 , and passes through the point $(0, 0)$, not $(0, 5)$.

Choice B is incorrect. This equation defines a function whose graph has a slope of 35 , not 7 , and passes through the point $(0, 0)$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a slope of 12 , not 7 .

Question Difficulty: Easy

Question ID c09dcebb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Systems of two linear equations in two variables	

ID: c09dcebb

$$10x = 110$$
$$6x - 63 = y$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. 63
- B. 11
- C. 10
- D. 3

ID: c09dcebb Answer

Correct Answer: D

Rationale

Choice D is correct. Dividing both sides of the equation $10x = 110$ by 10 yields $x = 11$. Substituting 11 for x in the equation $6x - 63 = y$ yields $6(11) - 63 = y$, which is equivalent to $66 - 63 = y$, or $3 = y$. Therefore, the value of y is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not y .

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 839db0f9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Algebra	Linear equations in two variables	

ID: 839db0f9

Line ℓ is defined by $3y + 12x = 5$. Line n is perpendicular to line ℓ in the xy -plane. What is the slope of line n ?

ID: 839db0f9 Answer

Correct Answer: 0.25, 1/4

Rationale

The correct answer is $\frac{1}{4}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line in the xy -plane defined by this equation. It's given that line ℓ is defined by $3y + 12x = 5$. Subtracting $12x$ from both sides of this equation yields $3y = -12x + 5$. Dividing both sides of this equation by 3 yields $y = -\frac{12}{3}x + \frac{5}{3}$, or $y = -4x + \frac{5}{3}$. Thus, the slope of line ℓ in the xy -plane is -4 . Since line n is perpendicular to line ℓ in the xy -plane, the slope of line n is the negative reciprocal of the slope of line ℓ . The negative reciprocal of -4 is $-\frac{1}{(-4)} = \frac{1}{4}$. Note that 1/4 and .25 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID e3aee324

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: e3aee324

$4x^2 - 5x - 5 = 0$

What is the greatest solution to the given equation?

- A. $\frac{5}{8} - \frac{\sqrt{105}}{8}$
- B. $\frac{5}{8} + \frac{\sqrt{105}}{8}$
- C. $\frac{5}{4} - \frac{\sqrt{105}}{4}$
- D. $\frac{5}{4} + \frac{\sqrt{105}}{4}$

ID: e3aee324 Answer

Correct Answer: B

Rationale

Choice B is correct. The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The given equation is of this form, where $a = 4$, $b = -5$, and $c = -5$. Substituting these values into the quadratic formula yields $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(4)(-5)}}{2(4)}$, or $x = \frac{5 \pm \sqrt{25 + 80}}{8}$. These solutions can be rewritten as $x = \frac{5 \pm \sqrt{105}}{8}$, or $x = \frac{5}{8} \pm \frac{\sqrt{105}}{8}$. Since $\frac{5}{8} + \frac{\sqrt{105}}{8}$ is greater than $\frac{5}{8} - \frac{\sqrt{105}}{8}$, it follows that the greatest solution to the given equation is $\frac{5}{8} + \frac{\sqrt{105}}{8}$.

Choice A is incorrect. This is the least, not the greatest, solution to the given equation.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 93931b53

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 93931b53

Which expression is equivalent to $19(x^2 - 7)$?

- A. $19x^2 - 133$
- B. $19x^2 - 26$
- C. $19x^2 - 7$
- D. $19x^2 + 12$

ID: 93931b53 Answer

Correct Answer: A

Rationale

Choice A is correct. The expression $19(x^2 - 7)$ can be rewritten as $19(x^2) - 19(7)$, which is equivalent to $19x^2 - 133$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 8d3089b0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 8d3089b0

Bacteria are growing in a liquid growth medium. There were **300,000** cells per milliliter during an initial observation. The number of cells per milliliter doubles every **3** hours. How many cells per milliliter will there be **15** hours after the initial observation?

- A. **1,500,000**
- B. **2,400,000**
- C. **4,500,000**
- D. **9,600,000**

ID: 8d3089b0 Answer

Correct Answer: D

Rationale

Choice D is correct. Let y represent the number of cells per milliliter x hours after the initial observation. Since the number of cells per milliliter doubles every **3** hours, the relationship between x and y can be represented by an exponential equation of the form $y = a(b)^{\frac{x}{k}}$, where a is the number of cells per milliliter during the initial observation and the number of cells per milliliter increases by a factor of b every k hours. It's given that there were **300,000** cells per milliliter during the initial observation. Therefore, $a = 300,000$. It's also given that the number of cells per milliliter doubles, or increases by a factor of **2**, every **3** hours. Therefore, $b = 2$ and $k = 3$. Substituting **300,000** for a , **2** for b , and **3** for k in the equation $y = a(b)^{\frac{x}{k}}$ yields $y = 300,000(2)^{\frac{x}{3}}$. The number of cells per milliliter there will be **15** hours after the initial observation is the value of y in this equation when $x = 15$. Substituting **15** for x in the equation $y = 300,000(2)^{\frac{x}{3}}$ yields $y = 300,000(2)^{\frac{15}{3}}$, or $y = 300,000(2)^5$. This is equivalent to $y = 300,000(32)$, or $y = 9,600,000$. Therefore, **15** hours after the initial observation, there will be **9,600,000** cells per milliliter.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 89160f9c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 89160f9c

$$f(x) = x^2 - 48x + 2,304$$

What is the minimum value of the given function?

ID: 89160f9c Answer

Correct Answer: 1728

Rationale

The correct answer is **1,728**. The given function can be rewritten in the form $f(x) = a(x - h)^2 + k$, where a is a positive constant and the minimum value, k , of the function occurs when the value of x is h . By completing the square, $f(x) = x^2 - 48x + 2,304$ can be written as $f(x) = x^2 - 48x + \left(\frac{48}{2}\right)^2 + 2,304 - \left(\frac{48}{2}\right)^2$, or $f(x) = (x - 24)^2 + 1,728$. This equation is in the form $f(x) = a(x - h)^2 + k$, where $a = 1$, $h = 24$, and $k = 1,728$. Therefore, the minimum value of the given function is **1,728**.

Question Difficulty: Hard

Question ID 9e74d905

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 9e74d905

When the quadratic function f is graphed in the xy -plane, where $y = f(x)$, its vertex is $(-3, 6)$. One of the x -intercepts of this graph is $(-\frac{17}{4}, 0)$. What is the other x -intercept of the graph?

- A. $(-\frac{29}{4}, 0)$
- B. $(-\frac{7}{4}, 0)$
- C. $(\frac{5}{4}, 0)$
- D. $(\frac{17}{4}, 0)$

ID: 9e74d905 Answer

Correct Answer: B

Rationale

Choice B is correct. Since the line of symmetry for the graph of a quadratic function contains the vertex of the graph, the x -coordinate of the vertex of the graph of $y = f(x)$ is the x -coordinate of the midpoint of its two x -intercepts. The midpoint of two points with x -coordinates x_1 and x_2 has x -coordinate x_m , where $x_m = \frac{x_1+x_2}{2}$. It's given that the vertex is $(-3, 6)$ and one of the x -intercepts is $(-\frac{17}{4}, 0)$. Substituting -3 for x_m and $-\frac{17}{4}$ for x_1 in the equation $x_m = \frac{x_1+x_2}{2}$ yields $-3 = \frac{-\frac{17}{4}+x_2}{2}$. Multiplying each side of this equation by 2 yields $-6 = -\frac{17}{4} + x_2$. Adding $\frac{17}{4}$ to each side of this equation yields $-\frac{7}{4} = x_2$. Therefore, the other x -intercept is $(-\frac{7}{4}, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID d7bf5261

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: d7bf5261

$$\begin{aligned}y &= -1.5 \\ y &= x^2 + 8x + a\end{aligned}$$

In the given system of equations, a is a positive constant. The system has exactly one distinct real solution. What is the value of a ?

ID: d7bf5261 Answer

Correct Answer: 14.5, 29/2

Rationale

The correct answer is $\frac{29}{2}$. According to the first equation in the given system, the value of y is -1.5 . Substituting -1.5 for y in the second equation in the given system yields $-1.5 = x^2 + 8x + a$. Adding 1.5 to both sides of this equation yields $0 = x^2 + 8x + a + 1.5$. If the given system has exactly one distinct real solution, it follows that $0 = x^2 + 8x + a + 1.5$ has exactly one distinct real solution. A quadratic equation in the form $0 = px^2 + qx + r$, where p, q , and r are constants, has exactly one distinct real solution if and only if the discriminant, $q^2 - 4pr$, is equal to 0 . The equation $0 = x^2 + 8x + a + 1.5$ is in this form, where $p = 1, q = 8$, and $r = a + 1.5$. Therefore, the discriminant of the equation $0 = x^2 + 8x + a + 1.5$ is $(8)^2 - 4(1)(a + 1.5)$, or $58 - 4a$. Setting the discriminant equal to 0 to solve for a yields $58 - 4a = 0$. Adding $4a$ to both sides of this equation yields $58 = 4a$. Dividing both sides of this equation by 4 yields $\frac{58}{4} = a$, or $\frac{29}{2} = a$. Therefore, if the given system of equations has exactly one distinct real solution, the value of a is $\frac{29}{2}$. Note that 29/2 and 14.5 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 2434c419

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2434c419

The y -intercept of the graph of $y = x^2 + 31$ in the xy -plane is $(0, y)$. What is the value of y ?

ID: 2434c419 Answer

Correct Answer: 31

Rationale

The correct answer is **31**. It's given that the y -intercept of the graph of $y = x^2 + 31$ in the xy -plane is $(0, y)$. Substituting **0** for x in the given equation yields $y = (0)^2 + 31$, or $y = 31$. Thus, the value of y is **31**.

Question Difficulty: Medium

Question ID 81121307

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 81121307

$5|x| = 45$

What is the positive solution to the given equation?

ID: 81121307 Answer

Correct Answer: 9

Rationale

The correct answer is **9**. Dividing both sides of the given equation by **5** yields $|x| = 9$. By the definition of absolute value, if $|x| = 9$, then $x = 9$ or $x = -9$. Therefore, the two solutions to the equation $5|x| = 45$ are **9** and **-9**. It follows that the positive solution to the given equation is **9**.

Question Difficulty: Easy

Question ID b2fde9d3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b2fde9d3

$$\begin{aligned}x^2 + y + 7 &= 7 \\ 20x + 100 - y &= 0\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

ID: b2fde9d3 Answer

Correct Answer: -10

Rationale

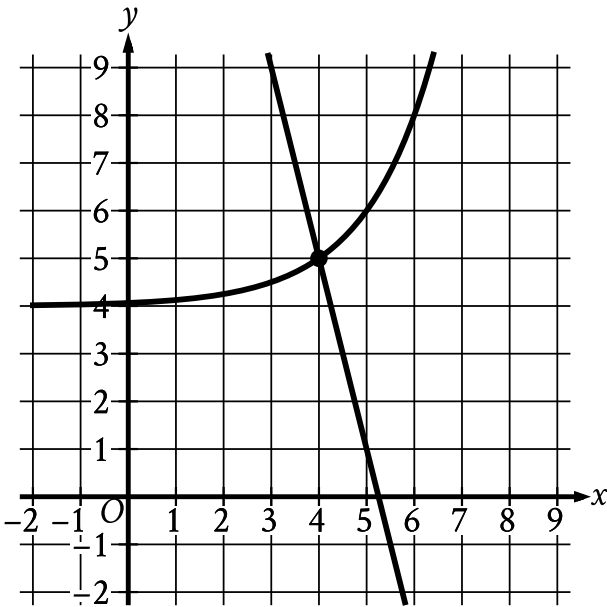
The correct answer is -10 . Adding y to both sides of the second equation in the given system yields $20x + 100 = y$. Substituting $20x + 100$ for y in the first equation in the given system yields $x^2 + 20x + 100 + 7 = 7$. Subtracting 7 from both sides of this equation yields $x^2 + 20x + 100 = 0$. Factoring the left-hand side of this equation yields $(x + 10)(x + 10) = 0$, or $(x + 10)^2 = 0$. Taking the square root of both sides of this equation yields $x + 10 = 0$. Subtracting 10 from both sides of this equation yields $x = -10$. Therefore, the value of x is -10 .

Question Difficulty: Hard

Question ID 78c1bf85

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 78c1bf85



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

- A. $(0, 0)$
- B. $(0, 4)$
- C. $(4, 5)$
- D. $(5, 0)$

ID: 78c1bf85 Answer

Correct Answer: C

Rationale

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the nonlinear equation shown intersect at the point $(4, 5)$. Thus, the solution to the system is $(4, 5)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 7a0870ed

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7a0870ed

The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. What is the value of p ?

- A. 19
- B. 44
- C. 56
- D. 81

ID: 7a0870ed Answer

Correct Answer: D

Rationale

Choice D is correct. For a function of the form $f(x) = a(b)^x$, where a and b are constants and $b < 1$, the value of $f(x)$ decreases by $100(1 - b)\%$ for every increase of x by 1. In the given function, $a = 56$ and $b = 0.19$. Therefore, for the given function, the value of $f(x)$ decreases by $100(1 - 0.19)\%$, or 81% , for every increase of x by 1. It's given that the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. The increase from $x = n - 1$ to $x = n$ can be expressed as $n - (n - 1)$, which is equivalent to $n - n + 1$, or 1. Since there is an increase by 1 from $n - 1$ to n , it follows that the value of $f(n)$ is 81% less than the value of $f(n - 1)$. Thus, the value of p is 81.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 498850a9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 498850a9

$y = 76$

$y = x^2 - 5$

The graphs of the given equations in the xy -plane intersect at the point (x, y) . What is a possible value of x ?

- A. $-\frac{76}{5}$
- B. -9
- C. 5
- D. 76

ID: 498850a9 Answer

Correct Answer: B

Rationale

Choice B is correct. Since the point (x, y) is an intersection point of the graphs of the given equations in the xy -plane, the pair (x, y) should satisfy both equations, and thus is a solution of the given system. According to the first equation, $y = 76$. Substituting 76 in place of y in the second equation yields $x^2 - 5 = 76$. Adding 5 to both sides of this equation yields $x^2 = 81$. Taking the square root of both sides of this equation yields two solutions: $x = 9$ and $x = -9$. Of these two solutions, only -9 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of coordinate y , rather than x , of the intersection point (x, y) .

Question Difficulty: Medium

Question ID d1b3d70e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: d1b3d70e

Square P has a side length of x inches. Square Q has a perimeter that is **176** inches greater than the perimeter of square P. The function f gives the area of square Q, in square inches. Which of the following defines f ?

- A. $f(x) = (x + 44)^2$
- B. $f(x) = (x + 176)^2$
- C. $f(x) = (176x + 44)^2$
- D. $f(x) = (176x + 176)^2$

ID: d1b3d70e Answer

Correct Answer: A

Rationale

Choice A is correct. Let x represent the side length, in inches, of square P. It follows that the perimeter of square P is **4x** inches. It's given that square Q has a perimeter that is **176** inches greater than the perimeter of square P. Thus, the perimeter of square Q is **176** inches greater than **4x** inches, or **4x + 176** inches. Since the perimeter of a square is **4** times the side length of the square, each side length of Q is $\frac{4x+176}{4}$, or $x + 44$ inches. Since the area of a square is calculated by multiplying the length of two sides, the area of square Q is $(x + 44)(x + 44)$, or $(x + 44)^2$ square inches. It follows that function f is defined by $f(x) = (x + 44)^2$.

Choice B is incorrect. This function represents a square with side lengths $(x + 176)$ inches.

Choice C is incorrect. This function represents a square with side lengths $(176x + 44)$ inches.

Choice D is incorrect. This function represents a square with side lengths $(176x + 176)$ inches.

Question Difficulty: Hard

Question ID b11955c5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b11955c5

The functions f and g are defined by the given equations.

$$f(x) = 3 + \left| -2x - x^2 \right|$$

$$g(w) = \left| \frac{-w}{w-1} \right| - w + 5$$

If $f(-4) = c$, where c is a constant, what is the value of $g(c)$?

ID: b11955c5 Answer

Correct Answer: -4.9, -49/10

Rationale

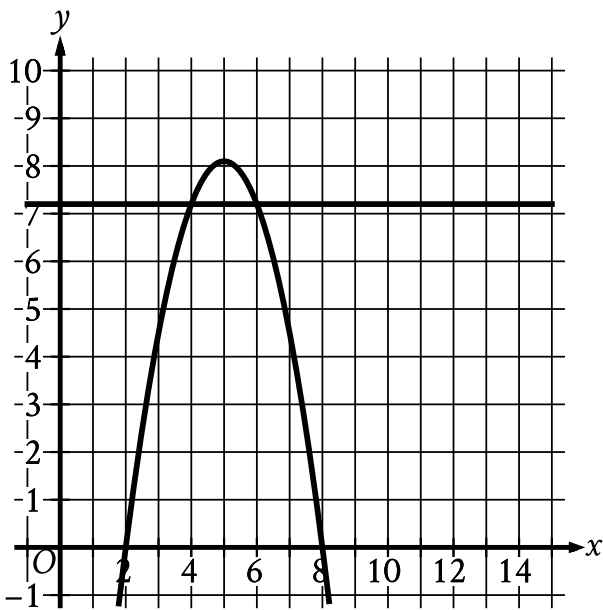
The correct answer is -4.9 . The value of $f(-4)$ is the value of $f(x)$ when $x = -4$. Substituting -4 for x in the equation $f(x) = 3 + \left| -2x - x^2 \right|$ yields $f(-4) = 3 + \left| -2(-4) - (-4)^2 \right|$, or $f(-4) = 3 + \left| -8 \right|$, which is equivalent to $f(-4) = 3 + 8$, or $f(-4) = 11$. Since it's given that $f(-4) = c$, it follows that $c = 11$ and the value of $g(c)$ is the value of $g(11)$. Substituting 11 for w in the equation $g(w) = \left| \frac{-w}{w-1} \right| - w + 5$ yields $g(11) = \left| \frac{-11}{11-1} \right| - 11 + 5$, or $g(11) = \left| -1.1 \right| - 6$, which is equivalent to $g(11) = 1.1 - 6$, or $g(11) = -4.9$. Note that -4.9 and -49/10 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 2c5e7c3b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 2c5e7c3b



The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y) . What is a possible value of x ?

- A. 4
- B. 5
- C. 7.2
- D. 8

ID: 2c5e7c3b Answer

Correct Answer: A

Rationale

Choice A is correct. The solutions to the system correspond to the points where the graphs of the equations intersect. The graphs of the linear equation and the quadratic equation shown intersect at approximately $(4, 7.2)$ and $(6, 7.2)$. It follows that the possible values of x are 4 and 6. Of these two possible values, only 4 is given as a choice.

Choice B is incorrect. This is the x -value of the vertex of the graph of the quadratic equation, not the x -value of a solution to the system.

Choice C is incorrect. This is the approximate y -value, not the x -value, of a solution to the system.

Choice D is incorrect. This is the x -value of an x -intercept of the graph of the quadratic equation.

Question ID 7fa6752e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 7fa6752e

$$v = -\frac{w}{150x}$$

The given equation relates the distinct positive numbers v , w , and x . Which equation correctly expresses w in terms of v and x ?

- A. $w = -150vx$
- B. $w = -\frac{150v}{x}$
- C. $w = -\frac{x}{150v}$
- D. $w = v + 150x$

ID: 7fa6752e Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that x is positive. Therefore, multiplying each side of the given equation by $-150x$ yields $-150xv = w$, which is equivalent to $w = -150vx$. Thus, the equation $w = -150vx$ correctly expresses w in terms of v and x .

Choice B is incorrect. This equation is equivalent to $v = -\frac{wx}{150}$.

Choice C is incorrect. This equation is equivalent to $v = -\frac{x}{150w}$.

Choice D is incorrect. This equation is equivalent to $v = w - 150x$.

Question Difficulty: Medium

Question ID b2142aba

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b2142aba

For the exponential function f , the value of $f(1)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

- A. $f(x) = 50(2)^{x+1}$
- B. $f(x) = 80(2)^x$
- C. $f(x) = 128(2)^{x-1}$
- D. $f(x) = 205(2)^{x-2}$

ID: b2142aba Answer

Correct Answer: C

Rationale

Choice C is correct. For the form of the function in choice C, $f(x) = 128(1.6)^{x-1}$, the value of $f(1)$ can be found as $128(1.6)^{1-1}$, which is equivalent to $128(1.6)^0$, or 128 . Therefore, $k = 128$, which is shown in $f(x) = 128(1.6)^{x-1}$ as the coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

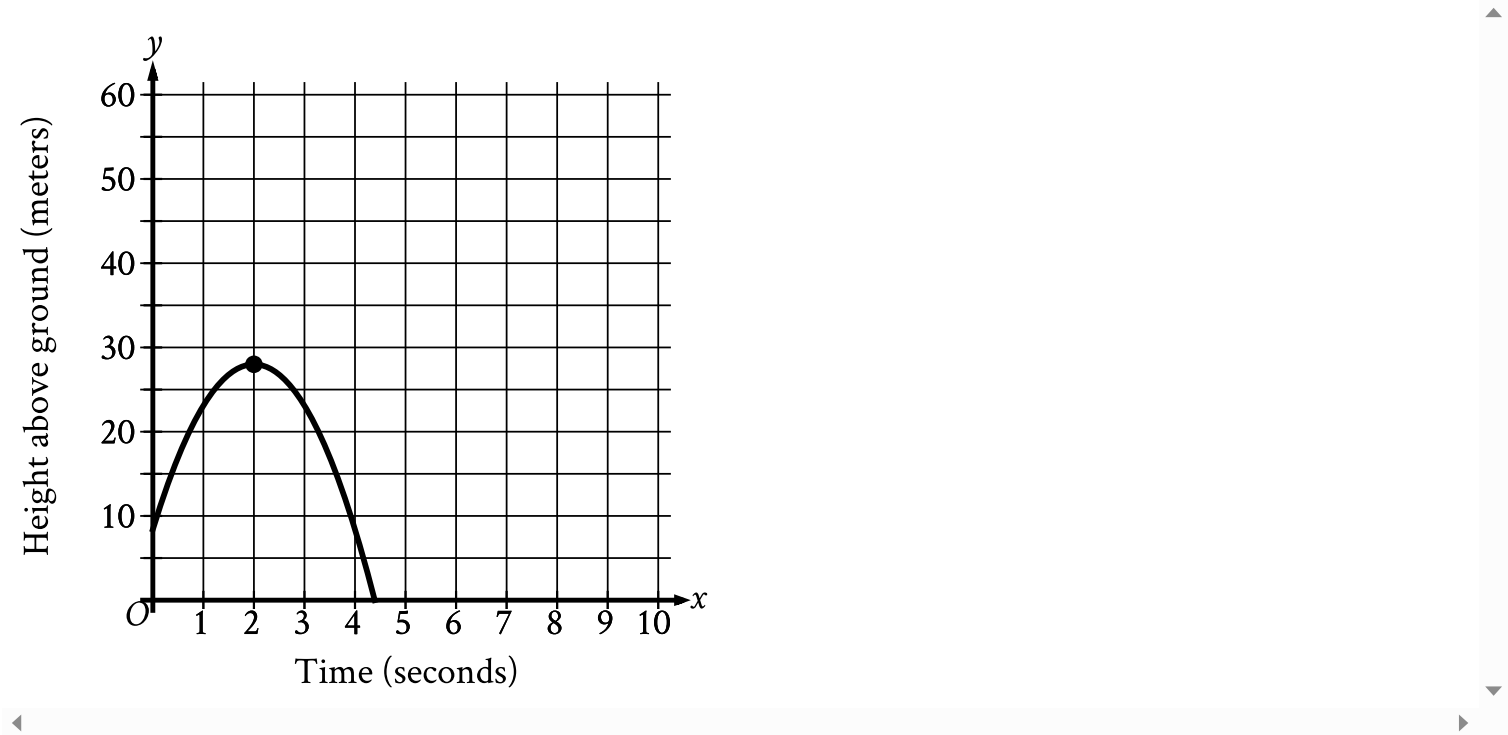
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 98579160

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 98579160



An object was launched upward from a platform. The graph shown models the height above ground, y , in meters, of the object x seconds after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

- A. From $x = 0$ to $x = 2$
- B. From $x = 0$ to $x = 4$
- C. From $x = 2$ to $x = 3$
- D. From $x = 3$ to $x = 4$

ID: 98579160 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the variable y represents the height, in meters, of the object above the ground. The graph shows that the height of the object was increasing from $x = 0$ to $x = 2$, and decreasing from $x = 2$ to $x = 4$. Therefore, the height of the object was increasing for the entire interval of time from $x = 0$ to $x = 2$.

Choice B is incorrect. The height of the object wasn't increasing for this entire interval of time, as it was decreasing from $x = 2$ to $x = 4$.

Choice C is incorrect. The height of the object was decreasing, not increasing, for this entire interval of time.

Choice D is incorrect. The height of the object was decreasing, not increasing, for this entire interval of time.

Question ID c36ff321

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: c36ff321

Which expression is equivalent to $x^2 + 3x - 40$?

- A. $(x - 4)(x + 10)$
- B. $(x - 5)(x + 8)$
- C. $(x - 8)(x + 5)$
- D. $(x - 10)(x + 4)$

ID: c36ff321 Answer

Correct Answer: B

Rationale

Choice B is correct. The given expression may be rewritten as $x^2 + 8x - 5x - 40$. Since the first two terms of this expression have a common factor of x and the last two terms of this expression have a common factor of -5 , this expression may be rewritten as $x(x) + x(8) - 5(x) - 5(8)$, or $x(x + 8) - 5(x + 8)$. Since each term of this expression has a common factor of $(x + 8)$, it may be rewritten as $(x - 5)(x + 8)$.

Alternate approach: An expression of the form $x^2 + bx + c$, where b and c are constants, can be factored if there are two values that add to give b and multiply to give c . In the given expression, $b = 3$ and $c = -40$. The values of -5 and 8 add to give 3 and multiply to give -40 , so the expression can be factored as $(x - 5)(x + 8)$.

Choice A is incorrect. This expression is equivalent to $x^2 + 6x - 40$, not $x^2 + 3x - 40$.

Choice C is incorrect. This expression is equivalent to $x^2 - 3x - 40$, not $x^2 + 3x - 40$.

Choice D is incorrect. This expression is equivalent to $x^2 - 6x - 40$, not $x^2 + 3x - 40$.

Question Difficulty: Easy

Question ID 6b7c82dd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 6b7c82dd

The function $f(x) = \frac{1}{9}(x - 7)^2 + 3$ gives a metal ball's height above the ground $f(x)$, in inches, x seconds after it started moving on a track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- A. The metal ball's minimum height was 3 inches above the ground.
- B. The metal ball's minimum height was 7 inches above the ground.
- C. The metal ball's height was 3 inches above the ground when it started moving.
- D. The metal ball's height was 7 inches above the ground when it started moving.

ID: 6b7c82dd Answer

Correct Answer: A

Rationale

Choice A is correct. The graph of a quadratic equation in the form $y = a(x - h)^2 + k$, where a , h , and k are positive constants, is a parabola that opens upward with vertex (h, k) . The given function $f(x) = \frac{1}{9}(x - 7)^2 + 3$ is in the form $y = a(x - h)^2 + k$, where $y = f(x)$, $a = \frac{1}{9}$, $h = 7$, and $k = 3$. Therefore, the graph of $y = f(x)$ is a parabola that opens upward with vertex $(7, 3)$. Since the parabola opens upward, the vertex is the lowest point on the graph. It follows that the y -coordinate of the vertex of the graph of $y = f(x)$ is the minimum value of $f(x)$. Therefore, the minimum value of $f(x)$ is 3. It's given that $f(x) = \frac{1}{9}(x - 7)^2 + 3$ represents the metal ball's height above the ground, in inches, x seconds after it started moving on a track. Therefore, the best interpretation of the vertex of the graph of $y = f(x)$ is that the metal ball's minimum height was 3 inches above the ground.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 71875bb0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 71875bb0

For the function q , the value of $q(x)$ decreases by 45% for every increase in the value of x by 1. If $q(0) = 14$, which equation defines q ?

- A. $q(x) = 0.55(14)^x$
- B. $q(x) = 1.45(14)^x$
- C. $q(x) = 14(0.55)^x$
- D. $q(x) = 14(1.45)^x$

ID: 71875bb0 Answer

Correct Answer: C

Rationale

Choice C is correct. Since the value of $q(x)$ decreases by a fixed percentage, 45%, for every increase in the value of x by 1, the function q is a decreasing exponential function. A decreasing exponential function can be written in the form $q(x) = a\left(1 - \frac{p}{100}\right)^x$, where a is the value of $q(0)$ and the value of $q(x)$ decreases by $p\%$ for every increase in the value of x by 1. If $q(0) = 14$, then $a = 14$. Since the value of $q(x)$ decreases by 45% for every increase in the value of x by 1, $p = 45$. Substituting 14 for a and 45 for p in the equation $q(x) = a\left(1 - \frac{p}{100}\right)^x$ yields $q(x) = 14\left(1 - \frac{45}{100}\right)^x$, which is equivalent to $q(x) = 14(1 - 0.45)^x$, or $q(x) = 14(0.55)^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. For this function, the value of $q(x)$ increases, rather than decreases, by 45% for every increase in the value of x by 1.

Question Difficulty: Hard

Question ID 0f268f46

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 0f268f46

Which expression is equivalent to $(9x^2 + 6) - (8x^2)$?

- A. 7
- B. $17x^2 + 6$
- C. $x^2 - 6$
- D. $x^2 + 6$

ID: 0f268f46 Answer

Correct Answer: D

Rationale

Choice D is correct. The given expression is equivalent to $9x^2 + 6 - 8x^2$. The terms $9x^2$ and $-8x^2$ are like terms. Combining these like terms yields x^2 . It follows that the expression $(9x^2 + 6) - (8x^2)$ is equivalent to $x^2 + 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is equivalent to $(9x^2 + 6) + (8x^2)$, not the given expression.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 8e481921

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 8e481921

$64x^2 + bx + 25 = 0$

In the given equation, b is a constant. For which of the following values of b will the equation have more than one real solution?

- A. -91
- B. -80
- C. 5
- D. 40

ID: 8e481921 Answer

Correct Answer: A

Rationale

Choice A is correct. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has either no real solutions, exactly one real solution, or exactly two real solutions. That is, for the given equation to have more than one real solution, it must have exactly two real solutions. When the value of the discriminant, or $b^2 - 4ac$, is greater than 0, the given equation has exactly two real solutions. In the given equation, $64x^2 + bx + 25 = 0$, $a = 64$ and $c = 25$. Therefore, the given equation has exactly two real solutions when $(b)^2 - 4(64)(25) > 0$, or $b^2 - 6,400 > 0$. Adding 6,400 to both sides of this inequality yields $b^2 > 6,400$. Taking the square root of both sides of $b^2 > 6,400$ yields two possible inequalities: $b < -80$ or $b > 80$. Of the choices, only choice A satisfies $b < -80$ or $b > 80$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 17690f09

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 17690f09

If $|4x - 4| = 112$, what is the positive value of $x - 1$?

ID: 17690f09 Answer

Correct Answer: 28

Rationale

The correct answer is **28**. The given absolute value equation can be rewritten as two linear equations: $4x - 4 = 112$ and $-(4x - 4) = 112$, or $4x - 4 = -112$. Adding **4** to both sides of the equation $4x - 4 = 112$ results in $4x = 116$. Dividing both sides of this equation by **4** results in $x = 29$. Adding **4** to both sides of the equation $4x - 4 = -112$ results in $4x = -108$. Dividing both sides of this equation by **4** results in $x = -27$. Therefore, the two values of $x - 1$ are $29 - 1$, or **28**, and $-27 - 1$, or **-28**. Thus, the positive value of $x - 1$ is **28**.

Alternate approach: The given equation can be rewritten as $|4(x - 1)| = 112$, which is equivalent to $4|x - 1| = 112$. Dividing both sides of this equation by **4** yields $|x - 1| = 28$. This equation can be rewritten as two linear equations: $x - 1 = 28$ and $-(x - 1) = 28$, or $x - 1 = -28$. Therefore, the positive value of $x - 1$ is **28**.

Question Difficulty: Medium

Question ID 264cd14a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 264cd14a

The function f gives the product of a number, x , and a number that is **91** more than x . Which equation defines f ?

- A. $f(x) = x^2 + x + 91$
- B. $f(x) = x^2 + 91$
- C. $f(x) = x^2 + 91x$
- D. $f(x) = x^2 + 91x + 91$

ID: 264cd14a Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the function f gives the product of a number, x , and a number that is **91** more than x . A number that is **91** more than x can be represented by the expression $x + 91$. Therefore, f can be defined by the equation $f(x) = x(x + 91)$, or $f(x) = x^2 + 91x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f11bc376

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: f11bc376

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39$$

In the given equation, c is a positive constant. Which of the following is one of the solutions to the given equation?

- A. $-c$
- B. $-c^2 - 39^2$
- C. $-\sqrt{39^2 - c^2}$
- D. $-\sqrt{c^2 + 39^2}$

ID: f11bc376 Answer

Correct Answer: D

Rationale

Choice D is correct. If $x^2 - c^2 \leq 0$, then neither side of the given equation is defined and there can be no solution. Therefore, $x^2 - c^2 > 0$. Subtracting $\frac{c^2}{\sqrt{x^2 - c^2}}$ from both sides of the given equation yields $\frac{x^2}{\sqrt{x^2 - c^2}} - \frac{c^2}{\sqrt{x^2 - c^2}} = 39$, or $\frac{x^2 - c^2}{\sqrt{x^2 - c^2}} = 39$. Squaring both sides of this equation yields $\left(\frac{x^2 - c^2}{\sqrt{x^2 - c^2}}\right)^2 = 39^2$, or $\frac{(x^2 - c^2)(x^2 - c^2)}{x^2 - c^2} = 39^2$. Since $x^2 - c^2$ is positive and, therefore, nonzero, the expression $\frac{x^2 - c^2}{x^2 - c^2}$ is defined and equivalent to 1. It follows that the equation $\frac{(x^2 - c^2)(x^2 - c^2)}{x^2 - c^2} = 39^2$ can be rewritten as $\left(\frac{x^2 - c^2}{x^2 - c^2}\right)(x^2 - c^2) = 39^2$, or $(1)(x^2 - c^2) = 39^2$, which is equivalent to $x^2 - c^2 = 39^2$. Adding c^2 to both sides of this equation yields $x^2 = c^2 + 39^2$. Taking the square root of both sides of this equation yields two solutions: $x = \sqrt{c^2 + 39^2}$ and $x = -\sqrt{c^2 + 39^2}$. Therefore, of the given choices, $-\sqrt{c^2 + 39^2}$ is one of the solutions to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 5cda58e0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5cda58e0

$(5x + 4)(2x - 5) = 0$

Which of the following is a solution to the given equation?

- A. $-\frac{5}{2}$
- B. $-\frac{5}{4}$
- C. $-\frac{4}{5}$
- D. $-\frac{2}{5}$

ID: 5cda58e0 Answer

Correct Answer: C

Rationale

Choice C is correct. Since a product of two factors is equal to **0** if and only if at least one of the factors is **0**, either **$5x + 4 = 0$** or **$2x - 5 = 0$** . Subtracting **4** from each side of the equation **$5x + 4 = 0$** yields **$5x = -4$** . Dividing each side of this equation by **5** yields **$x = -\frac{4}{5}$** . Adding **5** to each side of the equation **$2x - 5 = 0$** yields **$2x = 5$** . Dividing each side of this equation by **2** yields **$x = \frac{5}{2}$** . It follows that the solutions to the given equation are $-\frac{4}{5}$ and $\frac{5}{2}$. Therefore, $-\frac{4}{5}$ is a solution to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID bd16a3f5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: bd16a3f5

The function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. What is the product of $g(15a)$ and $g(7a)$?

ID: bd16a3f5 Answer

Correct Answer: 609

Rationale

The correct answer is **609**. It's given that the function g is defined by $g(x) = \frac{|x|}{a} - 14$, where $a < 0$. Substituting $15a$ for x in function g yields $g(15a) = \frac{|15a|}{a} - 14$. This function can be rewritten as $g(15a) = \frac{15|a|}{a} - 14$, or $g(15a) = 15\left(\frac{|a|}{a}\right) - 14$. Since $a < 0$, it follows that $\frac{|a|}{a} = -1$. Substituting -1 for $\frac{|a|}{a}$ in $g(15a) = 15\left(\frac{|a|}{a}\right) - 14$ yields $g(15a) = 15(-1) - 14$, or $g(15a) = -29$. Similarly, substituting $7a$ for x in function g yields $g(7a) = \frac{|7a|}{a} - 14$. This function can be rewritten as $g(7a) = \frac{7|a|}{a} - 14$, or $g(7a) = 7\left(\frac{|a|}{a}\right) - 14$. Since $a < 0$, it again follows that $\frac{|a|}{a} = -1$. Substituting -1 for $\frac{|a|}{a}$ in $g(7a) = 7\left(\frac{|a|}{a}\right) - 14$ yields $g(7a) = 7(-1) - 14$, or $g(7a) = -21$. Therefore, $g(15a) = -29$ and $g(7a) = -21$. Thus, the product of $g(15a)$ and $g(7a)$ is $(-29)(-21)$, or **609**.

Question Difficulty: Hard

Question ID 319cb885

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 319cb885

The function h is defined by $h(x) = \frac{8}{5x+6}$. What is the value of $h(2)$?

ID: 319cb885 Answer

Correct Answer: .5, 1/2

Rationale

The correct answer is $\frac{1}{2}$. The value of $h(2)$ is the value of $h(x)$ when $x = 2$. Substituting 2 for x in the given equation yields $h(2) = \frac{8}{5(2)+6}$, which is equivalent to $h(2) = \frac{8}{16}$, or $h(2) = \frac{1}{2}$. Therefore, the value of $h(2)$ is $\frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 74c25d42

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 74c25d42

$$(x - 47)^2 = 1$$

What is the sum of the solutions to the given equation?

ID: 74c25d42 Answer

Correct Answer: 94

Rationale

The correct answer is **94**. Taking the square root of each side of the given equation yields $x - 47 = 1$ or $x - 47 = -1$. Adding **47** to both sides of the equation $x - 47 = 1$ yields $x = 48$. Adding **47** to both sides of the equation $x - 47 = -1$ yields $x = 46$. Therefore, the sum of the solutions to the given equation is **48 + 46**, or **94**.

Question Difficulty: Hard

Question ID b2aee94a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b2aee94a

The function g is defined by $g(x) = |x + 18|$. What is the value of $g(4)$?

- A. -18
- B. -4
- C. 14
- D. 22

ID: b2aee94a Answer

Correct Answer: D

Rationale

Choice D is correct. The value of $g(4)$ is the value of $g(x)$ when $x = 4$. Substituting 4 for x in the given equation yields $g(4) = |4 + 18|$, which is equivalent to $g(4) = |22|$, or $g(4) = 22$. Therefore, the value of $g(4)$ is 22 .

Choice A is incorrect. This would be the value of $g(4)$ if function g was defined by $g(x) = -|18|$, not $g(x) = |x + 18|$.

Choice B is incorrect. This would be the value of $g(4)$ if function g was defined by $g(x) = -|x|$, not $g(x) = |x + 18|$.

Choice C is incorrect. This would be the value of $g(4)$ if function g was defined by $g(x) = |-x + 18|$, not $g(x) = |x + 18|$.

Question Difficulty: Easy

Question ID e1d81de8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: e1d81de8

Which expression is equivalent to $5x^5 - 6x^4 + 8x^3$?

- A. $x^4(5x - 6)$
- B. $x^3(5x^2 - 6x + 8)$
- C. $8x^3(5x^2 - 6x + 1)$
- D. $6x^5(-6x^4 + 8x^3 + 1)$

ID: e1d81de8 Answer

Correct Answer: B

Rationale

Choice B is correct. Since x^3 is a common factor of each term in the given expression, the expression can be rewritten as $x^3(5x^2 - 6x + 8)$.

Choice A is incorrect. This expression is equivalent to $5x^5 - 6x^4$.

Choice C is incorrect. This expression is equivalent to $40x^5 - 48x^4 + 8x^3$.

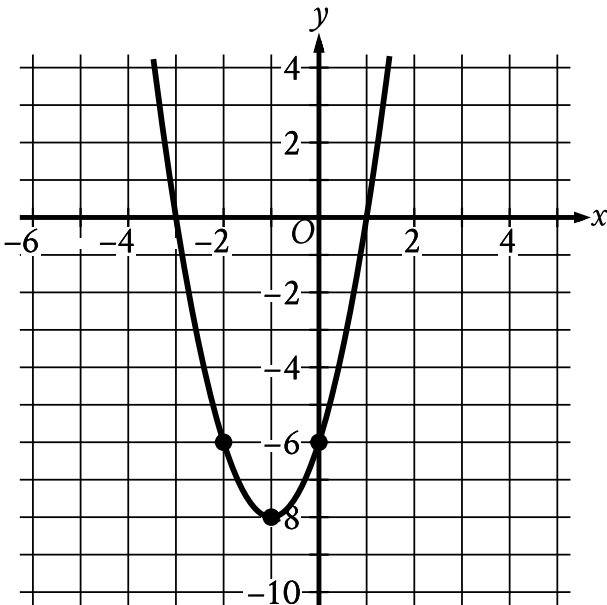
Choice D is incorrect. This expression is equivalent to $-36x^9 + 48x^8 + 6x^5$.

Question Difficulty: Easy

Question ID d1cf915b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: d1cf915b



The graph of $y = 2x^2 + bx + c$ is shown, where b and c are constants. What is the value of bc ?

ID: d1cf915b Answer

Correct Answer: -24

Rationale

The correct answer is -24 . Since the graph passes through the point $(0, -6)$, it follows that when the value of x is 0 , the value of y is -6 . Substituting 0 for x and -6 for y in the given equation yields $-6 = 2(0)^2 + b(0) + c$, or $-6 = c$. Therefore, the value of c is -6 . Substituting -6 for c in the given equation yields $y = 2x^2 + bx - 6$. Since the graph passes through the point $(-1, -8)$, it follows that when the value of x is -1 , the value of y is -8 . Substituting -1 for x and -8 for y in the equation $y = 2x^2 + bx - 6$ yields $-8 = 2(-1)^2 + b(-1) - 6$, or $-8 = 2 - b - 6$, which is equivalent to $-8 = -4 - b$. Adding 4 to each side of this equation yields $-4 = -b$. Dividing each side of this equation by -1 yields $4 = b$. Since the value of b is 4 and the value of c is -6 , it follows that the value of bc is $(4)(-6)$, or -24 .

Alternate approach: The given equation represents a parabola in the xy -plane with a vertex at $(-1, -8)$. Therefore, the given equation, $y = 2x^2 + bx + c$, which is written in standard form, can be written in vertex form, $y = a(x - h)^2 + k$, where (h, k) is the vertex of the parabola and a is the value of the coefficient on the x^2 term when the equation is written in standard form. It follows that $a = 2$. Substituting 2 for a , -1 for h , and -8 for k in this equation yields $y = 2(x - (-1))^2 + (-8)$, or $y = 2(x + 1)^2 - 8$. Squaring the binomial on the right-hand side of this equation yields $y = 2(x^2 + 2x + 1) - 8$. Multiplying each term inside the parentheses on the right-hand side of this equation by 2 yields $y = 2x^2 + 4x + 2 - 8$, which is equivalent to $y = 2x^2 + 4x - 6$. From the given equation $y = 2x^2 + bx + c$, it follows that the value of b is 4 and the value of c is -6 . Therefore, the value of bc is $(4)(-6)$, or -24 .

Question Difficulty: Hard

Question ID af172557

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: af172557

$f(x) = 9(4)^x$

The function f is defined by the given equation. If $g(x) = f(x + 2)$, which of the following equations defines the function g ?

- A. $g(x) = 18(4)^x$
- B. $g(x) = 144(4)^x$
- C. $g(x) = 18(8)^x$
- D. $g(x) = 81(16)^x$

ID: af172557 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = 9(4)^x$ and $g(x) = f(x + 2)$. Substituting $x + 2$ for x in $f(x) = 9(4)^x$ gives $f(x + 2) = 9(4)^{x+2}$. Rewriting this equation using properties of exponents gives $f(x + 2) = 9(4)^x(4)^2$, which is equivalent to $f(x + 2) = 9(4)^x(16)$. Multiplying 9 and 16 in this equation gives $f(x + 2) = 144(4)^x$. Since $g(x) = f(x + 2)$, $g(x) = 144(4)^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 011b5078

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 011b5078

$$5(x + 7) = 15(x - 17)(x + 7)$$

What is the sum of the solutions to the given equation?

ID: 011b5078 Answer

Correct Answer: 10.33, 31/3

Rationale

The correct answer is $\frac{31}{3}$. Subtracting $5(x + 7)$ from each side of the given equation yields $0 = 15(x - 17)(x + 7) - 5(x + 7)$. Since $5(x + 7)$ is a common factor of each of the terms on the right-hand side of this equation, it can be rewritten as $0 = 5(x + 7)(3(x - 17) - 1)$. This is equivalent to $0 = 5(x + 7)(3x - 51 - 1)$, or $0 = 5(x + 7)(3x - 52)$. Dividing both sides of this equation by 5 yields $0 = (x + 7)(3x - 52)$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $x + 7 = 0$ or $3x - 52 = 0$. Subtracting 7 from both sides of the equation $x + 7 = 0$ yields $x = -7$. Adding 52 to both sides of the equation $3x - 52 = 0$ yields $3x = 52$. Dividing both sides of this equation by 3 yields $x = \frac{52}{3}$. Therefore, the solutions to the given equation are -7 and $\frac{52}{3}$. It follows that the sum of the solutions to the given equation is $-7 + \frac{52}{3}$, which is equivalent to $-\frac{21}{3} + \frac{52}{3}$, or $\frac{31}{3}$. Note that 31/3 and 10.33 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID e9d19bb3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: e9d19bb3

$$\begin{aligned}x^2 + y^2 &= 36 \\ y &= mx + \frac{b}{4}\end{aligned}$$

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

- A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$
- B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$
- C. $-20m + 4\sqrt{11}$
- D. $20m - 4\sqrt{11}$

ID: e9d19bb3 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the graphs of $x^2 + y^2 = 36$ and $y = mx + \frac{b}{4}$, where m and b are negative constants, intersect at the point $(-5, y)$ in the xy -plane. It follows that this intersection point represents a solution to the given system of equations. Substituting -5 for x in the equation $x^2 + y^2 = 36$ yields $(-5)^2 + y^2 = 36$, or $25 + y^2 = 36$. Subtracting 25 from each side of this equation yields $y^2 = 11$. It's also given that $y < 0$, so taking the negative square root of each side of this equation yields $y = -\sqrt{11}$. Substituting -5 for x and $-\sqrt{11}$ for y in the equation $y = mx + \frac{b}{4}$ yields $-\sqrt{11} = m(-5) + \frac{b}{4}$, or $-\sqrt{11} = -5m + \frac{b}{4}$. Adding $5m$ to each side of this equation yields $5m - \sqrt{11} = \frac{b}{4}$. Multiplying each side of this equation by 4 yields $4(5m - \sqrt{11}) = b$. Applying the distributive property to the left-hand side of this equation yields $20m - 4\sqrt{11} = b$. Therefore, the expression $20m - 4\sqrt{11}$ represents the value of b .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b5e91945

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b5e91945

$$y = -\frac{1}{4}x^2 + 2x + 29$$

The given equation models a company’s scheduled deliveries over 8 months, where y is the estimated number of scheduled deliveries x months after the end of May 2012, where $0 \leq x \leq 8$. Which statement is the best interpretation of the y-intercept of the graph of this equation in the xy-plane?

- A. At the end of May 2012, the estimated number of scheduled deliveries was 0.
- B. At the end of May 2012, the estimated number of scheduled deliveries was 29.
- C. At the end of June 2012, the estimated number of scheduled deliveries was 0.
- D. At the end of June 2012, the estimated number of scheduled deliveries was 29.

ID: b5e91945 Answer

Correct Answer: B

Rationale

Choice B is correct. The y-intercept of a graph in the xy-plane is the point where $x = 0$. For the given equation, the y-intercept of the graph in the xy-plane can be found by substituting 0 for x in the equation, which yields $y = -\frac{1}{4}(0)^2 + 2(0) + 29$, or $y = 29$. Therefore, the y-intercept of the graph is (0, 29). It’s given that y is the estimated number of scheduled deliveries x months after the end of May 2012. Therefore, $x = 0$ represents 0 months after the end of May 2012, or the end of May 2012. Thus, the best interpretation of the y-intercept of the graph of this equation in the xy-plane is that at the end of May 2012, the estimated number of scheduled deliveries was 29.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 242d1f14

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 242d1f14

$$p + 34 = q + r$$

The given equation relates the variables p , q , and r . Which equation correctly expresses p in terms of q and r ?

- A. $p = q + r + 34$
- B. $p = q + r - 34$
- C. $p = -q - r + 34$
- D. $p = -q - r - 34$

ID: 242d1f14 Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting 34 from each side of the given equation yields $p = q + r - 34$. Thus, the equation $p = q + r - 34$ correctly expresses p in terms of q and r .

Choice A is incorrect. This equation can be rewritten as $p - 34 = q + r$.

Choice C is incorrect. This equation can be rewritten as $p - 34 = -q - r$.

Choice D is incorrect. This equation can be rewritten as $p + 34 = -q - r$.

Question Difficulty: Medium

Question ID ec792518

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: ec792518

$f(x) = |59 - 2x|$

The function f is defined by the given equation. For which of the following values of k does $f(k) = 3k$?

- A. $\frac{59}{5}$
- B. $\frac{59}{2}$
- C. $\frac{177}{5}$
- D. 59

ID: ec792518 Answer

Correct Answer: A

Rationale

Choice A is correct. The value of k for which $f(k) = 3k$ can be found by substituting k for x and $3k$ for $f(x)$ in the given equation, $f(x) = |59 - 2x|$, which yields $3k = |59 - 2k|$. For this equation to be true, either $-3k = 59 - 2k$ or $3k = 59 - 2k$. Adding $2k$ to both sides of the equation $-3k = 59 - 2k$ yields $-k = 59$. Dividing both sides of this equation by -1 yields $k = -59$. To check whether -59 is the value of k , substituting -59 for k in the equation $3k = |59 - 2k|$ yields $3(-59) = |59 - 2(-59)|$, which is equivalent to $-177 = |177|$, or $-177 = 177$, which isn't a true statement. Therefore, -59 isn't the value of k . Adding $2k$ to both sides of the equation $3k = 59 - 2k$ yields $5k = 59$. Dividing both sides of this equation by 5 yields $k = \frac{59}{5}$. To check whether $\frac{59}{5}$ is the value of k , substituting $\frac{59}{5}$ for k in the equation $3k = |59 - 2k|$ yields $3(\frac{59}{5}) = |59 - 2(\frac{59}{5})|$, which is equivalent to $\frac{177}{5} = |\frac{177}{5}|$, or $\frac{177}{5} = \frac{177}{5}$, which is a true statement. Therefore, the value of k for which $f(k) = 3k$ is $\frac{59}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6310780b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 6310780b

$$h(t) = -16t^2 + b$$

The function h estimates an object’s height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is **3,364** feet above the ground when it is dropped at $t = 0$. Approximately how many seconds after being dropped does the function estimate the object will hit the ground?

- A. 7.25
- B. 14.50
- C. 105.13
- D. 210.25

ID: 6310780b Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the function h estimates that the object is **3,364** feet above the ground when it's dropped at $t = 0$. Substituting **3,364** for $h(t)$ and **0** for t in the function h yields $3,364 = -16(0)^2 + b$, or $3,364 = b$. Substituting **3,364** for b in the function h yields $h(t) = -16t^2 + 3,364$. When the object hits the ground, its height will be **0** feet above the ground. Substituting **0** for $h(t)$ in $h(t) = -16t^2 + 3,364$ yields $0 = -16t^2 + 3,364$. Adding $16t^2$ to each side of this equation yields $16t^2 = 3,364$. Dividing each side of this equation by **16** yields $t^2 = 210.25$. Since the object will hit the ground at a positive number of seconds after it's dropped, the value of t can be found by taking the positive square root of each side of this equation, which yields $t = 14.50$. It follows that the function estimates the object will hit the ground approximately **14.50** seconds after being dropped.

Choice A is incorrect. The function estimates that **7.25** seconds after being dropped, the object's height will be $-16(7.25)^2 + 3,364$ feet, or **2,523** feet, above the ground.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID d4b3c7eb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: d4b3c7eb

$$f(x) = 272(2)^x$$

The function f is defined by the given equation. If $h(x) = f(x - 4)$, which of the following equations defines function h ?

- A. $h(x) = 17(2)^x$
- B. $h(x) = 68(2)^x$
- C. $h(x) = 272(16)^x$
- D. $h(x) = 272(8)^x$

ID: d4b3c7eb Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $f(x) = 272(2)^x$ and $h(x) = f(x - 4)$. Substituting $x - 4$ for x in $f(x) = 272(2)^x$ yields $f(x - 4) = 272(2)^{x-4}$. Substituting $h(x)$ for $f(x - 4)$ in this equation yields $h(x) = 272(2)^{x-4}$. Using the properties of exponents, the function $h(x) = 272(2)^{x-4}$ can be rewritten as $h(x) = \frac{272(2)^x}{2^4}$, which is equivalent to $h(x) = \frac{272(2)^x}{16}$, or $h(x) = 17(2)^x$. Therefore, of the given choices, an equation that defines function h is $h(x) = 17(2)^x$.

Choice B is incorrect. This equation defines function h if $h(x) = f(x - 2)$, not $h(x) = f(x - 4)$.

Choice C is incorrect. This equation defines function h if $h(x) = f(4x)$, not $h(x) = f(x - 4)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e581653e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: e581653e

Which expression is equivalent to $(x)^{\frac{1}{14}}$, where $x > 0$?

- A. $\frac{1}{14} \cdot x$
- B. $\sqrt[14]{x}$
- C. $14 \cdot x$
- D. $(x)^{14}$

ID: e581653e Answer

Correct Answer: B

Rationale

Choice B is correct. An expression in the form $x^{\frac{1}{k}}$, where $x > 0$ and $k > 0$, is equivalent to $\sqrt[k]{x}$. It follows that $x^{\frac{1}{14}}$, where $x > 0$, is equivalent to $\sqrt[14]{x}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 81ebd1e5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 81ebd1e5

$q(x) = 32(2^x)$

Which table gives three values of x and their corresponding values of $q(x)$ for function q ?

- A.

x	-1	0	1
$q(x)$	-64	0	64
- B.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	2	64
- C.

x	-1	0	1
$q(x)$	$\frac{1}{16}$	32	64
- D.

x	-1	0	1
$q(x)$	16	32	64

ID: 81ebd1e5 Answer

Correct Answer: D

Rationale

Choice D is correct. Substituting -1 for x in the given function yields $q(-1) = 32(2)^{-1}$, which is equivalent to $q(-1) = 32(\frac{1}{2})$, or $q(-1) = 16$. Therefore, when $x = -1$, the corresponding value of $q(x)$ for function q is 16 . Substituting 0 for x in the given function yields $q(0) = 32(2)^0$, which is equivalent to $q(0) = 32(1)$, or $q(0) = 32$. Therefore, when $x = 0$, the corresponding value of $q(x)$ for function q is 32 . Substituting 1 for x in the given function yields $q(1) = 32(2)^1$, which is equivalent to $q(1) = 32(2)$, or $q(1) = 64$. Therefore, when $x = 1$, the corresponding value of $q(x)$ for function q is 64 . Of the choices given, only the table in choice D gives these three values of x and their corresponding values of $q(x)$ for function q .

Choice A is incorrect. This table gives three values of x and their corresponding values of $q(x)$ for the function $q(x) = 32(2x)$.

Choice B is incorrect. This table gives three values of x and their corresponding values of $q(x)$ for the function $q(x) = 2(32)^x$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID de3eaf05

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: de3eaf05

$$P(t) = 290(1.04)^{(\frac{4}{6})t}$$

The function P models the population, in thousands, of a certain city t years after **2005**. According to the model, the population is predicted to increase by $n\%$ every **18** months. What is the value of n ?

- A. **0.38**
- B. **1.04**
- C. **4**
- D. **6**

ID: de3eaf05 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the function P models the population of the city t years after **2005**. Since there are **12** months in a year, **18** months is equivalent to $\frac{18}{12}$ years. Therefore, the expression $\frac{18}{12}x$ can represent the number of years in x **18**-month periods. Substituting $\frac{18}{12}x$ for t in the given equation yields $P(\frac{18}{12}x) = 290(1.04)^{(\frac{4}{6})(\frac{18}{12}x)}$, which is equivalent to $P(\frac{18}{12}x) = 290(1.04)^x$. Therefore, for each **18**-month period, the predicted population of the city is **1.04** times, or **104%** of, the previous population. This means that the population is predicted to increase by **4%** every **18** months.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. Each year, the predicted population of the city is **1.04** times the previous year's predicted population, which is not the same as an increase of **1.04%**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 851220ef

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 851220ef

$$\frac{14x}{7y} = 2\sqrt{w + 19}$$

The given equation relates the distinct positive real numbers w , x , and y . Which equation correctly expresses w in terms of x and y ?

- A. $w = \sqrt{\frac{x}{y}} - 19$
- B. $w = \sqrt{\frac{28x}{14y}} - 19$
- C. $w = \left(\frac{x}{y}\right)^2 - 19$
- D. $w = \left(\frac{28x}{14y}\right)^2 - 19$

ID: 851220ef Answer

Correct Answer: C

Rationale

Choice C is correct. Dividing each side of the given equation by 2 yields $\frac{14x}{14y} = \frac{2\sqrt{w+19}}{2}$, or $\frac{x}{y} = \sqrt{w + 19}$. Because it's given that each of the variables is positive, squaring each side of this equation yields the equivalent equation $\left(\frac{x}{y}\right)^2 = w + 19$. Subtracting 19 from each side of this equation yields $\left(\frac{x}{y}\right)^2 - 19 = w$, or $w = \left(\frac{x}{y}\right)^2 - 19$.

Choice A is incorrect. This equation isn't equivalent to the given equation.

Choice B is incorrect. This equation isn't equivalent to the given equation.

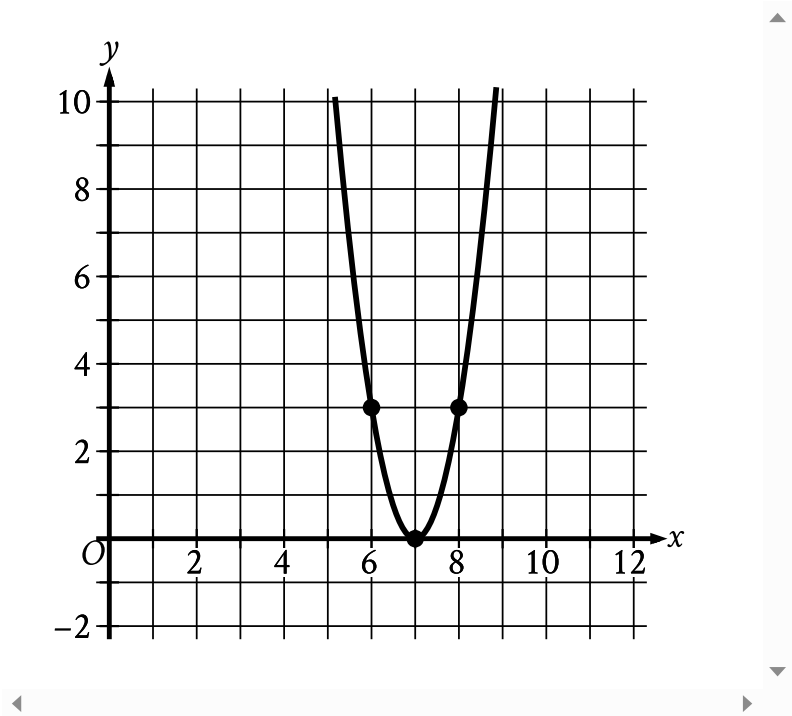
Choice D is incorrect. This equation isn't equivalent to the given equation.

Question Difficulty: Hard

Question ID 6217cb62

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 6217cb62



The x-intercept of the graph shown is $(x, 0)$. What is the value of x ?

ID: 6217cb62 Answer

Correct Answer: 7

Rationale

The correct answer is **7**. It's given that the x-intercept of the graph shown is $(x, 0)$. The graph passes through the point $(7, 0)$. Therefore, the value of x is **7**.

Question Difficulty: Easy

Question ID 91d0c03d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 91d0c03d

$$2|4 - x| + 3|4 - x| = 25$$

What is the positive solution to the given equation?

ID: 91d0c03d Answer

Correct Answer: 9

Rationale

The correct answer is **9**. The given equation can be rewritten as $5|4 - x| = 25$. Dividing each side of this equation by **5** yields $|4 - x| = 5$. By the definition of absolute value, if $|4 - x| = 5$, then $4 - x = 5$ or $4 - x = -5$. Subtracting **4** from each side of the equation $4 - x = 5$ yields $-x = 1$. Dividing each side of this equation by -1 yields $x = -1$. Similarly, subtracting **4** from each side of the equation $4 - x = -5$ yields $-x = -9$. Dividing each side of this equation by -1 yields $x = 9$. Therefore, since the two solutions to the given equation are -1 and **9**, the positive solution to the given equation is **9**.

Question Difficulty: Hard

Question ID 067262c7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 067262c7

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(7, 0)$ and $(-3, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

- A. -6
- B. -3
- C. 4
- D. 5

ID: 067262c7 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph of $y = f(x)$ in the xy -plane passes through the points $(7, 0)$ and $(-3, 0)$. It follows that when the value of x is either 7 or -3 , the value of $f(x)$ is 0. It's also given that the function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. It follows that the function f is a quadratic function and, therefore, may be written in factored form as $f(x) = a(x - u)(x - v)$, where the value of $f(x)$ is 0 when x is either u or v . Since the value of $f(x)$ is 0 when the value of x is either 7 or -3 , and the value of $f(x)$ is 0 when the value of x is either u or v , it follows that u and v are equal to 7 and -3 . Substituting 7 for u and -3 for v in the equation $f(x) = a(x - u)(x - v)$ yields $f(x) = a(x - 7)(x - (-3))$, or $f(x) = a(x - 7)(x + 3)$. Distributing the right-hand side of this equation yields $f(x) = a(x^2 - 7x + 3x - 21)$, or $f(x) = ax^2 - 4ax - 21a$. Since it's given that $f(x) = ax^2 + bx + c$, it follows that $b = -4a$. Adding a to each side of this equation yields $a + b = -3a$. Since $a + b = -3a$, if a is an integer, the value of $a + b$ must be a multiple of 3. If a is an integer greater than 1, it follows that $a \geq 2$. Therefore, $-3a \leq -3(2)$. It follows that the value of $a + b$ is less than or equal to $-3(2)$, or -6 . Of the given choices, only -6 is a multiple of 3 that's less than or equal to -6 .

Choice B is incorrect. This is the value of $a + b$ if a is equal to, not greater than, 1.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 1ebf5c78

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 1ebf5c78

Which expression is equivalent to $\frac{42a}{k} + 42ak$, where $k > 0$?

- A. $\frac{84a}{k}$
- B. $\frac{84ak^2}{k}$
- C. $\frac{42a(k+1)}{k}$
- D. $\frac{42a(k^2+1)}{k}$

ID: 1ebf5c78 Answer

Correct Answer: D

Rationale

Choice D is correct. Two fractions can be added together when they have a common denominator. Since $k > 0$, multiplying the second term in the given expression by $\frac{k}{k}$ yields $\frac{(42ak)k}{k}$, which is equivalent to $\frac{42ak^2}{k}$. Therefore, the expression $\frac{42a}{k} + 42ak$ can be written as $\frac{42a}{k} + \frac{42ak^2}{k}$ which is equivalent to $\frac{42a+42ak^2}{k}$. Since each term in the numerator of this expression has a factor of $42a$, the expression $\frac{42a+42ak^2}{k}$ can be rewritten as $\frac{42a(1)+42a(k^2)}{k}$, or $\frac{42a(1+k^2)}{k}$, which is equivalent to $\frac{42a(k^2+1)}{k}$.

Choice A is incorrect. This expression is equivalent to $\frac{42a}{k} + \frac{42a}{k}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $\frac{42a}{k} + 42a$.

Question Difficulty: Hard

Question ID dd63d964

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: dd63d964

$$c - 7 = 25p + k$$

The given equation relates the positive numbers c , p , and k . Which equation correctly expresses c in terms of p and k ?

- A. $c = 25p + k + 7$
- B. $c = 25p + k - 7$
- C. $c = 7(25p + k)$
- D. $c = \frac{25p+k}{7}$

ID: dd63d964 Answer

Correct Answer: A

Rationale

- Choice A is correct. Adding 7 to each side of the given equation yields $c = 25p + k + 7$.
- Choice B is incorrect. This equation is equivalent to $c + 7 = 25p + k$, not $c - 7 = 25p + k$.
- Choice C is incorrect. This equation is equivalent to $\frac{c}{7} = 25p + k$, not $c - 7 = 25p + k$.
- Choice D is incorrect. This equation is equivalent to $7c = 25p + k$, not $c - 7 = 25p + k$.

Question Difficulty: Medium

Question ID ca38bb3c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: ca38bb3c

Which of the following expressions is equivalent to $8x^{10} - 8x^9 + 88x$?

- A. $x(7x^{10} - 7x^9 + 87x)$
- B. $x(8^{10} - 8^9 + 88)$
- C. $8x(x^{10} - x^9 + 11x)$
- D. $8x(x^9 - x^8 + 11)$

ID: ca38bb3c Answer

Correct Answer: D

Rationale

Choice D is correct. Since $8x$ is a common factor of each term in the given expression, the expression can be rewritten as $8x(x^9 - x^8 + 11)$.

Choice A is incorrect. This expression is equivalent to $7x^{11} - 7x^{10} + 87x^2$.

Choice B is incorrect. This expression is equivalent to $8^{10}x - 8^9x + 88x$.

Choice C is incorrect. This expression is equivalent to $8x^{11} - 8x^{10} + 88x^2$.

Question Difficulty: Medium

Question ID 94549745

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 94549745

Which expression is equivalent to $20w - (4w + 3w)$?

- A. $10w$
- B. $13w$
- C. $19w$
- D. $21w$

ID: 94549745 Answer

Correct Answer: B

Rationale

Choice B is correct. Combining like terms inside the parentheses of the given expression, $20w - (4w + 3w)$, yields $20w - (7w)$. Combining like terms in this resulting expression yields $13w$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c2f84cf6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: c2f84cf6

There are **240** players in a tennis competition that includes **4** rounds of matches. Each player in the competition will play a match against another player in round **1**. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p , eliminated at the end of round r , where $r \leq 4$?

- A. $p = 15\left(\frac{1}{2}\right)^r$
- B. $p = 15(2)^r$
- C. $p = 240\left(\frac{1}{2}\right)^r$
- D. $p = 240(2)^r$

ID: c2f84cf6 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that one of the two players in a match is eliminated during each round. Therefore, the number of players decreases by half at the end of each round, and so the number of players eliminated at the end of a round also decreases by half. A decreasing exponential equation can be written in the form $p = a(1 - b)^r$, where b is the value of the decrease each round, r , and a is the value of p , the number of players, when $r = 0$. It's given that there are **240** players in the tennis competition. Therefore, $a = 240$. Since the number of players eliminated at the end of a round decreases by half, it follows that $b = \frac{1}{2}$. Substituting **240** for a and $\frac{1}{2}$ for b in the equation $p = a(1 - b)^r$ yields $p = 240\left(1 - \frac{1}{2}\right)^r$, or $p = 240\left(\frac{1}{2}\right)^r$.

Choice A is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are **15**, not **240**, players.

Choice B is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are **15**, not **240**, players, and for which the number of players doubles, rather than decreases by half, after each round.

Choice D is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which the number of players doubles, rather than decreases by half, after each round.

Question Difficulty: Easy

Question ID a3e4550c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a3e4550c

The function f is defined by $f(x) = \frac{1}{6x}$. What is the value of $f(x)$ when $x = 3$?

- A. $\frac{1}{3}$
- B. $\frac{1}{6}$
- C. $\frac{1}{9}$
- D. $\frac{1}{18}$

ID: a3e4550c Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $f(x) = \frac{1}{6x}$. Substituting 3 for x in this equation yields $f(3) = \frac{1}{6(3)}$, or $f(3) = \frac{1}{18}$. Therefore, when $x = 3$, the value of $f(x)$ is $\frac{1}{18}$.

Choice A is incorrect. This is the value of $f(x)$ when $x = 0.5$.

Choice B is incorrect. This is the value of $f(x)$ when $x = 1$.

Choice C is incorrect. This is the value of $f(x)$ when $x = 1.5$.

Question Difficulty: Easy

Question ID b1912d90

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b1912d90

A physics class is planning an experiment about a toy rocket. The equation $y = -16(x - 5.6)^2 + 502$ gives the estimated height y , in feet, of the toy rocket x seconds after it is launched into the air. Which of the following is the best interpretation of the vertex of the graph of the equation in the xy -plane?

- A. This toy rocket reaches an estimated maximum height of **502** feet **16** seconds after it is launched into the air.
- B. This toy rocket reaches an estimated maximum height of **502** feet **5.6** seconds after it is launched into the air.
- C. This toy rocket reaches an estimated maximum height of **16** feet **502** seconds after it is launched into the air.
- D. This toy rocket reaches an estimated maximum height of **5.6** feet **502** seconds after it is launched into the air.

ID: b1912d90 Answer

Correct Answer: B

Rationale

Choice B is correct. The vertex of the graph of a quadratic equation is where it reaches its minimum or maximum value. When a quadratic equation is written in the form $y = a(x - h)^2 + k$, the vertex of the parabola represented by the equation is $(x, y) = (h, k)$. In the given equation $y = -16(x - 5.6)^2 + 502$, the value of h is **5.6** and the value of k is **502**. It follows that the vertex of the graph of this equation in the xy -plane is $(x, y) = (5.6, 502)$. Additionally, since $a = -16$ in the given equation, the graph of the quadratic equation opens down, and the vertex represents a maximum. It's given that the value of y represents the estimated height, in feet, of the toy rocket x seconds after it is launched into the air. Therefore, this toy rocket reaches an estimated maximum height of **502** feet **5.6** seconds after it is launched into the air.

Choice A is incorrect. The **16** in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice C is incorrect. The **16** in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice D is incorrect. This is an interpretation of the vertex of the graph of the equation $y = -16(x - 502)^2 + 5.6$, not of the equation $y = -16(x - 5.6)^2 + 502$.

Question Difficulty: Medium

Question ID e1b3f286

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: e1b3f286

Which expression is equivalent to $100x^2 + 2,600$?

- A. $100(26x^2)$
- B. $100(x^2 + 26)$
- C. $2,600(100x^2)$
- D. $2,600(100x^2 + 1)$

ID: e1b3f286 Answer

Correct Answer: B

Rationale

Choice B is correct. Each term in the given expression, $100x^2 + 2,600$, has a common factor of **100**. Therefore, the expression can be rewritten as $100(x^2) + 100(26)$, or $100(x^2 + 26)$.

Choice A is incorrect. This expression is equivalent to $2,600x^2$, not $100x^2 + 2,600$.

Choice C is incorrect. This expression is equivalent to $260,000x^2$, not $100x^2 + 2,600$.

Choice D is incorrect. This expression is equivalent to $260,000x^2 + 2,600$, not $100x^2 + 2,600$.

Question Difficulty: Easy

Question ID 9f4b5268

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 9f4b5268

The function f is defined by $f(x) = 8\sqrt{x}$. For what value of x does $f(x) = 48$?

- A. 6
- B. 8
- C. 36
- D. 64

ID: 9f4b5268 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $f(x) = 8\sqrt{x}$. Substituting 48 for $f(x)$ in this equation yields $48 = 8\sqrt{x}$. Dividing both sides of this equation by 8 yields $6 = \sqrt{x}$. This can be rewritten as $\sqrt{x} = 6$. Squaring both sides of this equation yields $x = 36$. Therefore, the value of x for which $f(x) = 48$ is 36.

Choice A is incorrect. If $x = 6$, $f(x) = 8\sqrt{6}$, not 48.

Choice B is incorrect. If $x = 8$, $f(x) = 8\sqrt{8}$, not 48.

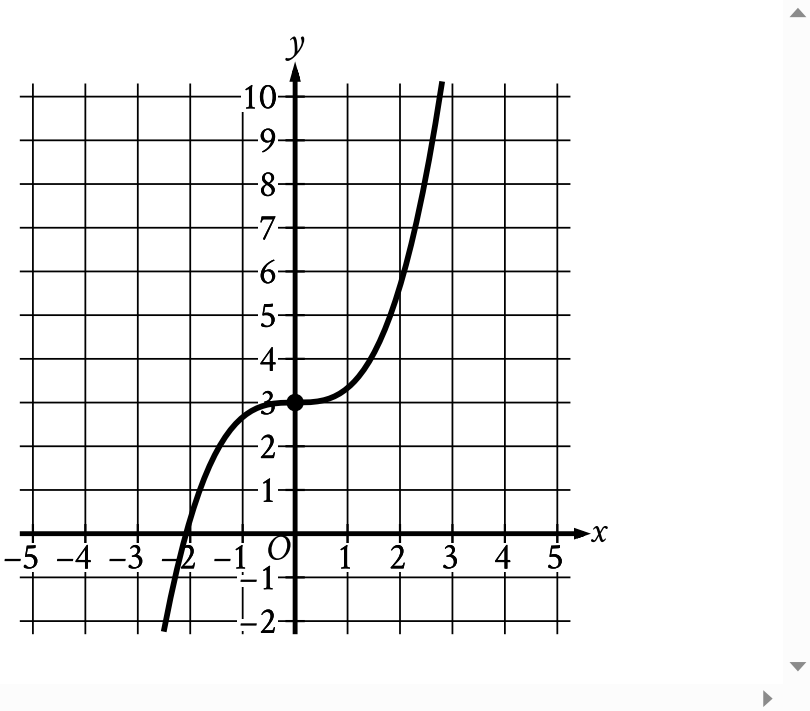
Choice D is incorrect. If $x = 64$, $f(x) = 8\sqrt{64}$, which is equivalent to 64, not 48.

Question Difficulty: Medium

Question ID 48235790

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 48235790



The graph of $y = f(x)$ is shown. What is the y -intercept of $y = f(x) + 5$?

- A. $(0, 8)$
- B. $(0, 5)$
- C. $(0, 3)$
- D. $(0, 0)$

ID: 48235790 Answer

Correct Answer: A

Rationale

Choice A is correct. The y -intercept of a graph is the point at which the graph intersects the y -axis. The graph of $y = f(x)$ shown intersects the y -axis at $(0, 3)$, so the y -intercept of the graph of $y = f(x)$ is $(0, 3)$. The graph of $y = f(x) + 5$ is the result of shifting the graph of $y = f(x)$ up 5 units. Therefore, the y -intercept of the graph of $y = f(x) + 5$ is 5 units above the y -intercept of the graph of $y = f(x)$. It follows that the y -intercept of the graph of $y = f(x) + 5$ is $(0, 3 + 5)$, which is equivalent to $(0, 8)$.

Choice B is incorrect. The point $(0, 5)$ is 5 units above the point $(0, 0)$, not the point $(0, 3)$.

Choice C is incorrect. This is the y -intercept of the graph of $y = f(x)$, not the y -intercept of the graph of $y = f(x) + 5$.

Choice D is incorrect. This is the origin of the coordinate plane, not the y -intercept of the graph of $y = f(x) + 5$.

Question Difficulty: Hard

Question ID 41227fe9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 41227fe9

$$\begin{aligned}y &= x^2 + 14x + 48 \\x + 8 &= 11\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: 41227fe9 Answer

Correct Answer: 99

Rationale

The correct answer is **99**. In the given system of equations, the second equation is $x + 8 = 11$. Subtracting **8** from both sides of this equation yields $x = 3$. In the given system of equations, the first equation is $y = x^2 + 14x + 48$. Substituting **3** for x in this equation yields $y = (3)^2 + 14(3) + 48$, or $y = 99$. Therefore, the solution to the given system of equations is $(x, y) = (3, 99)$. Thus, the value of y is **99**.

Question Difficulty: Medium

Question ID 239499fb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 239499fb

$|p| + 61 = 65$

Which value is a solution to the given equation?

- A. $\frac{65}{61}$
- B. 4
- C. 126
- D. 130

ID: 239499fb Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting 61 from each side of the given equation yields $|p| = 4$. By the definition of absolute value, if $|p| = 4$, then $p = 4$ or $p = -4$. Of the given choices, 4 is a solution to the given equation.

Choice A is incorrect. This is the quotient, not the difference, of 65 and 61.

Choice C is incorrect. This is the sum, not the difference, of 65 and 61.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 53445aef

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 53445aef

$$f(x) = 3,000(0.75)^x$$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after **2008**, where $x \leq 8$. Which of the following is the best interpretation of **3,000** in this context?

- A. The estimated percent decrease in the population for this species and area every **8** years after **2008**
- B. The estimated percent decrease in the population for this species and area each year after **2008**
- C. The estimated population for this species and area **8** years after **2008**
- D. The estimated initial population for this species and area in **2008**

ID: 53445aef Answer

Correct Answer: D

Rationale

Choice D is correct. Substituting **0** for x in the given equation yields $f(0) = 3,000(0.75)^0$, which is equivalent to $f(0) = 3,000(1)$, or $f(0) = 3,000$. It's given that the function estimates the species' population x years after **2008**, so it follows that the estimated population of the species is **3,000** in **2008**. Therefore, the best interpretation of **3,000** in this context is the estimated initial population for this species and area in **2008**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. The estimated percent decrease in the population for this species and area each year after **2008** is **25%**, not **3,000**.

Choice C is incorrect. The estimated population for this species and area **8** years after **2008** is $3,000(0.75)^8$, or approximately **300**, not **3,000**.

Question Difficulty: Medium

Question ID b26d01de

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b26d01de

The functions p and r are defined by the equations shown, where a and b are integer constants such that $a < b$ and $b < 0$. If $y = p(x)$ and $y = r(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- I. $p(x) = a(4.1)^x + b$
- II. $r(x) = a(4.1)^{x+b}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: b26d01de Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that a and b are integer constants such that $a < b$ and $b < 0$. The y -coordinate of the y -intercept of the graph of a function is the value of the function when $x = 0$. Equation I is $p(x) = a(4.1)^x + b$. Substituting 0 for x in this equation yields $p(0) = a(4.1)^0 + b$, which can be rewritten as $p(0) = a(1) + b$, or $p(0) = a + b$. Therefore, the y -coordinate of the y -intercept of the graph of $y = p(x)$ is $a + b$. Equation I has a coefficient of a and a constant b . Thus, the value of the y -coordinate of the y -intercept, $a + b$, is not displayed as a constant or coefficient in equation I. Equation II is $r(x) = a(4.1)^{x+b}$. Similarly, substituting 0 for x in this equation yields $r(0) = a(4.1)^{0+b}$, which can be rewritten as $r(0) = a(4.1)^b$. Since $b < 0$, the value of $(4.1)^b$ is not equal to 1. Thus, the value of $a(4.1)^b$ is not equal to a . The value of $a(4.1)^b$ is also not equal to b , since there is no pair of integers a and b such that $a(4.1)^b = b$. Thus, the value of the y -coordinate of the y -intercept for this equation is $a(4.1)^b$, which is not displayed in equation II as a constant or coefficient. The coefficient in equation II is a , and the only constant, b , appears in the exponent. Therefore, neither equation I nor equation II displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a78f860c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: a78f860c

$$x^2 - 34x + c = 0$$

In the given equation, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

ID: a78f860c Answer

Correct Answer: 289

Rationale

The correct answer is **289**. A quadratic equation of the form $ax^2 + bx + c = 0$, where a, b , and c are constants, has no real solutions when the value of the discriminant, $b^2 - 4ac$, is less than 0. In the given equation, $x^2 - 34x + c = 0$, $a = 1$ and $b = -34$. Therefore, the discriminant of the given equation can be expressed as $(-34)^2 - 4(1)(c)$, or $1,156 - 4c$. It follows that the given equation has no real solutions when $1,156 - 4c < 0$. Adding $4c$ to both sides of this inequality yields $1,156 < 4c$. Dividing both sides of this inequality by 4 yields $289 < c$, or $c > 289$. It's given that the equation $x^2 - 34x + c = 0$ has no real solutions when $c > n$. Therefore, the least possible value of n is **289**.

Question Difficulty: Hard

Question ID 2ebe1df5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2ebe1df5

$h(x) = x^2 - 3$

Which table gives three values of x and their corresponding values of $h(x)$ for the given function h ?

A.

x	1	2	3
$h(x)$	4	5	6

B.

x	1	2	3
$h(x)$	-2	1	6

C.

x	1	2	3
$h(x)$	-1	1	3

D.

x	1	2	3
$h(x)$	-2	1	3

ID: 2ebe1df5 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $h(x) = x^2 - 3$. Each table gives 1, 2, and 3 as the three given values of x . Substituting 1 for x in the equation $h(x) = x^2 - 3$ yields $h(1) = (1)^2 - 3$, or $h(1) = -2$. Substituting 2 for x in the equation $h(x) = x^2 - 3$ yields $h(2) = (2)^2 - 3$, or $h(2) = 1$. Finally, substituting 3 for x in the equation $h(x) = x^2 - 3$ yields $h(3) = (3)^2 - 3$, or $h(3) = 6$. Therefore, $h(x)$ is -2 when x is 1, $h(x)$ is 1 when x is 2, and $h(x)$ is 6 when x is 3. Choice B is a table with these values of x and their corresponding values of $h(x)$.

Choice A is incorrect. This is a table of values for the function $h(x) = x + 3$, not $h(x) = x^2 - 3$.

Choice C is incorrect. This is a table of values for the function $h(x) = 2x - 3$, not $h(x) = x^2 - 3$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID aa7efe94

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: aa7efe94

$$h(t) = -16t^2 + b$$

The function h estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. The function estimates that the ball is **19.36** feet above the floor when it is released at $t = 0$. How many seconds after being released does the function estimate the ball will reach the floor?

ID: aa7efe94 Answer

Correct Answer: 1.1, 11/10

Rationale

The correct answer is **1.1**. It's given that $h(t) = -16t^2 + b$ estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. It's also given that the function estimates that the ball is **19.36** feet above the floor when it is released at $t = 0$. Substituting **19.36** for $h(t)$ and **0** for t in the given function yields $19.36 = -16(0)^2 + b$, or $19.36 = b$. Substituting **19.36** for b in function h yields $h(t) = -16t^2 + 19.36$. When the ball reaches the floor, its height is **0** feet above the floor. Substituting **0** for $h(t)$ yields $0 = -16t^2 + 19.36$. Adding $16t^2$ to both sides of this equation yields $16t^2 = 19.36$. Dividing both sides of this equation by **16** yields $t^2 = 1.21$. Taking the positive square root of both sides of this equation yields $t = 1.1$. Therefore, the function estimates that the ball will reach the floor **1.1** seconds after being released. Note that 1.1 and 11/10 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 7904edea

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7904edea

The function f is defined by $f(x) = 4x^{-1}$. What is the value of $f(21)$?

- A. -84
- B. $\frac{1}{84}$
- C. $\frac{4}{21}$
- D. $\frac{21}{4}$

ID: 7904edea Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that function f is defined by the equation $f(x) = 4x^{-1}$. The value of $f(21)$ is the value of $f(x)$ when $x = 21$. Substituting 21 for x in the given equation yields $f(21) = 4(21)^{-1}$, which is equivalent to $f(21) = 4(\frac{1}{21})$, or $f(21) = \frac{4}{21}$.

Choice A is incorrect. This is the value of $f(21)$ when $f(x) = -4x$, rather than $f(x) = 4x^{-1}$.

Choice B is incorrect. This is the value of $f(21)$ when $f(x) = (4x)^{-1}$, rather than $f(x) = 4x^{-1}$.

Choice D is incorrect. This is the value of $f(21)$ when $f(x) = (4^{-1})x$, rather than $f(x) = 4x^{-1}$.

Question Difficulty: Medium

Question ID d2203268

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: d2203268

$(x + 2)(x - 5)(x + 9) = 0$

What is a positive solution to the given equation?

- A. 3
- B. 4
- C. 5
- D. 18

ID: d2203268 Answer

Correct Answer: C

Rationale

Choice C is correct. Applying the zero product property to the given equation yields three equations: $x + 2 = 0$, $x - 5 = 0$, and $x + 9 = 0$. Subtracting 2 from both sides of the equation $x + 2 = 0$ yields $x = -2$. Adding 5 to both sides of the equation $x - 5 = 0$ yields $x = 5$. Subtracting 9 from both sides of the equation $x + 9 = 0$ yields $x = -9$. Therefore, the solutions to the given equation are -2 , 5 , and -9 . It follows that a positive solution to the given equation is 5 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 3d0f48a9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 3d0f48a9

$$P(t) = 260(1.04)^{(\frac{6}{4})t}$$

The function P models the population, in thousands, of a certain city t years after **2003**. According to the model, the population is predicted to increase by **4%** every n months. What is the value of n ?

- A. 8
- B. 12
- C. 18
- D. 72

ID: 3d0f48a9 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the function P models the population, in thousands, of a certain city t years after **2003**. The value of the base of the given exponential function, **1.04**, corresponds to an increase of **4%** for every increase of **1** in the exponent, $(\frac{6}{4})t$. If the exponent is equal to **0**, then $(\frac{6}{4})t = 0$. Multiplying both sides of this equation by $(\frac{4}{6})$ yields $t = 0$. If the exponent is equal to **1**, then $(\frac{6}{4})t = 1$. Multiplying both sides of this equation by $(\frac{4}{6})$ yields $t = \frac{4}{6}$, or $t = \frac{2}{3}$. Therefore, the population is predicted to increase by **4%** every $\frac{2}{3}$ of a year. It’s given that the population is predicted to increase by **4%** every n months. Since there are **12** months in a year, $\frac{2}{3}$ of a year is equivalent to $(\frac{2}{3})(12)$, or **8**, months. Therefore, the value of n is **8**.

Choice B is incorrect. This is the number of months in which the population is predicted to increase by **4%** according to the model $P(t) = 260(1.04)^t$, not $P(t) = 260(1.04)^{(\frac{6}{4})t}$.

Choice C is incorrect. This is the number of months in which the population is predicted to increase by **4%** according to the model $P(t) = 260(1.04)^{(\frac{4}{6})t}$, not $P(t) = 260(1.04)^{(\frac{6}{4})t}$.

Choice D is incorrect. This is the number of months in which the population is predicted to increase by **4%** according to the model $P(t) = 260(1.04)^{(\frac{1}{6})t}$, not $P(t) = 260(1.04)^{(\frac{6}{4})t}$.

Question Difficulty: Hard

Question ID 16b1a141

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 16b1a141

Which expression is equivalent to $34x + 34y$?

- A. $34xy$
- B. $34(x + y)$
- C. $68y$
- D. $68x$

ID: 16b1a141 Answer

Correct Answer: B

Rationale

Choice B is correct. Since **34** is a common factor of each term in the given expression, the expression can be rewritten as **$34(x + y)$** .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to **$34y + 34y$** .

Choice D is incorrect. This expression is equivalent to **$34x + 34x$** .

Question Difficulty: Easy

Question ID 6f2658d4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 6f2658d4

The function f is defined by $f(x) = 10x^2 - 32x - 152$. What is the value of $f(0)$?

- A. -152
- B. -32
- C. 0
- D. 10

ID: 6f2658d4 Answer

Correct Answer: A

Rationale

Choice A is correct. The value of $f(0)$ is the value of $f(x)$ when $x = 0$. The function f is defined by $f(x) = 10x^2 - 32x - 152$. Substituting 0 for x in this equation yields $f(0) = 10(0)^2 - 32(0) - 152$. This equation can be rewritten as $f(0) = 10(0) - 0 - 152$, or $f(0) = -152$. Therefore, the value of $f(0)$ is -152 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 3599c6f2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 3599c6f2

The function $f(t) = 40,000(2)^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

- A. 2
- B. 790
- C. 1,580
- D. 40,000

ID: 3599c6f2 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that t minutes after an initial observation, the number of bacteria in a population is $40,000(2)^{\frac{t}{790}}$. This expression consists of the initial number of bacteria, **40,000**, multiplied by the expression $2^{\frac{t}{790}}$. The time, in minutes, it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{790}}$ to double. Since the base is **2**, the expression $2^{\frac{t}{790}}$ will double when the exponent increases by **1**. Since the exponent of this expression is $\frac{t}{790}$, the exponent will increase by **1** when t increases by **790**. Therefore, the time, in minutes, it takes for the number of bacteria in the population to double is **790**.

Choice A is incorrect. This is the base of the exponent, not the time it takes for the number of bacteria in the population to double.

Choice C is incorrect. This is the number of minutes it takes for the population to double twice.

Choice D is incorrect. This is the number of bacteria that are initially observed, not the time it takes for the number of bacteria in the population to double.

Question Difficulty: Hard

Question ID 840857f0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 840857f0

$(d - 30)(d + 30) - 7 = -7$

What is a solution to the given equation?

ID: 840857f0 Answer

Correct Answer: 30, -30

Rationale

The correct answer is either -30 or 30 . Adding 7 to each side of the given equation yields $(d - 30)(d + 30) = 0$. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0 , either $d - 30 = 0$ or $d + 30 = 0$. Adding 30 to each side of the equation $d - 30 = 0$ yields $d = 30$. Subtracting 30 from each side of the equation $d + 30 = 0$ yields $d = -30$. Therefore, the solutions to the given equation are -30 and 30 . Note that -30 and 30 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID b71042f8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b71042f8

$$7m = 2(n + p)$$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives m in terms of n and p ?

- A. $m = \frac{2(n+p)}{7}$
- B. $m = 2(n + p)$
- C. $m = 2(n + p) - 7$
- D. $m = 2 - n - p - 7$

ID: b71042f8 Answer

Correct Answer: A

Rationale

Choice A is correct. Dividing each side of the given equation by 7 yields $m = \frac{2(n+p)}{7}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation is equivalent to $7 + m = 2(n + p)$, not $7m = 2(n + p)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 8e87347e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 8e87347e

$$f(x) = 9,000(0.66)^x$$

The given function f models the number of advertisements a company sent to its clients each year, where x represents the number of years since **1997**, and $0 \leq x \leq 5$. If $y = f(x)$ is graphed in the xy -plane, which of the following is the best interpretation of the y -intercept of the graph in this context?

- A. The minimum estimated number of advertisements the company sent to its clients during the **5** years was **1,708**.
- B. The minimum estimated number of advertisements the company sent to its clients during the **5** years was **9,000**.
- C. The estimated number of advertisements the company sent to its clients in **1997** was **1,708**.
- D. The estimated number of advertisements the company sent to its clients in **1997** was **9,000**.

ID: 8e87347e Answer

Correct Answer: D

Rationale

Choice D is correct. The y -intercept of a graph in the xy -plane is the point where $x = 0$. For the given function f , the y -intercept of the graph of $y = f(x)$ in the xy -plane can be found by substituting **0** for x in the equation $y = 9,000(0.66)^x$, which gives $y = 9,000(0.66)^0$. This is equivalent to $y = 9,000(1)$, or $y = 9,000$. Therefore, the y -intercept of the graph of $y = f(x)$ is **(0, 9,000)**. It's given that the function f models the number of advertisements a company sent to its clients each year. Therefore, $f(x)$ represents the estimated number of advertisements the company sent to its clients each year. It's also given that x represents the number of years since **1997**. Therefore, $x = 0$ represents **0** years since **1997**, or **1997**. Thus, the best interpretation of the y -intercept of the graph of $y = f(x)$ is that the estimated number of advertisements the company sent to its clients in **1997** was **9,000**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID c0a5b1af

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: c0a5b1af

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by **24%** every **22.96** minutes. Which of the following equations could represent this model, where ***M*** is the estimated mass, in grams, of the sample ***t*** minutes after the lab analyst began observing the sample?

- A. $M = 100(0.24)^{t+22.96}$
- B. $M = 100(0.24)^{\frac{t}{22.96}}$
- C. $M = 100(0.76)^{t+22.96}$
- D. $M = 100(0.76)^{\frac{t}{22.96}}$

ID: c0a5b1af Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the mass of a sample of a substance decreases by **24%** every **22.96** minutes. Therefore, after each **22.96**-minute interval the mass is multiplied by **1 – 0.24**, or **0.76**. It’s also given that ***t*** represents the number of minutes after the lab analyst began observing the sample; therefore, $\frac{t}{22.96}$ represents the number of **22.96**-minute intervals. The equation that could represent this model is **$M = 100(0.76)^{\frac{t}{22.96}}$** , where ***M*** is the estimated mass, in grams, of the sample ***t*** minutes after the lab analyst began observing it.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a25d39ec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a25d39ec

The product of two positive integers is **462**. If the first integer is **5** greater than twice the second integer, what is the smaller of the two integers?

ID: a25d39ec Answer

Correct Answer: 14

Rationale

The correct answer is **14**. Let x represent the first integer and y represent the second integer. If the first integer is **5** greater than twice the second integer, then $x = 2y + 5$. It's given that the product of the two integers is **462**; therefore $xy = 462$. Substituting $2y + 5$ for x in this equation yields $(2y + 5)(y) = 462$, which can be written as $2y^2 + 5y = 462$. Subtracting **462** from each side of this equation yields $2y^2 + 5y - 462 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-462)$, or -924 , and whose sum is **5**. The two values whose product is -924 and whose sum is **5** are **33** and -28 . Thus, the equation $2y^2 + 5y - 462 = 0$ can be rewritten as $2y^2 - 28y + 33y - 462 = 0$, which is equivalent to $2y(y - 14) + 33(y - 14) = 0$, or $(2y + 33)(y - 14) = 0$. By the zero product property, it follows that $2y + 33 = 0$ or $y - 14 = 0$. Subtracting **33** from both sides of the equation $2y + 33 = 0$ yields $2y = -33$. Dividing both sides of this equation by **2** yields $y = -\frac{33}{2}$. Since y is a positive integer, the value of y isn't $-\frac{33}{2}$. Adding **14** to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting **14** for y in the equation $xy = 462$ yields $x(14) = 462$. Dividing both sides of this equation by **14** yields $x = 33$. Therefore, the two integers are **14** and **33**, so the smaller of the two integers is **14**.

Question Difficulty: Hard

Question ID cb2f4a8a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: cb2f4a8a

The function f is defined by $f(x) = 4 + \sqrt{x}$. What is the value of $f(144)$?

- A. 0
- B. 16
- C. 40
- D. 76

ID: cb2f4a8a Answer

Correct Answer: B

Rationale

Choice B is correct. The value of $f(144)$ is the value of $f(x)$ when $x = 144$. It's given that the function f is defined by $f(x) = 4 + \sqrt{x}$. Substituting 144 for x in this equation yields $f(144) = 4 + \sqrt{144}$. Since the positive square root of 144 is 12, it follows that this equation can be rewritten as $f(144) = 4 + 12$, or $f(144) = 16$. Therefore, the value of $f(144)$ is 16.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f(1,296)$, not $f(144)$.

Choice D is incorrect. This is the value of $f(5,184)$, not $f(144)$.

Question Difficulty: Easy

Question ID 54014ba0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 54014ba0

If $3x^2 - 18x - 15 = 0$, what is the value of $x^2 - 6x$?

ID: 54014ba0 Answer

Correct Answer: 5

Rationale

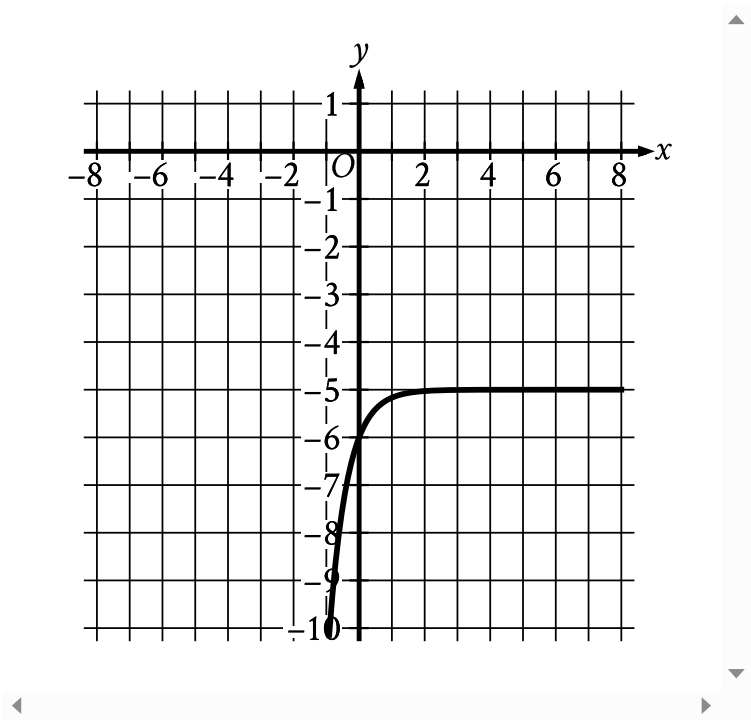
The correct answer is **5**. Dividing each side of the given equation by **3** yields $x^2 - 6x - 5 = 0$. Adding **5** to each side of this equation yields $x^2 - 6x = 5$. Therefore, if $3x^2 - 18x - 15 = 0$, the value of $x^2 - 6x$ is **5**.

Question Difficulty: Hard

Question ID 1b1b1cbf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 1b1b1cbf



What is the y-intercept of the graph shown?

- A. $(0, -6)$
- B. $(-6, 0)$
- C. $(0, 0)$
- D. $(-5, -5)$

ID: 1b1b1cbf Answer

Correct Answer: A

Rationale

Choice A is correct. The y-intercept of a graph in the xy -plane is the point (x, y) on the graph where $x = 0$. For the graph shown, at $x = 0$, the corresponding value of y is -6 . Therefore, the y-intercept of the graph shown is $(0, -6)$.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 45962172

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 45962172

The area A , in square centimeters, of a rectangular painting can be represented by the expression $w(w + 29)$, where w is the width, in centimeters, of the painting. Which expression represents the length, in centimeters, of the painting?

- A. w
- B. 29
- C. $(w + 29)$
- D. $w(w + 29)$

ID: 45962172 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the expression $w(w + 29)$ represents the area, in square centimeters, of a rectangular painting, where w is the width, in centimeters, of the painting. The area of a rectangle can be calculated by multiplying its length by its width. It follows that the length, in centimeters, of the painting is represented by the expression $(w + 29)$.

Choice A is incorrect. This expression represents the width, in centimeters, of the painting, not its length, in centimeters.

Choice B is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the painting, not its length, in centimeters.

Choice D is incorrect. This expression represents the area, in square centimeters, of the painting, not its length, in centimeters.

Question Difficulty: Medium

Question ID e89e2a0f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: e89e2a0f

The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

- I. $f(x) = 33(0.4)^{x+3}$
- II. $g(x) = 33(0.16)(0.4)^{x-2}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: e89e2a0f Answer

Correct Answer: B

Rationale

Choice B is correct. Functions f and g are both exponential functions with a base of 0.40 . Since 0.40 is less than 1 , functions f and g are both decreasing exponential functions. This means that $f(x)$ and $g(x)$ decrease as x increases. Since $f(x)$ and $g(x)$ decrease as x increases, the maximum value of each function occurs at the least value of x for which the function is defined. It's given that functions f and g are defined for $x \geq 0$. Therefore, the maximum value of each function occurs at $x = 0$. Substituting 0 for x in the equation defining f yields $f(0) = 33(0.4)^{0+3}$, which is equivalent to $f(0) = 33(0.4)^3$, or $f(0) = 2.112$. Therefore, the maximum value of f is 2.112 . Since the equation $f(x) = 33(0.4)^{x+3}$ doesn't display the value 2.112 , the equation defining f doesn't display the maximum value of f . Substituting 0 for x in the equation defining g yields $g(0) = 33(0.16)(0.4)^{0-2}$, which can be rewritten as $g(0) = 33(0.16)\left(\frac{1}{0.4^2}\right)$, or $g(0) = 33(0.16)\left(\frac{1}{0.16}\right)$, which is equivalent to $g(0) = 33$. Therefore, the maximum value of g is 33 . Since the equation $g(x) = 33(0.16)(0.4)^{x-2}$ displays the value 33 , the equation defining g displays the maximum value of g . Thus, only equation II displays, as a constant or coefficient, the maximum value of the function it defines.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

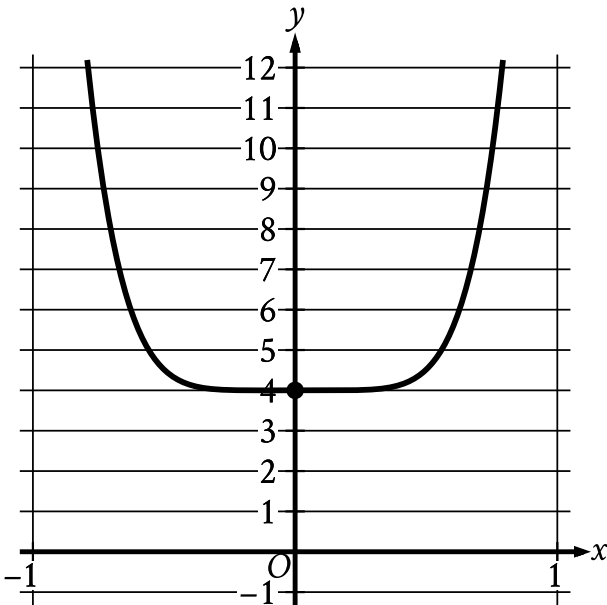
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 48551b37

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 48551b37



The graph of the polynomial function f , where $y = f(x)$, is shown. The y -intercept of the graph is $(0, y)$. What is the value of y ?

ID: 48551b37 Answer

Correct Answer: 4

Rationale

The correct answer is **4**. The y -intercept of the graph of a function in the xy -plane is the point where the graph intersects the y -axis. The graph of the polynomial function shown intersects the y -axis at the point $(0, 4)$. It's given that the y -intercept of the graph is $(0, y)$. Therefore, the value of y is **4**.

Question Difficulty: Easy

Question ID 8eec350c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 8eec350c

$$f(t) = 500(0.5)^{\frac{t}{12}}$$

The function f models the intensity of an X-ray beam, in number of particles in the X-ray beam, t millimeters below the surface of a sample of iron. According to the model, what is the estimated number of particles in the X-ray beam when it is at the surface of the sample of iron?

- A. 500
- B. 12
- C. 5
- D. 2

ID: 8eec350c Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function f models the intensity of an X-ray beam, in number of particles in the X-ray beam, t millimeters below the surface of a sample of iron. When the X-ray beam is at the surface of the sample of iron, it is **0** millimeters below the surface, so the value of t is **0**. Substituting **0** for t in the function $f(t) = 500(0.5)^{\frac{t}{12}}$ yields $f(0) = 500(0.5)^{\frac{0}{12}}$. Since any positive number raised to the power of **0** is equal to **1**, it follows that $f(0) = 500(1)$, or $f(0) = 500$. Therefore, the estimated number of particles in the X-ray beam at the surface of the sample of iron is **500**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 1c104baa

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 1c104baa

The function f is defined by $f(x) = 8x^3 + 4$. What is the value of $f(2)$?

ID: 1c104baa Answer

Correct Answer: 68

Rationale

The correct answer is **68**. It's given that the function f is defined by $f(x) = 8x^3 + 4$. Substituting **2** for x in this equation yields $f(2) = 8(2)^3 + 4$, or $f(2) = 8(8) + 4$, which is equivalent to $f(2) = 68$. Therefore, the value of $f(2)$ is **68**.

Question Difficulty: Medium

Question ID 97ecfd2a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 97ecfd2a

The function g is defined by $g(x) = \sqrt{x} + 300$. What is the value of $g(x)$ when $x = 81$?

- A. 9
- B. 300
- C. 309
- D. 381

ID: 97ecfd2a Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that function g is defined by $g(x) = \sqrt{x} + 300$. Substituting 81 for x in function g yields $g(81) = \sqrt{81} + 300$, which is equivalent to $g(81) = 9 + 300$, or $g(81) = 309$. Therefore, the value of $g(x)$ when $x = 81$ is 309 .

Choice A is incorrect. This is the value of $\sqrt{81}$, not $\sqrt{81} + 300$.

Choice B is incorrect and may result from conceptual or calculation errors.

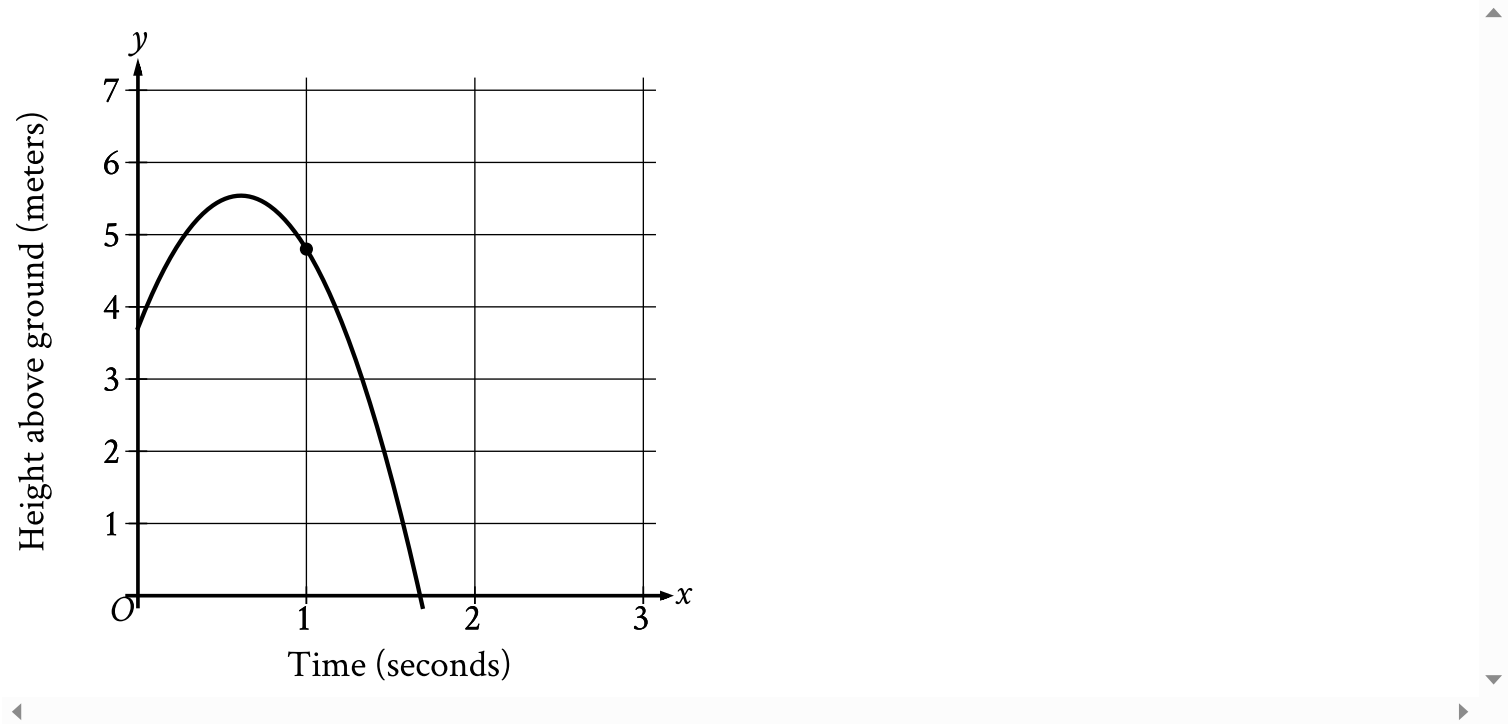
Choice D is incorrect. This is the value of $81 + 300$, not $\sqrt{81} + 300$.

Question Difficulty: Medium

Question ID 75a64b6b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 75a64b6b



The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point $(1.0, 4.8)$ in this context?

- A. 1.0 second after being launched, the ball's height above ground is 4.8 meters.
- B. 4.8 seconds after being launched, the ball's height above ground is 1.0 meter.
- C. The ball was launched from an initial height of 1.0 meter with an initial velocity of 4.8 meters per second.
- D. The ball was launched from an initial height of 4.8 meters with an initial velocity of 1.0 meter per second.

ID: 75a64b6b Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. In the graph shown, the x-axis represents time, in seconds, and the y-axis represents the height of the ball above ground, in meters. It follows that for the marked point $(1.0, 4.8)$, 1.00 represents the time, in seconds, after the ball was launched upward from a platform and 4.80 represents the height of the ball above ground, in meters. Therefore, the best interpretation of the marked point $(1.0, 4.8)$ is 1.00 second after being launched, the ball's height above ground is 4.80 meters.

Choice B is incorrect and may result from conceptual errors.

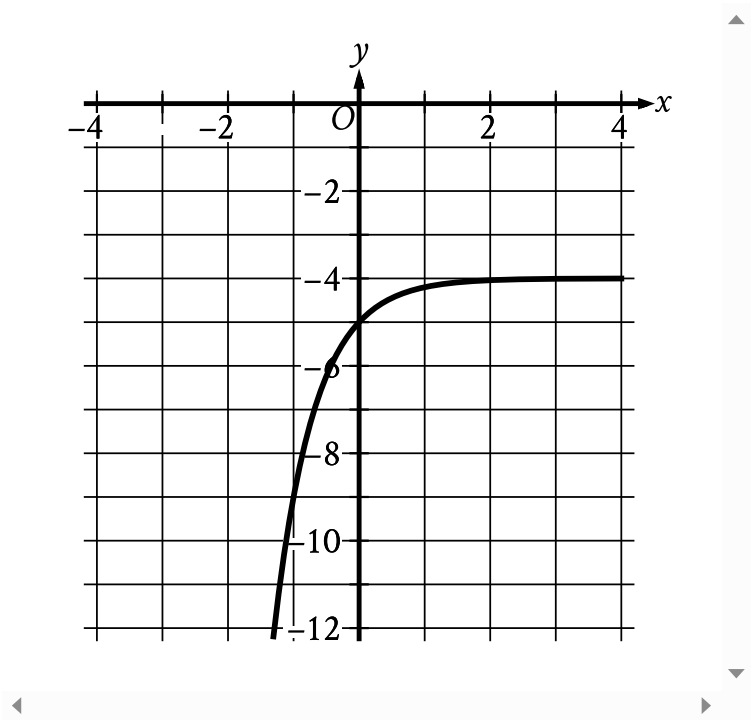
Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID 606edc58

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 606edc58



What is the y-intercept of the graph shown?

- A. $(-1, -9)$
- B. $(0, -5)$
- C. $(0, -4)$
- D. $(0, 0)$

ID: 606edc58 Answer

Correct Answer: B

Rationale

Choice B is correct. The y-intercept of a graph in the xy -plane is the point (x, y) on the graph where $x = 0$. At $x = 0$, the corresponding value of y is -5 . Therefore, the y-intercept of the graph shown is $(0, -5)$.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the y-intercept of a graph in the xy -plane that intersects the y -axis at $y = -4$, not $y = -5$.

Choice D is incorrect. This is the y-intercept of a graph in the xy -plane that intersects the y -axis at $y = 0$, not $y = -5$.

Question Difficulty: Easy

Question ID 49a4de69

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 49a4de69

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

- A. $x = \frac{b-72}{y}$
- B. $x = by - 72$
- C. $x = \frac{by-72}{y}$
- D. $x = by - 72y$

ID: 49a4de69 Answer

Correct Answer: D

Rationale

Choice D is correct. Multiplying both sides of the given equation by y yields $y(b - 72) = x$. Distributing on the left-hand side of this equation yields $by - 72y = x$, or $x = by - 72y$. Therefore, the equation $x = by - 72y$ correctly expresses x in terms of b and y .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 7e46fa03

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7e46fa03

The function f is defined by $f(x) = 6 + \sqrt{x}$. What is the value of $f(36)$?

ID: 7e46fa03 Answer

Correct Answer: 12

Rationale

The correct answer is **12**. The value of $f(36)$ is the value of $f(x)$ when $x = 36$. Substituting **36** for x in the given equation yields $f(36) = 6 + \sqrt{36}$, which is equivalent to $f(36) = 6 + 6$, or $f(36) = 12$. Thus, the value of $f(36)$ is **12**.

Question Difficulty: Medium

Question ID 60aca572

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 60aca572

Which expression is equivalent to $8 + d^2 + 3$?

- A. $d^2 + 24$
- B. $d^2 + 11$
- C. $d^2 + 5$
- D. $d^2 - 11$

ID: 60aca572 Answer

Correct Answer: B

Rationale

Choice B is correct. The given expression can be rewritten as $d^2 + 8 + 3$. Adding 8 and 3 in this expression yields $d^2 + 11$.

Choice A is incorrect. This expression is equivalent to $d^2 + 8(3)$.

Choice C is incorrect. This expression is equivalent to $8 + d^2 - 3$.

Choice D is incorrect. This expression is equivalent to $-8 + d^2 - 3$.

Question Difficulty: Easy

Question ID 48cea247

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 48cea247

Which expression is equivalent to $\frac{h^{15}q^7}{h^5q^{21}}$, where $h > 0$ and $q > 0$?

- A. $\frac{h^{10}}{q^{14}}$
- B. $\frac{h^3}{q^3}$
- C. $h^{10}q^{14}$
- D. h^3q^3

ID: 48cea247 Answer

Correct Answer: A

Rationale

Choice A is correct. For positive values of a , $\frac{a^m}{a^n} = a^{(m-n)}$, where m and n are integers. Since it's given that $h > 0$ and $q > 0$, this property can be applied to rewrite the given expression as $(h^{(15-5)})(q^{(7-21)})$, which is equivalent to $h^{10}q^{-14}$. For positive values of a , $a^{-n} = \frac{1}{a^n}$. This property can be applied to rewrite the expression $h^{10}q^{-14}$ as $(h^{10})\left(\frac{1}{q^{14}}\right)$, which is equivalent to $\frac{h^{10}}{q^{14}}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID fb399cf3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: fb399cf3

$$p = 20 + \frac{16}{n}$$

The given equation relates the numbers p and n , where n is not equal to 0 and $p > 20$. Which equation correctly expresses n in terms of p ?

- A. $n = \frac{p-20}{16}$
- B. $n = \frac{p}{16} + 20$
- C. $n = \frac{p}{16} - 20$
- D. $n = \frac{16}{p-20}$

ID: fb399cf3 Answer

Correct Answer: D

Rationale

Choice D is correct. To express n in terms of p , the given equation must be solved for n . Subtracting 9 from both sides of the given equation yields $p - 9 = \frac{14}{n}$. Since n is not equal to 0, multiplying both sides of this equation by n yields $(p - 9)(n) = 14$. It's given that $p > 9$, which means $p - 9$ is not equal to 0. Therefore, dividing both sides of $(p - 9)(n) = 14$ by $(p - 9)$ yields $\frac{(p-9)(n)}{p-9} = \frac{14}{p-9}$, or $n = \frac{14}{p-9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 2e2b270a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2e2b270a

$f(x) = (1.84)^{\frac{x}{4}}$

The function f is defined by the given equation. The equation can be rewritten as $f(x) = \left(1 + \frac{p}{100}\right)^x$, where p is a constant. Which of the following is closest to the value of p ?

- A. 16
- B. 21
- C. 46
- D. 96

ID: 2e2b270a Answer

Correct Answer: A

Rationale

Choice A is correct. The equation $f(x) = (1.84)^{\frac{x}{4}}$ can be rewritten as $f(x) = (1.84)^{(\frac{1}{4})(x)}$, which is equivalent to $f(x) = \left(1.84^{\frac{1}{4}}\right)^x$, or approximately $f(x) = (1.16467)^x$. Since it's given that $f(x) = (1.84)^{\frac{x}{4}}$ can be rewritten as $f(x) = \left(1 + \frac{p}{100}\right)^x$, where p is a constant, it follows that $1 + \frac{p}{100}$ is approximately equal to **1.16467**. Therefore, $\frac{p}{100}$ is approximately equal to **0.16467**. It follows that the value of p is approximately equal to **16.467**. Of the given choices, **16** is closest to the value of p .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

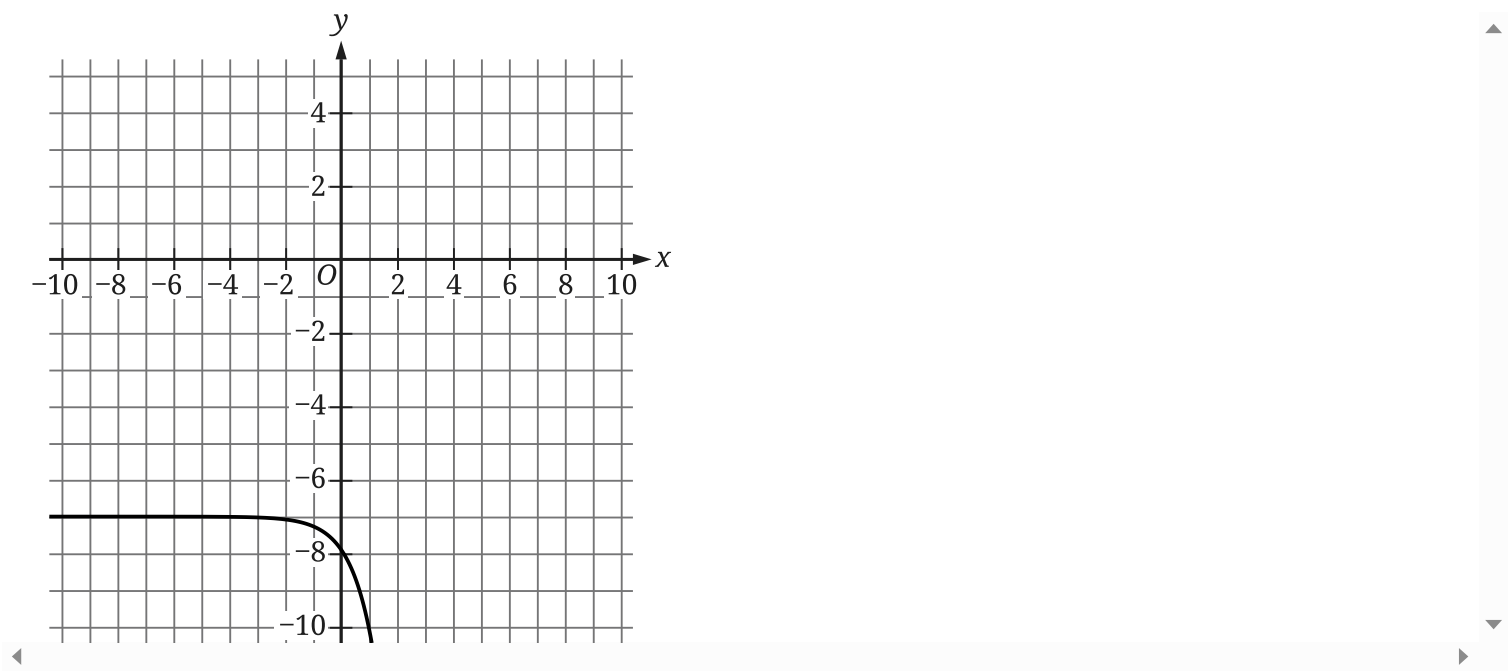
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID ab70ed73

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: ab70ed73



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a , b , and c are constants. For how many values of x does $f(x) = 0$?

- A. Three
- B. Two
- C. One
- D. Zero

ID: ab70ed73 Answer

Correct Answer: D

Rationale

Choice D is correct. Each point (x, y) on the graph of $y = f(x)$ in the xy -plane gives a value of x and its corresponding value of $f(x)$, or y . For any value of x for which $f(x) = 0$, there is a corresponding point on the graph of $y = f(x)$ with a y -coordinate of 0 . A point with a y -coordinate of 0 is a point where the graph intersects the x -axis. It's given that $f(x) = ab^x + c$, where a , b , and c are constants. In the xy -plane, the graph of an equation of this form will lie entirely either above or below the horizontal line defined by $y = c$. The part of the graph of $y = f(x)$ shown lies entirely below the horizontal line defined by $y = -7$, and thus the entire graph of $y = f(x)$ must lie below the line defined by $y = -7$. It follows that the graph of $y = f(x)$ will never intersect the x -axis. Therefore, there are zero values of x for which $f(x) = 0$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID 0d4b88df

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 0d4b88df

$$m(t) = -0.0274\left(\frac{t}{7}\right)^2 + 7.3873\left(\frac{t}{7}\right) + 75.032$$

The function m gives the predicted body mass $m(t)$, in **kilograms (kg)**, of a certain animal t days after it was born in a wildlife reserve, where $t \leq 390$. Which of the following is the best interpretation of the statement " $m(330)$ is approximately equal to **362**" in this context?

- A. The predicted body mass of the animal was approximately **330 kg 362** days after it was born.
- B. The predicted body mass of the animal was approximately **362 kg 330** days after it was born.
- C. The predicted body mass of the animal was approximately **362 kg $\frac{330}{7}$** days after it was born.
- D. The predicted body mass of the animal was approximately **$\frac{330}{7}$ kg 362** days after it was born.

ID: 0d4b88df Answer

Correct Answer: B

Rationale

Choice B is correct. In the statement " $m(330)$ is approximately equal to **362**," the input of the function, **330**, is the value of t , the elapsed time, in days, since the animal was born. The approximate value of the function, **362**, is the predicted body mass, in kilograms, of the animal after that time has elapsed. Therefore, the predicted body mass of the animal was approximately **362 kg 330** days after it was born.

Choice A is incorrect. This would be the best interpretation of the statement " $m(362)$ is approximately equal to **330**."

Choice C is incorrect. The number $\frac{330}{7}$ is the number of weeks, not the number of days, after the animal was born.

Choice D is incorrect. This would be the best interpretation of the statement " $m(362)$ is approximately equal to $\frac{330}{7}$."

Question Difficulty: Hard

Question ID b500ed25

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b500ed25

$-4x^2 - 7x = -36$

What is the positive solution to the given equation?

- A. $\frac{7}{4}$
- B. $\frac{9}{4}$
- C. 4
- D. 7

ID: b500ed25 Answer

Correct Answer: B

Rationale

Choice B is correct. Multiplying each side of the given equation by -16 yields $64x^2 + 112x = 576$. To complete the square, adding 49 to each side of this equation yields $64x^2 + 112x + 49 = 576 + 49$, or $(8x + 7)^2 = 625$. Taking the square root of each side of this equation yields two equations: $8x + 7 = 25$ and $8x + 7 = -25$. Subtracting 7 from each side of the equation $8x + 7 = 25$ yields $8x = 18$. Dividing each side of this equation by 8 yields $x = \frac{18}{8}$, or $x = \frac{9}{4}$. Therefore, $\frac{9}{4}$ is a solution to the given equation. Subtracting 7 from each side of the equation $8x + 7 = -25$ yields $8x = -32$. Dividing each side of this equation by 8 yields $x = -4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and -4 . Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Alternate approach: Adding $4x^2$ and $7x$ to each side of the given equation yields $0 = 4x^2 + 7x - 36$. The right-hand side of this equation can be rewritten as $4x^2 + 16x - 9x - 36$. Factoring out the common factor of $4x$ from the first two terms of this expression and the common factor of -9 from the second two terms yields $4x(x + 4) - 9(x + 4)$. Factoring out the common factor of $(x + 4)$ from these two terms yields the expression $(4x - 9)(x + 4)$. Since this expression is equal to 0 , it follows that either $4x - 9 = 0$ or $x + 4 = 0$. Adding 9 to each side of the equation $4x - 9 = 0$ yields $4x = 9$. Dividing each side of this equation by 4 yields $x = \frac{9}{4}$. Therefore, $\frac{9}{4}$ is a positive solution to the given equation. Subtracting 4 from each side of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and -4 . Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Choice A is incorrect. Substituting $\frac{7}{4}$ for x in the given equation yields $-\frac{49}{2} = -36$, which is false.

Choice C is incorrect. Substituting 4 for x in the given equation yields $-92 = -36$, which is false.

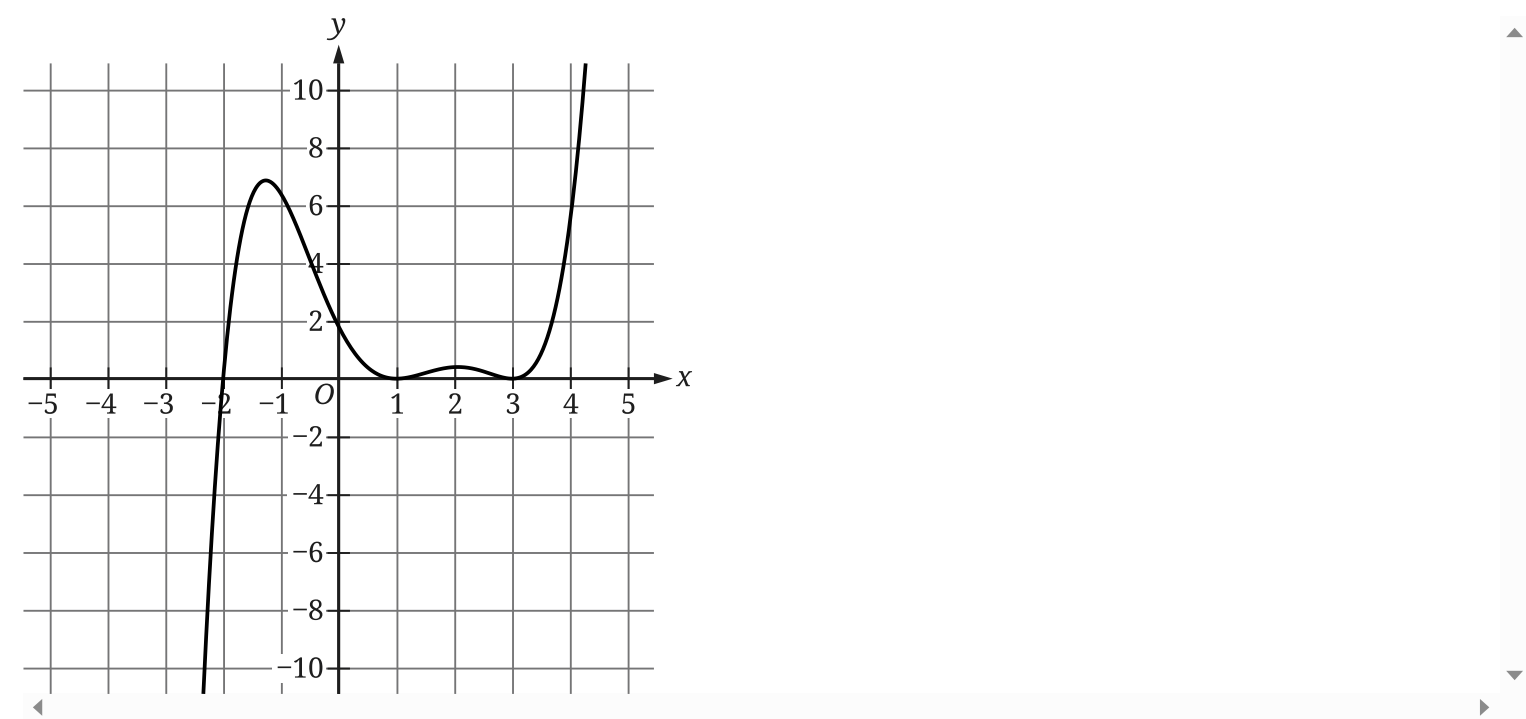
Choice D is incorrect. Substituting 7 for x in the given equation yields $-245 = -36$, which is false.

Question Difficulty: Medium

Question ID a67210ab

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a67210ab



The graph of $y = g(x)$ is shown. If $g(x + 28) = \frac{1}{10}(x - a)^2(x - b)^2(x - c)$, where a , b , and c are constants, which of the following could be the value of b ?

- A. 31
- B. 3
- C. -2
- D. -25

ID: a67210ab Answer

Correct Answer: D

Rationale

Choice D is correct. Substituting b for x in the given equation yields $g(b + 28) = \frac{1}{10}(b - a)^2(b - b)^2(b - c)$, which is equivalent to $g(b + 28) = \frac{1}{10}(b - a)^2(0)^2(b - c)$, or $g(b + 28) = 0$. It follows that when $x = b + 28$, $g(x) = 0$. There are three points where the graph of $y = g(x)$ intersects the x -axis: $(-2, 0)$, $(1, 0)$, and $(3, 0)$. Therefore, $g(-2) = 0$, $g(1) = 0$, and $g(3) = 0$. It follows that either $b + 28 = -2$, $b + 28 = 1$, or $b + 28 = 3$. Subtracting 28 from each side of the equation $b + 28 = -2$ yields $b = -30$. Subtracting 28 from each side of the equation $b + 28 = 1$ yields $b = -27$. Subtracting 28 from each side of the equation $b + 28 = 3$ yields $b = -25$. Thus, the value of b must be either -30 , -27 , or -25 . Of these values, only -25 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This would be a possible value of b if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

Choice C is incorrect. This would be the value of c , not the value of b , if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

Question Difficulty: Hard

Question ID 7e84c276

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7e84c276

x	$g(x)$
-1	25
0	1
1	$\frac{1}{25}$
2	$\frac{1}{625}$

For the exponential function g , the table shows four values of x and their corresponding values of $g(x)$. Which equation defines g ?

- A. $g(x) = -25^x$
- B. $g(x) = -\left(\frac{1}{25}\right)^x$
- C. $g(x) = 25^x$
- D. $g(x) = \left(\frac{1}{25}\right)^x$

ID: 7e84c276 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that function g is exponential. Therefore, an equation defining g can be written in the form $g(x) = a(b)^x$, where a and b are constants. The table shows that when $x = 0$, $g(x) = 1$. Substituting 0 for x and 1 for $g(x)$ in the equation $g(x) = a(b)^x$ yields $1 = a(b)^0$, which is equivalent to $1 = a$. Substituting 1 for a in the equation $g(x) = a(b)^x$ yields $g(x) = (b)^x$. The table also shows that when $x = 1$, $g(x) = \frac{1}{25}$. Substituting 1 for x and $\frac{1}{25}$ for $g(x)$ in the equation $g(x) = (b)^x$ yields $\frac{1}{25} = (b)^1$, which is equivalent to $\frac{1}{25} = b$. Substituting $\frac{1}{25}$ for b in the equation $g(x) = (b)^x$ yields $g(x) = \left(\frac{1}{25}\right)^x$.

Choice A is incorrect. For this function, $g(1)$ is equal to -25 , not $\frac{1}{25}$.

Choice B is incorrect. For this function, $g(1)$ is equal to $-\frac{1}{25}$, not $\frac{1}{25}$.

Choice C is incorrect. For this function, $g(1)$ is equal to 25 , not $\frac{1}{25}$.

Question Difficulty: Medium

Question ID 92fecab6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 92fecab6

- The function $f(x) = 206(1.034)^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. Which of the following is the best interpretation of “ $f(5)$ is approximately equal to 243” in this context?
- A. The value of the bank account is estimated to be approximately 5 dollars greater in 1962 than in 1957.
 - B. The value of the bank account is estimated to be approximately 243 dollars in 1962.
 - C. The value, in dollars, of the bank account is estimated to be approximately 5 times greater in 1962 than in 1957.
 - D. The value of the bank account is estimated to increase by approximately 243 dollars every 5 years between 1957 and 1972.

ID: 92fecab6 Answer

Correct Answer: B

Rationale

- Choice B is correct. It’s given that the function $f(x) = 206(1.034)^x$ models the value, in dollars, of a certain bank account by the end of each year from 1957 through 1972, where x is the number of years after 1957. It follows that $f(x)$ represents the estimated value, in dollars, of the bank account x years after 1957. Since the value of $f(5)$ is the value of $f(x)$ when $x = 5$, it follows that “ $f(5)$ is approximately equal to 243” means that $f(x)$ is approximately equal to 243 when $x = 5$. In the given context, this means that the value of the bank account is estimated to be approximately 243 dollars 5 years after 1957. Therefore, the best interpretation of the statement “ $f(5)$ is approximately equal to 243” in this context is the value of the bank account is estimated to be approximately 243 dollars in 1962.
- Choice A is incorrect and may result from conceptual errors.
- Choice C is incorrect and may result from conceptual errors.
- Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 34cf94ec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 34cf94ec

The function f is defined by $f(x) = x^3 + 9$. What is the value of $f(2)$?

- A. 14
- B. 15
- C. 17
- D. 18

ID: 34cf94ec Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $f(x) = x^3 + 9$. Substituting 2 for x in this equation yields $f(2) = (2)^3 + 9$. This is equivalent to $f(2) = 8 + 9$, or $f(2) = 17$.

Choice A is incorrect. This is the value of $2 + 3 + 9$, not $2^3 + 9$.

Choice B is incorrect. This is the value of $2(3) + 9$, not $2^3 + 9$.

Choice D is incorrect. This is the value of $3^2 + 9$, not $2^3 + 9$.

Question Difficulty: Easy

Question ID b16e8a1a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: b16e8a1a

$f(x) = 2x + 3$
 $g(x) = 7x - 2$
 $h(x) = 5x + 6$

The functions f , g , and h are defined as shown. If $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a , b , and c are constants, what is the value of b ?

- A. -5
- B. 12
- C. 20
- D. 22

ID: b16e8a1a Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = 2x + 3$, $g(x) = 7x - 2$, and $h(x) = 5x + 6$. It's also given that $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a , b , and c are constants. Substituting $2x + 3$ for $f(x)$, $7x - 2$ for $g(x)$, and $5x + 6$ for $h(x)$ in this equation yields $(2x + 3)(7x - 2) - (5x + 6) = ax^2 + bx + c$. Applying the distributive property to the left-hand side of this equation yields $(2x)(7x) + (2x)(-2) + (3)(7x) + (3)(-2) + (-1)(5x) + (-1)(6) = ax^2 + bx + c$, or $14x^2 - 4x + 21x - 6 - 5x - 6 = ax^2 + bx + c$. Combining like terms on the left-hand side of this equation yields $14x^2 + 12x - 12 = ax^2 + bx + c$. It follows that the value of b is 12 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID e3b52d28

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: e3b52d28

Two variables, x and y , are related such that for each increase of 1 in the value of x , the value of y increases by a factor of 4 . When $x = 0$, $y = 200$. Which equation represents this relationship?

- A. $y = 4(x)^{200}$
- B. $y = 4(200)^x$
- C. $y = 200(x)^4$
- D. $y = 200(4)^x$

ID: e3b52d28 Answer

Correct Answer: D

Rationale

Choice D is correct. Since the value of y increases by a constant factor, 4 , for each increase of 1 in the value of x , the relationship between x and y is exponential. An exponential relationship between x and y can be represented by an equation of the form $y = a(b)^x$, where a is the value of x when $y = 0$ and y increases by a factor of b for each increase of 1 in the value of x . Since $y = 200$ when $x = 0$, $a = 200$. Since y increases by a factor of 4 for each increase of 1 in the value of x , $b = 4$. Substituting 200 for a and 4 for b in the equation $y = a(b)^x$ yields $y = 200(4)^x$. Thus, the equation $y = 200(4)^x$ represents the relationship between x and y .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This equation represents a relationship where for each increase of 1 in the value of x , the value of y increases by a factor of 200 , not 4 , and when $x = 0$, y is equal to 4 , not 200 .

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 1d24b236

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 1d24b236

$$y = (x - 2)(x + 4)$$
$$y = 6x - 12$$

Which ordered pair (x, y) is the solution to the given system of equations?

- A. $(0, 2)$
- B. $(-4, 2)$
- C. $(2, 0)$
- D. $(2, -4)$

ID: 1d24b236 Answer

Correct Answer: C

Rationale

Choice C is correct. The second equation in the given system of equations is $y = 6x - 12$. Substituting $6x - 12$ for y in the first equation of the given system yields $6x - 12 = (x - 2)(x + 4)$. Factoring **6** out of the left-hand side of this equation yields $6(x - 2) = (x - 2)(x + 4)$. An expression with a factor of the form $(x - a)$ is equal to zero when $x = a$. Each side of this equation has a factor of $(x - 2)$, so each side of the equation is equal to zero when $x = 2$. Substituting **2** for x into the equation $6(x - 2) = (x - 2)(x + 4)$ yields $6(2 - 2) = (2 - 2)(2 + 4)$, or $0 = 0$, which is true. Substituting **2** for x into the second equation in the given system of equations yields $y = 6(2) - 12$, or $y = 0$. Therefore, the solution to the system of equations is the ordered pair $(2, 0)$.

Choice A is incorrect and may result from switching the order of the solutions for x and y .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 34050917

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 34050917

Which expression is equivalent to $9x^2 + 7x^2 + 9x$?

- A. $63x^4 + 9x$
- B. $9x^2 + 16x$
- C. $25x^5$
- D. $16x^2 + 9x$

ID: 34050917 Answer

Correct Answer: D

Rationale

Choice D is correct. In the given expression, the first two terms, $9x^2$ and $7x^2$, are like terms. Combining these like terms yields $9x^2 + 7x^2$, or $16x^2$. It follows that the expression $9x^2 + 7x^2 + 9x$ is equivalent to $16x^2 + 9x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 6481c302

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 6481c302

$\frac{x^2}{25} = 36$

What is a solution to the given equation?

- A. 6
- B. 30
- C. 450
- D. 900

ID: 6481c302 Answer

Correct Answer: B

Rationale

Choice B is correct. Multiplying the left- and right-hand sides of the given equation by **25** yields $x^2 = 900$. Taking the square root of the left- and right-hand sides of this equation yields $x = 30$ or $x = -30$. Of these two solutions, only **30** is given as a choice.

Choice A is incorrect. This is a solution to the equation $x^2 = 36$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 30f23a70

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 30f23a70

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. What is the value of r ?

ID: 30f23a70 Answer

Correct Answer: -12

Rationale

The correct answer is -12 . It's given that function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. It follows that the graph of $y = f(x)$ in the xy -plane passes through the points $(-20, 0)$ and $(-4, 0)$. When the graph of a quadratic function contains two points $(a, 0)$ and $(b, 0)$, the x -coordinate of the vertex of the graph is the average of a and b . Therefore, the x -coordinate of the vertex of the graph of $y = f(x)$ is $\frac{-20+(-4)}{2}$, or -12 . It's given that the graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. It follows that the value of r is -12 .

Question Difficulty: Hard

Question ID 9d4d6c68

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 9d4d6c68

$-2x^2 + 20x + c = 0$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

- A. -68
- B. -50
- C. -32
- D. 0

ID: 9d4d6c68 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $-2x^2 + 20x + c = 0$, where c is a constant, has exactly one solution. A quadratic equation of the form $ax^2 + bx + c = 0$ has exactly one solution if and only if its discriminant, $b^2 - 4ac$, is equal to zero. It follows that for the given equation, $a = -2$ and $b = 20$. Substituting -2 for a and 20 for b in $b^2 - 4ac$ yields $20^2 - 4(-2)(c)$, or $400 + 8c$. Since the discriminant must equal zero, it follows that $400 + 8c = 0$. Subtracting 400 from both sides of this equation yields $8c = -400$. Dividing each side of this equation by 8 yields $c = -50$. Therefore, the value of c is -50 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

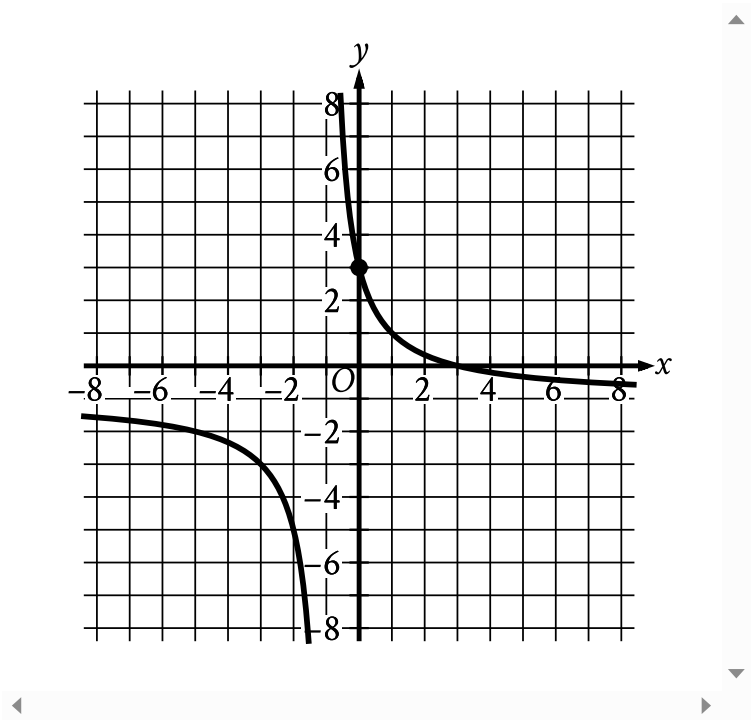
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e10502ca

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: e10502ca



What is the y -coordinate of the y -intercept of the graph shown?

ID: e10502ca Answer

Correct Answer: 3

Rationale

The correct answer is **3**. A y -intercept of a graph in the xy -plane is a point (x, y) on the graph where $x = 0$. For the graph shown, at $x = 0$, the corresponding value of y is **3**. Therefore, the y -coordinate of the y -intercept of the graph shown is **3**.

Question Difficulty: Easy

Question ID a4baf8ba

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: a4baf8ba

$38x^2 = 38(9)$

What is the negative solution to the given equation?

ID: a4baf8ba Answer

Correct Answer: -3

Rationale

The correct answer is -3 . Dividing both sides of the given equation by 38 yields $x^2 = 9$. Taking the square root of both sides of this equation yields the solutions $x = 3$ and $x = -3$. Therefore, the negative solution to the given equation is -3 .

Question Difficulty: Hard

Question ID 39cbb638

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 39cbb638

$y = 18$

$y = -3(x - 18)^2 + 15$

If the given equations are graphed in the xy -plane, at how many points do the graphs of the equations intersect?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

ID: 39cbb638 Answer

Correct Answer: D

Rationale

Choice D is correct. A point (x, y) is a solution to a system of equations if it lies on the graphs of both equations in the xy -plane. In other words, a solution to a system of equations is a point (x, y) at which the graphs intersect. It's given that the first equation is $y = 18$. Substituting 18 for y in the second equation yields $18 = -3(x - 18)^2 + 15$. Subtracting 15 from each side of this equation yields $3 = -3(x - 18)^2$. Dividing each side of this equation by -3 yields $-1 = (x - 18)^2$. Since the square of a real number is at least 0 , this equation can't have any real solutions. Therefore, the graphs of the equations intersect at zero points.

Alternate approach: The graph of the second equation is a parabola that opens downward and has a vertex at $(18, 15)$. Therefore, the maximum value of this parabola occurs when $y = 15$. The graph of the first equation is a horizontal line at 18 on the y -axis, or $y = 18$. Since 18 is greater than 15 , or the horizontal line is above the vertex of the parabola, the graphs of these equations intersect at zero points.

Choice A is incorrect. The graph of $y = 15$, not $y = 18$, and the graph of the second equation intersect at exactly one point.

Choice B is incorrect. The graph of any horizontal line such that the value of y is less than 15 , not greater than 15 , and the graph of the second equation intersect at exactly two points.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID cb62af75

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: cb62af75

$$6r = 7s + t$$

The given equation relates the variables r , s , and t . Which equation correctly expresses s in terms of r and t ?

- A. $s = 42r - t$
- B. $s = 7(6r - t)$
- C. $s = \frac{6}{7}r - t$
- D. $s = \frac{6r - t}{7}$

ID: cb62af75 Answer

Correct Answer: D

Rationale

Choice D is correct. Subtracting t from both sides of the given equation yields $6r - t = 7s$. Dividing both sides of this equation by 7 yields $\frac{6r - t}{7} = s$. Therefore, the equation $s = \frac{6r - t}{7}$ correctly expresses s in terms of r and t .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8e1f9e3a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 8e1f9e3a

The function f is defined by $f(x) = a(2.2^x + 2.2^b)$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f , where k and m are constants. Which of the following equations displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient?

- I. $g(x) = a(2.2^x + k)$
- II. $h(x) = a(2.2)^x + m$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: 8e1f9e3a Answer

Correct Answer: D

Rationale

Choice D is correct. A y -intercept of a graph in the xy -plane is a point where the graph intersects the y -axis, or a point where $x = 0$. Substituting 0 for x in the equation defining function f yields $f(0) = a(2.2^0 + 2.2^b)$, or $f(0) = a(1 + 2.2^b)$. So, the y -coordinate of the y -intercept of the graph is $a(1 + 2.2^b)$, or equivalently, $a + a(2.2)^b$. It's given that function g is equivalent to function f , where $0 < a < b$. It follows that $k = 2.2^b$. Since $a(2.2)^b$ can't be equal to 0 , the coefficient a can't be equal to $a + a(2.2)^b$. Since $0 < a$, the constant k , which is equal to 2.2^b , can't be equal to $a + a(2.2)^b$. Therefore, function g doesn't display the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient. It's also given that function h is equivalent to function f , where $0 < a < b$. The equation defining f can be rewritten as $f(x) = a(2.2)^x + a(2.2)^b$. It follows that $m = a(2.2)^b$. Since $a(2.2)^b$ can't be equal to 0 , the coefficient a can't be equal to $a + a(2.2)^b$. Since $0 < a$, the constant m , which is equal to $a(2.2)^b$, can't be equal to $a + a(2.2)^b$. Therefore, function h doesn't display the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient. Thus, neither function g nor function h displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 36d4ae4c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 36d4ae4c

The area of a triangle is equal to x^2 square centimeters. The length of the base of the triangle is $2x + 22$ centimeters, and the height of the triangle is $x - 10$ centimeters. What is the value of x ?

ID: 36d4ae4c Answer

Correct Answer: 110

Rationale

The correct answer is **110**. The area of a triangle is equal to one half of the product of the length of the base of the triangle and the height of the triangle. It's given that the length of the base of the triangle is $2x + 22$ centimeters and the height of the triangle is $x - 10$ centimeters; therefore, its area is $\frac{1}{2}(2x + 22)(x - 10)$ square centimeters. It's also given that the area of the triangle is equal to x^2 square centimeters. Therefore, it follows that $\frac{1}{2}(2x + 22)(x - 10) = x^2$. This equation can be rewritten as $(x + 11)(x - 10) = x^2$, or $x^2 + x - 110 = x^2$. Subtracting x^2 from both sides of this equation yields $x - 110 = 0$. Adding **110** to both sides of this equation yields $x = 110$. Therefore, the value of x is **110**.

Question Difficulty: Hard

Question ID ecbfeafb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: ecbfeafb

The function g is defined by $g(x) = x(x - 2)(x + 6)^2$. The value of $g(7 - w)$ is 0 , where w is a constant. What is the sum of all possible values of w ?

ID: ecbfeafb Answer

Correct Answer: 25

Rationale

The correct answer is **25**. The value of $g(7 - w)$ is the value of $g(x)$ when $x = 7 - w$, where w is a constant. Substituting $7 - w$ for x in the given equation yields $g(7 - w) = (7 - w)(7 - w - 2)(7 - w + 6)^2$, which is equivalent to $g(7 - w) = (7 - w)(5 - w)(13 - w)^2$. It's given that the value of $g(7 - w)$ is 0 . Substituting 0 for $g(7 - w)$ in the equation $g(7 - w) = (7 - w)(5 - w)(13 - w)^2$ yields $0 = (7 - w)(5 - w)(13 - w)^2$. Since the product of the three factors on the right-hand side of this equation is equal to 0 , at least one of these three factors must be equal to 0 . Therefore, the possible values of w can be found by setting each factor equal to 0 . Setting the first factor equal to 0 yields $7 - w = 0$. Adding w to both sides of this equation yields $7 = w$. Therefore, **7** is one possible value of w . Setting the second factor equal to 0 yields $5 - w = 0$. Adding w to both sides of this equation yields $5 = w$. Therefore, **5** is a second possible value of w . Setting the third factor equal to 0 yields $(13 - w)^2 = 0$. Taking the square root of both sides of this equation yields $13 - w = 0$. Adding w to both sides of this equation yields $13 = w$. Therefore, **13** is a third possible value of w . Adding the three possible values of w yields $7 + 5 + 13$, or **25**. Therefore, the sum of all possible values of w is **25**.

Question Difficulty: Hard

Question ID 30e765cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 30e765cc

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(2, 13)$. What is the value of ab ?

- A. 13
- B. 18
- C. 20
- D. 26

ID: 30e765cc Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that function h is defined by $h(x) = a^x + b$, where a and b are positive constants. It's also given that the graph of $y = h(x)$ in the xy -plane passes through the point $(0, 10)$. Substituting 0 for x and 10 for $h(x)$ in the equation $h(x) = a^x + b$ yields $10 = a^0 + b$. Since $a^0 = 1$, this equation can be rewritten as $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting 9 for b in the equation $h(x) = a^x + b$ yields $h(x) = a^x + 9$. It's also given that the graph of $y = h(x)$ in the xy -plane passes through the point $(2, 13)$. Substituting 2 for x and 13 for $h(x)$ in the equation $h(x) = a^x + 9$ yields $13 = a^2 + 9$. Subtracting 9 from both sides of this equation yields $4 = a^2$. Taking the square root of both sides of this equation yields $2 = a$ and $-2 = a$. Since a is positive, it follows that $2 = a$. Since $9 = b$ and $2 = a$, it follows that the value of ab is $(2)(9)$, or 18 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID bf9cec0d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: bf9cec0d

$-x^2 + bx - 676 = 0$

In the given equation, b is a positive integer. The equation has no real solution. What is the greatest possible value of b ?

ID: bf9cec0d Answer

Correct Answer: 51

Rationale

The correct answer is **51**. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solution if and only if its discriminant, $-4ac + b^2$, is negative. In the given equation, $a = -1$ and $c = -676$. Substituting -1 for a and -676 for c in this expression yields a discriminant of $b^2 - 4(-1)(-676)$, or $b^2 - 2,704$. Since this value must be negative, $b^2 - 2,704 < 0$, or $b^2 < 2,704$. Taking the positive square root of each side of this inequality yields $b < 52$. Since b is a positive integer, and the greatest integer less than **52** is **51**, the greatest possible value of b is **51**.

Question Difficulty: Hard

Question ID 782fbe68

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 782fbe68

Which expression is equivalent to $50x^2 + 5x^2$?

- A. $250x^2$
- B. $10x^2$
- C. $45x^2$
- D. $55x^2$

ID: 782fbe68 Answer

Correct Answer: D

Rationale

Choice D is correct. The given expression shows addition of two like terms. Therefore, the given expression is equivalent to $(50 + 5)x^2$, or $55x^2$.

Choice A is incorrect. This expression is equivalent to $(50)(5)x^2$, not $(50 + 5)x^2$.

Choice B is incorrect. This expression is equivalent to $(\frac{50}{5})x^2$, not $(50 + 5)x^2$.

Choice C is incorrect. This expression is equivalent to $(50 - 5)x^2$, not $(50 + 5)x^2$.

Question Difficulty: Easy

Question ID 8eccbb09

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 8eccbb09

Which expression is equivalent to $(9x^3 + 5x + 7) + (6x^3 + 5x^2 - 5)$?

- A. $15x^6 + 5x^2 - 5x - 35$
- B. $15x^3 + 10x^2 + 2$
- C. $15x^6 + 5x^2 + 5x + 2$
- D. $15x^3 + 5x^2 + 5x + 2$

ID: 8eccbb09 Answer

Correct Answer: D

Rationale

Choice D is correct. The given expression can be rewritten as $(9x^3 + 6x^3) + 5x^2 + 5x + (7 - 5)$. Combining like terms in this expression yields $15x^3 + 5x^2 + 5x + 2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 2009ef32

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2009ef32

The product of two positive integers is **546**. If the first integer is **11** greater than twice the second integer, what is the smaller of the two integers?

- A. **7**
- B. **14**
- C. **39**
- D. **78**

ID: 2009ef32 Answer

Correct Answer: B

Rationale

Choice B is correct. Let x be the first integer and let y be the second integer. If the first integer is **11** greater than twice the second integer, then $x = 2y + 11$. If the product of the two integers is **546**, then $xy = 546$. Substituting $2y + 11$ for x in this equation results in $(2y + 11)y = 546$. Distributing the y to both terms in the parentheses results in $2y^2 + 11y = 546$. Subtracting **546** from both sides of this equation results in $2y^2 + 11y - 546 = 0$. The left-hand side of this equation can be factored by finding two values whose product is $2(-546)$, or $-1,092$, and whose sum is **11**. The two values whose product is $-1,092$ and whose sum is **11** are **39** and -28 . Thus, the equation $2y^2 + 11y - 546 = 0$ can be rewritten as $2y^2 + 28y - 39y - 546 = 0$, which is equivalent to $2y(y - 14) + 39(y - 14) = 0$, or $(2y + 39)(y - 14) = 0$. By the zero product property, it follows that $2y + 39 = 0$ and $y - 14 = 0$. Subtracting **39** from both sides of the equation $2y + 39 = 0$ yields $2y = -39$. Dividing both sides of this equation by **2** yields $y = -\frac{39}{2}$. Since y is a positive integer, the value of y is not $-\frac{39}{2}$. Adding **14** to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting **14** for y in the equation $xy = 546$ yields $x(14) = 546$. Dividing both sides of this equation by **14** results in $x = 39$. Therefore, the two integers are **14** and **39**, so the smaller of the two integers is **14**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the larger of the two integers.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 459712c4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 459712c4

One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

ID: 459712c4 Answer

Correct Answer: 8

Rationale

The correct answer is **8**. Since each term of the given expression, $2x^3 + 42x^2 + 208x$, has a factor of $2x$, the expression can be rewritten as $2x(x^2) + 2x(21x) + 2x(104)$, or $2x(x^2 + 21x + 104)$. Since the values **8** and **13** have a sum of **21** and a product of **104**, the expression $x^2 + 21x + 104$ can be factored as $(x + 8)(x + 13)$. Therefore, the given expression can be factored as $2x(x + 8)(x + 13)$. It follows that the factors of the given expression are **2**, x , $x + 8$, and $x + 13$. Of these factors, only $x + 8$ and $x + 13$ are of the form $x + b$, where b is a positive constant. Therefore, the possible values of b are **8** and **13**. Thus, the smallest possible value of b is **8**.

Question Difficulty: Hard

Question ID a66d2c0f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: a66d2c0f

In the xy -plane, the graph of the equation $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c ?

- A. $-\frac{481}{4}$
- B. -100
- C. $-\frac{319}{4}$
- D. $-\frac{9}{2}$

ID: a66d2c0f Answer

Correct Answer: C

Rationale

Choice C is correct. In the xy -plane, the graph of the line $y = c$ is a horizontal line that crosses the y -axis at $y = c$ and the graph of the quadratic equation $y = -x^2 + 9x - 100$ is a parabola. A parabola can intersect a horizontal line at exactly one point only at its vertex. Therefore, the value of c should be equal to the y -coordinate of the vertex of the graph of the given equation. For a quadratic equation in vertex form, $y = a(x - h)^2 + k$, the vertex of its graph in the xy -plane is (h, k) . The given quadratic equation, $y = -x^2 + 9x - 100$, can be rewritten as $y = -\left(x^2 - 2\left(\frac{9}{2}\right)x + \left(\frac{9}{2}\right)^2\right) + \left(\frac{9}{2}\right)^2 - 100$, or $y = -\left(x - \frac{9}{2}\right)^2 + \left(-\frac{319}{4}\right)$. Thus, the value of c is equal to $-\frac{319}{4}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID d1673278

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: d1673278

If $k - x$ is a factor of the expression $-x^2 + \frac{1}{29}nk^2$, where n and k are constants and $k > 0$, what is the value of n ?

- A. -29
- B. $-\frac{1}{29}$
- C. $\frac{1}{29}$
- D. 29

ID: d1673278 Answer

Correct Answer: D

Rationale

Choice D is correct. If $k - x$ is a factor of the expression $-x^2 + \left(\frac{1}{29}\right)nk^2$, then the expression can be written as $(k - x)(ax + b)$, where a and b are constants. This expression can be rewritten as $akx + bk - ax^2 - bx$, or $-ax^2 + (ak - b)x + bk$. Since this expression is equivalent to $-x^2 + \left(\frac{1}{29}\right)nk^2$, it follows that $-a = -1$, $ak - b = 0$, and $bk = \left(\frac{1}{29}\right)nk^2$. Dividing each side of the equation $-a = -1$ by -1 yields $a = 1$. Substituting 1 for a in the equation $ak - b = 0$ yields $k - b = 0$. Adding b to each side of this equation yields $k = b$. Substituting k for b in the equation $bk = \left(\frac{1}{29}\right)nk^2$ yields $k^2 = \left(\frac{1}{29}\right)nk^2$. Since k is positive, dividing each side of this equation by k^2 yields $1 = \left(\frac{1}{29}\right)n$. Multiplying each side of this equation by 29 yields $29 = n$.

Alternate approach: The expression $x^2 - y^2$ can be written as $(x - y)(x + y)$, which is a difference of two squares. It follows that $\left(\frac{1}{29}\right)nk^2 - x^2$ is equivalent to $\left(\left(\sqrt{\frac{1}{29}n}\right)k - x\right)\left(\left(\sqrt{\frac{1}{29}n}\right)k + x\right)$. It's given that $k - x$ is a factor of $-x^2 + \left(\frac{1}{29}\right)nk^2$, so the factor $\left(\sqrt{\frac{1}{29}n}\right)k - x$ is equal to $k - x$. Adding x to both sides of the equation $\left(\sqrt{\frac{1}{29}n}\right)k - x = k - x$ yields $\left(\sqrt{\frac{1}{29}n}\right)k = k$. Since k is positive, dividing both sides of this equation by k yields $\sqrt{\frac{1}{29}n} = 1$. Squaring both sides of this equation yields $\frac{1}{29}n = 1$. Multiplying both sides of this equation by 29 yields $n = 29$.

Choice A is incorrect. This value of n gives the expression $-x^2 + \left(\frac{1}{29}\right)(-29)k^2$, or $-x^2 - k^2$. This expression doesn't have $k - x$ as a factor.

Choice B is incorrect. This value of n gives the expression $-x^2 + \left(\frac{1}{29}\right)\left(-\frac{1}{29}\right)k^2$, or $-x^2 + \left(-\frac{1}{841}\right)k^2$. This expression doesn't have $k - x$ as a factor.

Choice C is incorrect. This value of n gives the expression $-x^2 + \left(\frac{1}{29}\right)\left(\frac{1}{29}\right)k^2$, or $-x^2 + \left(\frac{1}{841}\right)k^2$. This expression doesn't have $k - x$ as a factor.

Question Difficulty: Hard

Question ID 733e3cb1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 733e3cb1

$$f(x) = x^5 + 9x + 17$$

For the given function f , the graph of $y = f(x)$ in the xy -plane passes through the point $(0, b)$, where b is a constant. What is the value of b ?

ID: 733e3cb1 Answer

Correct Answer: 17

Rationale

The correct answer is **17**. It's given that the graph of $y = f(x)$ in the xy -plane passes through the point $(0, b)$, where b is a constant. It follows that $f(0)$ equals b . Substituting **0** for x in the given function yields $f(0) = 0^5 + 9(0) + 17$, or $f(0) = 17$. Therefore, the value of b is **17**.

Question Difficulty: Medium

Question ID 926e7d00

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 926e7d00

$2x^2 - 8x - 7 = 0$

One solution to the given equation can be written as $\frac{8-\sqrt{k}}{4}$, where k is a constant. What is the value of k ?

ID: 926e7d00 Answer

Correct Answer: 120

Rationale

The correct answer is **120**. The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$ can be calculated using the quadratic formula and are given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -8$, and $c = -7$. It follows that the solutions to the given equation are $x = \frac{8 \pm \sqrt{(-8)^2 - 4(2)(-7)}}{2(2)}$, which is equivalent to $x = \frac{8 \pm \sqrt{64 + 56}}{4}$, or $x = \frac{8 \pm \sqrt{120}}{4}$. It's given that one solution to the equation $2x^2 - 8x - 7 = 0$ can be written as $\frac{8 - \sqrt{k}}{4}$. The solution $\frac{8 - \sqrt{120}}{4}$ is in this form. Therefore, the value of k is **120**.

Question Difficulty: Hard

Question ID de4fecb4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: de4fecb4

Which expression is equivalent to $(7x^3 + 7x) - (6x^3 - 3x)$?

- A. $x^3 + 10x$
- B. $-13x^3 + 10x$
- C. $-13x^3 + 4x$
- D. $x^3 + 4x$

ID: de4fecb4 Answer

Correct Answer: A

Rationale

Choice A is correct. Applying the distributive property, the given expression can be written as $7x^3 + 7x - 6x^3 + 3x$. Grouping like terms in this expression yields $(7x^3 - 6x^3) + (7x + 3x)$. Combining like terms in this expression yields $x^3 + 10x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 2ddf6e2e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 2ddf6e2e

Which expression is equivalent to $64t^2s^3 - 56t^3s$?

- A. $4ts(16s^2 - 14ts)$
- B. $4ts(16t^2s^2 - 14t)$
- C. $4t^2s(16ts - 14s)$
- D. $4t^2s(16s^2 - 14t)$

ID: 2ddf6e2e Answer

Correct Answer: D

Rationale

Choice D is correct. For the expression $64t^2s^3 - 56t^3s$, 4 is a common factor of 64 and 56 , t^2 is a common factor of t^2 and t^3 , and s is a common factor of s^3 and s . It follows that $4t^2s$ is a common factor of each term in the given expression. Factoring out $4t^2s$ from each term in the expression $64t^2s^3 - 56t^3s$ yields $4t^2s(16s^2 - 14t)$. Therefore, $4t^2s(16s^2 - 14t)$ is equivalent to $64t^2s^3 - 56t^3s$.

Choice A is incorrect. This expression is equivalent to $64ts^3 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

Choice B is incorrect. This expression is equivalent to $64t^3s^3 - 56t^2s$, not $64t^2s^3 - 56t^3s$.

Choice C is incorrect. This expression is equivalent to $64t^3s^2 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

Question Difficulty: Medium

Question ID 2c94a7ab

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2c94a7ab

$f(x) = (x + 6)(x + 5)(x - 4)$

The function f is given. Which table of values represents $y = f(x) - 3$?

A.

x	y
-6	-9
-5	-8
4	1

B.

x	y
-6	-3
-5	-3
4	-3

C.

x	y
-6	-3
-5	-2
4	7

D.

x	y
-6	3
-5	3
4	3

ID: 2c94a7ab Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = (x + 6)(x + 5)(x - 4)$ and $y = f(x) - 3$. Substituting $(x + 6)(x + 5)(x - 4)$ for $f(x)$ in the equation $y = f(x) - 3$ yields $y = (x + 6)(x + 5)(x - 4) - 3$. Substituting -6 for x in this equation yields $y = (-6 + 6)(-6 + 5)(-6 - 4) - 3$, or $y = -3$. Substituting -5 for x in the equation $y = (x + 6)(x + 5)(x - 4) - 3$ yields $y = (-5 + 6)(-5 + 5)(-5 - 4) - 3$, or $y = -3$. Substituting 4 for x in the equation $y = (x + 6)(x + 5)(x - 4) - 3$ yields $y = (4 + 6)(4 + 5)(4 - 4) - 3$, or $y = -3$. Therefore, when $x = -6$ then $y = -3$, when $x = -5$ then $y = -3$, and when $x = 4$ then $y = -3$. Thus, the table of values in choice B represents $y = f(x) - 3$.

Choice A is incorrect. This table represents $y = x - 3$ rather than $y = f(x) - 3$.

Choice C is incorrect. This table represents $y = x + 3$ rather than $y = f(x) - 3$.

Choice D is incorrect. This table represents $y = f(x) + 3$ rather than $y = f(x) - 3$.

Question Difficulty: Hard

Question ID 9064989a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 9064989a

Which expression is equivalent to $17(x^2 - 100y^2)$?

- A. $17(x - 2y)(x - 50y)$
- B. $17(x - 2y)(x + 50y)$
- C. $17(x - 10y)(x - 10y)$
- D. $17(x - 10y)(x + 10y)$

ID: 9064989a Answer

Correct Answer: D

Rationale

Choice D is correct. Expressions in the form $a^2 - b^2$ follow the difference of two squares pattern and can be factored as $(a - b)(a + b)$. In the given expression, $17(x^2 - 100y^2)$, the expression $x^2 - 100y^2$ follows the difference of two squares pattern. It follows that the expression $x^2 - 100y^2$ can be rewritten as $(x - 10y)(x + 10y)$. Therefore, the expression $17(x - 10y)(x + 10y)$ is equivalent to $17(x^2 - 100y^2)$.

Choice A is incorrect. This expression is equivalent to $17(x^2 - 52xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Choice B is incorrect. This expression is equivalent to $17(x^2 + 48xy - 100y^2)$, not $17(x^2 - 100y^2)$.

Choice C is incorrect. This expression is equivalent to $17(x^2 - 20xy + 100y^2)$, not $17(x^2 - 100y^2)$.

Question Difficulty: Medium

Question ID b732b6c0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b732b6c0

$|x - 5| = 10$

What is one possible solution to the given equation?

ID: b732b6c0 Answer

Correct Answer: 15, -5

Rationale

The correct answer is **15** or **−5**. By the definition of absolute value, if $|x - 5| = 10$, then $x - 5 = 10$ or $x - 5 = -10$. Adding **5** to both sides of the first equation yields $x = 15$. Adding **5** to both sides of the second equation yields $x = -5$. Thus, the given equation has two possible solutions, **15** and **−5**. Note that 15 and -5 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID c359c731

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: c359c731

The kinetic energy, in joules, of an object with mass **9** kilograms traveling at a speed of v meters per second is given by the function K , where $K(v) = \frac{9}{2}v^2$. Which of the following is the best interpretation of $K(34) = 5,202$ in this context?

- A. The object traveling at **34** meters per second has a kinetic energy of **5,202** joules.
- B. The object traveling at **340** meters per second has a kinetic energy of **5,202** joules.
- C. The object traveling at **5,202** meters per second has a kinetic energy of **34** joules.
- D. The object traveling at **23,409** meters per second has a kinetic energy of **34** joules.

ID: c359c731 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the kinetic energy, in joules, of an object with a mass of **9** kilograms traveling at a speed of v meters per second is given by the function K , where $K(v) = \frac{9}{2}v^2$. It follows that in the equation $K(34) = 5,202$, **34** is the value of v , or the speed of the object, in meters per second, and **5,202** is the kinetic energy, in joules, of the object at that speed. Therefore, the best interpretation of $K(34) = 5,202$ in this context is the object traveling at **34** meters per second has a kinetic energy of **5,202** joules.

Choice B is incorrect. The object traveling at **340** meters per second has a kinetic energy of **520,200** joules.

Choice C is incorrect. The object traveling at **5,202** meters per second has a kinetic energy of **121,773,618** joules.

Choice D is incorrect. The object traveling at **23,409** meters per second has a kinetic energy of **2,465,915,764.5** joules.

Question Difficulty: Medium

Question ID 0566a3f6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 0566a3f6

Which expression represents the product of $(x^{-6}y^3z^5)$ and $(x^4z^5 + y^8z^{-7})$?

- A. $x^{-2}z^{10} + y^{11}z^{-2}$
- B. $x^{-2}z^{10} + x^{-6}z^{-2}$
- C. $x^{-2}y^3z^{10} + y^8z^{-7}$
- D. $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$

ID: 0566a3f6 Answer

Correct Answer: D

Rationale

Choice D is correct. The product of $(x^{-6}y^3z^5)$ and $(x^4z^5 + y^8z^{-7})$ can be represented by the expression $(x^{-6}y^3z^5)(x^4z^5 + y^8z^{-7})$. Applying the distributive property to this expression yields $(x^{-6}y^3z^5)(x^4z^5) + (x^{-6}y^3z^5)(y^8z^{-7})$, or $x^{-6}x^4y^3z^5z^5 + x^{-6}y^3y^8z^5z^{-7}$. This expression is equivalent to $x^{-6+4}y^3z^{5+5} + x^{-6}y^{3+8}z^{5-7}$, or $x^{-2}y^3z^{10} + x^{-6}y^{11}z^{-2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

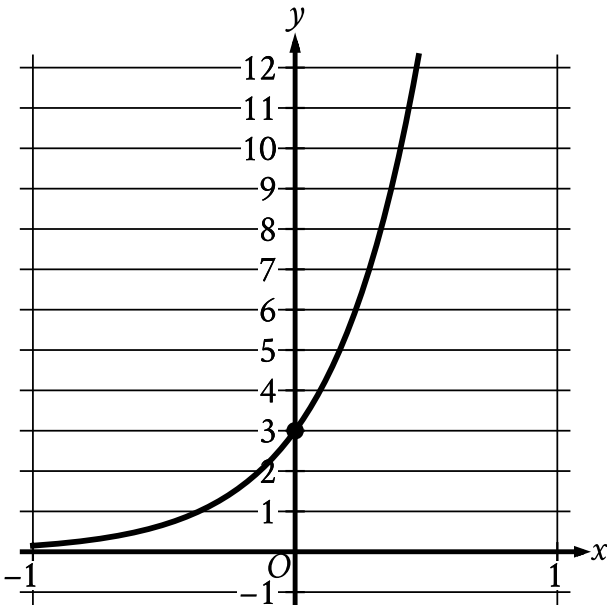
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 4aedd491

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 4aedd491



The graph of the exponential function f is shown, where $y = f(x)$. The y -intercept of the graph is $(0, y)$. What is the value of y ?

ID: 4aedd491 Answer

Correct Answer: 3

Rationale

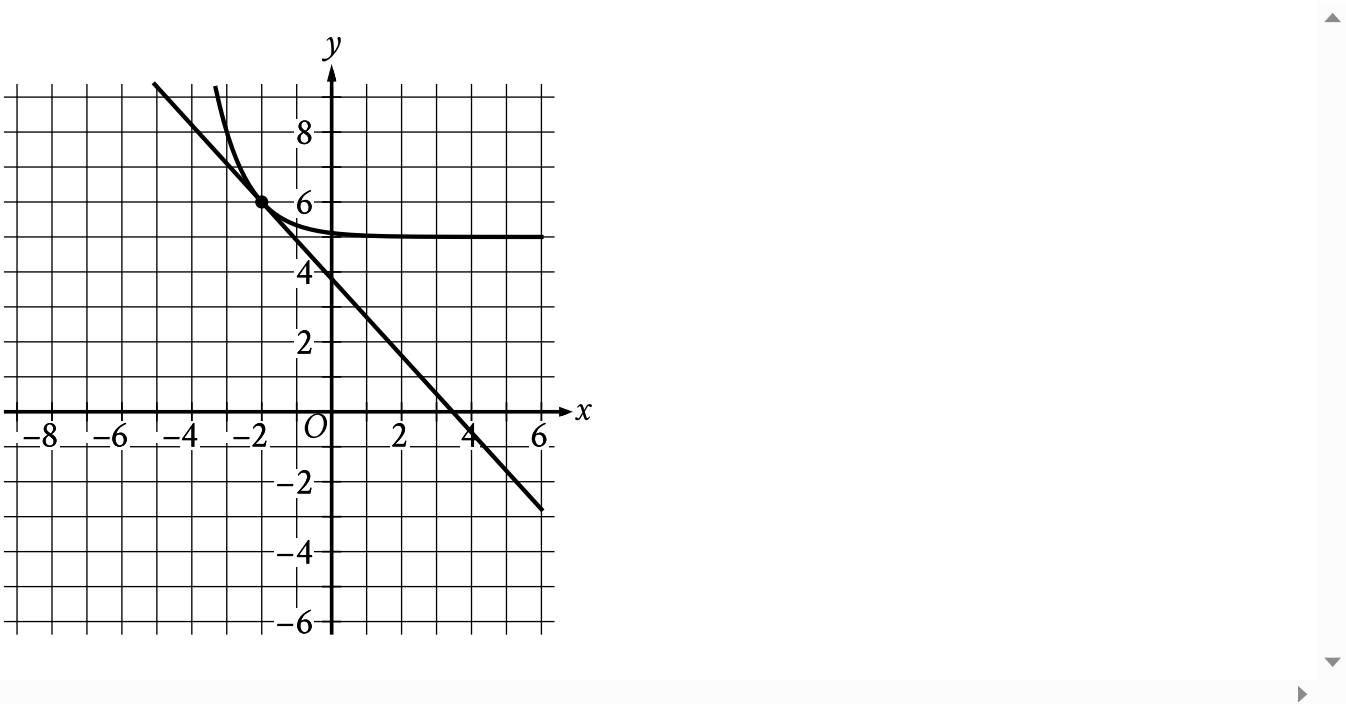
The correct answer is **3**. For the graph of the exponential function f shown, where $y = f(x)$, it's given that the y -intercept of the graph is $(0, y)$. The graph intersects the y -axis at the point $(0, 3)$. Therefore, the value of y is **3**.

Question Difficulty: Easy

Question ID 0eeb74f3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 0eeb74f3



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

- A. $(6, 0)$
- B. $(-2, 6)$
- C. $(0, -2)$
- D. $(0, 0)$

ID: 0eeb74f3 Answer

Correct Answer: B

Rationale

Choice B is correct. The solution (x, y) to the system of two equations corresponds to the point where the graphs of the equations intersect in the xy -plane. The graphs of the linear equation and the nonlinear equation shown intersect at the point $(-2, 6)$. Thus, the solution (x, y) to this system is $(-2, 6)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID cfe55495

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: cfe55495

A submersible device is used for ocean research. The function $g(x) = -\frac{1}{55}(x + 19)(x - 35)$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. How many minutes after collecting the sample did it take for the submersible device to reach the surface of the ocean?

ID: cfe55495 Answer

Correct Answer: 35

Rationale

The correct answer is **35**. It's given that the function $g(x) = -\frac{1}{55}(x + 19)(x - 35)$ gives the depth below the surface of the ocean, in meters, of the submersible device x minutes after collecting a sample, where $x > 0$. It follows that when the submersible device is at the surface of the ocean, the value of $g(x)$ is **0**. Substituting **0** for $g(x)$ in the equation $g(x) = -\frac{1}{55}(x + 19)(x - 35)$ yields $0 = -\frac{1}{55}(x + 19)(x - 35)$. Multiplying both sides of this equation by **-55** yields $0 = (x + 19)(x - 35)$. Since a product of two factors is equal to **0** if and only if at least one of the factors is **0**, either $x + 19 = 0$ or $x - 35 = 0$. Subtracting **19** from both sides of the equation $x + 19 = 0$ yields $x = -19$. Adding **35** to both sides of the equation $x - 35 = 0$ yields $x = 35$. Since $x > 0$, **35** minutes after collecting the sample the submersible device reached the surface of the ocean.

Question Difficulty: Hard

Question ID 5b5a9043

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 5b5a9043

Which expression is equivalent to $(2x^2 + x - 9) + (x^2 + 6x + 1)$?

- A. $2x^2 + 7x + 10$
- B. $2x^2 + 6x - 8$
- C. $3x^2 + 7x - 10$
- D. $3x^2 + 7x - 8$

ID: 5b5a9043 Answer

Correct Answer: D

Rationale

Choice D is correct. The given expression is equivalent to $(2x^2 + x + (-9)) + (x^2 + 6x + 1)$, which can be rewritten as $(2x^2 + x^2) + (x + 6x) + (-9 + 1)$. Adding like terms in this expression yields $3x^2 + 7x + (-8)$, or $3x^2 + 7x - 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 97fc27f8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 97fc27f8

Which expression is equivalent to $(8yz)(y)(7z)$?

- A. $56y^2z^2$
- B. $56y^2z$
- C. $56yz$
- D. $16yz$

ID: 97fc27f8 Answer

Correct Answer: A

Rationale

Choice A is correct. The given expression can be rewritten as $(8 \cdot 7)(y \cdot y)(z \cdot z)$, which is equivalent to $(56)(y^2)(z^2)$, or $56y^2z^2$.

Choice B is incorrect. This expression is equivalent to $(8yz)(y)(7)$.

Choice C is incorrect. This expression is equivalent to $(8z)(y)(7)$.

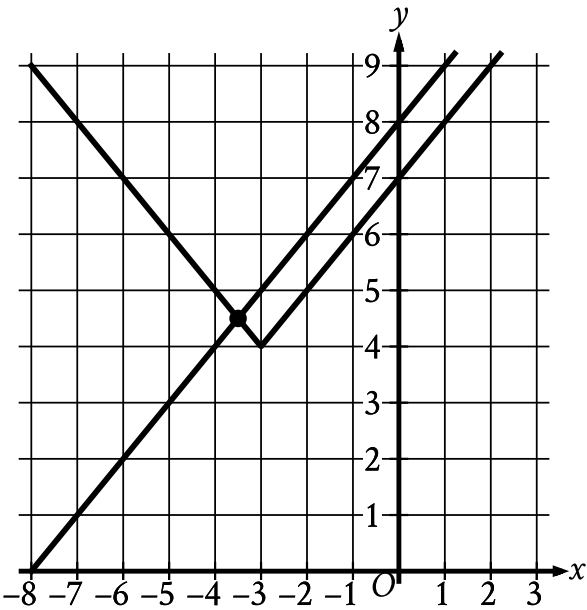
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID c5e39781

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: c5e39781



The graph of a system of an absolute value function and a linear function is shown. What is the solution (x, y) to this system of two equations?

- A. $(0, 8)$
- B. $(\frac{7}{2}, \frac{9}{2})$
- C. $(-\frac{7}{2}, \frac{9}{2})$
- D. $(-3, 4)$

ID: c5e39781 Answer

Correct Answer: C

Rationale

Choice C is correct. The solution to a system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at a point with an x -coordinate between -4 and -3 and a y -coordinate between 4 and 5 . Of the given choices, only $(-\frac{7}{2}, \frac{9}{2})$ has an x -coordinate between -4 and -3 and a y -coordinate between 4 and 5 .

Choice A is incorrect. This is the y -intercept of the graph of the linear function.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

Question ID 4398de28

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 4398de28

$f(x) = -6x^2 + 60x - 126$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

- A. $f(x) = -6x^2 + 42x + 18x - 126$
- B. $f(x) = -6x(x - 7) + 18(x - 7)$
- C. $f(x) = -6(x - 5)^2 + 24$
- D. $f(x) = -6(x - 7)(x - 3)$

ID: 4398de28 Answer

Correct Answer: C

Rationale

Choice C is correct. The given equation is quadratic. The maximum value of a function defined by a quadratic equation can be displayed as a constant in the vertex form of the equation, $f(x) = a(x - h)^2 + k$, where the maximum value of the function is k , which occurs when $x = h$, and a is a constant. The given equation can be rewritten in this form by completing the square. To complete the square, the given equation can be rewritten as $f(x) = -6(x^2 - 10x) - 126$, which is equivalent to $f(x) = -6(x^2 - 10x + 25) - 126 + 6(25)$, or $f(x) = -6(x - 5)^2 + 24$. This equation is in vertex form, where $a = -6$, $h = 5$, and $k = 24$. Therefore, $f(x) = -6(x - 5)^2 + 24$ displays the maximum value, 24 , of the function as a constant.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This form displays one of the zeros, not the maximum value, of the function as a constant.

Choice D is incorrect. This form displays the zeros, not the maximum value, of the function as constants.

Question Difficulty: Hard

Question ID 1c7fa590

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 1c7fa590

$$\sqrt{(x - 2)^2} = \sqrt{3x + 34}$$

What is the smallest solution to the given equation?

ID: 1c7fa590 Answer

Correct Answer: -3

Rationale

The correct answer is -3 . Squaring both sides of the given equation yields $(x - 2)^2 = 3x + 34$, which can be rewritten as $x^2 - 4x + 4 = 3x + 34$. Subtracting $3x$ and 34 from both sides of this equation yields $x^2 - 7x - 30 = 0$. This quadratic equation can be rewritten as $(x - 10)(x + 3) = 0$. According to the zero product property, $(x - 10)(x + 3)$ equals zero when either $x - 10 = 0$ or $x + 3 = 0$. Solving each of these equations for x yields $x = 10$ or $x = -3$. Therefore, the given equation has two solutions, 10 and -3 . Of these two solutions, -3 is the smallest solution to the given equation.

Question Difficulty: Hard

Question ID 7c505685

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7c505685

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(2, 0)$ and a y -intercept at $(0, -323)$. What is the value of b ?

ID: 7c505685 Answer

Correct Answer: -324

Rationale

The correct answer is -324 . It's given that the function f is defined by $f(x) = a^x + b$, where a and b are constants. It's also given that the graph of $y = f(x)$ has a y -intercept at $(0, -323)$. It follows that $f(0) = -323$. Substituting 0 for x and -323 for $f(x)$ in $f(x) = a^x + b$ yields $-323 = a^0 + b$, or $-323 = 1 + b$. Subtracting 1 from each side of this equation yields $-324 = b$. Therefore, the value of b is -324 .

Question Difficulty: Hard

Question ID 233ee25e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 233ee25e

The function g is defined by $g(x) = 10^{5x-8}$. What is the value of x when $g(x)$ is equal to 10,000?

- A. $\frac{11}{5}$
- B. $\frac{12}{5}$
- C. 3
- D. 4

ID: 233ee25e Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $g(x) = 10^{5x-8}$. Since 10,000 can be written as 10^4 , substituting 10^4 for $g(x)$ in this equation yields $10^4 = 10^{5x-8}$. Because the bases on the left-hand side and right-hand side of this equation are equal, it follows that the exponents must be equal. Therefore, $4 = 5x - 8$. Adding 8 to both sides of this equation yields $12 = 5x$. Dividing both sides of this equation by 5 yields $\frac{12}{5} = x$. Therefore, the value of x when $g(x)$ is equal to 10,000 is $\frac{12}{5}$.

Choice A is incorrect. This is the value of x when $g(x)$ is equal to 1,000, not 10,000.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $5x - 8$, not x , when $g(x)$ is equal to 10,000.

Question Difficulty: Medium

Question ID 14b5122c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 14b5122c

The equation $12t + b = c$ relates the variables t , b , and c . Which of the following correctly expresses the value of $c - b$ in terms of t ?

- A. $\frac{t}{12}$
- B. t
- C. $t + \frac{1}{12}$
- D. $12t$

ID: 14b5122c Answer

Correct Answer: D

Rationale

Choice D is correct. Subtracting b from each side of the given equation yields $12t = c - b$. Therefore, the expression $12t$ correctly expresses the value of $c - b$ in terms of t .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 873a9ba7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 873a9ba7

$8x^2 - 40 = 32$

What is the positive solution to the given equation?

- A. 3
- B. 4
- C. 9
- D. 72

ID: 873a9ba7 Answer

Correct Answer: A

Rationale

Choice A is correct. Adding 40 to both sides of the given equation yields $8x^2 = 72$. Dividing both sides of this equation by 8 yields $x^2 = 9$. Taking the square root of both sides of this equation yields $x = 3$ or $x = -3$. Therefore, the positive solution to the given equation is 3.

Choice B is incorrect. This is the solution to the equation $8x = 32$, not the given equation.

Choice C is incorrect. This is the solution to the equation $8x - 40 = 32$, not the given equation.

Choice D is incorrect. This is the solution to the equation $x - 40 = 32$, not the given equation.

Question Difficulty: Medium

Question ID 15b1df49

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 15b1df49

Which expression is equivalent to $256w^2 - 676$?

- A. $(16w - 26)(16w - 26)$
- B. $(8w - 13)(8w + 13)$
- C. $(8w - 13)(8w - 13)$
- D. $(16w - 26)(16w + 26)$

ID: 15b1df49 Answer

Correct Answer: D

Rationale

Choice D is correct. The given expression follows the difference of two squares pattern, $x^2 - y^2$, which factors as $(x - y)(x + y)$. Therefore, the expression $256w^2 - 676$ can be written as $(16w)^2 - 26^2$, or $(16w)(16w) - (26)(26)$, which factors as $(16w - 26)(16w + 26)$.

Choice A is incorrect. This expression is equivalent to $256w^2 - 832w + 676$.

Choice B is incorrect. This expression is equivalent to $64w^2 - 169$.

Choice C is incorrect. This expression is equivalent to $64w^2 - 208w + 169$.

Question Difficulty: Medium

Question ID a4f13e81

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a4f13e81

$f(x) = 4x^2 + 64x + 262$

The function g is defined by $g(x) = f(x + 5)$. For what value of x does $g(x)$ reach its minimum?

- A. -13
- B. -8
- C. -5
- D. -3

ID: a4f13e81 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $g(x) = f(x + 5)$. Since $f(x) = 4x^2 + 64x + 262$, it follows that $f(x + 5) = 4(x + 5)^2 + 64(x + 5) + 262$. Expanding the quantity $(x + 5)^2$ in this equation yields $f(x + 5) = 4(x^2 + 10x + 25) + 64(x + 5) + 262$. Distributing the 4 and the 64 yields $f(x + 5) = 4x^2 + 40x + 100 + 64x + 320 + 262$. Combining like terms yields $f(x + 5) = 4x^2 + 104x + 682$. Therefore, $g(x) = 4x^2 + 104x + 682$. For a quadratic function defined by an equation of the form $g(x) = a(x - h)^2 + k$, where a , h , and k are constants and a is positive, $g(x)$ reaches its minimum, k , when the value of x is h . The equation $g(x) = 4x^2 + 104x + 682$ can be rewritten in this form by completing the square. This equation is equivalent to $g(x) = 4(x^2 + 26x) + 682$, or $g(x) = 4(x^2 + 26x + 169 - 169) + 682$. This equation can be rewritten as $g(x) = 4((x + 13)^2 - 169) + 682$, or $g(x) = 4(x + 13)^2 - 4(169) + 682$, which is equivalent to $g(x) = 4(x + 13)^2 + 6$. This equation is in the form $g(x) = a(x - h)^2 + k$, where $a = 4$, $h = -13$, and $k = 6$. Therefore, $g(x)$ reaches its minimum when the value of x is -13 .

Choice B is incorrect. This is the value of x for which $f(x)$, rather than $g(x)$, reaches its minimum.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $f(x - 5)$, rather than $f(x + 5)$, reaches its minimum.

Question Difficulty: Hard

Question ID 45ed2e6b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 45ed2e6b

Time (years)	Total amount (dollars)
0	604.00
1	606.42
2	608.84

Rosa opened a savings account at a bank. The table shows the exponential relationship between the time t , in years, since Rosa opened the account and the total amount n , in dollars, in the account. If Rosa made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n ?

- A. $n = (1 + 604)^t$
- B. $n = (1 + 0.004)^t$
- C. $n = 604(1 + 0.004)^t$
- D. $n = 0.004(1 + 604)^t$

ID: 45ed2e6b Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the relationship between t and n is exponential. The table shows that the value of n increases as the value of t increases. Therefore, the relationship between t and n can be represented by an increasing exponential equation of the form $n = a(1 + b)^t$, where a and b are positive constants. The table shows that when $t = 0$, $n = 604$. Substituting 0 for t and 604 for n in the equation $n = a(1 + b)^t$ yields $604 = a(1 + b)^0$, which is equivalent to $604 = a(1)$, or $604 = a$. Substituting 604 for a in the equation $n = a(1 + b)^t$ yields $n = 604(1 + b)^t$. The table also shows that when $t = 1$, $n = 606.42$. Substituting 1 for t and 606.42 for n in the equation $n = 604(1 + b)^t$ yields $606.42 = 604(1 + b)^1$, or $606.42 = 604(1 + b)$. Dividing both sides of this equation by 604 yields approximately $1.004 = 1 + b$. Subtracting 1 from both sides of this equation yields that the value of b is approximately 0.004. Substituting 0.004 for b in the equation $n = 604(1 + b)^t$ yields $n = 604(1 + 0.004)^t$. Therefore, of the choices, choice C best represents the relationship between t and n .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 065cc8fc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 065cc8fc

The equation $E(t) = 5(1.8)^t$ gives the estimated number of employees at a restaurant, where t is the number of years since the restaurant opened. Which of the following is the best interpretation of the number 5 in this context?

- A. The estimated number of employees when the restaurant opened
- B. The increase in the estimated number of employees each year
- C. The number of years the restaurant has been open
- D. The percent increase in the estimated number of employees each year

ID: 065cc8fc Answer

Correct Answer: A

Rationale

Choice A is correct. For an exponential function of the form $E(t) = a(b)^t$, where a and b are constants, the initial value of the function—that is, the value of the function when $t = 0$ —is a and the value of the function increases by a factor of b each time t increases by 1. Since the function $E(t) = 5(1.8)^t$ gives the estimated number of employees at a restaurant and t is the number of years since the restaurant opened, the best interpretation of the number 5 in this context is the estimated number of employees when $t = 0$, or when the restaurant opened.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 914f0fb6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 914f0fb6

$$f(x) = 5,470(0.64)^{\frac{x}{12}}$$

The function f gives the value, in dollars, of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p ?

- A. 4
- B. 5
- C. 36
- D. 64

ID: 914f0fb6 Answer

Correct Answer: C

Rationale

Choice C is correct. For a function of the form $f(x) = a(r)^{\frac{x}{k}}$, where a , r , and k are constants and $r < 1$, the value of $f(x)$ decreases by $100(1 - r)\%$ for every increase of x by k . In the given function, $a = 5,470$, $r = 0.64$, and $k = 12$. Therefore, for the given function, the value of $f(x)$ decreases by $100(1 - 0.64)\%$, or 36% , for every increase of x by 12 . Since $f(x)$ represents the value, in dollars, of the equipment after x months of use, it follows that the value of the equipment decreases every 12 months by 36% of its value the preceding 12 months. Since there are 12 months in a year, the value of the equipment decreases each year by 36% of its value the preceding year. Thus, the value of p is 36 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 78bcfae7

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 78bcfae7

$$14j + 5k = m$$

The given equation relates the numbers j , k , and m . Which equation correctly expresses k in terms of j and m ?

- A. $k = \frac{m-14j}{5}$
- B. $k = \frac{1}{5}m - 14j$
- C. $k = \frac{14j-m}{5}$
- D. $k = 5m - 14j$

ID: 78bcfae7 Answer

Correct Answer: A

Rationale

Choice A is correct. Subtracting $14j$ from each side of the given equation results in $5k = m - 14j$. Dividing each side of this equation by 5 results in $k = \frac{m-14j}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 3c206d51

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 3c206d51

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$. What is the value of ab ?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 54
- D. 60

ID: 3c206d51 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the function h is defined by $h(x) = a^x + b$ and that the graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(-2, \frac{325}{36})$. Substituting 0 for x and 10 for $h(x)$ in the equation $h(x) = a^x + b$ yields $10 = a^0 + b$, or $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting -2 for x and $\frac{325}{36}$ for $h(x)$ in the equation $h(x) = a^x + 9$ yields $\frac{325}{36} = a^{-2} + 9$. Subtracting 9 from both sides of this equation yields $\frac{1}{36} = a^{-2}$, which can be rewritten as $a^2 = 36$. Taking the square root of both sides of this equation yields $a = 6$ and $a = -6$, but because it's given that a is a positive constant, a must equal 6. Because the value of a is 6 and the value of b is 9, the value of ab is $(6)(9)$, or 54.

Choice A is incorrect and may result from finding the value of $a^{-2}b$ rather than the value of ab .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from correctly finding the value of a as 6, but multiplying it by the y -value in the first ordered pair rather than by the value of b .

Question Difficulty: Hard

Question ID a1a27006

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a1a27006

The function $f(x) = 200,000(1.21)^x$ gives a company's predicted annual revenue, in dollars, x years after the company started selling light bulbs online, where $0 < x \leq 10$. What is the best interpretation of the statement " $f(5)$ is approximately equal to 518,748" in this context?

- A. 5 years after the company started selling light bulbs online, its predicted annual revenue is approximately 518,748 dollars.
- B. 5 years after the company started selling light bulbs online, its predicted annual revenue will have increased by a total of approximately 518,748 dollars.
- C. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5 times the predicted annual revenue for the previous year.
- D. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5% greater than the predicted annual revenue for the previous year.

ID: a1a27006 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the function $f(x) = 200,000(1.21)^x$ gives a company's predicted annual revenue, in dollars, x years after the company started selling light bulbs online. It follows that $f(x)$ represents the company's predicted annual revenue, in dollars, x years after the company started selling light bulbs online. Since the value of $f(5)$ is the value of $f(x)$ when $x = 5$, it follows that " $f(5)$ is approximately equal to 518,748" means that $f(x)$ is approximately equal to 518,748 when $x = 5$. Therefore, the best interpretation of the statement " $f(5)$ is approximately equal to 518,748" in this context is 5 years after the company started selling light bulbs online, its predicted annual revenue is approximately 518,748 dollars.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 1ca5e8ac

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 1ca5e8ac

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

- A. $(0, 14)$
- B. $(0, 2)$
- C. $(0, 22)$
- D. $(0, -8)$

ID: 1ca5e8ac Answer

Correct Answer: A

Rationale

Choice A is correct. The y-intercept of the graph of $y = f(x)$ in the xy-plane occurs at the point on the graph where $x = 0$. In other words, when $x = 0$, the corresponding value of $f(x)$ is the y-coordinate of the y-intercept. Substituting 0 for x in the given equation yields $f(0) = (-8)(2)^0 + 22$, which is equivalent to $f(0) = (-8)(1) + 22$, or $f(0) = 14$. Thus, when $x = 0$, the corresponding value of $f(x)$ is 14. Therefore, the y-intercept of the graph of $y = f(x)$ in the xy-plane is $(0, 14)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This could be the y-intercept for $f(x) = (-8)(2)^x$, not $f(x) = (-8)(2)^x + 22$.

Question Difficulty: Hard

Question ID 7be44fda

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 7be44fda

x	$f(x)$
-1	10
0	14
1	20

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- A. $f(x) = 3x^2 + 3x + 14$
- B. $f(x) = 5x^2 + x + 14$
- C. $f(x) = 9x^2 - x + 14$
- D. $f(x) = x^2 + 5x + 14$

ID: 7be44fda Answer

Correct Answer: D

Rationale

Choice D is correct. The equation of a quadratic function can be written in the form $f(x) = a(x - h)^2 + k$, where a , h , and k are constants. It's given in the table that when $x = -1$, the corresponding value of $f(x)$ is 10 . Substituting -1 for x and 10 for $f(x)$ in the equation $f(x) = a(x - h)^2 + k$ gives $10 = a(-1 - h)^2 + k$, which is equivalent to $10 = a(1 + 2h + h^2) + k$, or $10 = a + 2ah + ah^2 + k$. It's given in the table that when $x = 0$, the corresponding value of $f(x)$ is 14 . Substituting 0 for x and 14 for $f(x)$ in the equation $f(x) = a(x - h)^2 + k$ gives $14 = a(0 - h)^2 + k$, or $14 = ah^2 + k$. It's given in the table that when $x = 1$, the corresponding value of $f(x)$ is 20 . Substituting 1 for x and 20 for $f(x)$ in the equation $f(x) = a(x - h)^2 + k$ gives $20 = a(1 - h)^2 + k$, which is equivalent to $20 = a(1 - 2h + h^2) + k$, or $20 = a - 2ah + ah^2 + k$. Adding $20 = a - 2ah + ah^2 + k$ to the equation $10 = a + 2ah + ah^2 + k$ gives $30 = 2a + 2ah^2 + 2k$. Dividing both sides of this equation by 2 gives $15 = a + ah^2 + k$. Since $14 = ah^2 + k$, substituting 14 for $ah^2 + k$ into the equation $15 = a + ah^2 + k$ gives $15 = a + 14$. Subtracting 14 from both sides of this equation gives $a = 1$. Substituting 1 for a in the equations $14 = ah^2 + k$ and $20 = ah^2 - 2ah + a + k$ gives $14 = h^2 + k$ and $20 = 1 - 2h + h^2 + k$, respectively. Since $14 = h^2 + k$, substituting 14 for $h^2 + k$ in the equation $20 = 1 - 2h + h^2 + k$ gives $20 = 1 - 2h + 14$, or $20 = 15 - 2h$. Subtracting 15 from both sides of this equation gives $5 = -2h$. Dividing both sides of this equation by -2 gives $-\frac{5}{2} = h$. Substituting $-\frac{5}{2}$ for h into the equation $14 = h^2 + k$ gives $14 = (-\frac{5}{2})^2 + k$, or $14 = \frac{25}{4} + k$. Subtracting $\frac{25}{4}$ from both sides of this equation gives $\frac{31}{4} = k$. Substituting 1 for a , $-\frac{5}{2}$ for h , and $\frac{31}{4}$ for k in the equation $f(x) = a(x - h)^2 + k$ gives $f(x) = (x + \frac{5}{2})^2 + \frac{31}{4}$, which is equivalent to $f(x) = x^2 + 5x + \frac{25}{4} + \frac{31}{4}$, or $f(x) = x^2 + 5x + 14$. Therefore, $f(x) = x^2 + 5x + 14$ defines f .

Choice A is incorrect. If $f(x) = 3x^2 + 3x + 14$, then when $x = -1$, the corresponding value of $f(x)$ is 14 , not 10 .

Choice B is incorrect. If $f(x) = 5x^2 + x + 14$, then when $x = -1$, the corresponding value of $f(x)$ is 18 , not 10 .

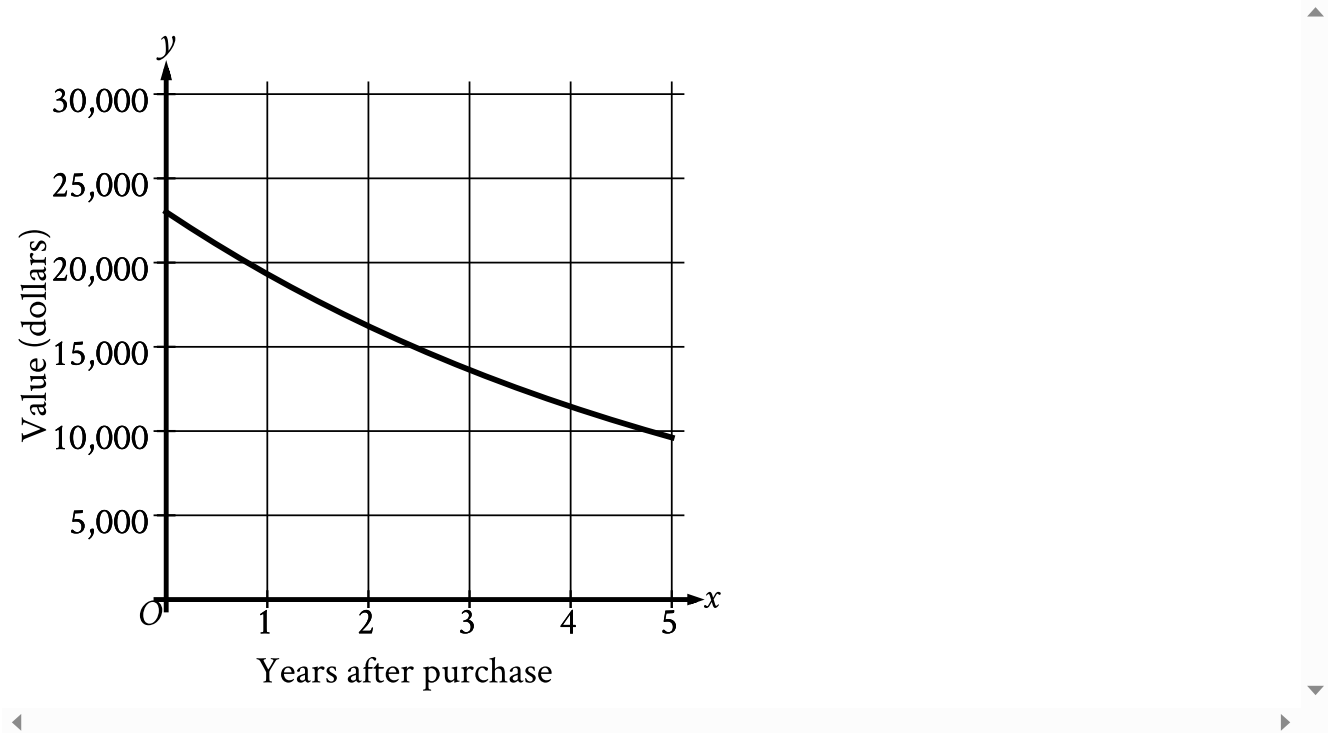
Choice C is incorrect. If $f(x) = 9x^2 - x + 14$, then when $x = -1$, the corresponding value of $f(x)$ is 24 , not 10 , and when $x = 1$, the corresponding value of $f(x)$ is 22 , not 20 .

Question ID be579e5f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: be579e5f

The graph shows the predicted value y , in dollars, of a certain sport utility vehicle x years after it is first purchased.



Which of the following is closest to the predicted value of the sport utility vehicle **3** years after it is first purchased?

- A. \$9,619
- B. \$13,632
- C. \$19,320
- D. \$23,000

ID: be579e5f Answer

Correct Answer: B

Rationale

Choice B is correct. For the graph shown, the horizontal axis represents the number of years after a certain sport utility vehicle is first purchased, and the vertical axis represents the predicted value, in dollars, of the sport utility vehicle. According to the graph, **3** years after the sport utility vehicle is purchased, the predicted value of the sport utility vehicle is between **\$10,000** and **\$15,000**. Of the given choices, only **\$13,632** is between **\$10,000** and **\$15,000**. Therefore, **\$13,632** is closest to the predicted value of the sport utility vehicle **3** years after it is first purchased.

Choice A is incorrect. This is closest to the predicted value of the sport utility vehicle **5** years after it is first purchased.

Choice C is incorrect. This is closest to the predicted value of the sport utility vehicle **1** year after it is first purchased.

Choice D is incorrect. This is closest to the predicted value of the sport utility vehicle when it is first purchased.

Question ID f02c2b48

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: f02c2b48

Which quadratic equation has exactly one distinct real solution?

- A. $(x + 15)^2 = 0$
- B. $(x + 15)^2 = -45$
- C. $(x + 15)^2 = 45$
- D. $(x + 15)^2 = 135$

ID: f02c2b48 Answer

Correct Answer: A

Rationale

Choice A is correct. Taking the square root of both sides of $(x + 15)^2 = 0$ yields $x + 15 = 0$. Subtracting 15 from both sides of this equation yields $x = -15$. Therefore, this equation has exactly one distinct real solution.

Choice B is incorrect. This equation has no distinct real solutions, not exactly one distinct real solution.

Choice C is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Choice D is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Question Difficulty: Medium

Question ID bdc0fe85

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: bdc0fe85

The functions g and h are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

- I. $g(x) = 18(1.16)(1.4)^{x+2}$
- II. $h(x) = 18(1.4)^{x+4}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: bdc0fe85 Answer

Correct Answer: D

Rationale

Choice D is correct. A function defined by an equation in the form $f(x) = a(b)^{x+h}$, where a , b , and h are positive constants and $x \geq 0$, has a minimum value of $f(0)$. It's given that function g is defined by $g(x) = 18(1.16)(1.4)^{x+2}$, which is equivalent to $g(x) = 20.88(1.4)^{x+2}$. Substituting 0 for x in this equation yields $g(0) = 20.88(1.4)^{0+2}$, or $g(0) = 40.9248$. Therefore, the minimum value of $g(x)$ is 40.9248, so $g(x) = 18(1.16)(1.4)^{x+2}$ doesn't display its minimum value as a constant or coefficient. It's also given that function h is defined by $h(x) = 18(1.4)^{x+4}$. Substituting 0 for x in this equation yields $h(0) = 18(1.4)^{0+4}$, or $h(0) = 69.1488$. Therefore, the minimum value of $h(x)$ is 69.1488, so $h(x) = 18(1.4)^{x+4}$ doesn't display its minimum value as a constant or coefficient. Therefore, neither I nor II displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 1b7e9db0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 1b7e9db0

$$y = 2(x - d)(x + d)(x + g)(x - d)$$

In the given equation, d and g are unique positive constants. When the equation is graphed in the xy -plane, how many distinct x -intercepts does the graph have?

- A. 4
- B. 3
- C. 2
- D. 1

ID: 1b7e9db0 Answer

Correct Answer: B

Rationale

Choice B is correct. An x -intercept of a graph in the xy -plane is a point on the graph where the value of y is 0. Substituting 0 for y in the given equation yields $0 = 2(x - d)(x + d)(x + g)(x - d)$. By the zero product property, the solutions to this equation are $x = d$, $x = -d$, $x = -g$, and $x = d$. However, $x = d$ and $x = d$ are identical. It's given that d and g are unique positive constants. It follows that the equation $0 = 2(x - d)(x + d)(x + g)(x - d)$ has 3 unique solutions: $x = d$, $x = -d$, and $x = -g$. Thus, the graph of the given equation has 3 distinct x -intercepts.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

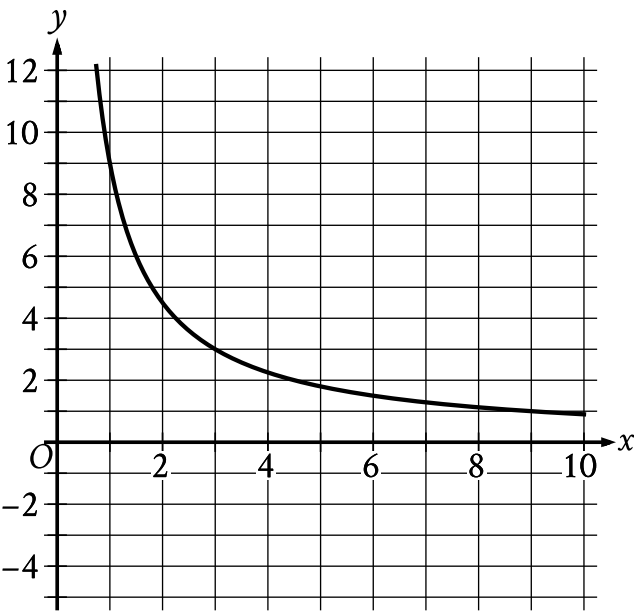
Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID f8c4065f

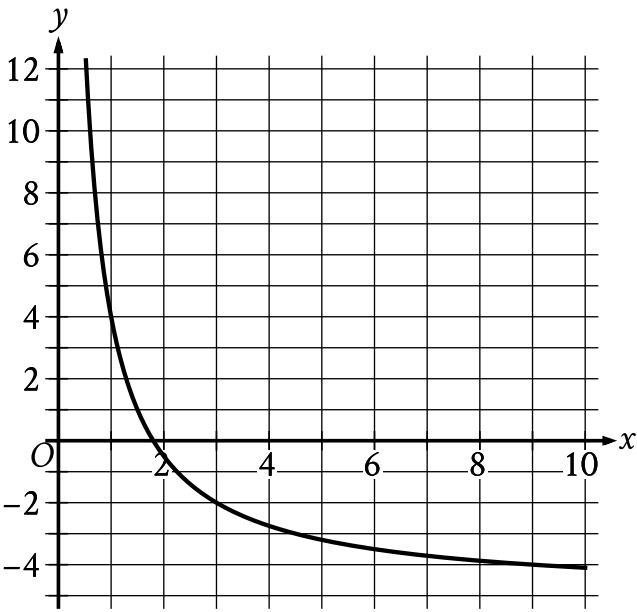
Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: f8c4065f

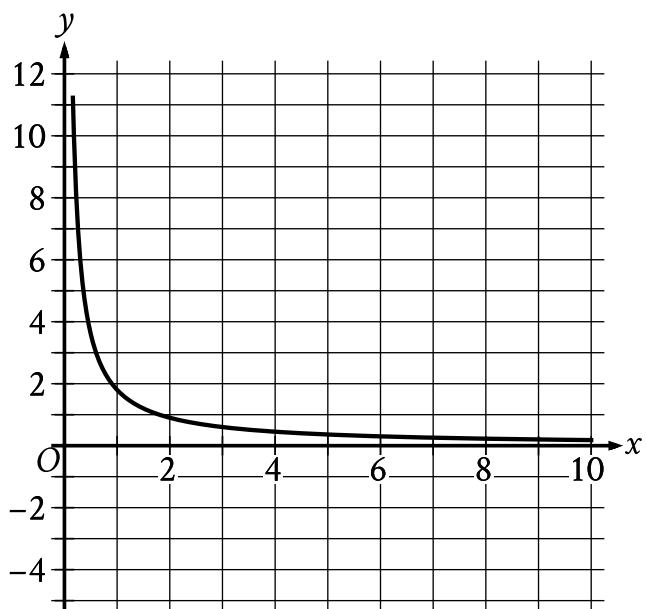


The graph of the rational function f is shown, where $y = f(x)$ and $x \geq 0$. Which of the following is the graph of $y = f(x) + 5$, where $x \geq 0$?

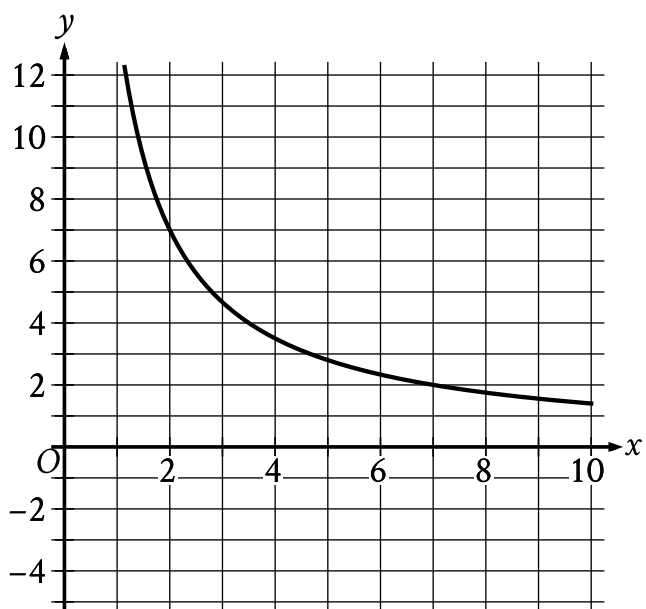
A.



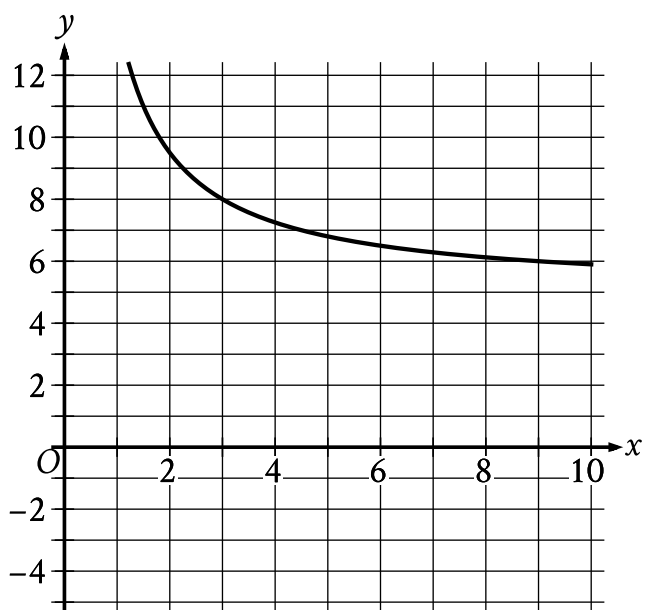
B.



C.



D.



Correct Answer: D

Rationale

Choice D is correct. It's given that the graph of the rational function f is shown, where $y = f(x)$ and $x \geq 0$. The graph shown passes through the point $(3, 3)$. It follows that when the value of x is 3 , the value of $f(x)$ is 3 . When the value of $f(x)$ is 3 , the value of $f(x) + 5$ is $3 + 5$, or 8 . Therefore, the graph of $y = f(x) + 5$ passes through the point $(3, 8)$. Of the given choices, choice D is the only graph that passes through the point $(3, 8)$ and is therefore the graph of $y = f(x) + 5$.

Choice A is incorrect. This is the graph of $y = f(x) - 5$, rather than $y = f(x) + 5$.

Choice B is incorrect. This is the graph of $y = \frac{f(x)}{5}$, rather than $y = f(x) + 5$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 65751f2d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 65751f2d

On April 1, there were 233 views of an advertisement posted on a website. Every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. The function f gives the predicted number of views x days after April 1. Which equation defines f ?

- A. $f(x) = 233(0.70)^{\frac{x}{2}}$
- B. $f(x) = 233(0.70)^{2x}$
- C. $f(x) = 233(1.70)^{\frac{x}{2}}$
- D. $f(x) = 233(1.70)^{2x}$

ID: 65751f2d Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that on April 1, there were 233 views of the advertisement. It’s also given that every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. This situation can be represented by an exponential function of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{c}}$, where a is the number of views on April 1 and every c days after April 1, the number of views had increased by $r\%$ of the number of views c days earlier. It follows that $a = 233$, $r = 70$, and $c = 2$. Substituting 233 for a , 70 for r , and 2 for c in the equation $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{c}}$ yields $f(x) = 233\left(1 + \frac{70}{100}\right)^{\frac{x}{2}}$, or $f(x) = 233(1.70)^{\frac{x}{2}}$.

Choice A is incorrect. This function gives the predicted number of views for an advertisement for which every 2 days, the number of views was 70%, rather than increased by 70%, of the number of views 2 days earlier.

Choice B is incorrect. This function gives the predicted number of views for an advertisement for which every $\frac{1}{2}$ days, the number of views was 70% of the number of views $\frac{1}{2}$ days earlier, rather than an advertisement for which every 2 days, the number of views had increased by 70% of the number of views 2 days earlier.

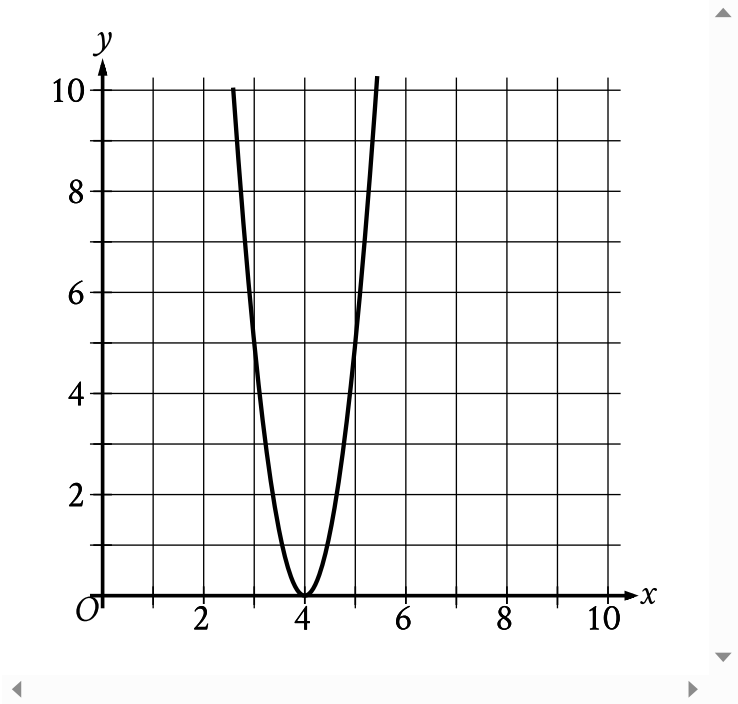
Choice D is incorrect. This function gives the predicted number of views for an advertisement for which every $\frac{1}{2}$ days, rather than every 2 days, the number of views had increased by 70% of the number of views $\frac{1}{2}$ days earlier, rather than 2 days earlier.

Question Difficulty: Hard

Question ID 53730316

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 53730316



What is the x-intercept of the graph shown?

- A. $(-5, 0)$
- B. $(5, 0)$
- C. $(-4, 0)$
- D. $(4, 0)$

ID: 53730316 Answer

Correct Answer: D

Rationale

Choice D is correct. The x-intercept of the graph shown is the point (x, y) on the graph where $y = 0$. At $y = 0$, the corresponding value of x is **4**. Therefore, the x-intercept of the graph shown is **$(4, 0)$** .

Choice A is incorrect. This is the x-intercept of a graph in the xy -plane that intersects the x -axis at $x = -5$, not $x = 4$.

Choice B is incorrect. This is the x-intercept of a graph in the xy -plane that intersects the x -axis at $x = 5$, not $x = 4$.

Choice C is incorrect. This is the x-intercept of a graph in the xy -plane that intersects the x -axis at $x = -4$, not $x = 4$.

Question Difficulty: Easy

Question ID c1a53bc9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: c1a53bc9

$x^2 - 40x - 10 = 0$

What is the sum of the solutions to the given equation?

- A. 0
- B. 5
- C. 10
- D. 40

ID: c1a53bc9 Answer

Correct Answer: D

Rationale

Choice D is correct. Adding 10 to each side of the given equation yields $x^2 - 40x = 10$. To complete the square, adding $(\frac{40}{2})^2$, or 20^2 , to each side of this equation yields $x^2 - 40x + 20^2 = 10 + 20^2$, or $(x - 20)^2 = 410$. Taking the square root of each side of this equation yields $x - 20 = \pm\sqrt{410}$. Adding 20 to each side of this equation yields $x = 20 \pm \sqrt{410}$. Therefore, the solutions to the given equation are $x = 20 + \sqrt{410}$ and $x = 20 - \sqrt{410}$. The sum of these solutions is $(20 + \sqrt{410}) + (20 - \sqrt{410})$, or 40.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e97a0ffd

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: e97a0ffd

The population of the town of Smithville doubled every **75** years from **1659** to **1959**. The population of this town was **240,000** in **1959**. What was the population of this town in **1659**?

ID: e97a0ffd Answer

Correct Answer: 15000

Rationale

The correct answer is **15,000**. Let **y** represent the population of Smithville **x** years after **1659**. Since the population doubled every **75** years, the relationship between **x** and **y** can be represented by an exponential equation in the form $y = a(b)^{\frac{x}{k}}$, where **a** is the population of Smithville in **1659** and the population increases by a factor of **b** every **k** years. It's given that the population doubled, or increased by a factor of **2**, every **75** years. Therefore, **b = 2** and **k = 75**. Substituting **2** for **b** and **75** for **k** in the equation $y = a(b)^{\frac{x}{k}}$ yields $y = a(2)^{\frac{x}{75}}$. It's also given that the population of this town was **240,000** in **1959**. The number of years from **1659** to **1959** is **1959 - 1659**, or **300** years. Therefore, when **x = 300**, the value of **y** is **240,000**. Substituting **300** for **x** and **240,000** for **y** in the equation $y = a(2)^{\frac{x}{75}}$ yields $240,000 = a(2)^{\frac{300}{75}}$, or $240,000 = a(2)^4$, which is equivalent to $240,000 = a(16)$. Dividing both sides of this equation by **16** yields **15,000 = a**. Therefore, the population of this town in **1659** was **15,000**.

Question Difficulty: Hard

Question ID b4b27aa2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: b4b27aa2

Which expression is equivalent to $(m^4q^4z^{-1})(mq^5z^3)$, where m , q , and z are positive?

- A. $m^4q^{20}z^{-3}$
- B. $m^5q^9z^2$
- C. $m^6q^8z^{-1}$
- D. $m^{20}q^{12}z^{-2}$

ID: b4b27aa2 Answer

Correct Answer: B

Rationale

Choice B is correct. Applying the commutative property of multiplication, the expression $(m^4q^4z^{-1})(mq^5z^3)$ can be rewritten as $(m^4m)(q^4q^5)(z^{-1}z^3)$. For positive values of x , $(x^a)(x^b) = x^{a+b}$. Therefore, the expression $(m^4m)(q^4q^5)(z^{-1}z^3)$ can be rewritten as $(m^{4+1})(q^{4+5})(z^{-1+3})$, or $m^5q^9z^2$.

Choice A is incorrect and may result from multiplying, not adding, the exponents.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 2b85b3c1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 2b85b3c1

Which expression is equivalent to $11x^3 - 5x^3$?

- A. $16x^3$
- B. $6x^3$
- C. $6x^6$
- D. $16x^6$

ID: 2b85b3c1 Answer

Correct Answer: B

Rationale

Choice B is correct. The given expression can be rewritten as $11x^3 + (-5)x^3$. Since the two terms of this expression are both constant multiples of x^3 , they are like terms and can, therefore, be combined through addition. Adding like terms in the expression $11x^3 + (-5)x^3$ yields $6x^3$.

Choice A is incorrect. This is equivalent to $11x^3 + 5x^3$, not $11x^3 - 5x^3$.

Choice C is incorrect. This is equivalent to $11x^6 - 5x^6$, not $11x^3 - 5x^3$.

Choice D is incorrect. This is equivalent to $11x^6 + 5x^6$, not $11x^3 - 5x^3$.

Question Difficulty: Easy

Question ID 5cd94924

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5cd94924

$$\frac{r}{38} = s + t$$

The given equation relates the quantities r , s , and t . Which equation correctly expresses r in terms of s and t ?

- A. $r = s + t + 38$
- B. $r = s + t - 38$
- C. $r = 38(s + t)$
- D. $r = \frac{s+t}{38}$

ID: 5cd94924 Answer

Correct Answer: C

Rationale

Choice C is correct. Multiplying both sides of the given equation by 38 yields $r = 38(s + t)$.

Choice A is incorrect. This equation is equivalent to $r - 38 = s + t$, not $\frac{r}{38} = s + t$.

Choice B is incorrect. This equation is equivalent to $r + 38 = s + t$, not $\frac{r}{38} = s + t$.

Choice D is incorrect. This equation is equivalent to $38r = s + t$, not $\frac{r}{38} = s + t$.

Question Difficulty: Medium

Question ID 91b6a09f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 91b6a09f

$x^2 = (22)(22)$

What is the positive solution to the given equation?

ID: 91b6a09f Answer

Correct Answer: 22

Rationale

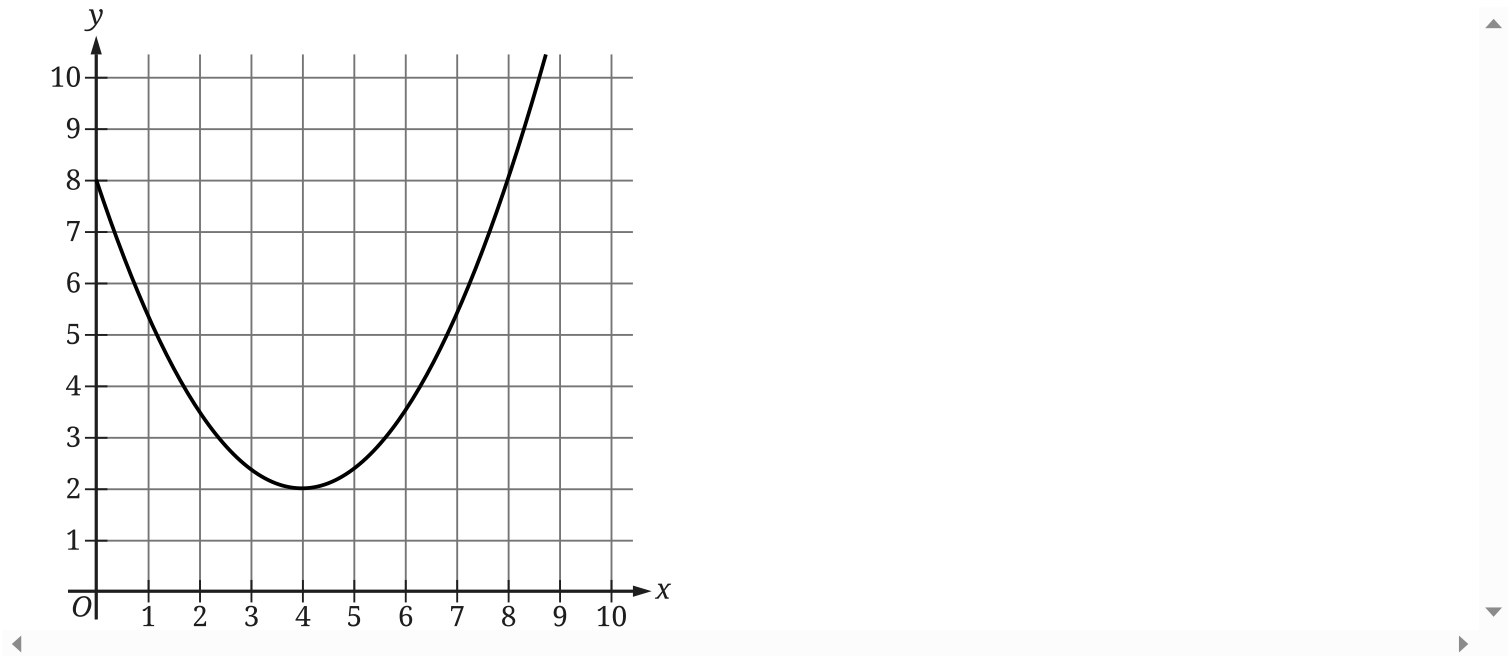
The correct answer is **22**. The given equation, $x^2 = (22)(22)$, is equivalent to $x^2 = (22)^2$. Taking the square root of each side of this equation yields $x = \pm 22$. Thus, the positive solution to the given equation is **22**.

Question Difficulty: Medium

Question ID c37efb0d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: c37efb0d



The graph shows a marble's height above the ground y , in inches, x seconds after it started moving on an elevated track of a marble run. Which of the following is the best interpretation of the y -intercept of the graph?

- A. The marble's height was 0 inches above the ground 8 seconds after it started moving.
- B. The marble's height was 8 inches above the ground when it started moving.
- C. The marble's minimum height was 0 inches above the ground.
- D. The marble's minimum height was 8 inches above the ground.

ID: c37efb0d Answer

Correct Answer: B

Rationale

Choice B is correct. The y -intercept of a graph is the point at which the graph intersects the y -axis. The graph shown intersects the y -axis at the point $(0, 8)$. Therefore, the y -intercept of the graph is $(0, 8)$. It's given that y is the height of the marble above the ground, in inches, and x is the number of seconds after the marble started moving. It follows that the marble's height was 8 inches above the ground 0 seconds after it started moving. Therefore, the best interpretation of the y -intercept of the graph is that the marble's height was 8 inches above the ground when it started moving.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 2f73623c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 2f73623c

x	$g(x)$
-27	3
-9	0
21	5

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+3}$ and f is a linear function. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

- A. $(0, 36)$
- B. $(0, 12)$
- C. $(0, 4)$
- D. $(0, -9)$

ID: 2f73623c Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the table shows values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+3}$. It's also given that f is a linear function. It follows that an equation that defines f can be written in the form $f(x) = mx + b$, where m represents the slope and b represents the y-coordinate of the y-intercept $(0, b)$ of the graph of $y = f(x)$ in the xy-plane. The slope of the graph of $y = f(x)$ can be found using two points, (x_1, y_1) and (x_2, y_2) , that are on the graph of $y = f(x)$, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since the table shows values of x and their corresponding values of $g(x)$, substituting values of x and $g(x)$ in the equation $g(x) = \frac{f(x)}{x+3}$ can be used to define function f . Using the first pair of values from the table, $x = -27$ and $g(x) = 3$, yields $3 = \frac{f(-27)}{-27+3}$, or $3 = \frac{f(-27)}{-24}$. Multiplying each side of this equation by -24 yields $-72 = f(-27)$, so the point $(-27, -72)$ is on the graph of $y = f(x)$. Using the second pair of values from the table, $x = -9$ and $g(x) = 0$, yields $0 = \frac{f(-9)}{-9+3}$, or $0 = \frac{f(-9)}{-6}$. Multiplying each side of this equation by -6 yields $0 = f(-9)$, so the point $(-9, 0)$ is on the graph of $y = f(x)$. Substituting $(-27, -72)$ and $(-9, 0)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{0 - (-72)}{-9 - (-27)}$, or $m = 4$. Substituting 4 for m in the equation $f(x) = mx + b$ yields $f(x) = 4x + b$. Since $0 = f(-9)$, substituting -9 for x and 0 for $f(x)$ in the equation $f(x) = 4x + b$ yields $0 = 4(-9) + b$, or $0 = -36 + b$. Adding 36 to both sides of this equation yields $36 = b$. It follows that 36 is the y-coordinate of the y-intercept $(0, b)$ of the graph of $y = f(x)$. Therefore, the y-intercept of the graph of $y = f(x)$ is $(0, 36)$.

Choice B is incorrect. 12 is the y-coordinate of the y-intercept of the graph of $y = g(x)$.

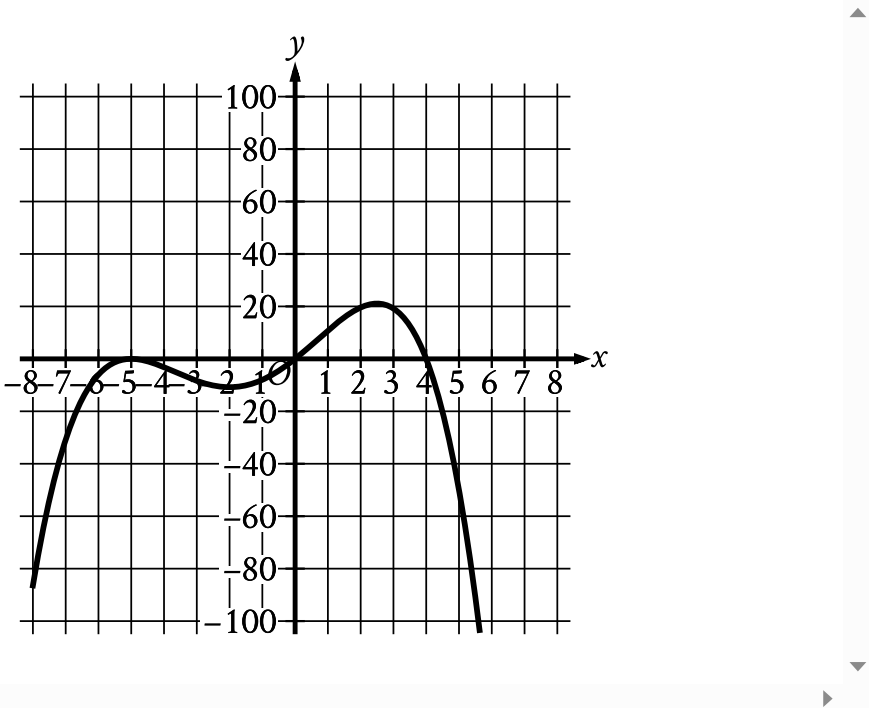
Choice C is incorrect. 4 is the slope of the graph of $y = f(x)$.

Choice D is incorrect. -9 is the x-coordinate of the x-intercept of the graph of $y = f(x)$.

Question ID ed951a6b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: ed951a6b



Which of the following could be the equation of the graph shown in the xy -plane?

- A. $y = -\frac{1}{10}x(x - 4)(x + 5)$
- B. $y = -\frac{1}{10}x(x - 4)(x + 5)^2$
- C. $y = -\frac{1}{10}x(x - 5)(x + 4)$
- D. $y = -\frac{1}{10}x(x - 5)^2(x + 4)$

ID: ed951a6b Answer

Correct Answer: B

Rationale

Choice B is correct. Each of the given choices is an equation of the form $y = -\frac{1}{10}x(x - a)^m(x + b)^n$, where a , b , m , and n are positive constants. In the xy -plane, the graph of an equation of this form has x -intercepts at $x = 0$, $x = a$, and $x = -b$. The graph shown has x -intercepts at $x = 0$, $x = 4$, and $x = -5$. Therefore, $a = 4$ and $b = 5$. Of the given choices, only choices A and B have $a = 4$ and $b = 5$. For an equation in the form $y = -\frac{1}{10}x(x - a)^m(x + b)^n$, if all values of x that are less than $-b$ or greater than a correspond to negative y -values, then the sum of all the exponents of the factors on the right-hand side of the equation is even. In the graph shown, all values of x less than -5 or greater than 4 correspond to negative y -values. Therefore, the sum of all the exponents of the factors on the right-hand side of the equation $y = -\frac{1}{10}x(x - 4)^m(x + 5)^n$ must be even. For choice A, the sum of these exponents is $1 + 1 + 1$, or 3 , which is odd. For choice B, the sum of these exponents is $1 + 1 + 2$, or 4 , which is even. Therefore, $y = -\frac{1}{10}x(x - 4)(x + 5)^2$ could be the equation of the graph shown.

Choice A is incorrect. For the graph of this equation, all values of x less than -5 correspond to positive, not negative, y -values.

Choice C is incorrect. The graph of this equation has x -intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x -intercepts at $x = -5$, $x = 0$, and $x = 4$.

Choice D is incorrect. The graph of this equation has x -intercepts at $x = -4$, $x = 0$, and $x = 5$, rather than x -intercepts at $x = -5$, $x = 0$, and $x = 4$.

Question Difficulty: Hard

Question ID b4712366

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: b4712366

An auditorium has seats for 1,800 people. Tickets to attend a show at the auditorium currently cost \$4.00. For each \$1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. If this equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

- A. 4
- B. 7
- C. 14
- D. 18

ID: b4712366 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the situation can be modeled by the equation $y = -100x^2 + 1,400x + 7,200$, where x represents the increase in ticket price, in dollars, and y represents the revenue, in dollars, from ticket sales. Since the coefficient of the x^2 term is negative, the graph of this equation in the xy -plane opens downward and reaches its maximum value at its vertex. If a quadratic equation in the form $y = ax^2 + bx + c$, where a , b , and c are constants, is graphed in the xy -plane, the x -coordinate of the vertex is equal to $-\frac{b}{2a}$. For the equation $y = -100x^2 + 1,400x + 7,200$, $a = -100$, $b = 1,400$, and $c = 7,200$. It follows that the x -coordinate of the vertex is $-\frac{1,400}{2(-100)}$, or 7. Therefore, if the given equation is graphed in the xy -plane, the maximum of the graph occurs at an x -value of 7.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 2d2e29ba

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 2d2e29ba

$$x^2 + 7x + 5 = 0$$

One solution to the given equation can be written as $x = \frac{-7+\sqrt{k}}{2}$, where k is a constant. What is the value of k ?

ID: 2d2e29ba Answer

Correct Answer: 29

Rationale

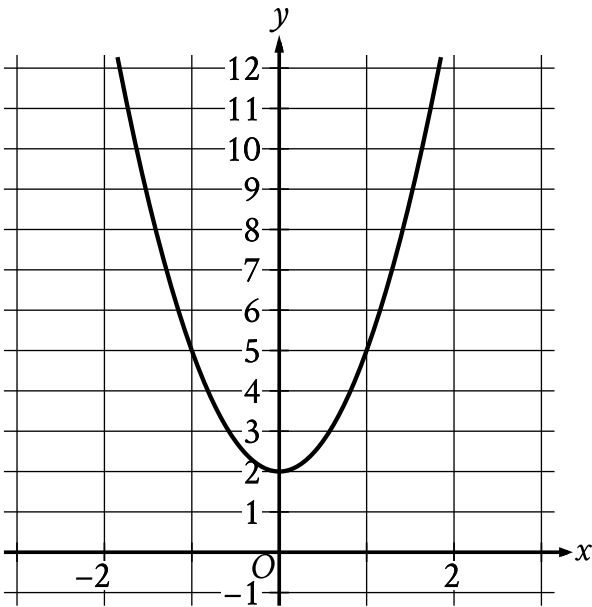
The correct answer is **29**. The solutions to a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula, which gives two solutions: $x = \frac{-b+\sqrt{b^2-4ac}}{2a}$ and $x = \frac{-b-\sqrt{b^2-4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 1$, $b = 7$, and $c = 5$. It follows that one solution to the given quadratic equation is $x = \frac{-7+\sqrt{7^2-4(1)(5)}}{2(1)}$, or $x = \frac{-7+\sqrt{49-20}}{2}$, which is equivalent to $x = \frac{-7+\sqrt{29}}{2}$. It's given that one solution to the given equation can be written as $x = \frac{-7+\sqrt{k}}{2}$, where k is a constant. It follows that the value of k is **29**.

Question Difficulty: Hard

Question ID 994cedb8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 994cedb8



The graph of the quadratic function $y = f(x)$ is shown. What is the vertex of the graph?

- A. $(0, -2)$
- B. $(0, -3)$
- C. $(0, 2)$
- D. $(0, 3)$

ID: 994cedb8 Answer

Correct Answer: C

Rationale

Choice C is correct. The vertex of the graph of a quadratic function in the xy -plane is the point at which the graph is either at its minimum or maximum y -value. In the graph shown, the minimum y -value occurs at the point $(0, 2)$.

Choice A is incorrect. The graph shown doesn't pass through the point $(0, -2)$.

Choice B is incorrect. The graph shown doesn't pass through the point $(0, -3)$.

Choice D is incorrect. The graph shown doesn't pass through the point $(0, 3)$.

Question Difficulty: Easy

Question ID 685e4f6b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 685e4f6b

The function $f(t) = 60,000(2)^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

ID: 685e4f6b Answer

Correct Answer: 410

Rationale

The correct answer is **410**. It's given that t minutes after an initial observation, the number of bacteria in a population is $60,000(2)^{\frac{t}{410}}$. This expression consists of the initial number of bacteria, **60,000**, multiplied by the expression $2^{\frac{t}{410}}$. The time it takes for the number of bacteria to double is the increase in the value of t that causes the expression $2^{\frac{t}{410}}$ to double. Since the base of the expression $2^{\frac{t}{410}}$ is **2**, the expression $2^{\frac{t}{410}}$ will double when the exponent increases by **1**. Since the exponent of the expression $2^{\frac{t}{410}}$ is $\frac{t}{410}$, the exponent will increase by **1** when t increases by **410**. Therefore the time, in minutes, it takes for the number of bacteria in the population to double is **410**.

Question Difficulty: Hard

Question ID 83a6080a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 83a6080a

Which expression is equivalent to $(x^2 + 11)^2 + (x - 5)(x + 5)$?

- A. $x^4 + 23x^2 - 14$
- B. $x^4 + 23x^2 + 96$
- C. $x^4 + 12x^2 + 121$
- D. $x^4 + x^2 + 146$

ID: 83a6080a Answer

Correct Answer: B

Rationale

Choice B is correct. The expression $(x^2 + 11)^2$ can be written as $(x^2 + 11)(x^2 + 11)$, which is equivalent to $x^2(x^2 + 11) + 11(x^2 + 11)$. Distributing x^2 and 11 to $(x^2 + 11)$ yields $x^4 + 11x^2 + 11x^2 + 121$, or $x^4 + 22x^2 + 121$. The expression $(x - 5)(x + 5)$ is equivalent to $(x - 5)x + (x - 5)5$. Distributing x and 5 to $(x - 5)$ yields $x^2 - 5x + 5x - 25$, or $x^2 - 25$. Therefore, the expression $(x^2 + 11)^2 + (x - 5)(x + 5)$ is equivalent to $(x^4 + 22x^2 + 121) + (x^2 - 25)$, or $x^4 + 22x^2 + 121 + x^2 - 25$. Combining like terms in this expression yields $x^4 + 23x^2 + 96$.

Choice A is incorrect. Equivalent expressions must be equivalent for any value of x . Substituting 0 for x in this expression yields -14 , whereas substituting 0 for x in the given expression yields 96 .

Choice C is incorrect. Equivalent expressions must be equivalent for any value of x . Substituting 0 for x in this expression yields 121 , whereas substituting 0 for x in the given expression yields 96 .

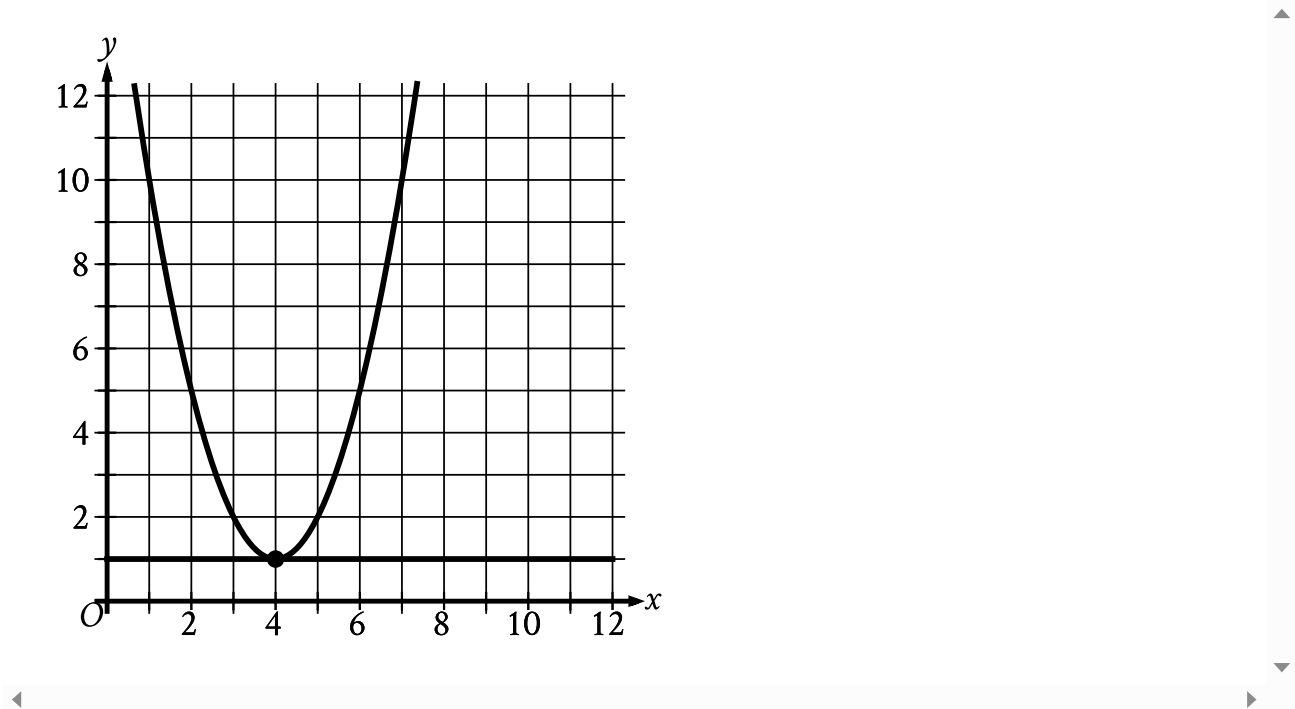
Choice D is incorrect. Equivalent expressions must be equivalent for any value of x . Substituting 0 for x in this expression yields 146 , whereas substituting 0 for x in the given expression yields 96 .

Question Difficulty: Medium

Question ID 84aaa912

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 84aaa912



The graph of a system of a linear and a quadratic equation is shown. What is the solution (x, y) to this system?

- A. $(0, 0)$
- B. $(-4, 1)$
- C. $(4, -1)$
- D. $(4, 1)$

ID: 84aaa912 Answer

Correct Answer: D

Rationale

Choice D is correct. The solution to the system corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the quadratic equation shown intersect at the point $(4, 1)$. Therefore, $(4, 1)$ is the solution to this system.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 886111f9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 886111f9

$\frac{1}{4xy} + xyz = \frac{1}{3yz}$

In the given equation, x , y , and z are positive numbers. Which expression is equivalent to y ?

- A. $\frac{4x-3z}{12x^2z^2}$
- B. $\sqrt{\frac{4x-3z}{12x^2z^2}}$
- C. $\frac{1}{3xz^2-4x^2z}$
- D. $\sqrt{\frac{1}{3xz^2-4x^2z}}$

ID: 886111f9 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that in the equation $\frac{1}{4xy} + xyz = \frac{1}{3yz}$, x , y , and z are positive numbers. Multiplying both sides of this equation by the least common denominator of $12xyz$ yields $12xyz\left(\frac{1}{4xy} + xyz\right) = 12xyz\left(\frac{1}{3yz}\right)$, which gives $\frac{12xyz}{4xy} + 12x^2y^2z^2 = \frac{12xyz}{3yz}$, or $3z + 12x^2y^2z^2 = 4x$. Subtracting $3z$ from both sides of this equation yields $12x^2y^2z^2 = 4x - 3z$. Dividing both sides of this equation by $12x^2z^2$ yields $y^2 = \frac{4x-3z}{12x^2z^2}$. Taking the square root of both sides of this equation yields $y = \pm\sqrt{\frac{4x-3z}{12x^2z^2}}$. Since y is a positive number, the expression $\sqrt{\frac{4x-3z}{12x^2z^2}}$ is equivalent to y .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a31203f3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: a31203f3

The function f is defined by $f(x) = x^2 + x + 71$. What is the value of $f(2)$?

ID: a31203f3 Answer

Correct Answer: 77

Rationale

The correct answer is 77. It's given that the function f is defined by $f(x) = x^2 + x + 71$. Substituting 2 for x in function f yields $f(2) = (2)^2 + 2 + 71$, which is equivalent to $f(2) = 4 + 2 + 71$, or $f(2) = 77$. Therefore, the value of $f(2)$ is 77.

Question Difficulty: Medium

Question ID e3571ba2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: e3571ba2

The number of bacteria in a liquid medium doubles every day. There are **44,000** bacteria in the liquid medium at the start of an observation. Which represents the number of bacteria, y , in the liquid medium t days after the start of the observation?

- A. $y = \frac{1}{2}(44,000)^t$
- B. $y = 2(44,000)^t$
- C. $y = 44,000\left(\frac{1}{2}\right)^t$
- D. $y = 44,000(2)^t$

ID: e3571ba2 Answer

Correct Answer: D

Rationale

Choice D is correct. Since the number of bacteria doubles every day, the relationship between t and y can be represented by an exponential equation of the form $y = a(b)^t$, where a is the number of bacteria at the start of the observation and the number of bacteria increases by a factor of b every day. It's given that there are **44,000** bacteria at the start of the observation. Therefore, $a = 44,000$. It's also given that the number of bacteria doubles, or increases by a factor of **2**, every day. Therefore, $b = 2$. Substituting **44,000** for a and **2** for b in the equation $y = a(b)^t$ yields $y = 44,000(2)^t$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the number of bacteria is decreasing by half, not doubling, every day.

Question Difficulty: Medium

Question ID 244c26d8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 244c26d8

$$\begin{aligned}x + 7 &= 10 \\ (x + 7)^2 &= y\end{aligned}$$

Which ordered pair (x, y) is a solution to the given system of equations?

- A. $(3, 100)$
- B. $(3, 3)$
- C. $(3, 10)$
- D. $(3, 70)$

ID: 244c26d8 Answer

Correct Answer: A

Rationale

Choice A is correct. The solution to a system of equations is the ordered pair (x, y) that satisfies all equations in the system. It's given by the first equation in the system that $x + 7 = 10$. Substituting **10** for $x + 7$ into the second equation yields $10^2 = y$, or $y = 100$. The x-coordinate of the solution to the system of equations can be found by subtracting **7** from both sides of the equation $x + 7 = 10$, which yields $x = 3$. Therefore, the ordered pair **$(3, 100)$** is a solution to the given system of equations.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID e4c5e586

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: e4c5e586

$$57x^2 + (57b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab , where k is a constant. What is the value of k ?

- A. $\frac{1}{57}$
- B. $\frac{1}{19}$
- C. 1
- D. 57

ID: e4c5e586 Answer

Correct Answer: A

Rationale

Choice A is correct. The left-hand side of the given equation is the expression $57x^2 + (57b + a)x + ab$. Applying the distributive property to this expression yields $57x^2 + 57bx + ax + ab$. Since the first two terms of this expression have a common factor of $57x$ and the last two terms of this expression have a common factor of a , this expression can be rewritten as $57x(x + b) + a(x + b)$. Since the two terms of this expression have a common factor of $(x + b)$, it can be rewritten as $(x + b)(57x + a)$. Therefore, the given equation can be rewritten as $(x + b)(57x + a) = 0$. By the zero product property, it follows that $x + b = 0$ or $57x + a = 0$. Subtracting b from both sides of the equation $x + b = 0$ yields $x = -b$. Subtracting a from both sides of the equation $57x + a = 0$ yields $57x = -a$. Dividing both sides of this equation by 57 yields $x = \frac{-a}{57}$. Therefore, the solutions to the given equation are $-b$ and $\frac{-a}{57}$. It follows that the product of the solutions of the given equation is $(-b)(\frac{-a}{57})$, or $\frac{ab}{57}$. It's given that the product of the solutions of the given equation is kab . It follows that $\frac{ab}{57} = kab$, which can also be written as $ab(\frac{1}{57}) = ab(k)$. It's given that a and b are positive constants. Therefore, dividing both sides of the equation $ab(\frac{1}{57}) = ab(k)$ by ab yields $\frac{1}{57} = k$. Thus, the value of k is $\frac{1}{57}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID e8c30041

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: e8c30041

$w^2 + 12w - 40 = 0$

Which of the following is a solution to the given equation?

- A. $6 - 2\sqrt{19}$
- B. $2\sqrt{19}$
- C. $\sqrt{19}$
- D. $-6 + 2\sqrt{19}$

ID: e8c30041 Answer

Correct Answer: D

Rationale

Choice D is correct. Adding **40** to both sides of the given equation yields $w^2 + 12w = 40$. To complete the square, adding $\left(\frac{12}{2}\right)^2$, or **6²**, to both sides of this equation yields $w^2 + 12w + 6^2 = 40 + 6^2$, or $(w + 6)^2 = 76$. Taking the square root of both sides of this equation yields $w + 6 = \pm\sqrt{76}$, or $w + 6 = \pm2\sqrt{19}$. Subtracting **6** from both sides of this equation yields $w = -6 \pm 2\sqrt{19}$. Therefore, the solutions to the given equation are $-6 + 2\sqrt{19}$ and $-6 - 2\sqrt{19}$. Of these two solutions, only $-6 + 2\sqrt{19}$ is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 4bcc3513

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 4bcc3513

The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

- I. $f(x) = 18(1.25)^x + 41$
- II. $g(x) = 9(0.73)^x$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: 4bcc3513 Answer

Correct Answer: B

Rationale

Choice B is correct. For the function f , since the base of the exponent, 1.25 , is greater than 1 , the value of $(1.25)^x$ increases as x increases. Therefore, the value of $18(1.25)^x$ and the value of $18(1.25)^x + 41$ also increase as x increases. Since f is therefore an increasing function where $x \geq 0$, the function f has no maximum value. For the function g , since the base of the exponent, 0.73 , is less than 1 , the value of $(0.73)^x$ decreases as x increases. Therefore, the value of $9(0.73)^x$ also decreases as x increases. It follows that the maximum value of $g(x)$ for $x \geq 0$ occurs when $x = 0$. Substituting 0 for x in the function g yields $g(0) = 9(0.73)^0$, which is equivalent to $g(0) = 9(1)$, or $g(0) = 9$. Therefore, the maximum value of $g(x)$ for $x \geq 0$ is 9 , which appears as a coefficient in equation II. So, of the two equations given, only II displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

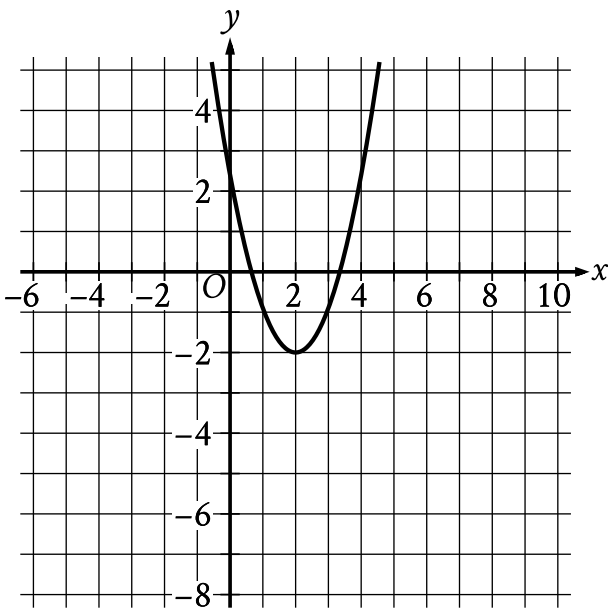
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 703dd32e

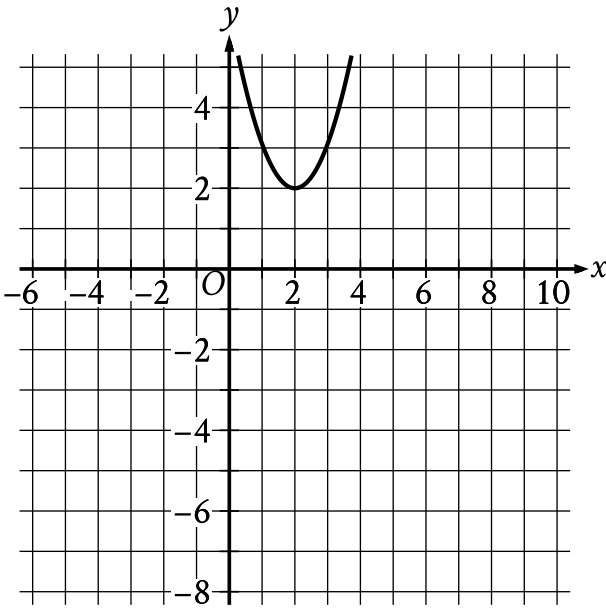
Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 703dd32e

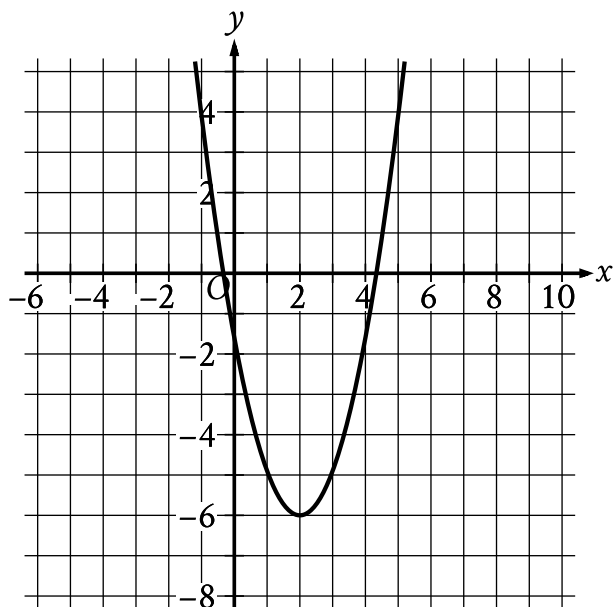


The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

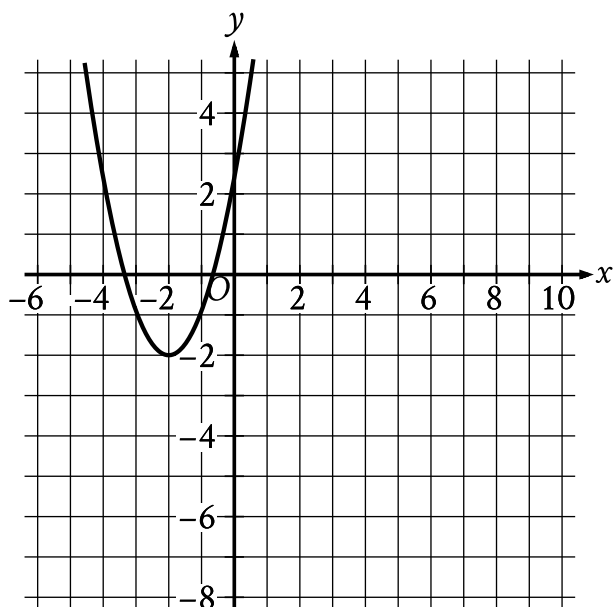
A.



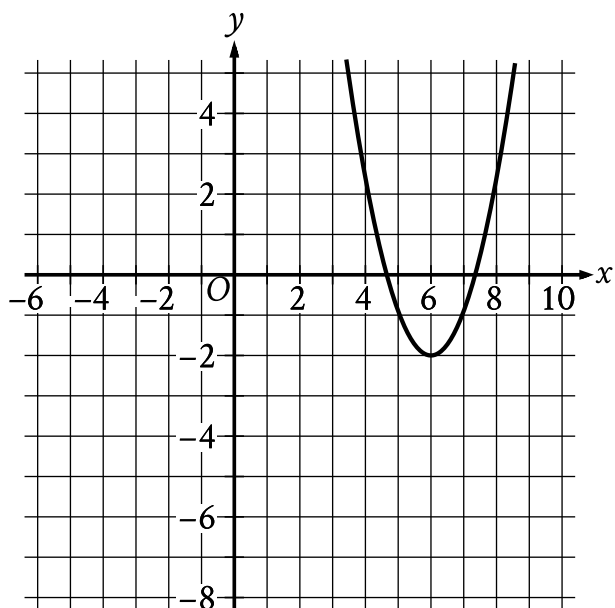
B.



C.



D.



Correct Answer: A

Rationale

Choice A is correct. When a graph is translated up **4** units, each point on the resulting graph is **4** units above the point on the original graph. In other words, the y -value of each point on the graph increases by **4**. The graph shown passes through the points **(1, -1)**, **(2, -2)**, and **(3, -1)**. It follows that when the graph shown is translated up **4** units, the resulting graph will pass through the points **(1, -1 + 4)**, **(2, -2 + 4)**, and **(3, -1 + 4)**. These points are **(1, 3)**, **(2, 2)**, and **(3, 3)**, respectively. Of the given choices, only the graph in choice A passes through the points **(1, 3)**, **(2, 2)**, and **(3, 3)**.

Choice B is incorrect. This is the result of translating the graph down, rather than up, **4** units.

Choice C is incorrect. This is the result of translating the graph left, rather than up, **4** units.

Choice D is incorrect. This is the result of translating the graph right, rather than up, **4** units.

Question Difficulty: Easy

Question ID 52bceaab

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 52bceaab

An object is kicked from a platform. The equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

- A. 0
- B. 4.9
- C. 7
- D. 9

ID: 52bceaab Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height, in meters, of the object t seconds after it is kicked. It follows that the height, in meters, from which the object was kicked is the value of h when $t = 0$. Substituting 0 for t in the equation $h = -4.9t^2 + 7t + 9$ yields $h = -4.9(0)^2 + 7(0) + 9$, or $h = 9$. Therefore, the object was kicked from a height of 9 meters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

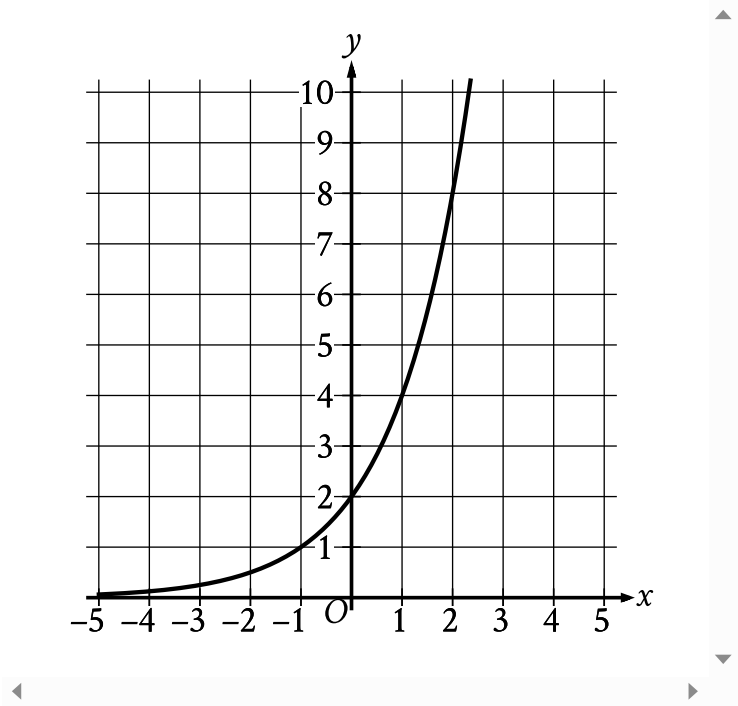
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID f92896a6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: f92896a6



What is the y-intercept of the graph shown?

- A. $(0, 0)$
- B. $(0, 2)$
- C. $(2, 0)$
- D. $(2, 2)$

ID: f92896a6 Answer

Correct Answer: B

Rationale

Choice B is correct. The y -intercept of a graph in the xy -plane is the point at which the graph crosses the y -axis. The graph shown crosses the y -axis at the point $(0, 2)$. Therefore, the y -intercept of the graph shown is $(0, 2)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 8de64a72

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 8de64a72

$y = -2.5$

$y = x^2 + 8x + k$

In the given system of equations, k is a positive integer constant. The system has no real solutions. What is the least possible value of k ?

ID: 8de64a72 Answer

Correct Answer: 14

Rationale

The correct answer is **14**. It's given by the first equation of the system of equations that $y = -2.5$. Substituting -2.5 for y in the second given equation, $y = x^2 + 8x + k$, yields $-2.5 = x^2 + 8x + k$. Adding 2.5 to both sides of this equation yields $0 = x^2 + 8x + k + 2.5$. A quadratic equation of the form $0 = ax^2 + bx + c$, where a , b , and c are constants, has no real solutions if and only if its discriminant, $b^2 - 4ac$, is negative. In the equation $0 = x^2 + 8x + k + 2.5$, where k is a positive integer constant, $a = 1$, $b = 8$, and $c = k + 2.5$. Substituting 1 for a , 8 for b , and $k + 2.5$ for c in $b^2 - 4ac$ yields $8^2 - 4(1)(k + 2.5)$, or $64 - 4(k + 2.5)$. Since this value must be negative, $64 - 4(k + 2.5) < 0$. Adding $4(k + 2.5)$ to both sides of this inequality yields $64 < 4(k + 2.5)$. Dividing both sides of this inequality by 4 yields $16 < k + 2.5$. Subtracting 2.5 from both sides of this inequality yields $13.5 < k$. Since k is a positive integer constant, the least possible value of k is **14**.

Question Difficulty: Hard

Question ID 72127a15

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 72127a15

The expression $4x^2 + bx - 45$, where b is a constant, can be rewritten as $(hx + k)(x + j)$, where h , k , and j are integer constants. Which of the following must be an integer?

- A. $\frac{b}{h}$
- B. $\frac{b}{k}$
- C. $\frac{45}{h}$
- D. $\frac{45}{k}$

ID: 72127a15 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that $4x^2 + bx - 45$ can be rewritten as $(hx + k)(x + j)$. The expression $(hx + k)(x + j)$ can be rewritten as $hx^2 + jhx + kx + kj$, or $hx^2 + (jh + k)x + kj$. Therefore, $hx^2 + (jh + k)x + kj$ is equivalent to $4x^2 + bx - 45$. It follows that $kj = -45$. Dividing each side of this equation by k yields $j = \frac{-45}{k}$. Since j is an integer, $-\frac{45}{k}$ must be an integer. Therefore, $\frac{45}{k}$ must also be an integer.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID eeebee8d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: eeebee8d

$$b = 42cf$$

The given equation relates the positive numbers b , c , and f . Which equation correctly expresses c in terms of b and f ?

- A. $c = \frac{b}{42f}$
- B. $c = \frac{b-42}{f}$
- C. $c = 42bf$
- D. $c = 42 - b - f$

ID: eeebee8d Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the equation $b = 42cf$ relates the positive numbers b , c , and f . Dividing each side of the given equation by $42f$ yields $\frac{b}{42f} = c$, or $c = \frac{b}{42f}$. Thus, the equation $c = \frac{b}{42f}$ correctly expresses c in terms of b and f .

Choice B is incorrect. This equation can be rewritten as $b = cf + 42$.

Choice C is incorrect. This equation can be rewritten as $b = \frac{c}{42f}$.

Choice D is incorrect. This equation can be rewritten as $b = 42 - c - f$.

Question Difficulty: Medium

Question ID fd736602

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: fd736602

$$f(x) = 4,000(0.75)^x$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after **2012**, where $x \leq 7$. Which of the following is the best interpretation of **4,000** in this context?

- A. The estimated initial beetle population for this species and area in **2012**
- B. The estimated beetle population for this species and area **7** years after **2012**
- C. The estimated percent decrease in the beetle population for this species and area each year after **2012**
- D. The estimated percent decrease in the beetle population for this species and area every **7** years after **2012**

ID: fd736602 Answer

Correct Answer: A

Rationale

Choice A is correct. For an exponential function in the form $f(x) = a(b)^x$, where a and b are positive constants and $b < 1$, the initial value of $f(x)$, or the value of $f(x)$ when $x = 0$, is a and the value of $f(x)$ decreases by $100(1 - b)\%$ each time x increases by 1 . Therefore, the initial value of the function $f(x) = 4,000(0.75)^x$, or the value of $f(x)$ when $x = 0$, is **4,000**. Therefore, the best interpretation of **4,000** in this context is the estimated initial beetle population for this species and area in **2012**.

Choice B is incorrect. The estimated beetle population for this species and area **7** years after **2012** is $4,000(0.75)^7$, or approximately **534**, not **4,000**.

Choice C is incorrect. The estimated percent decrease in the beetle population for this species and area each year after **2012** is $100(1 - 0.75)$, or **25**, not **4,000**.

Choice D is incorrect. The estimated percent decrease in the beetle population for this species and area every **7** years after **2012** is $100(1 - 0.75^7)$, or approximately **87**, not **4,000**.

Question Difficulty: Medium

Question ID 8eafbf55

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 8eafbf55

$$\begin{aligned} g(x) &= \frac{3}{5}x + \frac{7}{6} \\ h(x) &= 6x - 5 \end{aligned}$$

The functions g and h are defined by the equations shown. Which expression is equivalent to $g(x) \cdot h(x)$?

- A. $\frac{18x^2}{5} - \frac{35}{6}$
- B. $\frac{18x^2}{5} + \frac{27x}{11} - \frac{35}{6}$
- C. $\frac{18x^2}{5} - 4x - \frac{35}{6}$
- D. $\frac{18x^2}{5} + 4x - \frac{35}{6}$

ID: 8eafbf55 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that $g(x) = \frac{3}{5}x + \frac{7}{6}$ and $h(x) = 6x - 5$. Substituting $\frac{3}{5}x + \frac{7}{6}$ for $g(x)$ and $6x - 5$ for $h(x)$ in the expression $g(x) \cdot h(x)$ yields $(\frac{3}{5}x + \frac{7}{6})(6x - 5)$. This expression can be rewritten as $\frac{3}{5}x(6x - 5) + \frac{7}{6}(6x - 5)$, or $\frac{18x^2}{5} - 3x + 7x - \frac{35}{6}$, which is equivalent to $\frac{18x^2}{5} + 4x - \frac{35}{6}$.

Choice A is incorrect. This expression is equivalent to $\frac{3}{5}x(6x) + \frac{7}{6}(-5)$, not $(\frac{3}{5}x + \frac{7}{6})(6x - 5)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to $(\frac{3}{5}x - \frac{7}{6})(6x + 5)$, not $(\frac{3}{5}x + \frac{7}{6})(6x - 5)$.

Question Difficulty: Hard

Question ID 90206d7a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 90206d7a

$x(x + 1) - 56 = 4x(x - 7)$

What is the sum of the solutions to the given equation?

ID: 90206d7a Answer

Correct Answer: 29/3, 9.666, 9.667

Rationale

The correct answer is $\frac{29}{3}$. Applying the distributive property to the left-hand side of the given equation, $x(x + 1) - 56$, yields $x^2 + x - 56$. Applying the distributive property to the right-hand side of the given equation, $4x(x - 7)$, yields $4x^2 - 28x$. Thus, the equation becomes $x^2 + x - 56 = 4x^2 - 28x$. Combining like terms on the left- and right-hand sides of this equation yields $0 = (4x^2 - x^2) + (-28x - x) + 56$, or $3x^2 - 29x + 56 = 0$. For a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, the quadratic formula gives the solutions to the equation in the form $x = \frac{(-b \pm \sqrt{b^2 - 4ac})}{2a}$. Substituting 3 for a , -29 for b , and 56 for c from the equation $3x^2 - 29x + 56 = 0$ into the quadratic formula yields $x = \frac{(29 \pm \sqrt{(-29)^2 - 4(3)(56)})}{2(3)}$, or $x = \frac{29}{6} \pm \frac{13}{6}$. It follows that the solutions to the given equation are $\frac{29}{6} + \frac{13}{6}$ and $\frac{29}{6} - \frac{13}{6}$. Adding these two solutions gives the sum of the solutions: $\frac{29}{6} + \frac{13}{6} + \frac{29}{6} - \frac{13}{6}$, which is equivalent to $\frac{29}{6} + \frac{29}{6}$, or $\frac{29}{3}$. Note that 29/3, 9.666, and 9.667 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID cc1ceb8b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: cc1ceb8b

$| -5x + 13 | = 73$

What is the sum of the solutions to the given equation?

- A. $-\frac{146}{5}$
- B. -12
- C. 0
- D. $\frac{26}{5}$

ID: cc1ceb8b Answer

Correct Answer: D

Rationale

Choice D is correct. By the definition of absolute value, if $| -5x + 13 | = 73$, then $-5x + 13 = 73$ or $-5x + 13 = -73$. Subtracting **13** from both sides of the equation $-5x + 13 = 73$ yields $-5x = 60$. Dividing both sides of this equation by -5 yields $x = -12$. Subtracting **13** from both sides of the equation $-5x + 13 = -73$ yields $-5x = -86$. Dividing both sides of this equation by -5 yields $x = \frac{86}{5}$. Therefore, the solutions to the given equation are -12 and $\frac{86}{5}$, and it follows that the sum of the solutions to the given equation is $-12 + \frac{86}{5}$, or $\frac{26}{5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is a solution, not the sum of the solutions, to the given equation.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID aacaf7f1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: aacaf7f1

$3x(x - 4)(x + 5) = 0$

What is one of the solutions to the given equation?

- A. **-4**
- B. **0**
- C. **3**
- D. **5**

ID: aacaf7f1 Answer

Correct Answer: B

Rationale

Choice B is correct. Applying the zero product property to the given equation yields $3x = 0$, $x - 4 = 0$, and $x + 5 = 0$. Dividing each side of the equation $3x = 0$ by 3 yields $x = 0$. Adding 4 to each side of the equation $x - 4 = 0$ yields $x = 4$. Subtracting 5 from each side of the equation $x + 5 = 0$ yields $x = -5$. Therefore, the solutions to the given equation are 0 , 4 , and -5 . Thus, one of the solutions to the given equation is 0 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID fa2d55ad

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: fa2d55ad

$$f(x) = (x - 44)(x - 46)$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its minimum?

- A. 46
- B. 45
- C. 44
- D. -1

ID: fa2d55ad Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = (x - 44)(x - 46)$, which can be rewritten as $f(x) = x^2 - 90x + 2,024$. Since the coefficient of the x^2 -term is positive, the graph of $y = f(x)$ in the xy -plane opens upward and reaches its minimum value at its vertex. For an equation in the form $f(x) = ax^2 + bx + c$, where a , b , and c are constants, the x -coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 - 90x + 2,024$, $a = 1$, $b = -90$, and $c = 2,024$. It follows that the x -coordinate of the vertex is $-\frac{(-90)}{2(1)}$, or 45. Therefore, $f(x)$ reaches its minimum when the value of x is 45.

Choice A is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice C is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice D is incorrect. This is the y -coordinate of the vertex of the graph of $y = f(x)$ in the xy -plane.

Question Difficulty: Hard

Question ID d53586d6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: d53586d6

$$y = 7,400(0.87)^x$$

The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

- A. 740
- B. 6,438
- C. 7,400
- D. 8,700

ID: d53586d6 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $y = 7,400(0.87)^x$ estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. The estimated value of the piece of equipment at the time it was purchased corresponds to the value of y when $x = 0$. Substituting 0 for x in the equation $y = 7,400(0.87)^x$ yields $y = 7,400(0.87)^0$, which is equivalent to $y = 7,400(1)$, or $y = 7,400$. Therefore, the estimated value, in dollars, of the piece of equipment at the time it was purchased was 7,400.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This was the estimated value, in dollars, of the piece of equipment 1 year after it was purchased.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 6f761b44

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 6f761b44

$$\frac{1}{7b} = \frac{11x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

- A. $x = \frac{7by}{11}$
- B. $x = y - 77b$
- C. $x = \frac{y}{77b}$
- D. $x = 77by$

ID: 6f761b44 Answer

Correct Answer: C

Rationale

Choice C is correct. Multiplying each side of the given equation by y yields the equivalent equation $\frac{y}{7b} = 11x$. Dividing each side of this equation by 11 yields $\frac{y}{77b} = x$, or $x = \frac{y}{77b}$.

Choice A is incorrect. This equation is not equivalent to the given equation.

Choice B is incorrect. This equation is not equivalent to the given equation.

Choice D is incorrect. This equation is not equivalent to the given equation.

Question Difficulty: Medium

Question ID 5e48ccb6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 5e48ccb6

The function C defined by $C(t) = 6(1.5)^t$ gives the estimated number of clients at a certain business, where t is the number of weeks since the business opened. Which of the following is the best interpretation of the number **6** in this context?

- A. The estimated number of clients when the business opened
- B. The increase in the estimated number of clients each week
- C. The number of weeks the business has been open
- D. The percent increase in the estimated number of clients each week

ID: 5e48ccb6 Answer

Correct Answer: A

Rationale

Choice A is correct. For an exponential function of the form $C(t) = a(b)^t$, where a and b are constants, the initial value of the function, or the value of the function when $t = 0$, is a , and the value of the function changes by a factor of b each time t increases by 1 . It follows that in the given function, the initial value is **6** and the value of the function changes by a factor of **1.5** each time t increases by 1 . It's given that the function C gives the estimated number of clients at a certain business and t is the number of weeks since the business opened. Therefore, the best interpretation of the number **6** in this context is the estimated number of clients when $t = 0$, or when the business opened.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID db85220f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: db85220f

Time (years)	Total amount (dollars)
0	670.00
1	674.02
2	678.06

Sara opened a savings account at a bank. The table shows the exponential relationship between the time t , in years, since Sara opened the account and the total amount d , in dollars, in the account. If Sara made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and d ?

- A. $d = 0.006(1 + 670)^t$
- B. $d = 670(1 + 0.006)^t$
- C. $d = (1 + 0.006)^t$
- D. $d = (1 + 670)^t$

ID: db85220f Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the relationship between t and d is exponential. The table shows that the value of d increases as the value of t increases. Therefore, the relationship between t and d can be represented by an increasing exponential equation of the form $d = a(1 + b)^t$, where a and b are positive constants. The table shows that when $t = 0$, $d = 670$. Substituting 0 for t and 670 for d in the equation $d = a(1 + b)^t$ yields $670 = a(1 + b)^0$, which is equivalent to $670 = a(1)$, or $670 = a$. Substituting 670 for a in the equation $d = a(1 + b)^t$ yields $d = 670(1 + b)^t$. The table also shows that when $t = 1$, $d = 674.02$. Substituting 1 for t and 674.02 for d in the equation $d = 670(1 + b)^t$ yields $674.02 = 670(1 + b)^1$, or $674.02 = 670(1 + b)$. Dividing both sides of this equation by 670 yields $1.006 = 1 + b$. Subtracting 1 from both sides of this equation yields $b = 0.006$. Substituting 0.006 for b in the equation $d = 670(1 + b)^t$ yields $d = 670(1 + 0.006)^t$. Therefore, of the choices, choice B best represents the relationship between t and d .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 169a5cef

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 169a5cef

$$\frac{1}{x} + 8 = \frac{1}{6} + 8$$

What is the solution to the given equation?

- A. 1
- B. 6
- C. 8
- D. 14

ID: 169a5cef Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting 8 from each side of the given equation yields $\frac{1}{x} = \frac{1}{6}$. Multiplying each side of this equation by $6x$ yields $6 = x$. Thus, the solution to the given equation is 6.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 877585e8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 877585e8

A model estimates that at the end of each year from **2015** to **2020**, the number of squirrels in a population was **150%** more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of **2016**, there were **180** squirrels in the population. Which of the following equations represents this model, where n is the estimated number of squirrels in the population t years after the end of **2015** and $t \leq 5$?

- A. $n = 72(1.5)^t$
- B. $n = 72(2.5)^t$
- C. $n = 180(1.5)^t$
- D. $n = 180(2.5)^t$

ID: 877585e8 Answer

Correct Answer: B

Rationale

Choice B is correct. Since the model estimates that the number of squirrels in the population increased by a fixed percentage, **150%**, each year, the model can be represented by an exponential equation of the form $n = a(1 + \frac{p}{100})^t$, where a is the estimated number of squirrels in the population at the end of **2015**, and the model estimates that at the end of each year, the number is $p\%$ more than the number at the end of the previous year. Since the model estimates that at the end of each year, the number was **150%** more than the number at the end of the previous year, $p = 150$. Substituting **150** for p in the equation $n = a(1 + \frac{p}{100})^t$ yields $n = a(1 + \frac{150}{100})^t$, which is equivalent to $n = a(1 + 1.5)^t$, or $n = a(2.5)^t$. It's given that the estimated number of squirrels at the end of **2016** was **180**. This means that when $t = 1$, $n = 180$. Substituting **1** for t and **180** for n in the equation $n = a(2.5)^t$ yields $180 = a(2.5)^1$, or $180 = 2.5a$. Dividing each side of this equation by **2.5** yields $72 = a$. Substituting **72** for a in the equation $n = a(2.5)^t$ yields $n = 72(2.5)^t$.

Choice A is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was **150%** of, not **150%** more than, the estimated number at the end of the previous year.

Choice C is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was **150%** of, not **150%** more than, the estimated number at the end of the previous year, and the estimated number of squirrels at the end of **2015**, not the end of **2016**, was **180**.

Choice D is incorrect. This equation represents a model where the estimated number of squirrels at the end of **2015**, not the end of **2016**, was **180**.

Question Difficulty: Hard

Question ID f3665402

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: f3665402

$|x - 9| + 45 = 63$

What is the sum of the solutions to the given equation?

ID: f3665402 Answer

Correct Answer: 18

Rationale

The correct answer is **18**. Subtracting **45** from each side of the given equation yields $|x - 9| = 18$. By the definition of absolute value, if $|x - 9| = 18$, then $x - 9 = 18$ or $x - 9 = -18$. Adding **9** to each side of the equation $x - 9 = 18$ yields $x = 27$. Adding **9** to each side of the equation $x - 9 = -18$ yields $x = -9$. Therefore, the solutions to the given equation are **27** and **-9**, and it follows that the sum of the solutions to the given equation is $27 + (-9)$, or **18**.

Question Difficulty: Hard

Question ID 42bc647d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 42bc647d

A machine launches a softball from ground level. The softball reaches a maximum height of **51.84** meters above the ground at **1.8** seconds and hits the ground at **3.6** seconds. Which equation represents the height above ground h , in meters, of the softball t seconds after it is launched?

- A. $h = -t^2 + 3.6$
- B. $h = -t^2 + 51.84$
- C. $h = -16(t - 1.8)^2 - 3.6$
- D. $h = -16(t - 1.8)^2 + 51.84$

ID: 42bc647d Answer

Correct Answer: D

Rationale

Choice D is correct. An equation representing the height above ground h , in meters, of a softball t seconds after it is launched by a machine from ground level can be written in the form $h = -a(t - b)^2 + c$, where a , b , and c are positive constants. In this equation, b represents the time, in seconds, at which the softball reaches its maximum height of c meters above the ground. It's given that this softball reaches a maximum height of **51.84** meters above the ground at **1.8** seconds; therefore, $b = 1.8$ and $c = 51.84$. Substituting **1.8** for b and **51.84** for c in the equation $h = -a(t - b)^2 + c$ yields $h = -a(t - 1.8)^2 + 51.84$. It's also given that this softball hits the ground at **3.6** seconds; therefore, $h = 0$ when $t = 3.6$. Substituting **0** for h and **3.6** for t in the equation $h = -a(t - 1.8)^2 + 51.84$ yields $0 = -a(3.6 - 1.8)^2 + 51.84$, which is equivalent to $0 = -a(1.8)^2 + 51.84$, or $0 = -3.24a + 51.84$. Adding **3.24a** to both sides of this equation yields $3.24a = 51.84$. Dividing both sides of this equation by **3.24** yields $a = 16$. Substituting **16** for a in the equation $h = -a(t - 1.8)^2 + 51.84$ yields $h = -16(t - 1.8)^2 + 51.84$. Therefore, $h = -16(t - 1.8)^2 + 51.84$ represents the height above ground h , in meters, of this softball t seconds after it is launched.

Choice A is incorrect. This equation represents a situation where the maximum height is **3.6** meters above the ground at **0** seconds, not **51.84** meters above the ground at **1.8** seconds.

Choice B is incorrect. This equation represents a situation where the maximum height is **51.84** meters above the ground at **0** seconds, not **1.8** seconds.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6d44de27

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 6d44de27

$f(x) = ax^2 + 4x + c$

In the given quadratic function, a and c are constants. The graph of $y = f(x)$ in the xy -plane is a parabola that opens upward and has a vertex at the point (h, k) , where h and k are constants. If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

- I. $c < 0$

II. $a \geq 1$
- A. I only

B. II only

C. I and II

D. Neither I nor II

ID: 6d44de27 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the graph of $y = f(x)$ in the xy -plane is a parabola with vertex (h, k) . If $f(-9) = f(3)$, then for the graph of $y = f(x)$, the point with an x -coordinate of -9 and the point with an x -coordinate of 3 have the same y -coordinate. In the xy -plane, a parabola is a symmetric graph such that when two points have the same y -coordinate, these points are equidistant from the vertex, and the x -coordinate of the vertex is halfway between the x -coordinates of these two points. Therefore, for the graph of $y = f(x)$, the points with x -coordinates -9 and 3 are equidistant from the vertex, (h, k) , and h is halfway between -9 and 3 . The value that is halfway between -9 and 3 is $\frac{-9+3}{2}$, or -3 . Therefore, $h = -3$. The equation defining f can also be written in vertex form, $f(x) = a(x - h)^2 + k$. Substituting -3 for h in this equation yields $f(x) = a(x - (-3))^2 + k$, or $f(x) = a(x + 3)^2 + k$. This equation is equivalent to $f(x) = a(x^2 + 6x + 9) + k$, or $f(x) = ax^2 + 6ax + 9a + k$. Since $f(x) = ax^2 + 4x + c$, it follows that $6a = 4$ and $9a + k = c$. Dividing both sides of the equation $6a = 4$ by 6 yields $a = \frac{4}{6}$, or $a = \frac{2}{3}$. Since $\frac{2}{3} < 1$, it's not true that $a \geq 1$. Therefore, statement II isn't true. Substituting $\frac{2}{3}$ for a in the equation $9a + k = c$ yields $9(\frac{2}{3}) + k = c$, or $6 + k = c$. Subtracting 6 from both sides of this equation yields $k = c - 6$. If $k < 0$, then $c - 6 < 0$, or $c < 6$. Since c could be any value less than 6 , it's not necessarily true that $c < 0$. Therefore, statement I isn't necessarily true. Thus, neither I nor II must be true.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 043afcfc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 043afcfc

In the xy -plane, a line with equation $2y = c$ for some constant c intersects a parabola at exactly one point. If the parabola has equation $y = -2x^2 + 9x$, what is the value of c ?

ID: 043afcfc Answer

Correct Answer: 20.25, 81/4

Rationale

The correct answer is $\frac{81}{4}$. The given linear equation is $2y = c$. Dividing both sides of this equation by 2 yields $y = \frac{c}{2}$. Substituting $\frac{c}{2}$ for y in the equation of the parabola yields $\frac{c}{2} = -2x^2 + 9x$. Adding $2x^2$ and $-9x$ to both sides of this equation yields $2x^2 - 9x + \frac{c}{2} = 0$. Since it's given that the line and the parabola intersect at exactly one point, the equation $2x^2 - 9x + \frac{c}{2} = 0$ must have exactly one solution. An equation of the form $Ax^2 + Bx + C = 0$, where A , B , and C are constants, has exactly one solution when the discriminant, $B^2 - 4AC$, is equal to 0 . In the equation $2x^2 - 9x + \frac{c}{2} = 0$, where $A = 2$, $B = -9$, and $C = \frac{c}{2}$, the discriminant is $(-9)^2 - 4(2)(\frac{c}{2})$. Setting the discriminant equal to 0 yields $(-9)^2 - 4(2)(\frac{c}{2}) = 0$, or $81 - 4c = 0$. Adding $4c$ to both sides of this equation yields $81 = 4c$. Dividing both sides of this equation by 4 yields $c = \frac{81}{4}$. Note that 81/4 and 20.25 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 0eced2ad

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 0eced2ad

A function p estimates that there were 2,000 animals in a population in 1998. Each year from 1998 to 2010, the function estimates that the number of animals in this population increased by 3% of the number of animals in the population the previous year. Which equation defines this function, where $p(x)$ is the estimated number of animals in the population x years after 1998?

- A. $p(x) = 2,000(3)^x$
- B. $p(x) = 2,000(1.97)^x$
- C. $p(x) = 2,000(1.03)^x$
- D. $p(x) = 2,000(0.97)^x$

ID: 0eced2ad Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that a function p estimates that there were 2,000 animals in a population in 1998 and that each year from 1998 to 2010, the number of animals in this population increased by 3% of the number of animals in the population the previous year. It follows that this situation can be represented by the function $p(x) = a(1 + \frac{r}{100})^x$, where $p(x)$ is the estimated number of animals in the population x years after 1998, a is the estimated number of animals in the population in 1998, and each year the estimated number of animals increased by $r\%$. Substituting 2,000 for a and 3 for r in this function yields $p(x) = 2,000(1 + \frac{3}{100})^x$, or $p(x) = 2,000(1.03)^x$.

Choice A is incorrect. This function represents a population in which each year the number of animals increased by 200%, not 3%, of the number of animals in the population the previous year.

Choice B is incorrect. This function represents a population in which each year the number of animals increased by 97%, not 3%, of the number of animals in the population the previous year.

Choice D is incorrect. This function represents a population in which each year the number of animals decreased, rather than increased, by 3% of the number of animals in the population the previous year.

Question Difficulty: Medium

Question ID b57aa354

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: b57aa354

In the xy -plane, a line with equation $2y = 4.5$ intersects a parabola at exactly one point. If the parabola has equation $y = -4x^2 + bx$, where b is a positive constant, what is the value of b ?

ID: b57aa354 Answer

Correct Answer: 6

Rationale

The correct answer is **6**. It's given that a line with equation $2y = 4.5$ intersects a parabola with equation $y = -4x^2 + bx$, where b is a positive constant, at exactly one point in the xy -plane. It follows that the system of equations consisting of $2y = 4.5$ and $y = -4x^2 + bx$ has exactly one solution. Dividing both sides of the equation of the line by **2** yields $y = 2.25$. Substituting **2.25** for y in the equation of the parabola yields $2.25 = -4x^2 + bx$. Adding $4x^2$ and subtracting bx from both sides of this equation yields $4x^2 - bx + 2.25 = 0$. A quadratic equation in the form of $ax^2 + bx + c = 0$, where a , b , and c are constants, has exactly one solution when the discriminant, $b^2 - 4ac$, is equal to zero. Substituting **4** for a and **2.25** for c in the expression $b^2 - 4ac$ and setting this expression equal to **0** yields $b^2 - 4(4)(2.25) = 0$, or $b^2 - 36 = 0$. Adding **36** to each side of this equation yields $b^2 = 36$. Taking the square root of each side of this equation yields $b = \pm 6$. It's given that b is positive, so the value of b is **6**.

Question Difficulty: Hard

Question ID 64eb9d1e

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Equivalent expressions	

ID: 64eb9d1e

Which expression is equivalent to $15w^2 + 8w$?

- A. $w(15w + 8)$
- B. $8w(15w + 1)$
- C. $15w^2(8w + 1)$
- D. $23(w^2 + w)$

ID: 64eb9d1e Answer

Correct Answer: A

Rationale

Choice A is correct. Since each term of the given expression has a common factor of w , it may be rewritten as $w(15w + 8)$. Therefore, the expression $w(15w + 8)$ is equivalent to $15w^2 + 8w$.

Choice B is incorrect. This expression is equivalent to $120w^2 + 8w$, not $15w^2 + 8w$.

Choice C is incorrect. This expression is equivalent to $120w^3 + 15w^2$, not $15w^2 + 8w$.

Choice D is incorrect. This expression is equivalent to $23w^2 + 23w$, not $15w^2 + 8w$.

Question Difficulty: Easy

Question ID 6d84ee10

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 6d84ee10

An egg is thrown from a rooftop. The equation $h = -4.9t^2 + 9t + 18$ represents this situation, where h is the height of the egg above the ground, in meters, t seconds after it is thrown. According to the equation, what is the height, in meters, from which the egg was thrown?

ID: 6d84ee10 Answer

Correct Answer: 18

Rationale

The correct answer is **18**. It's given that an egg is thrown from a rooftop and that the equation $h = -4.9t^2 + 9t + 18$ represents this situation, where h is the height of the egg above the ground, in meters, t seconds after it is thrown. It follows that the height, in meters, from which the egg was thrown is the value of h when $t = 0$. Substituting **0** for t in the equation $h = -4.9t^2 + 9t + 18$ yields $h(0) = -4.9(0)^2 + 9(0) + 18$, or $h = 18$. Therefore, according to the equation, the height, in meters, from which the egg was thrown is **18**.

Question Difficulty: Medium

Question ID db14a20c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: db14a20c

The function f is defined by $f(x) = 270(0.1)^x$. What is the value of $f(0)$?

- A. 0
- B. 1
- C. 27
- D. 270

ID: db14a20c Answer

Correct Answer: D

Rationale

Choice D is correct. The value of $f(0)$ is the value of $f(x)$ when $x = 0$. Substituting 0 for x in the given function yields $f(0) = 270(0.1)^0$, or $f(0) = 270(1)$, which is equivalent to $f(0) = 270$. Therefore, the value of $f(0)$ is 270.

Choice A is incorrect. This is the value of x , not $f(x)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f(1)$, not $f(0)$.

Question Difficulty: Medium

Question ID 31041ee4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 31041ee4

The function g is defined by $g(x) = (x + 14)(t - x)$, where t is a constant. In the xy -plane, the graph of $y = g(x)$ passes through the point $(24, 0)$. What is the value of $g(0)$?

ID: 31041ee4 Answer

Correct Answer: 336

Rationale

The correct answer is **336**. By the zero product property, if $(x + 14)(t - x) = 0$, then $x + 14 = 0$, which gives $x = -14$, or $(t - x) = 0$, which gives $x = t$. Therefore, $g(x) = 0$ when $x = -14$ and when $x = t$. Since the graph of $y = g(x)$ passes through the point $(24, 0)$, it follows that $g(24) = 0$, so $t = 24$. Substituting **24** for t in the equation $g(x) = (x + 14)(t - x)$ yields $g(x) = (x + 14)(24 - x)$. The value of $g(0)$ can be calculated by substituting **0** for x in this equation, which yields $g(0) = (0 + 14)(24 - 0)$, or $g(0) = 336$.

Question Difficulty: Hard

Question ID e2a01f52

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: e2a01f52

$5x^2 - 37x - 24 = 0$

What is the positive solution to the given equation?

- A. $\frac{3}{5}$
- B. 3
- C. 8
- D. 37

ID: e2a01f52 Answer

Correct Answer: C

Rationale

Choice C is correct. The left-hand side of the given equation can be factored as $(5x + 3)(x - 8)$. Therefore, the given equation, $5x^2 - 37x - 24 = 0$, can be written as $(5x + 3)(x - 8) = 0$. Applying the zero product property to this equation yields $5x + 3 = 0$ and $x - 8 = 0$. Subtracting 3 from both sides of the equation $5x + 3 = 0$ yields $5x = -3$. Dividing both sides of this equation by 5 yields $x = -\frac{3}{5}$. Adding 8 to both sides of the equation $x - 8 = 0$ yields $x = 8$. Therefore, the two solutions to the given equation, $5x^2 - 37x - 24 = 0$, are $-\frac{3}{5}$ and 8. It follows that 8 is the positive solution to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 5e905ec0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5e905ec0

$$x^2 - 12x + 27 = 0$$

How many distinct real solutions does the given equation have?

- A. Exactly two
- B. Exactly one
- C. Zero
- D. Infinitely many

ID: 5e905ec0 Answer

Correct Answer: A

Rationale

Choice A is correct. The number of solutions of a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be determined by the value of the discriminant, $b^2 - 4ac$. If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation, $x^2 - 12x + 27 = 0$, $a = 1$, $b = -12$, and $c = 27$. Substituting these values for a , b , and c in $b^2 - 4ac$ yields $(-12)^2 - 4(1)(27)$, or 36 . Since the value of its discriminant is positive, the given equation has exactly two distinct real solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID d3042402

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: d3042402

$$7x^2 - 20x - 32 = 0$$

What is the positive solution to the given equation?

ID: d3042402 Answer

Correct Answer: 4

Rationale

The correct answer is **4**. The left-hand side of the given equation can be factored as $(7x + 8)(x - 4)$. Therefore, the given equation, $7x^2 - 20x - 32 = 0$, can be written as $(7x + 8)(x - 4) = 0$. Applying the zero product property to this equation yields $7x + 8 = 0$ and $x - 4 = 0$. Subtracting **8** from both sides of the equation $7x + 8 = 0$ yields $7x = -8$. Dividing both sides of this equation by **7** yields $x = -\frac{8}{7}$. Adding **4** to both sides of the equation $x - 4 = 0$ yields $x = 4$. Therefore, the two solutions to the given equation, $7x^2 - 20x - 32 = 0$, are $-\frac{8}{7}$ and **4**. It follows that **4** is the positive solution to the given equation.

Question Difficulty: Hard

Question ID 66015767

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 66015767

What is an x-coordinate of an x-intercept of the graph of $y = 3(x - 14)(x + 5)(x + 4)$ in the xy-plane?

ID: 66015767 Answer

Correct Answer: 14, -5, -4

Rationale

The correct answer is either **14**, **-5**, or **-4**. The x-intercepts of a graph in the xy-plane are the points at which the graph intersects the x-axis, or when the value of y is **0**. Substituting **0** for y in the given equation yields $0 = 3(x - 14)(x + 5)(x + 4)$. Dividing both sides of this equation by **3** yields $0 = (x - 14)(x + 5)(x + 4)$. Applying the zero product property to this equation yields three equations: $x - 14 = 0$, $x + 5 = 0$, and $x + 4 = 0$. Adding **14** to both sides of the equation $x - 14 = 0$ yields $x = 14$, subtracting **5** from both sides of the equation $x + 5 = 0$ yields $x = -5$, and subtracting **4** from both sides of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the x-coordinates of the x-intercepts of the graph of the given equation are **14**, **-5**, and **-4**. Note that 14, -5, and -4 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 0fb16f2f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear functions	

ID: 0fb16f2f

A model predicts that the population of Bergen was **15,000** in **2005**. The model also predicts that each year for the next **5** years, the population p increased by **4%** of the previous year's population. Which equation best represents this model, where x is the number of years after **2005**, for $x \leq 5$?

- A. $p = 0.96(15,000)^x$
- B. $p = 1.04(15,000)^x$
- C. $p = 15,000(0.96)^x$
- D. $p = 15,000(1.04)^x$

ID: 0fb16f2f Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that a model predicts the population of Bergen in **2005** was **15,000**. The model also predicts that each year for the next **5** years, the population increased by **4%** of the previous year's population. The predicted population in one of these years can be found by multiplying the predicted population from the previous year by **1.04**. Since the predicted population in **2005** was **15,000**, the predicted population **1** year later is **15,000(1.04)**. The predicted population **2** years later is this value times **1.04**, which is **15,000(1.04)(1.04)**, or **15,000(1.04)²**. The predicted population **3** years later is this value times **1.04**, or **15,000(1.04)³**. More generally, the predicted population, p , x years after **2005** is represented by the equation $p = 15,000(1.04)^x$.

Choice A is incorrect. Substituting **0** for x in this equation indicates the predicted population in **2005** was **0.96** rather than **15,000**.

Choice B is incorrect. Substituting **0** for x in this equation indicates the predicted population in **2005** was **1.04** rather than **15,000**.

Choice C is incorrect. This equation indicates the predicted population is decreasing, rather than increasing, by **4%** each year.

Question Difficulty: Medium

Question ID d4c19fa8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT & PSAT 10	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: d4c19fa8

$$x^2 - 5x - 24 = 0$$

What is the sum of the solutions to the given equation?

ID: d4c19fa8 Answer

Correct Answer: 5

Rationale

The correct answer is **5**. The given quadratic equation can be rewritten in factored form as $(x - 8)(x + 3) = 0$. Based on the zero product property, it follows that $x - 8 = 0$ or $x + 3 = 0$. Adding **8** to both sides of the equation $x - 8 = 0$ yields $x = 8$. Subtracting **3** from both sides of the equation $x + 3 = 0$ yields $x = -3$. Therefore, the solutions to the given equation are **8** and **-3**. It follows that the sum of the solutions to the given equation is $8 + (-3)$, or **5**.

Question Difficulty: Hard